

Disorders of Potassium Balance

David B. Mount

CHAPTER OUTLINE

POTASSIUM DISORDERS, 537
 NORMAL POTASSIUM BALANCE, 537
 RENAL POTASSIUM EXCRETION, 541

CONSEQUENCES OF HYPOKALEMIA AND
 HYPERKALEMIA, 547
 HYPOKALEMIA, 550
 HYPERKALEMIA, 564

POTASSIUM DISORDERS

The diagnosis and management of potassium disorders are central skills in clinical nephrology, relevant not only to consultative nephrology but also to dialysis and renal transplantation. An understanding of the underlying physiology is critical to the diagnostic and management approach to hyper- and hypokalemic patients. This chapter reviews those aspects of the physiology of potassium homeostasis judged to be relevant to the understanding of potassium disorders; a more detailed review of renal potassium transport is provided in Chapter 6.

The pathophysiology of potassium disorders continues to evolve. The expanding list of drugs with a potential to affect the plasma potassium concentration (K^+) has made clinical management more complex, yet has provided the opportunity to new insight into potassium homeostasis. The evolving molecular understanding of both common and rare disorders affecting plasma K^+ continues to uncover novel pathways of regulation.^{1–8} These advances can be incorporated into an increasingly mechanistic, molecular understanding of potassium disorders.

NORMAL POTASSIUM BALANCE

The dietary intake of potassium ranges from less than 35 to more than 110 mmol/day in US men and women. Despite this widespread variation in intake, homeostatic mechanisms serve to maintain plasma K^+ precisely between 3.5 and 5.0 mmol/L. In a healthy individual at steady state, the entire daily intake of potassium is excreted, approximately 90% in the urine and 10% in the stool. More than 98% of total body potassium is intracellular, chiefly in muscle (Fig. 17.1). Buffering of extracellular K^+ by this large intracellular pool plays a crucial role in the regulation of plasma K^+ .⁹ Thus, within 60 minutes of an intravenous load of 0.5 mmol/kg of K^+-Cl^- , only 41% appears in the urine, yet the serum K^+ level rises by no more than 0.6 mmol/L.⁹ Adding the

equivalent 35 mmol exclusively to the extracellular space of a 70-kg man would be expected to raise the serum K^+ level by ~2.5 mmol/L.¹⁰ Changes in cellular distribution also serve to defend plasma K^+ during K^+ depletion. For example, military recruits have been shown to maintain a normal serum K^+ level after 11 days of basic training in hot environments, despite a profound K^+ deficit generated by renal and extrarenal losses.¹¹ The rapid exchange of intracellular K^+ with extracellular K^+ plays a crucial role in maintaining plasma K^+ within such a narrow range; this is accomplished by overlapping and synergistic¹² regulation of a number of renal and extrarenal transport pathways.

POTASSIUM TRANSPORT MECHANISMS

The intracellular accumulation of K^+ against its electrochemical gradient is an energy-consuming process, mediated by the ubiquitous Na^+/K^+ -ATPase enzyme. The Na^+/K^+ -ATPase functions as an electrogenic pump, because the stoichiometry of transport is three intracellular Na^+ ions to two extracellular K^+ ions. The enzyme complex is made up of a tissue-specific combination of multiple alpha, beta, and gamma subunits, which are further subject to tissue-specific patterns of regulation.¹³ The Na^+/K^+ -ATPase proteins share significant homology with the corresponding subunits of the H^+/K^+ -ATPase enzymes (see “Potassium Transport in the Distal Nephron”). Cardiac glycosides, such as digoxin and ouabain, bind to the alpha subunits of Na^+/K^+ -ATPase at an exposed extracellular hairpin loop that also contains the major binding sites for extracellular K^+ .¹⁴ The binding of digoxin and K^+ to the Na^+/K^+ -ATPase complex is thus mutually antagonistic, explaining in part the potentiation of digoxin toxicity by hypokalemia.¹⁵ Although the four alpha subunits have equivalent affinity for ouabain, they differ significantly in intrinsic K^+ -ouabain antagonism.¹⁶ Ouabain binding to isozymes containing the ubiquitous alpha-1 subunit is relatively insensitive to K^+ concentrations within the physiologic range, such that this isozyme is protected from digoxin under conditions wherein cardiac alpha-2 and alpha-3 subunits, the probable therapeutic targets,¹⁷ are inhibited.¹⁶ Genetic reduction in cardiac alpha-1

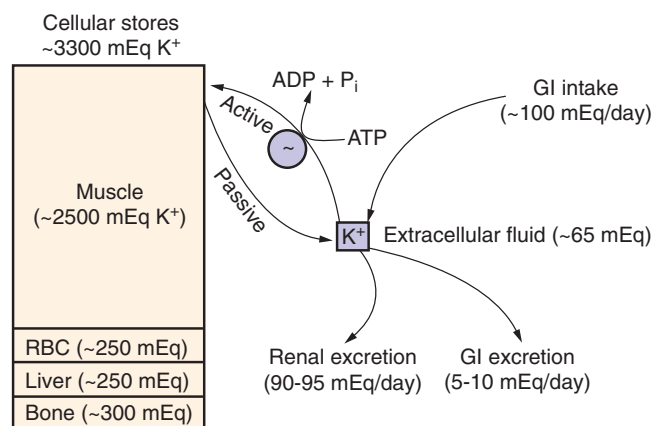


Fig. 17.1 Body K^+ distribution and cellular K^+ flux. ADP, Adenosine diphosphate; ATP, adenosine triphosphate; GI, gastrointestinal; RBC, red blood cell. (From Wingo CS, Weiner ID. Disorders of potassium balance. In Brenner BM ed. *The Kidney*. ed 6. Philadelphia: WB Saunders; 2000:998–1035.)

content has a negative inotropic effect,¹⁷ such that the relative resistance of this subunit to digoxin at physiologic plasma K^+ has an additional cardioprotective effect. Notably, the digoxin-ouabain binding site of α subunits is highly conserved, suggesting a potential role in the physiologic response to endogenous ouabain–digoxin-like compounds. Mice that express α -2 subunits with engineered resistance to ouabain are strikingly resistant to ouabain-induced hypertension and to adrenocorticotrophic hormone (ACTH)-dependent hypertension,¹⁸ the latter known to involve an increase in circulating ouabain-like glycosides. These provocative data lend credence to the controversial role of such ouabain-like molecules in hypertension and cardiovascular disease. Furthermore, modulation of the K^+ -dependent binding of circulating ouabain-like compounds to Na^+/K^+ -ATPase may underlie at least some of cardiovascular complications of hypokalemia¹⁹ (see “Effects of Hypokalemia”).

Skeletal muscle contains as much as 75% of body potassium (see Fig. 17.1) and exerts considerable influence on extracellular K^+ . Exercise is thus a well-described cause of transient hyperkalemia; interstitial K^+ in human muscle can reach levels as high as 10 mM after fatiguing exercise.²⁰ Not surprisingly, therefore, changes in skeletal muscle Na^+/K^+ -ATPase activity and abundance are major determinants of the capacity for extrarenal K^+ homeostasis. Hypokalemia induces a marked decrease in muscle K^+ content and Na^+/K^+ -ATPase activity,²¹ an altruistic²² mechanism to regulate plasma K^+ . This is primarily as a result of dramatic decreases in the protein abundance of the α -2 subunit of Na^+/K^+ -ATPase.²³ In contrast, hyperkalemia caused by potassium loading is associated with adaptive increases in muscle K^+ content and Na^+/K^+ -ATPase activity.²⁴ These interactions are reflected in the relationship between physical activity and the ability to regulate extracellular K^+ during exercise.²⁵ For example, exercise training is associated with increases in muscle Na^+/K^+ -ATPase concentration and activity, with reduced interstitial K^+ in trained muscles²⁶ and an enhanced recovery of plasma K^+ after defined amounts of exercise.²⁵

Potassium can also accumulate in cells by coupling to the gradient for Na^+ entry, entering via the electroneutral

$Na^+-K^+-2Cl^-$ cotransporters NKCC1 and NKCC2. The NKCC2 protein is found only at the apical membrane of the thick ascending limb (TAL) and macula densa cells (Fig. 17.2; see Fig. 17.10), where it functions in transepithelial salt transport and tubular regulation of renin release.²⁷ In contrast, NKCC1 is widely expressed in multiple tissues,²⁷ including muscle.²⁸ The cotransport of K^+-Cl^- by the four K^+-Cl^- cotransporters (KCC1-4) can also function in the transfer of K^+ across membranes; although the KCCs typically function as efflux pathways, they can mediate influx when extracellular K^+ increases.²⁷

The efflux of K^+ out of cells is largely accomplished by K^+ channels, which comprise the largest family of ion channels in the human genome. There are three major subclasses of mammalian K^+ channels—the six-transmembrane domain (TMD) family, which encompasses both the voltage-sensitive and Ca^{2+} -activated K^+ channels; the two-pore, four TMD family; and the two TMD family of inward rectifying K^+ (Kir) channels.²⁹ There is tremendous genomic variety in human K^+ channels, with at least 26 separate genes encoding principal subunits of the voltage-gated Kv channels and 17 genes encoding the principal Kir subunits. Further complexity is generated by the presence of multiple accessory subunits and alternative patterns of mRNA splicing. Not surprisingly, an increasing number and variety of K^+ channels have been implicated in the control of K^+ homeostasis and the membrane potential of excitable cells such as muscle and heart, with important, evolving roles in the pathophysiology of potassium disorders.^{2,30–32}

FACTORS AFFECTING INTERNAL DISTRIBUTION OF POTASSIUM

A number of hormones and physiologic conditions have acute effects on the distribution of K^+ between the intracellular and extracellular space (Table 17.1). Some of these factors are of particular clinical relevance and are therefore reviewed here in detail.

INSULIN

The hypokalemic effect of insulin has been known since the early 20th century.³³ The impact of insulin on plasma K^+ and plasma glucose is separable at multiple levels, suggesting independent mechanisms.^{21,34,35} For example, despite impaired glucose uptake, peripheral K^+ uptake is not impaired in humans with type 2 diabetes.³⁵ Notably, the hypokalemic effect of insulin is not renal-dependent.³⁶ Insulin and K^+ appear to form a feedback loop of sorts, in that increases in plasma K^+ have a marked stimulatory effect on insulin levels.^{21,37} Insulin-stimulated K^+ uptake, measured in rats using a K^+ clamp technique, is rapidly reduced by 2 days of K^+ depletion, before a modest drop in plasma K^+ ,³⁸ and in the absence of a change in plasma K^+ in rats subject to a lesser K^+ restriction for 14 days.¹² Insulin-mediated K^+ uptake is thus modulated by the factors that serve to preserve plasma K^+ in the setting of K^+ deprivation (see also “Control of Potassium Secretion: Effect of Potassium Intake”). Inhibition of basal insulin secretion in normal subjects by somatostatin infusion increases serum K^+ by up to 0.5 mmol/L, in the absence of a change in urinary excretion, emphasizing the crucial role of circulating insulin in the regulation of plasma K^+ .³⁹ Clinically, inhibition of insulin secretion by the

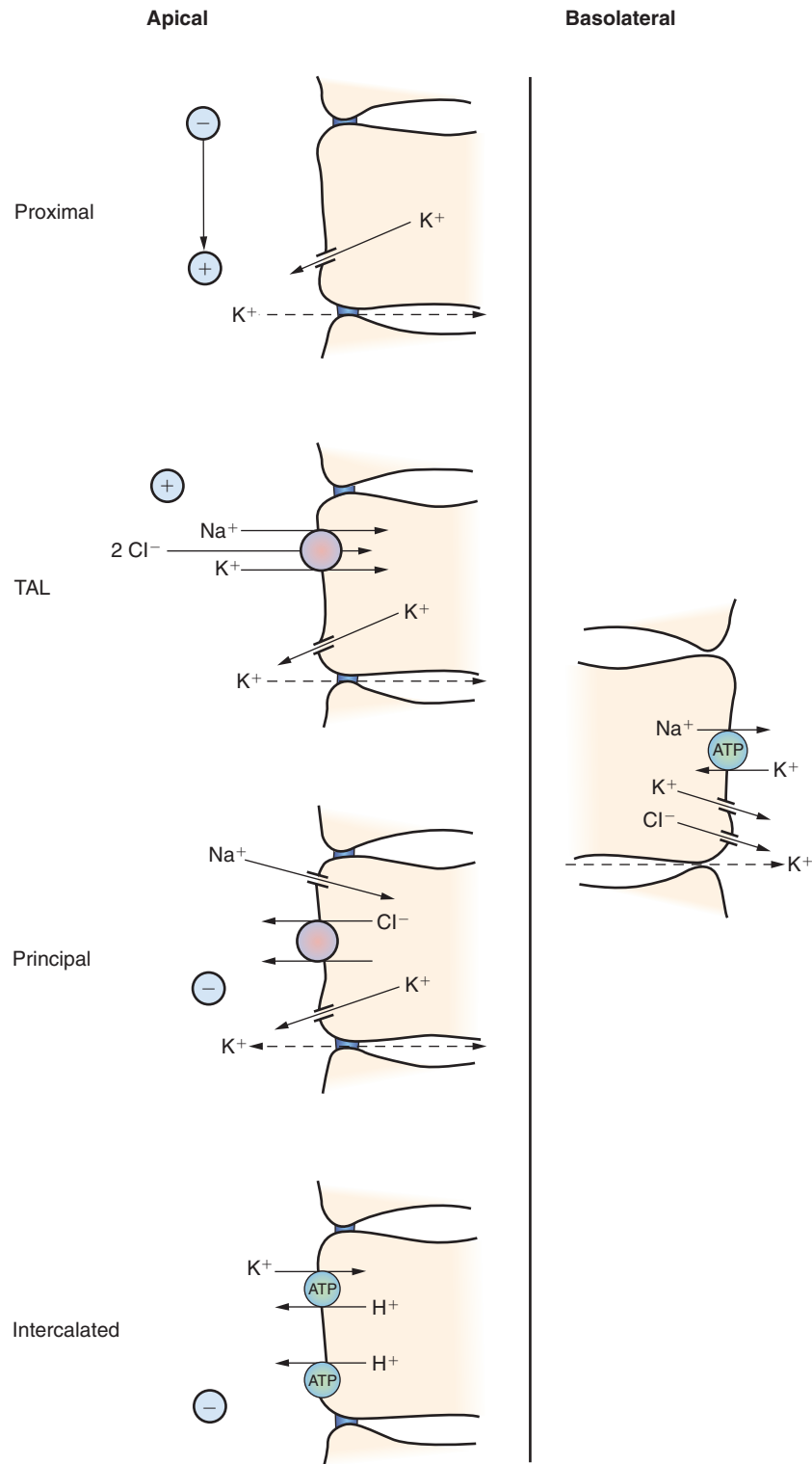


Fig. 17.2 Schematic cell models of potassium transport along the nephron. Cell types are as specified; Note the differences in luminal potential difference along the nephron. ATP, Adenosine triphosphate; TAL, thick ascending limb. (From Giebisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol.* 1998;274:F817–F833.)

somatostatin agonist octreotide can cause significant hyperkalemia in both anephric patients⁴⁰ and patients with normal renal function.⁴¹

Insulin stimulates the uptake of K^+ by several tissues, most prominently liver, skeletal muscle, cardiac muscle, and fat.^{21,42}

It does so by activating several K^+ transport pathways, with particularly well-documented effects on the Na^+/K^+ -ATPase.⁴³ Insulin activates Na^+-H^+ exchange and/or $Na^+-K^+-2Cl^-$ cotransport in several tissues. Although the ensuing increase in intracellular Na^+ was postulated to have a secondary activating

Table 17.1 Factors Affecting Distribution of Potassium Between Intracellular and Extracellular Compartments

Factor	Effect
Acute: Effect on Potassium	
Insulin	Enhanced cell uptake
• β -Catecholamines	Enhanced cell uptake
• α -Catecholamines	Impaired cell uptake
Acidosis	Impaired cell uptake
Alkalosis	Enhanced cell uptake
External potassium balance	Loose correlation
Cell damage	Impaired cell uptake
Hyperosmolality	Enhanced cell efflux
Chronic: Effect on ATP Pump Density	
Thyroid	Enhanced
Adrenal steroids	Enhanced
Exercise (training)	Enhanced
Growth	Enhanced
Diabetes	Impaired
Potassium deficiency	Impaired
Chronic renal failure	Impaired

From Giebisch G: Renal potassium transport: mechanisms and regulation. *Am J Physiol.* 1998;274:F817-F833.

Table 17.2 Sustained Effects of β - and α -Adrenergic Agonists and Antagonists on Serum Potassium Concentration

Catecholamine Specificity	Sustained Effect on Serum K^+
Beta-1 + beta-2 agonist (e.g., epinephrine, isoproterenol)	Decreased
Pure beta-1 agonist (e.g., ITP)	None
Pure beta-2 agonist (e.g., salbutamol, soterenol, terbutaline)	Decreased
Beta-1 + beta-2 antagonist (e.g., propranolol, sotalol)	Increased; blocks the effect of beta agonists
Beta-1 antagonist (e.g., practolol, metoprolol, atenolol)	None; does not block effect of beta agonists
Beta-2 antagonist (e.g., butoxamine, H 35/25)	Blocks hypokalemic effect of beta agonists
Alpha agonist (e.g., phenylephrine)	Increased
Alpha antagonist (e.g., phenoxybenzamine)	None; blocks effect of alpha agonists

ITP, Isopropylamino-3-(2-thiazoloxo)-2-propanol.

From Giebisch G: Renal potassium transport: mechanisms and regulation. *Am J Physiol.* 1998;274:F817-F833.

effect on Na^+/K^+ -ATPase,²⁹ it is clear that this is not the primary mechanism in most cell types.⁴⁴ Insulin induces translocation of the Na^+/K^+ -ATPase alpha-2 subunit to the plasma membrane of skeletal muscle cells, with a lesser effect on the alpha-1 subunit.⁴⁵ This translocation is dependent on the activity of phosphoinositide-3 (PI-3) kinase,⁴⁵ which itself also binds to a proline-rich motif in the N-terminus of the alpha subunit.⁴⁶ The activation of PI3 kinase by insulin thus induces phosphatase enzymes to dephosphorylate a specific serine residue adjacent to the PI3 kinase binding domain. Trafficking of Na^+/K^+ -ATPase to the cell surface also appears to require the phosphorylation of an adjacent tyrosine residue, perhaps catalyzed by the tyrosine kinase activity of the insulin receptor itself.⁴⁷ Finally, the serum and glucocorticoid-induced kinase-1 (SGK1) plays a critical role in insulin-stimulated K^+ uptake, presumably via the known stimulatory effects of this kinase on Na^+/K^+ -ATPase activity and/or $Na^+-K^+-2Cl^-$ cotransport.⁴⁸ The hypokalemic effect of insulin plus glucose is blunted in SGK1 knockout mice, with a marked reduction in hepatic insulin-stimulated K^+ uptake.⁴⁸

SYMPATHETIC NERVOUS SYSTEM

The sympathetic nervous system plays a prominent role in regulating the balance between extracellular and intracellular K^+ . Again, as is the case for insulin, the effect of catecholamines on plasma K^+ has been known for some time⁴⁹; however, a complicating issue is the differential effect of stimulating α - and β -adrenergic receptors (Table 17.2). Uptake of K^+ by liver and muscle, with resultant hypokalemia, is stimulated via beta-2 receptors.²⁹ The hypokalemic effect of catecholamines appears to be largely independent of changes in circulating insulin²⁹ and has been reported in nephrectomized animals.⁵⁰ The cellular mechanisms whereby

catecholamines induce K^+ uptake in muscle include an activation of the Na^+/K^+ -ATPase,⁵¹ likely via increases in cyclic adenosine monophosphate (cAMP).⁵² However, beta-adrenergic receptors in skeletal muscle also activate the inwardly directed $Na^+-K^+-2Cl^-$ cotransporter NKCC1, which may account for as much as one-third of the uptake response to catecholamines.^{21,28}

In contrast to beta-adrenergic stimulation, alpha-adrenergic agonists impair the ability to buffer increases in K^+ induced via intravenous loading or by exercise⁵³; the cellular mechanisms whereby this occurs are not known. It is thought that beta-adrenergic stimulation increases K^+ uptake during exercise to avoid hyperkalemia, whereas alpha-adrenergic mechanisms help blunt the ensuing postexercise nadir.⁵³ The clinical consequences of the sympathetic control of extrarenal K^+ homeostasis are reviewed elsewhere in this chapter.

ACID-BASE STATUS

The association between changes in pH and plasma K^+ was observed some time ago.⁵⁴ It has long been thought that acute disturbances in acid-base equilibrium result in changes in serum K^+ such that alkalemia shifts K^+ into cells, whereas acidemia is associated with K^+ release.^{55,56} It is believed that this apparent K^+-H^+ exchange and/or $K^+-HCO_3^-$ cotransport helps maintain extracellular pH.⁵⁷ Several different transport mechanisms together result in the net exchange of K^+ with H^+ , including functional coupling between Na^+-H^+ exchangers with the Na^+/K^+ -ATPase, coupling between $Na^+-2HCO_3^-$ cotransport with the Na^+/K^+ -ATPase, and coupling between $Cl^-HCO_3^-$ exchange and K^+-Cl^- cotransporters.⁵⁷

Rather limited data exist for the durable concept that a change of 0.1 unit in serum pH will result in a 0.6-mmol/L change in serum K^+ in the opposite direction.⁵⁸ However,

despite the complexities of changes in K^+ homeostasis associated with various acid-base disorders, a few general observations can be made. The induction of metabolic acidosis by the infusion of mineral acids ($NH_4^+-Cl^-$ or H^+-Cl^-) consistently increases serum K^+ ,^{55,56,58–60} whereas organic acidosis generally fails to increase serum K^+ .^{56,59,61,62} Notably, a more recent report failed to detect an increase in serum K^+ in normal human subjects with acute acidosis secondary to duodenal $NH_4^+-Cl^-$ infusion, in which a modest acidosis was accompanied by an increase in circulating insulin.⁶³ However, as noted by Adrogué and Madias,⁶⁴ the concomitant infusion of 350 mL of dextrose 5% in water (D5W) in these fasting subjects may have served to increase circulating insulin, thus blunting the potential hyperkalemic response to $NH_4^+-Cl^-$. Clinically, use of the oral phosphate binder sevelamer hydrochloride in patients with end-stage kidney disease (ESKD) is associated with acidosis caused by effective gastrointestinal absorption of H^+-Cl^- ; in hemodialysis patients, this acidosis has been associated with an increase in serum K^+ , which is ameliorated by an increase in dialysis bicarbonate concentration.⁶⁵ Of note, hyperkalemia is not an expected complication of sevelamer carbonate, which has started to supplant sevelamer hydrochloride as a phosphate binder.

Metabolic alkalosis induced by sodium bicarbonate infusion usually results in a modest reduction in serum K^+ .^{55,56,58,60,66} Respiratory alkalosis reduces plasma K^+ by a magnitude comparable to that of metabolic alkalosis.^{55,56,58,66} Finally, acute respiratory acidosis increases plasma K^+ ; the absolute increase is smaller than that induced by metabolic acidosis secondary to inorganic acids.^{55,56,58} Again, however, some studies have failed to show a change in serum K^+ following acute respiratory acidosis.^{56,67} The smaller increments in serum K^+ with respiratory acidosis are explained in part by the elevated serum HCO_3^- levels in respiratory acidosis, which blunt pH effects on the $Na^+-2HCO_3^-$ cotransporter, resulting in lesser apparent K^+-H^+ exchange than in metabolic acidosis.⁵⁷

RENAL POTASSIUM EXCRETION

POTASSIUM TRANSPORT IN THE DISTAL NEPHRON

The proximal tubule and loop of Henle mediate the bulk of potassium reabsorption, such that a considerable fraction of filtered potassium is reabsorbed before entry into the superficial distal tubules.⁶⁸ Renal potassium excretion is primarily determined by regulated secretion in the distal nephron, specifically within the connecting segment (CNT) and cortical collecting duct (CCD). The principal cells of the CNT and CCD play a dominant role in K^+ secretion; the relevant transport pathways are shown in Fig. 17.3 (see Fig. 17.2). Apical Na^+ entry via the amiloride-sensitive epithelial Na^+ channel (ENaC)⁶⁹ results in the generation of a lumen-negative potential difference in the CNT and CCD, which drives passive K^+ exit through apical K^+ channels. A critical, clinically relevant consequence of this relationship is that K^+ secretion is dependent on the delivery of adequate luminal Na^+ to the CNT and CCD;^{70,71} K^+ secretion by the CCD essentially ceases as the luminal Na^+ drops below 8 mmol/L.⁷² Selective increases in thiazide-sensitive Na^+-Cl^- cotransporter in the distal convoluted tubule (DCT), as seen in familial

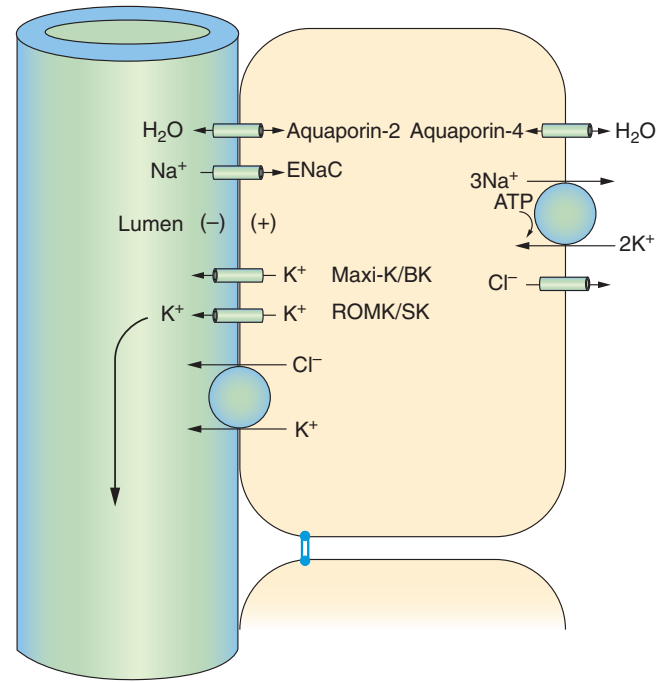


Fig. 17.3 K^+ secretory pathways in principal cells of the connecting segment and cortical collecting duct. The absorption of Na^+ via the amiloride-sensitive epithelial sodium channel (ENaC) generates a lumen-negative potential difference, which drives K^+ excretion through the apical secretory K^+ channel ROMK. Flow-dependent K^+ secretion is mediated by an apical voltage-gated, calcium-sensitive BK channel. Chloride-dependent, electroneutral K^+ secretion is likely mediated by a K^+-Cl^- cotransporter. Water transport in principal cells occurs via aquaporin-2 (Aqp-2) and aquaporins-3/4 (Aqp-3/4). ATP, Adenosine triphosphate.

hyperkalemia with hypertension (FHHt; see “Hereditary Tubular Defects and Potassium Excretion”), reduce Na^+ delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.⁷³ Dietary Na^+ intake also influences K^+ excretion, such that excretion is enhanced by excess Na^+ intake and reduced by Na^+ restriction (see Fig. 17.4).^{70,71} Basolateral exchange of Na^+ and K^+ is mediated by the $Na^+/K^+-ATPase$, providing the driving force for both Na^+ entry and K^+ exit at the apical membrane (see Figs. 17.2 and 17.3).

Under basal conditions of high Na^+-Cl^- and low K^+ intake, the bulk of aldosterone-stimulated Na^+ and K^+ transport occurs in the CNT, before the entry of tubular fluid into the CCD.⁷⁴ The density of both Na^+ and K^+ channels is considerably greater in the CNT than in the CCD.^{75,76} Thus, in the setting of replete Na^+-Cl^- intake, the capacity of the CNT for Na^+ reabsorption may be as much as 10 times greater than that of the CCD.⁷⁶ The recruitment of ENaC subunits in response to dietary Na^+ restriction begins in the CNT, with the progressive recruitment of subunits to the apical membrane of the CCD at lower levels of dietary Na^+ .⁷⁸ The activity of the secretory K^+ channels in the CNT is also influenced by changes in dietary K^+ ⁷⁹; again, this is consistent with progressive axial recruitment of transport capacity for the absorption of Na^+ and secretion of K^+ along the distal nephron.

Electrophysiologic characterization has documented the presence of several subpopulations of apical K^+ channels in

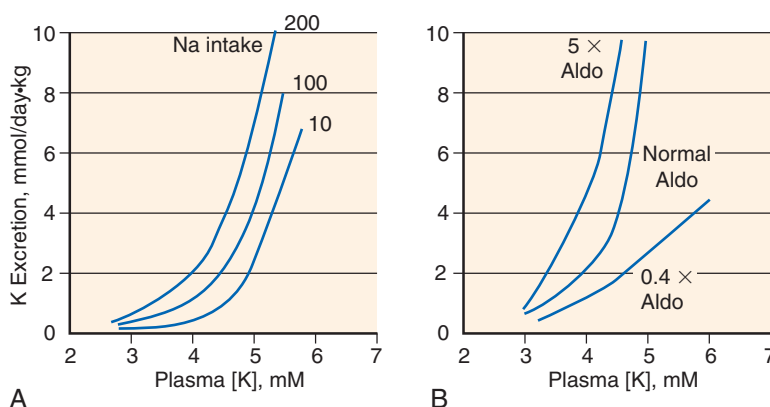


Fig. 17.4 (A) Relationship between steady-state plasma K⁺ and urinary K⁺ excretion in the dog, as a function of dietary Na⁺ intake (mmol/day). Animals were adrenalectomized and replaced with aldosterone, dietary K⁺ and Na⁺ content were varied as specified. (B) Relationship between steady-state plasma K⁺ and urinary K⁺ excretion as a function of circulating aldosterone. Animals were adrenalectomized and variably replaced with aldosterone (*Aldo*); dietary K⁺ content was varied. (A from Young DB, Jackson TE, Tipayamontri U, Scott RC. Effects of sodium intake on steady-state potassium excretion. *Am J Physiol.* 1984;246:F772–F778; B from Young DB. Quantitative analysis of aldosterone's role in potassium regulation. *Am J Physiol.* 1988;255:F811–F822.)

the CCD and CNT, most prominently a small-conductance (SK) 30-picosiemens (pS) channel^{75,77} and a large-conductance, Ca²⁺-activated 150-pS (BK) channel^{75,78} (see Fig. 17.3). The SK channel is thought to mediate K⁺ secretion under baseline conditions—hence, its designation as the “secretory” K⁺ channel. SK channel activity is mediated by the ROMK (renal outer medullary K⁺ channel) protein, encoded by the *Kcnj1* gene; targeted deletion of this gene in mice results in complete loss of SK activity within the CCD.⁷⁸ Increased distal flow has a significant stimulatory effect on K⁺ secretion, due in part to both enhanced delivery and absorption of Na⁺ and to increased removal of secreted K⁺.^{71,72} The apical Ca²⁺-activated BK channel plays a critical role in flow-dependent K⁺ secretion by the CNT and CCD.⁷⁷ BK channels have a heteromeric structure, with alpha subunits that form the ion channel pore and modulatory beta subunits.⁷⁷ The beta-1 subunits of BK channels are restricted to principal cells within the CNT,^{77,79} whereas beta-4 subunits are detectable at the apical membranes of the TAL, DCT, and intercalated cells.⁷⁹ Flow-dependent K⁺ secretion is reduced in mice with targeted deletion of the alpha-1 and beta-1 subunits,^{77,80,81} consistent with a dominant role for BK channels. The Ca²⁺-permeable TRPV4 channel increases intracellular Ca²⁺ concentrations in response to tubular flow, resulting in the activation of BK channels.⁸²

In addition to apical K⁺ channels, considerable evidence implicates apical K⁺-Cl[−] cotransport in distal K⁺ secretion.^{70,83,84} Pharmacologic studies of perfused tubules are consistent with K⁺-Cl[−] cotransport mediated by the KCC proteins.⁸³ A particularly provocative study has underlined the importance of ENaC-independent K⁺ excretion, whether it is mediated by apical K⁺-Cl[−] cotransport and/or by other mechanisms.⁸⁵ Rats were infused with amiloride via osmotic minipumps, generating urinary concentrations considered sufficient to inhibit more than 98% of ENaC activity. Whereas amiloride almost abolished K⁺ excretion in rats on a normal K⁺ intake, acute and long-term high-K⁺ diets led to an increasing fraction of K⁺ excretion that was independent of ENaC activity (~50% after 7–9 days on a high-K⁺ diet).⁸⁵ Therefore, increased K⁺

intake induces a K⁺ excretion pathway that is independent of ENaC activity.

In addition to secretion, the distal nephron is capable of considerable reabsorption of K⁺, particularly during restriction of dietary K⁺.^{21,68,86,87} This reabsorption is accomplished primarily by intercalated cells in the outer medullary collecting duct (OMCD), via the activity of apical H⁺/K⁺-ATPase pumps (see Fig. 17.2). Therefore, decreased K⁺ intake induces a K⁺ reabsorption pathway that is independent of ENaC activity.

CONTROL OF POTASSIUM SECRETION: ALDOSTERONE

Aldosterone is well established as an important regulatory factor in K⁺ excretion, and increases in plasma K⁺ are an important stimulus for aldosterone secretion (see also “Regulation of Renal Renin and Adrenal Aldosterone”). However, an important principle is that aldosterone plays a permissive, synergistic, but not essential role in K⁺ homeostasis.^{88–90} This is reflected clinically in the frequent absence of hyper- or hypokalemia in disorders associated with a deficiency or an overabundance of circulating aldosterone, respectively (see “Hyperaldosteronism” and “Hypoaldosteronism”). Regardless, it is clear that aldosterone and downstream effectors of this hormone have clinically relevant effects on plasma K⁺ levels and that the ability to excrete K⁺ is modulated by systemic aldosterone levels (Fig. 17.4).

Aldosterone has no effect on the density of apical SK channels in the CCD or CNT; rather, the hormone induces a marked increase in the density of apical Na⁺ channels,⁹¹ thus increasing the driving force for apical K⁺ excretion. The apical amiloride-sensitive epithelial Na⁺ channel (ENaC) is comprised of three subunits, alpha, beta, and gamma, that assemble together to traffic to the cell membrane synergistically and mediate Na⁺ transport. Aldosterone activates ENaC channel complexes by multiple mechanisms. First, it induces transcription of the alpha-ENaC subunit,^{92,93} increasing the availability for co-assembly with the more abundant beta and gamma subunits.⁹⁴ Second, aldosterone and dietary Na⁺-Cl[−] restriction stimulate a significant redistribution of ENaC

subunits in the CNT and early CCD, from a largely cytoplasmic location during dietary $\text{Na}^+\text{-Cl}^-$ excess to a purely apical distribution after aldosterone administration or $\text{Na}^+\text{-Cl}^-$ restriction.^{95–97} Third, aldosterone induces the expression of SGK1⁹⁸; coexpression of SGK1 with ENaC subunits results in increased expression at the plasma membrane.⁹⁶ SGK1 modulates membrane expression of ENaC by interfering with regulated endocytosis of its channel subunits. Specifically, the kinase interferes with interactions between ENaC subunits and the ubiquitin ligase Nedd4-2.⁹⁴ The so-called PPxY domains in the C-termini of all three ENaC subunits bind to WW domains of Nedd4-2⁹⁹; these PPxY domains are deleted, truncated, or mutated in patients with Liddle syndrome¹⁰⁰ (see “Liddle Syndrome”), leading to a gain of function in channel activity.¹⁰¹ Nedd4-2 ubiquitinates ENaC subunits, thus inducing removal of channel subunits from the cell membrane followed by degradation in lysosomes and the proteasome.⁹⁴ A PPxY domain in SGK1 also binds to Nedd4-2, which is a phosphorylation substrate for the kinase; phosphorylation of Nedd4-2 by SGK1 abrogates the inhibitory effect of this ubiquitin ligase on ENaC subunits³ (Fig. 17.5).

The importance of SGK1 in K^+ and Na^+ homeostasis is illustrated by the phenotype of SGK1 knockout mice.^{102,103} On a normal diet, homozygous SGK1^{-/-} mice exhibit normal blood pressure and a normal plasma K^+ , with only a mild elevation of circulating aldosterone. However, dietary $\text{Na}^+\text{-Cl}^-$ restriction of these mice results in relative Na^+ wasting and hypotension, marked weight loss, and a drop in the glomerular filtration rate (GFR), despite considerable increases in circulating aldosterone.¹⁰³ In addition, dietary K^+ loading over 6 days leads to a 1.5-mmol/L increase in serum K^+ , also accompanied by a considerable increase in circulating aldosterone (~fivefold greater than that of wild-type littermate controls).¹⁰² This hyperkalemia occurs despite evident increases in apical ROMK expression, compared with the normokalemic littermate controls. The amiloride-sensitive, lumen-negative potential difference generated by ENaC is reduced in these SGK1 knockout mice,¹⁰² resulting in a decreased driving force for distal K^+ secretion and the observed susceptibility to hyperkalemia. Notably, however, a

more recent characterization of this same knockout strain failed to detect a reduction in the increase in ENaC activity induced by dietary potassium loading, suggesting that the hyperkalemia in these mice is caused by a defect in ENaC-independent K^+ excretion.¹⁰⁴

Another mechanism whereby aldosterone activates ENaC involves proteolytic cleavage of the channel proteins by serine proteases. A so-called channel-activating protease that increases channel activity of ENaC was initially identified in *Xenopus laevis* A6 cells.¹⁰⁵ The mammalian ortholog, denoted CAP1 (channel-activating protease-1), or prostasin, is an aldosterone-induced protein in principal cells.¹⁰⁶ Urinary excretion of CAP1 is increased in hyperaldosteronism, with a reduction after adrenalectomy.¹⁰⁶ CAP1 is membrane-associated, via a glycosylphosphatidylinositol (GPI) linkage¹⁰⁵; mammalian principal cells also express two transmembrane proteases, denoted CAP2 and CAP3, with homology to CAP1.¹⁰⁷ These and other proteases (e.g., furin, plasmin) activate ENaC by excising extracellular inhibitory domains from the alpha and gamma subunits, increasing the open probability of channels at the plasma membrane.^{107,108} This proteolytic cleavage of ENaC appears to activate the channel by removing the self-inhibitory effect of external Na^+ ¹⁰⁹; in the case of furin-mediated proteolysis of $\alpha\text{-ENaC}$, this appears to involve the removal of an inhibitory domain from within the extracellular loop.¹¹⁰ Extracellular Na^+ appears to interact with a specific acidic cleft in the cleaved extracellular loop of $\alpha\text{-ENaC}$, causing inhibition of the channel.¹¹¹ Because SGK1 increases channel expression at the cell surface,⁹⁶ one would expect synergistic activation by coexpressed CAP1–3 and SGK; this is indeed the case.¹⁰⁷ Therefore, aldosterone activates ENaC by at least three separate synergistic mechanisms; induction of $\alpha\text{-ENaC}$, induction of SGK1 and repression of Nedd4-2, and induction of channel-activating proteases. Clinically, the inhibition of channel-activating proteases by the protease inhibitor nafamostat causes hyperkalemia as a result of the inhibition of ENaC activity.^{112,113} Nafamostat has been used as a regional anticoagulant for hemodialysis in patients at high risk for bleeding. In contrast with heparin, nafamostat prolongs clotting times only in the extracorporeal circuit.

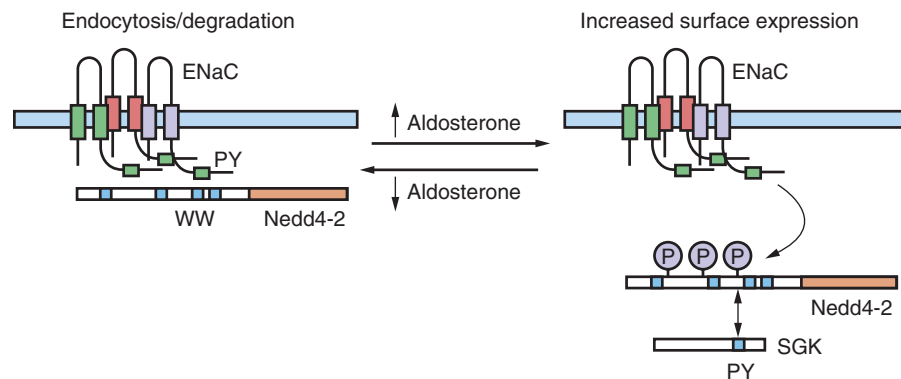


Fig. 17.5 Coordinated regulation of the epithelial sodium channel (ENaC) by the aldosterone-induced SGK kinase and the ubiquitin ligase Nedd4-2. Nedd4-2 binds via its WW domains to ENaC subunits via their PPXY domains (denoted PY here), ubiquitinating the channel subunits and targeting them for removal from the cell membrane and destruction in the proteasome. Aldosterone induces the SGK kinase, which phosphorylates and inactivates Nedd4-2, thus increasing the surface expression of ENaC channels. Mutations that cause Liddle syndrome affect the interaction between ENaC and Nedd4-2. (From Snyder PM, Olson DR, Thomas BC. Serum and glucocorticoid-regulated kinase modulates Nedd4-2-mediated inhibition of the epithelial Na^+ channel. *J Biol Chem*. 2002;277:5–8.)

CONTROL OF POTASSIUM SECRETION: EFFECT OF POTASSIUM INTAKE

Changes in K^+ intake strongly modulate K^+ channel activity in the CNT and CCD (secretory capacity) in addition to H^+ / K^+ -ATPase activity in the OMCD (reabsorptive capacity). Increased dietary K^+ rapidly increases the activity of SK channels in the CCD and CNT,^{114,115} along with a modest increase in Na^+ channel (ENaC) activity⁹¹; this is associated with an increase in apical expression of the ROMK channel protein.¹¹⁶ The increase in ENaC and SK channel density in the CCD occurs within hours of assuming a high- K^+ diet, with a minimal associated increase in circulating aldosterone.¹¹⁵ BK channels in the CNT and CCD are also activated by dietary K^+ loading. Trafficking of BK subunits is thus affected by dietary K^+ , with largely intracellular distribution of alpha subunits in K^+ -restricted rats and prominent apical expression in K^+ -loaded rats.¹¹⁷ Again, aldosterone does not directly contribute to the regulation of BK channel activity or expression in response to a high- K^+ diet.¹¹⁸ However, aldosterone does participate in marked upregulation of the TRPV4 channel in response to a high- K^+ diet; Ca^{2+} entry via TRPV4 increases intracellular Ca^{2+} concentrations in response to tubular flow, thus activating BK channels.⁸² TRPV4 knockout mice demonstrate an impaired adaptation to high- K^+ diet.⁸²

A complex synergistic mix of signaling pathways regulates K^+ channel activity in response to changes in dietary K^+ . In particular, the WNK (with no lysine) kinases play a critical role in modulating distal K^+ secretion. WNK1 and WNK4 were initially identified as the causative genes for FHHt (see also “Hyperkalemia: Hereditary Tubular Defects and Potassium Excretion”). ROMK expression at the membrane of *Xenopus* oocytes is reduced by coexpression of WNK4; FHHt-associated mutations increase this effect, suggesting a direct inhibition of SK channels in FHHt.¹¹⁹ Transcription of the WNK1 gene generates several different isoforms; the predominant intrarenal WNK1 isoform is generated by a distal nephron transcriptional site that bypasses the N-terminal exons that encode the kinase domain, yielding a kinase-deficient short isoform¹²⁰ (WNK1-S). Full-length WNK1 (WNK1-L) inhibits ROMK by inducing endocytosis of the channel protein; the shorter, kinase-deficient WNK1-S isoform inhibits this effect of WNK1-L,^{121,122} and ROMK activity is impaired in WNK1-S knockout mice.¹²³ The ratio of WNK1-S to WNK1-L transcripts is reduced by K^+ restriction (greater endocytosis of ROMK)^{122,124} and increased by K^+ loading (reduced endocytosis of ROMK),^{121,124} suggesting that this ratio between WNK1-S and WNK1-L functions as a molecular switch to regulate distal K^+ secretion. The BK channel is also regulated by the WNK kinases.^{125–127}

The membrane trafficking of ROMK is also modulated by tyrosine phosphorylation of the channel protein such that tyrosine phosphorylation stimulates endocytosis, and tyrosine dephosphorylation induces exocytosis.^{128,129} Intrarenal activity of the cytoplasmic tyrosine kinases c-src and c-yes is inversely related to dietary K^+ intake, with a decrease under high- K^+ conditions and a marked increase after several days of K^+ restriction.^{130,131} Several studies have implicated the intrarenal generation of superoxide anions in the activation of cytoplasmic tyrosine kinases by K^+ depletion.^{132–134} Potential candidates for the upstream hormonal signals include angiotensin II (Ang II) and growth factors such as insulin-like

growth factor-1 (IGF-1).¹³² In particular, Ang II inhibits ROMK activity in K^+ -restricted rats, but not rats on a normal K^+ diet.¹³⁵ This inhibition by Ang II involves downstream activation of superoxide production and c-src activity, such that the induction of Ang II by a low- K^+ diet appears to play a major role in reducing distal tubular K^+ secretion.¹³⁶

INTEGRATED REGULATION OF DISTAL SODIUM ABSORPTION AND POTASSIUM SECRETION

Under certain physiologic conditions associated with marked induction of aldosterone, such as dietary sodium restriction, Na^+ balance can be maintained without significant effects on K^+ excretion. Yet, by activating ENaC and generating a more lumen-negative potential difference, increases in aldosterone should lead to an obligatory kaliuresis; how is this physiologic consequence avoided? The mechanisms that underlie this so-called aldosterone paradox, the independent regulation of Na^+ and K^+ handling by the aldosterone-sensitive distal nephron, have been elucidated. The major factors that allow for integrated but independent control of Na^+ and K^+ transport appear to include electroneutral thiazide-sensitive Na^+ - Cl^- transport within the CCD,^{137–139} ENaC-independent K^+ excretion within the distal nephron,⁸⁵ and the differential regulation of various signaling pathways by aldosterone, Ang II, and dietary K^+ ^{141,142} (see also Chapter 6).

Electroneutral Na^+ - Cl^- transport in the CCD and ENaC-independent K^+ secretion may play important roles in disconnecting Na^+ and K^+ transport within the distal nephron. Electroneutral, thiazide-sensitive, and amiloride-resistant Na^+ - Cl^- transport within the CCD^{137–139} is mediated by the combined activity of the Na^+ -dependent SLC4A8 Cl^- - HCO_3^- exchanger and the SLC26A4 Cl^- - HCO_3^- exchanger¹³⁹ (see also Chapter 6). This transport mechanism is apparently responsible for as much 50% of Na^+ - Cl^- transport in mineralocorticoid-stimulated rat CCD,^{137,138} allowing for ENaC-independent, electroneutral Na^+ absorption that will not directly affect K^+ secretion. The converse effect emerges after dietary K^+ loading, which increases the fraction of ENaC-independent, amiloride-resistant K^+ excretion to about 50%.⁸⁵

NCC, the thiazide-sensitive Na^+ - Cl^- cotransporter in the DCT, plays a key role in K^+ homeostasis. Selective increases in DCT NCC activity, as seen in FHHt, reduce Na^+ delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.⁷³ The DCT also clearly functions as a potassium sensor, directly responding to changes in circulating potassium. Reduction in potassium intake and/or hypokalemia thus lead to reduced basolateral $[K^+]$ in the DCT; the subsequent hyperpolarization is dependent on basolateral KIR4.1-containing K^+ channels.⁴ Hyperpolarization leads to chloride exit, via basolateral CLC-NKB chloride channels, and a reduction in intracellular chloride; the reduction in intracellular chloride activates the WNK cascade, resulting in phosphorylation of NCC and activation of the transporter.⁴ Ang II also activates NCC via WNK-dependent activation of the SPAK kinase and phosphorylation of the transporter protein,^{140,141} reducing delivery of Na^+ to the CNT and limiting K^+ secretion. In contrast, Ang II inhibits ROMK activity via several mechanisms, including downstream activation of c-src tyrosine kinases (see previously).^{134–136} Whereas K^+ restriction induces renin and circulating Ang II levels (see “Effects of Hypokalemia”), increases in dietary K^+ are suppressive.^{136,142}

A high- K^+ diet also inactivates NCC¹⁴³ as a result of the associated decrease in Ang II, in addition to the increase in the ratio of WNK1-S to WNK1-L isoforms that occurs with increased K^+ intake.^{121,122,124} WNK1-S antagonizes the effect of WNK1-L on NCC, leading to the inhibition of NCC in conditions with a relative excess of WNK1-S.¹⁴⁴ Nedd4-2 also negatively regulates NCC and WNK1, in addition to ENaC; mice with homozygous deletion of the Nedd4-2 gene develop severe hypokalemia during K^+ restriction, indicating a key role in renal adaptation.¹⁴⁵

Finally, in principal cells, increases in aldosterone induce the SGK1 kinase, which phosphorylates WNK4 and attenuates the effect of WNK4 on ROMK,¹⁴⁶ while activating ENaC.^{94,96,98} However, when dietary K^+ intake is reduced, c-src tyrosine kinase activity increases under the influence of increased Ang II, causing direction inhibition of ROMK activity via tyrosine phosphorylation of the channel.^{128,147,148} The increase in c-src tyrosine kinase activity also abrogates the inhibitory effect of SGK1 on WNK4¹⁴⁹; src family tyrosine kinases also directly phosphorylate WNK4 and modulate its effects on ROMK.¹⁵⁰ Again, Ang II appears to mediate part of its inhibitory effect on ROMK through the activation of c-src,¹³⁶ such that c-src

serves as an important component of the switch that regulates K^+ secretion in response to changes in dietary K^+ .

To summarize these important physiologic principles, the differential effects of K^+ intake on Ang II versus aldosterone appear to be critical in resolving the aldosterone paradox; so too are the differential effects of K^+ intake on NCC-dependent Na^+Cl^- transport in the DCT and on secretory K^+ channels in the downstream CNT and CCD (Fig. 17.6). Under conditions of low Na^+ intake but moderate K^+ intake, Ang II and aldosterone are both strongly induced, leading to enhanced Na^+Cl^- transport via NCC, increased ENaC activity, and decreased secretory K^+ channel activity. Although ENaC is activated, the relative inhibition of ROMK by the increased Ang II prevents excessive kaliuresis. Ang II-dependent activation of c-src kinases has direct inhibitory effects on ROMK trafficking and also abrogates the inhibitory effect of SGK1 on WNK4,¹⁴⁹ leading to unopposed inhibition of ROMK by WNK4. In addition, the aldosterone-dependent induction of electroneutral Na^+Cl^- transport within the CCD^{137–139} increases Na^+Cl^- reabsorption but blunts the effect on the lumen-negative potential as a result of increased ENaC activity, thus limiting kaliuresis. When dietary K^+ increases,

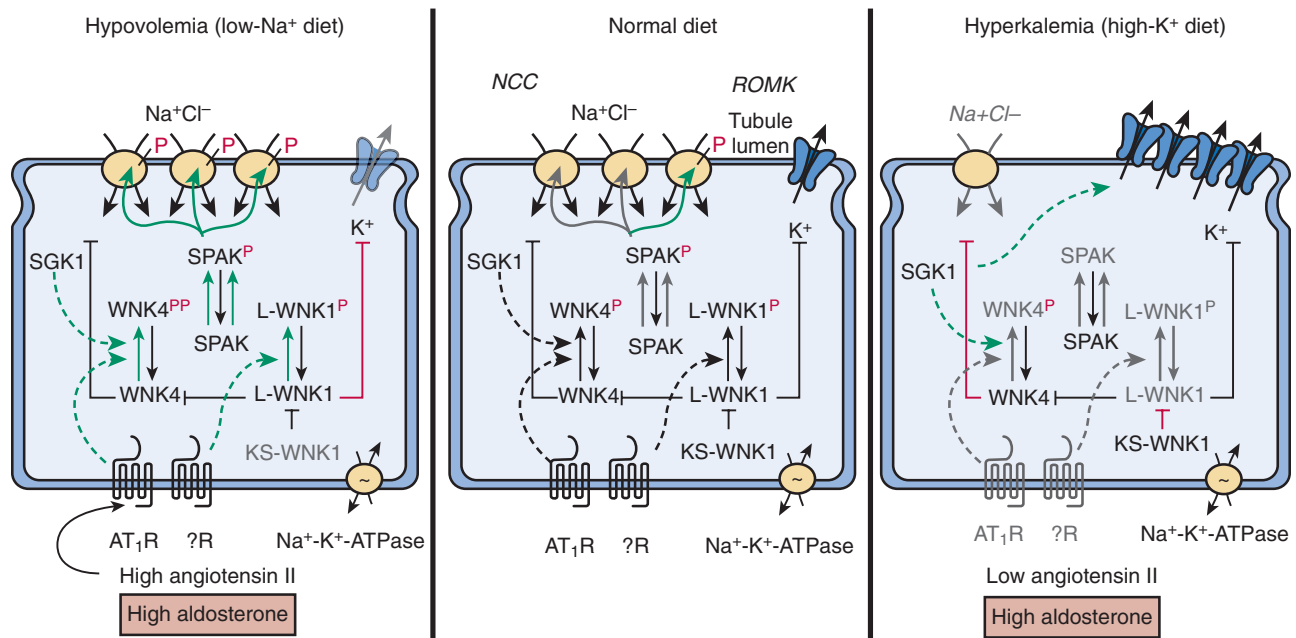


Fig. 17.6 Integrated regulation of Na^+Cl^- and K^+ transport in the distal convoluted tubule (DCT), connecting segment (CNT), and cortical collecting duct (CCD) (green arrowheads, activating pathways; red blunt end, inhibitory pathway). *Left*, Pathway in the setting of a low- Na^+ diet, wherein angiotensin II (Ang II) and SGK1 signaling leads to phosphorylation of WNK4. This stimulates the phosphorylation of SPAK, which in turn phosphorylates and activates thiazide-sensitive Na^+Cl^- cotransport in the DCT via the NCC. Stimulation of unknown receptors is hypothesized to cause phosphorylation of L-WNK1, which can also stimulate SPAK phosphorylation. L-WNK1 also has other functions: (1) It blocks the NCC inhibitory form of WNK4, thus activating NCC. (2) It inhibits the secretion of K^+ via ROMK channels. *Center*, Normal diet. *Right*, Pathway in the setting of high dietary K^+ intake, wherein aldosterone is stimulated and Ang II is low. In the absence of sufficient Ang II, the AT_1 receptor (AT_1R) cannot activate WNK4. This reduces SPAK activation and NCC phosphorylation. Dietary potassium loading also increases the level of the KS-WNK1 isoform to suppress the activity of L-WNK1. In consequence, the inhibitory effect of WNK4 on NCC dominates, blocking traffic of NCC to the apical membrane and thereby reducing NCC activity. KS-WNK1 also blocks the effect of L-WNK1 on ROMK endocytosis, causing ROMK to increase at the apical membrane. The net effect is that K^+ secretion in the DCT, CNT, and CCD is maximized, whereas NCC is suppressed. Aldosterone stimulation of ENaC (not shown) offsets the decreased Na^+ reabsorption by NCC, allowing robust potassium secretion without changes in sodium balance. The roles of WNK3, SGK1, and c-src cytoplasmic tyrosine kinases are not shown in the interest of clarity; see text for further details. NCC, $NaCl$ cotransporter; ROMK, renal outer medullary K^+ channel; SPAK, STE20/SPS1-related proline-alanine-rich kinase; WNK, with no (lysine) kinase. (From Welling PA, Chang YP, Delpire E, Wade JB. Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int.* 2010;77:1063–1069.)

circulating aldosterone is moderately induced but Ang II is suppressed. This leads to the inhibition of NCC and increased downstream delivery of Na^+ to principal cells in the CNT and CCD, where ENaC activity is increased and ROMK and BK channels are significantly activated. ENaC-independent K^+ secretion is also strongly induced by increased dietary K^+ intake,⁸⁵ contributing significantly to the ability to excrete K^+ in the urine.

REGULATION OF RENAL RENIN AND ADRENAL ALDOSTERONE

Modulation of the renin-angiotensin-aldosterone system (RAAS) axis has profound clinical effects on K^+ homeostasis. Although multiple tissues are capable of renin secretion, renin of renal origin has a dominant physiologic impact. Renin secretion by juxtaglomerular cells in the afferent arteriole is initiated in response to a signal from the macula densa,¹⁵¹ specifically a decrease in luminal chloride¹⁵² transported through the $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransporter (NKCC2) at the apical membrane of macula densa cells.²⁷ In addition to this macula densa signal, decreased renal perfusion pressure and renal sympathetic tone stimulate renal renin secretion.¹⁵³ The various inhibitors of renin release include Ang II, nitric oxide, aldosterone, endothelin,¹⁵⁴ adenosine,¹⁵⁵ atrial natriuretic peptide (ANP),¹⁵⁶ tumor necrosis factor- α (TNF- α),¹⁵⁷ and active vitamin D.¹⁵⁸ The cyclic guanosine monophosphate (cGMP)-dependent protein kinase type II (cGKII) tonically inhibits renin secretion, in that renin secretion in response to several stimuli is exaggerated in homozygous *cGKII* knockout mice.¹⁵⁹ Activation of cGKII by ANP and/or nitric oxide has a marked inhibitory effect on the release of renin from juxtaglomerular cells.¹⁵⁶ Local factors that stimulate renin release from juxtaglomerular cells include prostaglandins,¹⁶⁰ adrenomedullin,¹⁶¹ catecholamines (beta-1 receptors),¹⁶² and succinate (GPR91 receptor).¹⁶³

The relationship between renal renin release, the RAAS, and cyclooxygenase-2 (COX-2) is particularly complex.¹⁶⁰ COX-2 is heavily expressed in the macula densa,¹⁶⁰ with a significant recruitment of COX-2⁽⁺⁾ cells seen with salt restriction or furosemide treatment.^{21,160} Reduced intracellular chloride in macula densa cells appears to stimulate COX-2 expression via p38 MAP kinase,¹⁶⁴ whereas both aldosterone and Ang II reduce its expression.¹⁶⁰ Prostaglandins derived from COX-2 in the macula densa play a dominant role in the stimulation of renal renin release by salt restriction, furosemide, renal artery occlusion, or angiotensin-converting enzyme (ACE) inhibition.^{21,165} Specifically, COX-2-derived prostaglandins appear to play a role in the tonic expression of renin in juxtaglomerular (JG) cells via the modulation of intracellular cAMP and calcium, rather than functioning in the acute regulation of renin release.¹⁶⁰ Prostaglandins generated by the macula densa also participate in the recruitment during salt restriction of CD44⁺ mesenchymal stromal cells, which differentiate into renin-producing cells.¹⁶⁶

Renin released from the kidney ultimately stimulates aldosterone release from the adrenal via Ang II. Hyperkalemia per se is also an independent and synergistic stimulus (Fig. 17.7) for aldosterone release from the adrenal gland,^{21,167} although dietary K^+ loading is less potent than dietary Na^+-Cl^- restriction in increasing circulating aldosterone.⁸⁸ The resting membrane potential of adrenal glomerulosa cells is hyperpolarized as a result of the activity of the “leak” K^+

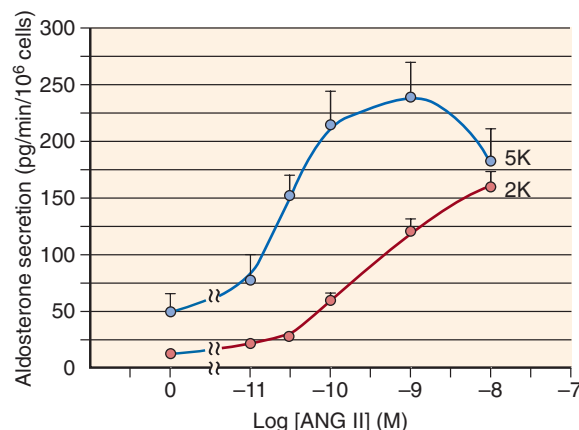


Fig. 17.7 Synergistic effect of increased extracellular K^+ and angiotensin II (ANG-II) in inducing aldosterone release from bovine adrenal glomerulosa cells. Dose response curves for ANG-II were performed at an extracellular K^+ of 2 mmol/L (red $\bullet$) and 5 mmol/L (blue $\bullet$). (From Chen XL, Bayliss DA, Fern RJ, Barrett PQ. A role for T-type Ca^{2+} channels in the synergistic control of aldosterone production by ANG II and K^+ . *Am J Physiol.* 1999;276:F674–F683.)

channels TASK-1 and TASK-3; combined deletion of genes encoding these channels leads to baseline depolarization of adrenal glomerulosa cells and an increase in serum aldosterone that is resistant to dietary sodium loading.¹⁶⁸ Ang II and K^+ both activate Ca^{2+} entry in glomerulosa cells via voltage-sensitive, T-type Ca^{2+} channels,^{21,169} primarily Cav3.2.¹⁷⁰ Elevations in extracellular K^+ thus depolarize glomerulosa cells and activate these Ca^{2+} channels, which are independently and synergistically activated by Ang II.¹⁶⁹ Calcium-dependent activation of calcium calmodulin (CaM)-dependent protein kinase in turn activates the synthesis and release of aldosterone via induction of aldosterone synthase.¹⁷¹ K^+ and Ang II also enhance transcription of the Cav3.2 Ca^{2+} channel by abrogating repression of this gene by the neuron restrictive silencing factor (NRS); this ultimately amplifies the induction of aldosterone synthase.¹⁷⁰

The role of adrenal K^+ sensing in aldosterone release has been dramatically underlined by the reports of both germline and somatic mutations, in aldosterone-producing adenomas, of transport proteins that control membrane excitability of adrenal zona glomerulosa cells (see also “Hyperaldosteronism”). For example, somatic mutations in the adrenal K^+ channel KCNJ5 (GIRK4) can be detected in about 40% of aldosterone-producing adrenal adenomas¹⁷²; these mutations endow the channel with a novel Na^+ conductance, leading to adrenal glomerulosa cell depolarization, Ca^{2+} influx, and aldosterone release.

The adrenal release of aldosterone caused by increased K^+ is dependent on an intact adrenal renin-angiotensin system,¹⁷³ particularly during Na^+ restriction. ACE inhibitors and angiotensin-receptor blockers (ARBs) thus completely abrogate the effect of high K^+ on salt-restricted adrenals.¹⁷⁴ Direct, G-protein-dependent activation of the TASK-1 and/or TASK-3 K^+ channels by Ang II AT_{1A} or AT_{1B} receptors is thought to underlie the effect of Ang II on adrenal aldosterone release,¹⁶⁸ with abrogation of this effect by ARBs or ACE inhibitors. Other clinically relevant activators of adrenal aldosterone release include prostaglandins¹⁷⁵ and

catecholamines¹⁷⁶ via increases in cAMP.^{177,178} Finally, ANP exerts a potent negative effect on aldosterone release induced by K⁺ and other stimuli,¹⁷⁹ at least in part by inhibiting early events in aldosterone synthesis.¹⁸⁰ ANP is therefore capable of inhibiting both renal renin release and adrenal aldosterone release, functions that may be central to the pathophysiology of hyporeninemic hypoaldosteronism.

URINARY INDICES OF POTASSIUM EXCRETION

A bedside test to measure distal tubular K⁺ secretion in humans directly would be ideal; however, for obvious reasons this is not technically feasible. A widely used surrogate is the transtubular K⁺ gradient (TTKG), which is defined as follows:

$$\text{TTKG} = ([\text{K}^+]_{\text{urine}} \times \text{Osm}_{\text{blood}}) / ([\text{K}^+]_{\text{blood}} \times \text{Osm}_{\text{urine}})$$

The expected values of the TTKG are largely based on historical data and are less than 3 to 4 in the presence of hypokalemia and more than 6 to 7 in the presence of hyperkalemia. Varying opinions regarding the physiologically appropriate TTKG in hyperkalemia have been reviewed multiple times.¹⁸¹ Clearly, water absorption in the CCD and medullary collecting duct is an important determinant of the absolute K⁺ concentration in the final urine—hence, the use of a ratio of urine-to-plasma osmolality. Indeed, water absorption may, in large part, determine the TTKG, such that it far exceeds the limiting K⁺ gradient.¹⁸² More recently, the originators of the TTKG have suggested that it fails to incorporate the putative effects of distal tubular urea reabsorption in K⁺ excretion; however, neither urea transporter knockout mice nor rats treated with a urea transporter inhibitor¹⁸³ have demonstrated abnormalities in K⁺ homeostasis. The TTKG may be less useful in patients ingesting diets of changing K⁺ and mineralocorticoid intake.¹⁸⁴ There is, however, a linear relationship between plasma aldosterone and the TTKG, suggesting that it provides a rough approximation of the ability to respond to aldosterone with a kaliuresis.¹⁸⁵ The response of the TTKG to mineralocorticoid administration, typically fludrocortisone, can thus be used in the diagnostic approach to hyperkalemia.¹⁸¹ In hypokalemic patients, a TTKG of less than 2 to 3 separates patients with redistributive hypokalemia from those with hypokalemia caused by renal potassium wasting, who will have TTKG values that are more than 4.¹⁸⁶

An alternative to the TTKG in hypokalemic patients is the measurement of the urine K⁺-to-creatinine ratio. This ratio is usually less than 13 mEq/g creatinine (1.5 mEq/mmol creatinine) when hypokalemia is caused by poor dietary intake, transcellular potassium shifts, gastrointestinal losses, or previous use of diuretics.¹⁸⁶ Higher values are indicative of ongoing renal potassium wasting. The utility of the K⁺-to-creatinine ratio has been evaluated in a study of 43 patients with severe hypokalemia (range, 1.5–2.6 mmol/L) associated with paralysis.¹⁸⁶ The urine K⁺-to-creatinine ratio reliably distinguished between the 30 patients with hypokalemic periodic paralysis and the 13 patients with hypokalemia due mostly to renal potassium wasting. The K⁺-to-creatinine ratio was thus significantly lower in the patients with periodic paralysis (11 vs. 36 mEq/g creatinine; 1.3 vs. 4.1 mEq/mmol creatinine). The cutoff value was approximately 22 mEq/g creatinine (2.5 mEq/mmol).

The determination of urinary electrolytes for calculation of the TTKG or urine K⁺-to-creatinine ratio provides the

opportunity to measure the urinary Na⁺, which will determine whether significant prerenal stimuli are limiting distal Na⁺ delivery and thus K⁺ excretion (see Fig. 17.4). Urinary electrolytes also afford the opportunity to calculate the urinary anion gap, an indirect index of urinary NH₄⁺ content and thus indicating the ability to respond to an acidemia.¹⁸⁷

CONSEQUENCES OF HYPERKALEMIA AND HYPOKALEMIA

Hyperkalemia and hypokalemia have a number of effects, as discussed here.

EFFECTS OF HYPOKALEMIA

EXCITABLE TISSUES: MUSCLE AND HEART

Hypokalemia is a well-described risk factor for both ventricular and atrial arrhythmias.^{29,188} For example, in patients undergoing cardiac surgery, a serum K⁺ level of less than 3.5 mmol/L is a predictor of serious intraoperative arrhythmia, perioperative arrhythmia, and postoperative atrial fibrillation.¹⁸⁹ Moderate hypokalemia does not, however, appear to increase the risk of serious arrhythmia during exercise stress testing.¹⁹⁰ Electrocardiographic changes in hypokalemia include broad flat T waves, ST depression, and QT prolongation; these are most marked when the serum K⁺ level is less than 2.7 mmol/L.¹⁹¹ Hypokalemia, often accompanied by hypomagnesemia, is an important cause of the long QT syndrome (LQTS) and torsades de pointes, either alone¹⁹² or in combination with drug toxicity¹⁹³ or with LQTS-associated mutations in cardiac K⁺ and Na⁺ channels.¹⁹⁴ Hypokalemia accelerates the clathrin-dependent internalization and degradation of the cardiac HERG (human ether-a-go-go) K⁺ channel protein.³¹ HERG encodes pore-forming subunits of the cardiac rapidly activating, delayed rectifier K⁺ channel (I_{Kr}); I_{Kr} is largely responsible for potassium efflux during phases 2 and 3 of the cardiac action potential.¹⁹⁵ Loss-of-function mutations in HERG reduce I_{Kr} and cause type II LQTS³¹; downregulation of HERG and I_{Kr} by hypokalemia provides an elegant explanation for the association with LQTS and torsades de pointes.

In accordance with the Nernst equation, the resting membrane potential is related to the ratio of the intracellular to extracellular potassium concentration. In skeletal muscle, a reduction in plasma K⁺ will increase this ratio and therefore hyperpolarize the cell membrane (i.e., make the resting potential more electronegative); this impairs the ability of the muscle to depolarize and contract, leading to weakness. However, in some human cardiac cells, particularly Purkinje fibers in the conducting system, hypokalemia results in a paradoxical depolarization¹⁹⁶; this paradoxical depolarization plays an important role in the genesis of hypokalemic cardiac arrhythmias.^{196,197} The resting membrane potential in excitable cells is largely determined by a large family of K²P1 K⁺ channels, so-named because of the presence of two pore-forming (P) loop domains in each subunit. Hypokalemia causes K²P1 channels, which are normally selective for potassium, to transport sodium suddenly into cells, causing the paradoxical depolarization.¹⁹⁸ Notably, rodent cardiac cells respond to hypokalemia with a Nernst equation–predicted hyperpolarization and, unlike human cardiomyocytes, do not express the K²P1 channel TWIK-1. Genetic manipulation has indicated that TWIK-1

expression confers this paradoxical depolarization behavior on human and mouse cardiomyocytes.¹⁹⁸

Another important mechanism for hypokalemia-induced arrhythmia involves downregulation of cardiac Na^+/K^+ -ATPase activity.^{199,200} The resulting increase in intracellular Na^+ , which impedes removal of intracellular Ca^{2+} by the $\text{Na}^+/\text{Ca}^{2+}$ exchanger and other mechanisms, leads to intracellular calcium overload. The ensuing increase in calmodulin kinase II activity reduces repolarization reserve by activating late Na^+ and Ca^{2+} currents.²⁰¹ This, in turn, predisposes the heart to early afterdepolarization-associated arrhythmias, such as torsades de pointes and polymorphic ventricular tachycardia (PVT).²⁰¹

In skeletal muscle, hypokalemia causes hyperpolarization, thus impairing the capacity to depolarize and contract. Weakness and paralysis are therefore not infrequent consequences of hypokalemia of diverse causes.^{202,203} On a historical note, the realization in 1946 that K^+ replacement reversed the hypokalemic diaphragmatic paralysis induced by treatment of diabetic ketoacidosis (DKA) was a milestone in diabetes care.²⁰⁴ Pathologically, muscle biopsies in hypokalemic myopathy demonstrate phagocytosis of degenerating muscle fibers, fiber regeneration, and atrophy of type 2 fibers.²⁰⁵ Most patients with significant myopathy will have elevations in creatine kinase levels, and hypokalemia of diverse causes predisposes to rhabdomyolysis with acute renal failure.

RENAL CONSEQUENCES

Hypokalemia causes a host of structural and functional changes in the kidney, which are reviewed in detail elsewhere.²⁰⁶ In humans, the renal pathology includes a relatively specific proximal tubular vacuolization,^{206–208} interstitial nephritis,²⁰⁹ and renal cysts.²¹⁰ Hypokalemic nephropathy can cause ESKD, mostly in patients with long-standing hypokalemia because of eating disorders and/or laxative abuse²¹¹; acute renal failure with proximal tubular vasculopathy has also been described.²¹² In animal models, hypokalemia increases susceptibility to acute renal failure induced by ischemia, gentamicin, and amphotericin.²¹ Potassium restriction in rats induces cortical Ang II and medullary endothelin-1 expression, with an ischemic pattern of renal injury.²¹³ Hypokalemic nephropathy in rats is associated with progressive capillary loss, with reduced angiogenesis because of reduced vascular endothelial growth factor (VEGF) expression.²¹⁴

The prominent functional changes in renal physiology that are induced by hypokalemia include Na^+/Cl^- retention, polyuria,²⁰⁷ phosphaturia,²⁹ hypocitraturia,²¹⁵ and increased ammoniogenesis.²⁰⁶ K^+ depletion in rats causes proximal tubular hyperabsorption of Na^+/Cl^- in association with an upregulation of Ang II,²¹³ AT_1 receptor,²⁹ and α_2 -adrenergic receptor²⁹ in this nephron segment. NHE3, the dominant apical Na^+ entry site in the proximal tubule, is massively (>700%) upregulated in K^+ -deficient rats,²¹⁶ which is consistent with the observed hyperabsorption of both Na^+/Cl^- and bicarbonate.²⁰⁶ Polyuria in hypokalemia is caused by polydipsia²¹⁷ and to a vasopressin-resistant defect in urinary concentrating ability.²⁰⁶ This renal concentrating defect is multifactorial, with evidence for both a reduced hydroosmotic response to vasopressin in the collecting duct²⁰⁶ and decreased Na^+/Cl^- absorption by the TAL.²⁹ K^+ restriction has been shown to result in a rapid reversible decrease in the expression of aquaporin-2 in the collecting duct,²¹⁸ beginning in the CCD

and extending to the medullary collecting duct within the first 24 hours.²¹⁹ Downregulation of aquaporin-2 and several other proteins in hypokalemic nephrogenic diabetes insipidus appears to occur via autophagy.²²⁰ In the TAL, the marked reductions seen during K^+ restriction in both the apical K^+ channel ROMK and the apical $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter NKCC2²¹⁶ reduce Na^+/Cl^- absorption and thus inhibit countercurrent multiplication and the driving force for water absorption by the collecting duct.

CARDIOVASCULAR CONSEQUENCES

A large body of experimental and epidemiologic evidence has implicated hypokalemia and/or reduced dietary K^+ in the genesis or worsening of hypertension, heart failure, and stroke.²²¹ K^+ depletion in young rats induces hypertension,²²² with a salt sensitivity that persists after K^+ levels are normalized; presumably this salt sensitivity is as a result of the significant tubulointerstitial injury induced by K^+ restriction.²¹³ Hypokalemia has also recently been linked to vascular calcification and arterial stiffness via induction of autophagy and promotion of vascular smooth muscle calcification.²²³ Elevated autophagy and cAMP response element-binding protein (CREB) signaling were demonstrated in calcified arteries from mice fed a low-potassium diet, as well as arteries exposed to low potassium *ex vivo*. This novel insight may prove to be a causative effect of dietary potassium intake on atherosclerotic vascular calcification and stiffness. Short-term K^+ restriction in healthy humans and patients with essential hypertension also induces Na^+/Cl^- retention and hypertension,²⁹ and abundant epidemiological data has linked dietary K^+ deficiency and/or hypokalemia with hypertension.^{221,224} Correction of hypokalemia is particularly important in hypertensive patients treated with diuretics; blood pressure in this setting is improved with the establishment of normokalemia,²²⁵ and the cardiovascular benefits of diuretic agents are blunted by hypokalemia.^{29,226} Hypokalemia reduces insulin secretion; this mechanism may play an important role in thiazide-associated diabetes.²²⁷ Finally, K^+ depletion may play important roles in the pathophysiology and progression of heart failure.²²¹

CONSEQUENCES OF HYPERKALEMIA

EXCITABLE TISSUES: MUSCLE AND HEART

Hyperkalemia constitutes a medical emergency, primarily because of its effect on the heart. In skeletal muscle, a reduction in plasma K^+ will hyperpolarize the cell membrane. Furthermore, as noted, in Purkinje fibers of the human conducting system, hypokalemia results in a paradoxical depolarization. In contrast, hyperkalemia depolarizes cardiac myocytes, reducing the membrane potential from -90 mV to about -80 mV. This brings the membrane potential closer to the threshold for generation of an action potential; mild and/or rapid onset hyperkalemia will initially increase cardiac excitability because a lesser depolarizing stimulus is required to generate an action potential. Mild increases in extracellular K^+ also affect the repolarization phase of the cardiac action potential via increases in I_{Kr} ; as discussed earlier (see “Effects of Hypokalemia”), I_{Kr} is highly sensitive to changes in extracellular K^+ .³¹ This effect on repolarization is thought to underlie the early signs of hyperkalemia,²²⁸ including ST-T segment depression, peaked T waves, and Q-T interval shortening.¹⁹⁵

Persistent and increasing depolarization inactivates cardiac sodium channels, thus reducing the rate of phase 0 of the action potential ($V_{\max}$); the decrease in $V_{\max}$ results in a reduction in myocardial conduction, with progressive prolongation of the P wave, PR interval, and QRS complex.¹⁹⁵ Severe hyperkalemia results in loss of the P wave and a progressive widening of the QRS complex; fusion with T waves causes a sine wave sinoventricular rhythm.

Cardiac arrhythmias associated with hyperkalemia include sinus bradycardia, sinus arrest, slow idioventricular rhythms, ventricular tachycardia, ventricular fibrillation, and asystole^{228,229}; a multitude of mechanisms are involved.²⁰¹ The differential diagnosis and treatment of a wide-complex tachycardia in hyperkalemia can be particularly problematic; moreover, hyperkalemia potentiates the blocking effect of lidocaine on the cardiac Na^+ channel, so that use of this agent may precipitate asystole or ventricular fibrillation in this setting.²³⁰ Hyperkalemia can also cause a type I Brugada pattern in the electrocardiogram (ECG), with a pseudo-right bundle branch block (RBBB) and persistent “coved” ST segment elevation in at least two precordial leads. This hyperkalemic Brugada sign occurs in critically ill patients with significant hyperkalemia (serum $\text{K}^+ > 7$ mmol/L), and can be differentiated from genetic Brugada syndrome by an absence of P waves, marked QRS widening, and an abnormal QRS axis.²³¹

Classically, the electrocardiographic manifestations in hyperkalemia progress as shown in Table 17.3. However, these changes are notoriously insensitive, so that only 55% of patients with serum K^+ more than 6.8 mmol/L in one case series manifested peaked T waves.²³² There is large interpatient variability in the absolute potassium level leading to electrocardiographic changes and cardiac toxicity of hyperkalemia. Relevant variables include the rapidity of the onset of hyperkalemia^{233,234} and the presence or absence of concomitant hypocalcemia, acidemia, and/or

hyponatremia.^{235,236} Hemodialysis patients²³⁶ and patients with chronic renal failure,²³⁷ in particular, may not demonstrate electrocardiographic changes. Care should also be taken to distinguish the symmetrically peaked, church steeple T waves induced by hyperkalemia adequately from T wave changes because of other causes.²³⁸ The ratio of the precordial T wave to R wave amplitude (T:R ratio) may be a more specific sign of hyperkalemia than T wave tenting.²³⁹

Hyperkalemia can also rarely present with ascending paralysis,²¹ denoted “secondary hyperkalemic paralysis” to differentiate it from familial hyperkalemic periodic paralysis (HYPP). This presentation of hyperkalemia can mimic Guillain-Barré syndrome and may include diaphragmatic paralysis and respiratory failure.²⁴⁰ Hyperkalemia from a diversity of causes can cause paralysis, as reviewed by Evers and associates.²⁴¹ The mechanism is not entirely clear; however, nerve conduction studies in one case have suggested a neurogenic mechanism, rather than a direct effect on muscle excitability.²⁴¹

In contrast with secondary hyperkalemic paralysis, HYPP is a primary myopathy. Patients with HYPP develop myopathic weakness during hyperkalemia induced by increased K^+ intake or rest after heavy exercise.²⁴² The hyperkalemic trigger in HYPP serves to differentiate this syndrome from hypokalemic periodic paralysis (HOKP); a further distinguishing feature is the presence of myotonia in HYPP.²⁴² Depolarization of skeletal muscle by hyperkalemia unmasks an inactivation defect in a tetrodotoxin-sensitive Na^+ channel in patients with HYPP, and autosomal dominant mutations in the *SCN4A* gene encoding this channel cause most forms of the disease.²⁴³ Mild muscle depolarization (5–10 mV) in HYPP results in a persistent inward Na^+ current through the mutant channel; the normal, allelic *SCN4* channels quickly recover from inactivation and can then be reactivated, resulting in myotonia. When muscle depolarization is more marked—that is, 20 to 30 mV—all the Na^+ channels are inactivated, rendering the muscle inexcitable and causing weakness (Fig. 17.8). Related disorders because of mutations within the large *SCN4A* channel protein include HOKP type II,²⁴⁴ paramyotonia congenita,²⁴³ and K^+ -aggravated myopathy.²⁴³ American quarter horses have a high incidence (4.4%) of HYPP due a mutation in equine *SCN4A* traced to the sire “Impressive” (see Fig. 17.8).²⁴³ Finally, loss-of-function mutations in the muscle-specific K^+ channel subunit called “MinK-related peptide 2” (MiRP2) have also been shown to cause HYPP; MiRP2 and the associated $\text{Kv}3.4$ K^+ channel play a role in setting the resting membrane potential of skeletal muscle.²⁴⁵

RENAL CONSEQUENCES

Hyperkalemia has a significant effect on the ability to excrete an acid urine because of interference with the urinary excretion of ammonium (NH_4^+). Potassium loading in humans results in modest reduction in urinary NH_4^+ excretion and an impaired response to acid loading.²⁴⁶ In rats, chronic potassium loading leads to hyperkalemia and a metabolic acidosis because of a 40% reduction in urinary NH_4^+ excretion.²⁴⁷ Proximal tubular ammonia generation falls, but without a significant effect on proximal tubular secretion of NH_4^+ .²⁴⁸ The TAL absorbs NH_4^+ from the tubular lumen, followed by countercurrent multiplication and ultimately excretion from the medullary interstitium.²⁴⁸ Hyperkalemia appears to inhibit renal acid excretion by competing with

Table 17.3 Approximate Relationship Between Hyperkalemic Electrocardiographic Changes and Serum Potassium Concentration

Serum K^+ Concentration (mmol/L)	Electrocardiographic Abnormality
5.5–6.5	Tall peaked T waves with narrow base, best seen in precordial leads
6.5–8.0	Peaked T waves; prolonged PR interval; decreased amplitude of P waves; widening of QRS complex
>8.0	Absence of P waves; Intraventricular blocks, fascicular blocks, bundle branch blocks, QRS axis shift; progressive widening of the QRS complex; sine wave pattern (sinoventricular rhythm), ventricular fibrillation, asystole

From Mattu A, Brady WJ, Robinson DA: Electrocardiographic manifestations of hyperkalemia. *Am J Emerg Med.* 2000; 18:721–729.

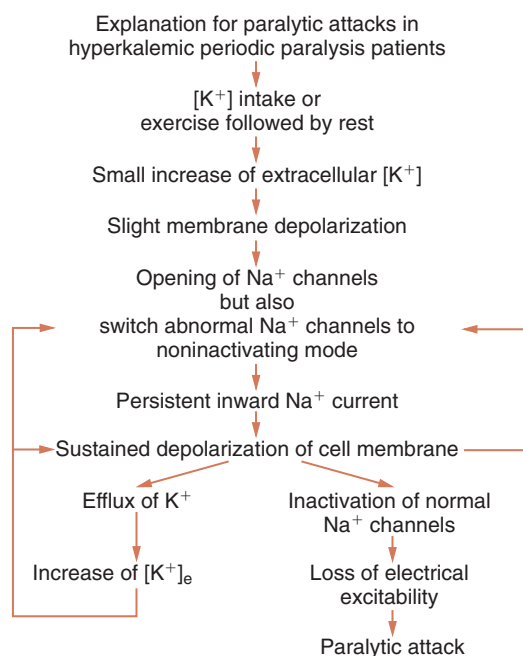


Fig. 17.8 Hyperkalemic periodic paralysis (HYPP) caused by mutations in the voltage-gated Na^+ channel of skeletal muscle. (A) This disorder is particularly common in thoroughbred quarter horses; an affected horse is shown during a paralytic attack, triggered by rest after heavy exercise. (B) Mechanistic explanation for muscle paralysis in HYPP. (A, courtesy Dr. Eric Hoffman; B from Lehmann-Horn F, Jurkat-Rott K. Voltage-gated ion channels and hereditary disease. *Physiol Rev.* 1999;79:1317–1372.)

NH_4^+ for reabsorption by the TAL, thus preventing augmented NH_4^+ levels in the medullary interstitium.²⁴⁹

The NH_4^+ ion has the same ionic radius as K^+ and can be transported in lieu of K^+ by NKCC2,²⁵⁰ the apical $\text{Na}^+\text{-K}^+/\text{NH}_4^+\text{-2Cl}^-$ cotransporter of the TAL; NH_4^+ exits the TAL via the basolateral Na^+/H^+ exchanger NHE4.²⁵¹ As is the case for other cations, countercurrent multiplication of NH_4^+ by the TAL greatly increases the concentration of $\text{NH}_4^+/\text{NH}_3$ available for secretion in the collecting duct. The NH_4^+ produced by the proximal tubule in response to acidosis is thus reabsorbed across the TAL, concentrated by countercurrent multiplication in the medullary interstitium and secreted in the collecting duct. The capacity of the TAL to reabsorb NH_4^+ is increased during acidosis because of an induction of NKCC2²⁵⁰ and NHE4 expression.²⁵¹ Hyperkalemia induces

acidosis in rats by reducing the NH_4^+ between the vasa recta (surrogate for interstitial fluid) and collecting duct²⁴⁹ caused by interference by K^+ with absorption of NH_4^+ by the TAL.

More recently, in a mouse model for hyporeninemic hypoaldosteronism with prominent, treatable hyperkalemic acidosis, hyperkalemia was correlated with reduced expression of ammonia-generating enzymes in the proximal tubule, combined with upregulation of the ammonia-recycling enzyme glutamine synthetase.²⁵² Hyperkalemia did not affect the expression of NKCC2 or NHE3. However, these mice demonstrated decreased expression of the ammonia transporter family member Rhcg and decreased apical polarization of H^+ -ATPase in the inner stripe of the OMCD, further compromising urinary NH_4^+ excretion. These various changes in enzyme and transporter expression were reversed by correcting the hyperkalemia.²⁵²

Clinically, patients with hyperkalemic acidosis caused by hyporeninemic hypoaldosteronism demonstrate an increase in urinary NH_4^+ excretion in response to normalization of plasma K^+ with cation exchange resins,^{253,254} indicating a significant role for hyperkalemia in the generation of the acidosis.

HYPOKALEMIA

CAUSES OF HYPOKALEMIA

A number of causative factors are involved in the development of hypokalemia.

EPIDEMIOLOGY

Hypokalemia is a relatively common finding in both outpatients and inpatients, perhaps the most common electrolyte abnormality encountered in clinical practice.²⁵⁵ When defined as a serum K^+ of less than 3.6 mmol/L, it is found in approximately 20% of hospitalized patients^{256,257}; defined as a serum K^+ of less than 3.4 mmol/L, it occurs in 16.8% of first-time hospital admissions.²⁵⁸ Hypokalemia is usually mild, with K^+ levels in the 3.0 to 3.5 mmol/L range, but in up to 25% of hypokalemic patients it can be moderate to severe (<3.0 mmol/L).^{256,259} The most common causative factors in hospitalized patients with hypokalemia are gastrointestinal losses of potassium, diuretic therapy, and hypomagnesemia.²⁶⁰ It is a particularly prominent problem in patients receiving thiazide diuretics for hypertension, with an incidence of up to 48% (average, 15%–30%).^{29,261} The thiazide-type diuretic metolazone is frequently used for the management of heart failure refractory to loop diuretics alone, causing moderate ($\text{K}^+ \leq 3.0$ mmol/L) or severe ($\text{K}^+ \leq 2.5$ mmol/L) hypokalemia in approximately 40% and 10% of patients, respectively.²⁶² Hypokalemia is also a common finding in patients undergoing peritoneal dialysis, with 10% to 20% requiring potassium supplementation.²⁶³ Hypokalemia per se can increase the in-hospital mortality rate up to tenfold,^{258,259} likely caused by the profound effects on arrhythmogenesis, blood pressure, and cardiovascular morbidity.^{221,264}

SPURIOUS HYPOKALEMIA

Delayed sample analysis is a well-recognized cause of spurious hypokalemia as a result of increased cellular uptake; this may become clinically relevant if the ambient temperature is increased.^{29,265,266} Very rarely, patients with profound

leukocytosis caused by acute leukemia present with artifactual hypokalemia caused by the time-dependent uptake of K^+ by the large white cell mass.²⁶⁵ Such patients do not develop clinical or electrocardiographic complications of hypokalemia, and the plasma K^+ level is normal if measured immediately after venipuncture.

REDISTRIBUTION AND HYPOKALEMIA

Manipulation of the factors affecting internal distribution of K^+ (see “[Factors Affecting Internal Distribution of Potassium](#)”) can cause hypokalemia as a result of redistribution of K^+ between the extracellular and intracellular compartments. Endogenous insulin is rarely a cause of hypokalemia; however, administered insulin is a frequent cause of iatrogenic hypokalemia²⁵⁶ and may be a factor in the so-called *dead in bed syndrome* associated with aggressive glycemic control.²⁶⁷ Insulin also may play a significant role in the hypokalemia associated with refeeding syndrome.²⁶⁸ Alterations in the activity of the endogenous sympathetic nervous system can cause hypokalemia in several settings, including alcohol withdrawal,²⁶⁹ acute myocardial infarction,^{271,270} and head injury.^{271,272} Redistributive hypokalemia after severe head injury can be truly profound, with reported serum K^+ levels of 1.2 mmol/L²⁷¹ and 1.9 mmol/L²⁷² and marked rebound hyperkalemia after repletion.

Because of their ability to activate both Na^+/K^+ -ATPase⁵¹ and the $Na^+-K^+-2Cl^-$ cotransporter NKCC1,^{21,28} beta-2 agonists are powerful activators of cellular K^+ uptake. These agents are chiefly encountered in the therapy of asthma; however, tocolytics such as ritodrine, can induce hypokalemia and arrhythmias during maternal labor.²⁷³ The long-acting beta-2 agonist clenbuterol, not approved for medical use in the United States, has caused hypokalemia in poisonings, including outbreaks of toxicity from clenbuterol-adulterated heroin.²⁷⁴ Occult sources of sympathomimetics, such as pseudoephedrine and ephedrine in cough syrup²⁰³ or dieting agents,²⁷⁵ can be overlooked causes of hypokalemia. Finally, downstream activation of cAMP by xanthines such as theophylline^{21,276} and dietary caffeine²⁷⁷ may induce hypokalemia and may synergize in this respect with beta-2 agonists.²⁷⁸

Whereas beta-2 agonists activate K^+ uptake via the Na^+/K^+ -ATPase, one would expect that inhibition of passive K^+ efflux would also lead to hypokalemia; this is accomplished by barium, a potent inhibitor of inward-rectifying K^+ channels.²⁷⁹ This rare cause of hypokalemia is usually because of ingestion of the rodenticide (colloquially known as rat poison) barium carbonate, either unintentionally or during a suicide attempt.²⁸⁰ Suicidal ingestion of barium-containing shaving powder²⁸¹ and hair remover²⁸² has also been described. Barium salts are widely used in industry, and poisoning has been described by various mechanisms in industrial accidents.^{21,283} Patients have a particularly prominent U wave, likely because of direct inhibition of cardiac inward-rectifying K^+ channels.²⁷⁹ Muscle paralysis can also occur²⁷⁹ because of inhibition of muscle KIR channels. Treatment of barium poisoning with K^+ serves to increase plasma K^+ and displace barium from affected K^+ channels²⁸⁰; hemodialysis is also an effective treatment.²⁸⁴

Cesium also inhibits multiple K^+ channels and can provoke hypokalemia and associated arrhythmias, with inappropriate kaliuresis presumably because of the inhibition of ROMK and BK secretory channels.²⁸⁵ Hypokalemia is also common with

chloroquine toxicity or overdose,²⁸⁶ although the mechanism is not entirely clear.

HYPOKALEMIC PERIODIC PARALYSIS

The periodic paralyses have both genetic and acquired causes, and are further subdivided into hyper- and hypokalemic forms.^{21,242–244} The genetic and secondary forms of hyperkalemic paralysis are discussed previously (see “[Consequences of Hyperkalemia](#)”). Autosomal dominant mutations in the *CACNA1S* gene encoding the alpha-1 subunit of L-type calcium channels are the most common genetic cause of hypokalemic periodic paralysis (HOKP type I), whereas type II HOKP is because of mutations in the *SCN4A* gene encoding the skeletal Na^+ channel.²⁸⁷ In Andersen syndrome, autosomal dominant mutations in the *KCNJ2* gene encoding the inwardly rectifying K^+ channel Kir2.1 cause periodic paralysis, cardiac arrhythmias, and dysmorphic features.²⁸⁸ Paralysis in Andersen syndrome can be normokalemic, hypokalemic, or hyperkalemic; however, the symptomatic trigger is consistent within individual kindreds.²⁸⁸

The pathophysiology of HOKP is complex. Structurally, about 90% of the HOKP-associated mutations result in loss of positively charged arginine residues in the S4, voltage sensor domains of L-type calcium channels and the skeletal Na^+ channel.²⁴⁴ This generates a so-called “gating current,” generated by a cation leak through an aberrant pore; this abnormal cation leak may directly lead to K^+ -dependent paradoxical depolarization and hypokalemic weakness.²⁸⁹ Muscles of an Na^+ channel knock-in mutant mouse ($Na_v1.4$ -R669H) also exhibit an anomalous inward current at hyperpolarized potentials, attributed to this gating pore current.²⁹⁰

Abnormalities in insulin-sensitive transport events may also contribute to the hypokalemic weakness in HOKP. Reversible attacks of paralysis with hypokalemia in HOKP are typically precipitated by rest after exercise and/or meals rich in carbohydrates.²⁴⁴ Although the induction of endogenous insulin by carbohydrate meals is thought to reduce plasma K^+ , thus triggering weakness, insulin can precipitate paralysis in HOKP in the absence of significant hypokalemia.²⁹¹ The generation of action potentials and muscle contraction are reduced in types I and II HOKP muscle fibers exposed to insulin *in vitro*^{287,292}; this effect is seen at an extracellular K^+ of 4.0 mmol/L and is potentiated as K^+ decreases.²⁹² Type I HOKP muscles have a reduced activity of ATP-sensitive, inward rectifying K^+ channels (K_{ATP}),^{293,294} which likely contributes to hypokalemia because of the resultant unopposed activity of muscle Na^+/K^+ -ATPase.²⁹⁵ Insulin inhibits the remaining K_{ATP} activity in muscle fibers of both type I HOKP patients²⁹² and hypokalemic rats,²⁹⁶ resulting in a depolarizing shift toward the equilibrium potential for the Cl^- ion (≈ 50 mV); at this potential, voltage-dependent Na^+ channels are largely inactivated, resulting in paralysis.

Paralysis is associated with multiple other causes of hypokalemia, both acquired and genetic.^{202,203,297} Renal causes of hypokalemia with paralysis include Fanconi syndrome,²⁹⁸ Gitelman syndrome,²⁹⁷ and the various causes of hypokalemic distal renal tubular acidosis.^{29,299} The activity and regulation of skeletal muscle K_{ATP} channels are aberrant in animal models of hypokalemia, suggesting a parallel muscle physiology to that of genetic HOKP (see previously). However, the pathophysiology of thyrotoxic periodic paralysis (TPP), a particularly important cause of hypokalemic paralysis, is distinctly different

from that of HOKP; for example, despite the clinical similarities between the two syndromes, thyroxine has no effect on HOKP.

TPP is classically seen in patients of Asian origin, but also occurs at higher frequencies in Hispanic patients.³⁰⁰ This shared predisposition has been linked to genetic variation in the *KCNJ18* gene encoding Kir2.6, a muscle-specific, thyroid hormone–responsive K⁺ channel.³⁰ Genome-wide association studies have also implicated variations in the *KCNJ2* gene, which encodes a related muscle K⁺ channel, Kir 2.1, with a predisposition to TPP.³⁰¹ Patients typically present with weakness of the extremities and limb girdles, with attacks occurring most frequently between 1 and 6 AM. As in HOKP, paralytic attacks in TPP may be precipitated by rest and/or by carbohydrate-rich meals. Clinical signs and symptoms of hyperthyroidism are not invariably present.^{300,302} Hypokalemia is profound, ranging between 1.1 and 3.4 mol/L, and is frequently accompanied by hypophosphatemia and hypomagnesemia³⁰⁰; all three abnormalities presumably contribute to the associated weakness. Diagnostically, a TTKG of less than 2 to 3 or urine K⁺-to-creatinine ratio of less than 2.5 mmol/mmol separates patients with TPP from those with hypokalemia because of renal potassium wasting, who will have TTKG values that are higher than 4.¹⁸⁶ This distinction is of considerable therapeutic relevance; patients with large potassium deficits require aggressive repletion with K⁺-Cl[−], which has a significant risk of rebound hyperkalemia in TPP and related disorders.^{186,303}

The hypokalemia in TPP is most likely because of both direct and indirect activation of the Na⁺/K⁺-ATPase, given the evidence for increased activity in erythrocytes and platelets in TPP patients.^{29,304} Thyroid hormone clearly induces expression of multiple subunits of the Na⁺/K⁺-ATPase in skeletal muscle.³⁰⁵ Increases in the β -adrenergic response because of hyperthyroidism also play an important role because high-dose propranolol (3 mg/kg) rapidly reverses the hypokalemia, hypophosphatemia, and paralysis seen in acute attacks.^{306,307} Of particular importance, no rebound hyperkalemia is associated with this treatment, whereas aggressive K⁺ replacement in TPP is associated with an incidence of about 25%³⁰³; repletion-associated rebound hyperkalemia in TPP can be fatal.³⁰⁸

Outward-directed, inward-rectifying K⁺ current, mediated by KIR channels (primarily Kir2.1 and Kir2.2 tetramers), is also reduced in skeletal muscles of patients with TPP,²⁹⁴ providing an additional mechanism for hypokalemia. Together with increased Na⁺/K⁺-ATPase activity and increased circulating insulin, this reduced KIR current may trigger a feed-forward cycle of hypokalemia, leading to inactivation of muscle Na⁺ channels, paradoxical depolarization, and paralysis.³⁰⁹ The role of TPP-associated sequence variants in the *KCNJ18* gene that encodes Kir 2.6 is not entirely clear at this point, given that some genetic studies have failed to find a convincing association.³¹⁰ As noted, the muscle Kir channel is largely formed by tetramers of Kir 2.1 with Kir 2.2 proteins, with robust expression of these channel proteins at the plasma membrane and T tubules of skeletal muscle.³¹¹ In contrast, wild-type Kir 2.6 protein appears to be restricted primarily to the endoplasmic reticulum (ER), with a dominant negative effect on expression and function of Kir 2.1 and Kir 2.2 subunits.³¹¹ It remains unclear how TPP-associated sequence variants, perhaps induced by high levels of thyroid hormone,³⁰ explain the proposed predisposition to TPP.

NONRENAL POTASSIUM LOSS

The loss of K⁺ from skin is typically low, with the exception of extremes in physical exertion.¹¹ Direct gastric loss of K⁺ because of vomiting or nasogastric suctioning is also typically minimal; however, the ensuing hypochloremic alkalosis results in persistent kaliuresis caused by secondary hyperaldosteronism and bicarbonaturia.^{312,313} Intestinal loss of K⁺ as a result of diarrhea is a quantitatively important cause of hypokalemia, given the worldwide prevalence of diarrheal disease, and may be associated with acute complications such as myopathy and flaccid paralysis.³¹⁴ The presence of a non-anion gap metabolic acidosis with a negative urinary anion gap¹⁸⁷ (consistent with an intact ability to increase NH₄⁺ excretion) should strongly suggest diarrhea as a cause of hypokalemia. Polyethylene glycol–based bowel preparation regimens for colonoscopy can also lead to hypokalemia in older patients³¹⁵ and/or patients on diuretics.³¹⁶ Noninfectious gastrointestinal processes such as celiac disease,³¹⁷ ileostomy, and chronic laxative abuse can present with acute hypokalemic syndromes or with chronic complications such as ESKD.²¹

There has been an appreciation of the role of intestinal BK K⁺ channels in hypokalemia associated with intestinal disease. Three reports initially identified a novel association between colonic pseudo-obstruction (Ogilvie syndrome) and hypokalemia as a result of secretory diarrhea with an abnormally high K⁺ content.^{318–320} In one patient with concomitant ESKD, immunohistochemistry revealed massive upregulation of the apical BK channel throughout the surface crypt axes³¹⁸; colonic BK channels may play a significant role in intestinal K⁺ secretion in a variety of pathologies, including ESKD.³²¹ Several hypotheses for the association between Ogilvie syndrome and enhanced intestinal K⁺ secretion have been postulated, including active stimulation by catecholamines induced by colonic pseudo-obstruction³²⁰; BK channels appear to mediate adrenaline-induced colonic K⁺ secretion.³²² Notably, aldosterone also induces colonic BK channel expression and activity³²³; consistent with a role for aldosterone, a dramatic response to spironolactone in Ogilvie syndrome–associated diarrhea and hypokalemia was recently reported.³²⁴

Increased fecal loss of K⁺ may play a broader role in hypokalemia associated with diarrhea.³²¹ Recruitment of colonic BK channels along intestinal crypts, similar to that seen in Ogilvie syndrome, has thus been demonstrated as a consistent feature of colonic biopsies in ulcerative colitis.³²⁵ Direct enhancement of intestinal K⁺ excretion has also been demonstrated in a hypokalemic patient with Crohn disease following treatment with budesonide.³²⁶

RENAL POTASSIUM LOSS

Drugs

Diuretics are an especially important cause of hypokalemia because of their ability to increase distal flow rate and distal delivery of Na⁺. Thiazides generally cause more hypokalemia^{21,225,261} than loop diuretics, despite their lower natriuretic efficacy. One potential explanation is the differential effect of loop diuretics and thiazides on calcium excretion. Whereas thiazides and loss-of-function mutations in the Na⁺-Cl[−] cotransporter decrease Ca²⁺ excretion,³²⁷ loop diuretics cause a significant calciuresis.³²⁸ Increases in luminal Ca²⁺ in the distal nephron serve to reduce the lumen-negative driving force

for K^+ excretion,³²⁹ perhaps by direct inhibition of ENaC in principal cells. A mechanistic explanation is provided by the presence of an apical calcium-sensing receptor (CaSR) in the collecting duct.³³⁰ Analogous to the evident decrease in the apical trafficking of aquaporin-2 induced by luminal Ca^{2+} , tubular Ca^{2+} may stimulate endocytosis of ENaC via the CaSR and thus limit generation of the lumen-negative potential difference that is so critical for distal K^+ excretion. Regardless of the underlying mechanism, the increase in distal delivery of Ca^{2+} induced by loop diuretics may serve to blunt kaliuresis; such a mechanism would not occur with thiazides, which reduce distal delivery of Ca^{2+} , with unopposed activity of ENaC and increased kaliuresis.

A substantial and growing body of evidence has indicated a key role for the NCC in K^+ homeostasis, the thiazide-sensitive Na^+-Cl^- cotransporter in the DCT, so that it is not surprising that thiazide treatment has such potent effects on serum K^+ . Selective increases in DCT NCC activity, as seen in FHHt, reduce Na^+ delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.⁷³ The DCT also clearly functions as a potassium sensor, directly responding to changes in circulating potassium.⁴ A high- K^+ diet also inactivates NCC, whereas NCC is activated in hypokalemia.¹⁴³

Other drugs associated with hypokalemia because of kaliuresis include toxic levels of acetaminophen, which causes dose-dependent hypokalemia.³³¹ High doses of penicillin-related antibiotics are another important cause of hypokalemia, increasing obligatory K^+ excretion by acting as nonreabsorbable anions in the distal nephron. In addition to penicillin, implicated antibiotics include nafcillin, dicloxacillin, ticarcillin, oxacillin, and carbenicillin.³³² Increased distal delivery of other anions such as SO_4^{2-} and HCO_3^- also induces a kaliuresis. The usual explanation is that K^+ excretion increases so as to balance the negative charge of these nonreabsorbable anions. However, increased delivery of such anions will also increase the electrochemical gradient for K^+-Cl^- exit via apical K^+-Cl^- cotransport or parallel K^+-H^+ and $Cl^- -HCO_3^-$ exchange^{70,83,84} (see also “Potassium Transport in the Distal Nephron”). Drugs are also an important cause of Fanconi syndrome,³³³ which is often associated with significant hypokalemia (see “Renal Tubular Acidosis”).

Several tubular toxins result in both K^+ and magnesium wasting. These include gentamicin, which can cause tubular toxicity with hypokalemia that can masquerade as Bartter syndrome.³³⁴ Other drugs that can cause mixed magnesium and K^+ wasting include amphotericin, foscarnet,³³⁵ cisplatin,^{21,336} and ifosfamide.³³⁷ One intriguing cause of hypomagnesemia and hypokalemia is cetuximab, a humanized monoclonal antibody specific for the receptor for epidermal growth factor (EGF)³³⁸, and often used in metastatic colorectal carcinoma. Paracrine EGF stimulates magnesium transport via the apical TRPM6 cation channel in the DCT, with magnesium wasting and hypomagnesemia in patients treated with cetuximab.³³⁹ Aggressive replacement of magnesium is obligatory in the treatment of combined hypokalemia and hypomagnesemia because successful K^+ replacement depends on treatment of the hypomagnesemia (see “Hypomagnesemia”).

Finally, a handful of drugs lead to hypokalemia through the inhibition of 11β -hydroxysteroid dehydrogenase-2 (11β -HSD-2), resulting in unopposed mineralocorticoid activity of circulating cortisol. The classic 11β -HSD-2 inhibitor is licorice through the active components glycyrrhetic,

glycyrrhizinic acid, and carbenoxolone. More recently, the antifungals itraconazole and posaconazole have been shown to inhibit 11β -HSD-2,³⁴⁰ with several reports of hypertension and hypokalemia in patients treated with these agents^{341,342} (see “Syndromes of Apparent Mineralocorticoid Excess”).

Hyperaldosteronism

Increases in circulating aldosterone (hyperaldosteronism) may be primary or secondary. Increased levels of circulating renin in secondary forms of hyperaldosteronism leads to increased Ang II and thus aldosterone and can be associated with hypokalemia. Causes include renal artery stenosis,³⁴³ Page kidney (renal compression by a subcapsular mass or hematoma, with hyperreninemia),³⁴⁴ a paraneoplastic process,³⁴⁵ or renin-secreting renal tumors.³⁴⁶ The incidence of hypokalemia in renal artery stenosis is thought to be less than 20%.³⁴³ An unusual presentation of renal artery stenosis and renal ischemia is the hyponatremic hypertensive syndrome, in which concurrent hypokalemia may be profound.³⁴⁷

Primary hyperaldosteronism (PA) may have a hereditary, genetic cause. Hypertension and hypokalemia, generally attributed to increases in circulating 11-deoxycorticosterone,³⁴⁸ are seen in patients with congenital adrenal hyperplasia because of defects in either steroid 11β -hydroxylase³⁴⁸ or steroid 17α -hydroxylase.³⁴⁹ Deficient 11β -hydroxylase results in virilization and other signs of androgen excess,³⁴⁸ whereas reduced sex steroids in 17α -hydroxylase deficiency result in hypogonadism.³⁴⁹

The two major hereditary forms of isolated PA are denoted familial hyperaldosteronism type I (FH-I, also known as glucocorticoid-remediable hyperaldosteronism [GRA])³⁵⁰ and familial hyperaldosteronism type II (FH-II), in which aldosterone production is not repressible by exogenous glucocorticoids. Patients with FH-II are clinically indistinguishable from sporadic forms of PA because of bilateral adrenal hyperplasia; a gene has been localized to chromosome 7p22 by linkage analysis, but has yet to be characterized.³⁵¹ A third form of familial hyperaldosteronism (FH-III) was initially described in 2008, with hyporeninemia, hyperaldosteronism resistant to dexamethasone, and very high levels of 18-oxocortisol and 18-hydroxycortisol.³⁵² FH-III is caused by somatic mutations in the adrenal K^+ channel KCNJ5, which endow the channel with a novel Na^+ conductance and activate adrenal glomerulosa proliferation and aldosterone release.² Somatic mutations in KCNJ5 are also found in spontaneous adrenal adenomas (see later). More recently, a fourth form of familial hyperaldosteronism (FH-IV), caused by autosomal dominant gain-of-function mutations in the *CACNA1H* gene, which encodes the $CaV3.2$ calcium channel.^{353,354} The $CaV3.2$ calcium channel protein is expressed in human adrenal glomerulosa cells,³⁵³ and the mutant channels demonstrate gain-of-function phenotypes, leading to increased induction of aldosterone synthase in cultured adrenal cells.³⁵⁴ The clinical characteristics of patients with FH-IV has not been fully described; however, the index patients had developed PA in childhood, and one subject had a multiplex developmental disorder.^{353,354}

Patients with FH-I/GRA are generally hypertensive, typically presenting at an early age; the severity of hypertension is, however, variable, so that some genetically affected individuals are actually normotensive.³⁵⁰ Aldosterone levels are modestly elevated and regulated solely by ACTH. The diagnosis can

be biochemically confirmed by a dexamethasone suppression test, with a suppression of aldosterone to less than 4 ng/dL, consistent with the diagnosis.³⁵⁵ Patients also have high levels of abnormal hybrid 18-hydroxylated steroids, generated by the transformation of steroids typically formed in the zona fasciculata by aldosterone synthase, an enzyme that is normally expressed in the zona glomerulosa.^{356,357} FH-I has been shown to be caused by a chimeric gene duplication between the homologous 11 β -hydroxylase (*CYP11B1*) and aldosterone synthase (*CYP11B2*) genes, fusing the ACTH-responsive 11 β -hydroxylase promoter to the coding region of aldosterone synthase. This chimeric gene is thus under the control of ACTH and is expressed in a glucocorticoid-repressible fashion.³⁵⁶ Ectopic expression of the hybrid *CYP11B1-CYP11B2* gene in the zona fasciculata has been reported in a single case where adrenal tissue became available for molecular analysis.³⁵⁸ Direct genetic testing for the hybrid *CYP11B1-CYP11B2* has largely supplanted biochemical screening for FH-I. Genetic testing for FH-I should be pursued in patients with PA and a family history of PA and/or of strokes at a young age, or in young patients with PA (<20 years of age).³⁵⁹ For reasons that are unclear, patients with FH-I have a high frequency of hemorrhagic stroke caused by cerebral aneurysms, roughly equivalent to the frequency in autosomal polycystic kidney disease.³⁶⁰

Although the initial patients reported with FH-I were hypokalemic, most were normokalemic,^{357,361} albeit perhaps with a propensity to develop hypokalemia while on thiazide diuretics.³⁵⁷ Patients with FH-I are able to increase K⁺ excretion appropriately in response to K⁺ loading or fludrocortisone, but fail to increase plasma aldosterone in response to hyperkalemia.³⁶² This may reflect the ectopic expression of the chimeric aldosterone synthase in the adrenal fasciculata, which likely lack the appropriate constellation of ion channels to respond to increases in extracellular K⁺ with an increase in aldosterone secretion.

Acquired causes of PA classically include aldosterone-producing adenomas (APAs; 35% of cases), primary or unilateral adrenal hyperplasia (PAH; 2% of cases), idiopathic hyperaldosteronism (IHA) caused by bilateral adrenal hyperplasia (60% of cases), and adrenal carcinoma (<1% of cases).³⁶³ A rare case involving paraneoplastic overexpression of aldosterone synthase in lymphoma has also been described.³⁶⁴

The molecular characterization of adrenal adenomas with whole-genome sequencing and related techniques has been remarkably fruitful. In particular, acquired somatic mutations of the adrenal K⁺ channel *KCNJ5* gene can be detected in about 40% of aldosterone-producing adrenal adenomas.¹⁷² As in FH-III (see previously), these somatic mutations endow the channel with a novel Na⁺ conductance, leading to adrenal glomerulosa cell depolarization, Ca²⁺ influx, and aldosterone release. Clinically, patients with adrenal *KCNJ5* mutations have a higher preoperative aldosterone level¹⁷² and higher lateralization index in adrenal vein sampling,³⁶⁵ without affecting the surgical response to adrenalectomy.¹⁷² Less frequently, somatic mutations in adrenal adenomas can be detected in the calcium channel *CACNA1D*³⁶⁶ or in a subunit (*ATP2B3*) of the Ca²⁺-ATPase pump,³⁶⁷ predicted also to lead to increased Ca²⁺ influx and aldosterone release.³⁶⁶ Acquired mutations in the ATP_{1A} alpha-1 subunit of the Na⁺/K⁺-ATPase are, in turn, thought to generate chronic depolarization,

leading also to exaggerated aldosterone release.³⁶⁷ A common consequence of the mutations in these so-called aldosterone driver genes is that they ultimately increase aldosterone production through increased expression of aldosterone synthase (*CYP11B2*).³⁶⁸

Recently, other pathways separate from aldosterone signaling have been implicated in the genesis of APAs. Mutations in the *CTNNB1* gene encoding β -catenin have also been described in APAs.^{369,370} The mutations described appear to stabilize β -catenin, suggesting a role in WNT signaling, which has previously been implicated in adrenal development and adrenal adenomas.³⁶⁹

Landmark studies with monoclonal antibodies to CYP11B2 (aldosterone synthase) and CYP11B1 (steroid 11 β -hydroxylase, which catalyzes the terminal step in cortisol production) have provided a molecular definition of the zonation of human adrenal cortices.^{368,371} Whereas CYP11B1 expression is uniformly expressed throughout the zona fasciculata, there is only sporadic expression of CYP11B2 in the zona glomerulosa of normal human adrenal glands.³⁷¹ However, a subcortical population of nonneoplastic clusters of cells can also be detected; given the coexpression of other steroidogenic enzymes that are required for aldosterone synthesis, these CYP11B2+ clusters have been designated “aldosterone-producing cell clusters” (APCCs).^{368,371,372} APCCs have also been demonstrated in adrenal tissues adjacent to APAs, suggesting that APCCs are a precursor to APA.^{368,371} Consistent with this hypothesis, mutations in the known aldosterone-driver genes can be seen in about 35% of APCCs in normal adrenal glands.³⁷²

Immunohistochemical characterization of APAs also indicates expression of cortisol-producing enzymes CYP17A1 and CYP11B1,^{371,373} providing a molecular explanation for the hypercortisolism associated with hyperaldosteronism.^{373,374} In hyperaldosteronism as a result of both bilateral adrenal hyperplasia and aldosterone-producing adenomas, there is significant urinary excretion of cortisol and other glucocorticoid metabolites, exceeded only in patients with clinically overt Cushing syndrome.³⁷⁴ Glucocorticoid excretion in primary aldosteronism appears to correlate better with metabolic risk (type 2 diabetes, osteoporosis, metabolic syndrome) than aldosterone excretion.³⁷⁴ To the extent that hypercortisolism is not affected by mineralocorticoid antagonists, these findings tend to underscore the preference for adrenalectomy in APAs. Increasing utilization of the plasma aldosterone (PAC)/plasma renin activity (PRA) ratio in hypertension clinics has led to reports of a much higher incidence of PA than previously appreciated, with incidence rates in hypertension ranging from 0% to 72%.³⁷⁵ However, the prevalence was 3.2% in a large multicenter study of patients with mild to moderate hypertension without hypokalemia.³⁷⁶

It should be noted that recent studies have revealed that there is a continuum in renin-independent hyperaldosteronism between normotensive individuals with suppressed renin and inappropriately “normal” or high aldosterone levels, individuals with unrecognized but biochemically overt PA, and individuals with severe hypertensive PA.^{377,378} The presumed pathophysiology behind this continuum of increasing autonomous aldosterone secretion is the age-dependent development of APCCs, the evident precursor to APAs.^{368,371,372} Regardless, the PAC:PRA ratio is a screening tool, which classically must be confirmed by aldosterone suppression testing, measuring

PAC or aldosterone secretion after loading with or salt or intravenous saline.³⁷⁹ After controlling hypertension and hypokalemia (serum $K^+ \geq 4.0$ mEq/L),³⁷⁹ oral salt loading over 3 days is followed by measurement of 24-hour urine aldosterone, sodium, and creatinine excretion. The 24-hour sodium excretion should exceed 200 mmol/day for adequate suppression, and a urinary aldosterone level of more than 33 nmol/day (12 μ g/day) is consistent with PA. Alternatively, in the saline infusion test, recumbent patients are infused with 2 L of isotonic saline over 4 hours, followed by measurement of PAC. In patients without PA, the measured PAC after saline infusion should decrease to less than 139 pmol/L. The measured PAC in patients with PA usually does not suppress to less than 277 pmol/L; indeterminate values between 139 and 277 pmol/L can be seen in patients with IHA.³⁶³ Patients with high probability features (hypokalemia, hypertension, high PAC:PRA ratio, abnormalities) may not necessarily require confirmatory testing, proceeding instead directly to imaging and adrenal venous sampling.³⁸⁰

Because surgery can be curative in APA, adequate differentiation of APA from IHA is critical; this requires both

adrenal imaging and adrenal venous sampling (Fig. 17.9). Contemporary reports and recommendations have thus emphasized the continued importance of adrenal vein sampling in subtype differentiation.³⁷⁹ Laparoscopic adrenalectomy is increasingly the preferred surgical management for APA or PAH.³⁷⁹ Radiofrequency ablation appears to be an effective alternative in patients judged to be inappropriate for surgery.³⁸¹ Mineralocorticoid receptor antagonists are indicated for medical therapy of PA, with carefully monitored use of glucocorticoid to suppress ACTH in some patients with FH-I/GRA.³⁷⁹

The true incidence of hypokalemia in patients with acquired forms of PA remains difficult to evaluate because of a variety of factors:

1. Historically, patients have only been screened for hyperaldosteronism when hypokalemia is present; hence, even recent case series from clinics with such a referral pattern may suffer from a selection bias. Other recent series have concentrated on hypertensive patients, also a selection bias.

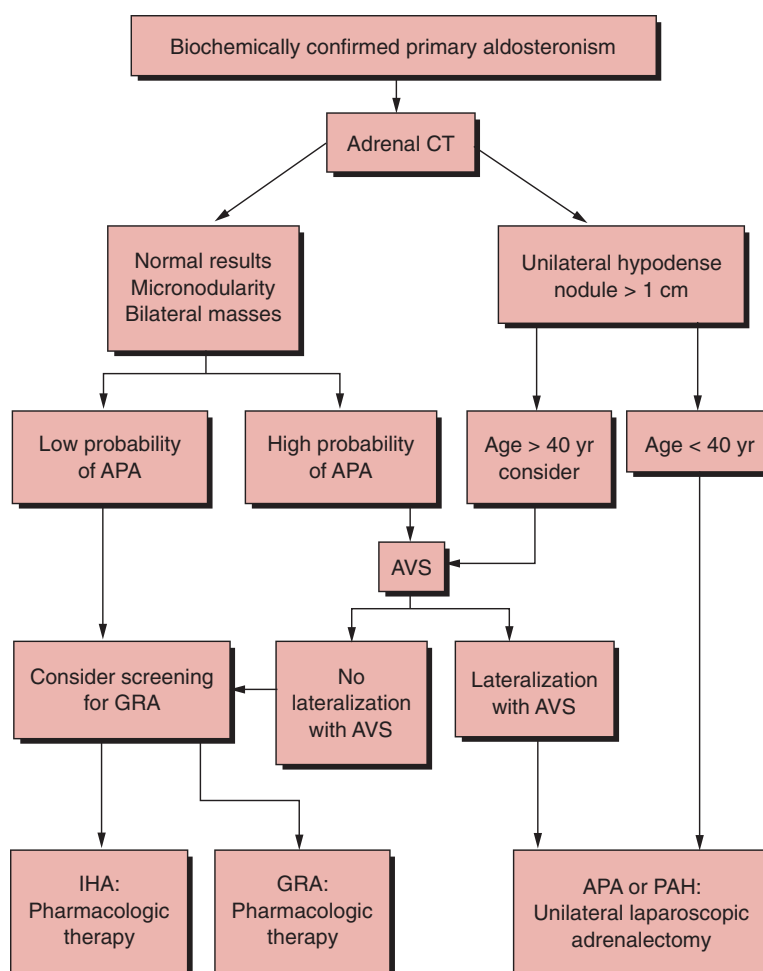


Fig. 17.9 Diagnostic algorithm in patients with primary hyperaldosteronism. Adrenal adenoma (APA) must be distinguished from glucocorticoid remediable hyperaldosteronism (FH-I [or GRA]), primary or unilateral adrenal hyperplasia (PAH), and idiopathic hyperaldosteronism (IHA). This requires computed axial tomography (CT), adrenal venous sampling (AVS), and the relevant diagnostic biochemical and hormonal assays (see text). (From Young WF Jr. Adrenalectomy for primary aldosteronism. *Ann Intern Med.* 2003;138(2):157–159.)

2. The incidence of hypokalemia is higher in adrenal adenomas than in IHA, likely because of higher average levels of aldosterone.³⁸²
3. Because increased kaliuresis in hyperaldosteronism can be induced by dietary $\text{Na}^+\text{-Cl}^-$ loading or diuretics, dietary factors and/or medications may play a role in the incidence of hypokalemia at presentation.
4. As noted earlier, there is a continuum in renin-independent hyperaldosteronism between normotensive individuals with suppressed renin and inappropriately “normal” or high aldosterone levels, individuals with unrecognized but biochemically overt PA, and individuals with severe hypertensive PA.^{377,378}

Nevertheless, it is clear that hypokalemia is not a universal feature of PA; this is perhaps not unexpected, because aldosterone does not appear to affect the hypokalemic response of $\text{H}^+/\text{K}^+\text{-ATPase}$,³⁸³ the major reabsorptive pathway for K^+ in the distal nephron. A related issue is whether PA is underdiagnosed when hypokalemia is used as a criterion for further investigation.

Finally, hypokalemia may also occur with systemic increases in glucocorticoids.^{384,385} In actual Cushing syndrome caused by increases in pituitary ACTH, the incidence of hypokalemia is only 10%, whereas it is 57% to 100% in patients with ectopic ACTH,^{384,385} despite a similar incidence of hypertension. Ectopic ACTH expression is associated primarily with neuroendocrine malignancies, most commonly bronchial carcinoid tumors, small lung cancer, and other neuroendocrine tumors.³⁸⁶ Indirect evidence has suggested that the activity of renal $11\beta\text{-HSD-2}$ is reduced in patients with ectopic ACTH compared with Cushing syndrome,³⁸⁷ resulting in a syndrome of apparent mineralocorticoid excess (see later). Whether this reflects a greater degree of saturation of the enzyme by circulating cortisol or direct inhibition of $11\beta\text{-HSD-2}$ by ACTH is not entirely clear, and there is evidence for both mechanisms.³⁸⁵ However, indirect indices of $11\beta\text{-HSD-2}$ activity in patients with ectopic ACTH expression correlate with hypokalemia and other measures of mineralocorticoid activity.³⁸⁸ Similar mechanisms likely underlie the severe hypokalemia reported in patients with familial glucocorticoid resistance, in which loss-of-function mutations in the glucocorticoid receptor result in marked hypercortisolism without cushingoid features, accompanied by very high ACTH levels.³⁸⁹

Syndromes of Apparent Mineralocorticoid Excess

The syndromes of apparent mineralocorticoid excess (AME) have a self-explanatory label. In the classic form of AME, recessive loss-of-function mutations in the $11\beta\text{-HSD-2}$ gene ($11\beta\text{-HSD-2}$) cause a defect in the peripheral conversion of cortisol to the inactive glucocorticoid cortisone; the resulting increase in the half-life of cortisol is associated with a marked decrease in synthesis, so that plasma levels of cortisol are normal and patients are not cushingoid.³⁹⁰ The $11\beta\text{-HSD-2}$ protein is expressed in epithelial cells that are targets for aldosterone; in the kidney, these include cells of the DCT, CNT, and CCD.³⁹¹ Because the mineralocorticoid receptor (MR) has equivalent affinity for aldosterone and cortisol, generation of cortisone by $11\beta\text{-HSD-2}$ serves to protect mineralocorticoid-responsive cells from illicit activation by cortisol.³⁹² In patients with AME, the unregulated mineralocorticoid effect of glucocorticoids results in hypertension,

hypokalemia, and metabolic alkalosis, with suppressed PRA and aldosterone.³⁹⁰ Biochemical diagnosis entails measuring the urinary-free cortisol to the urinary free cortisone ratio on a 24-hour urine collection. Biochemical studies of mutant enzymes usually indicate a complete loss of function; lesser enzymatic defects in patients with AME are associated with altered ratios of urinary cortisone/cortisol metabolites,³⁹³ lesser impairment in the peripheral conversion of cortisol to cortisone,³⁹⁴ and/or older age at presentation.³⁹⁵

Mice with a homozygous targeted deletion of $11\beta\text{-HSD-2}$ exhibit hypertension, hypokalemia, and polyuria; the polyuria is likely secondary to the hypokalemia (see “Hypokalemia: Renal Consequences”), which reaches 2.4 mmol/mL in $11\beta\text{-HSD-2}$ -null mice.³⁹⁶ As expected, both PRA and plasma aldosterone in the $11\beta\text{-HSD-2}$ -null mice are profoundly suppressed, with a decreased urinary $\text{Na}^+:\text{K}^+$ ratio that is increased by dexamethasone (given to suppress endogenous cortisol). These knockout mice have significant nephromegaly because of a massive hypertrophy and hyperplasia of DCTs. The relative effect of genotype on the morphology of cells in the DCT, CNT, and CCD has not been determined by the appropriate phenotypic studies³⁹⁷; however, it is known that both the DCT and CCD are target cells for aldosterone,^{398,399} and both cell types express $11\beta\text{-HSD-2}$. The induction of ENaC activity by unregulated glucocorticoid, now having mineralocorticoid-like activity, causes the Na^+ retention and the marked increase in K^+ excretion in $11\beta\text{-HSD-2}$ -null mice; distal tubular micropuncture studies in rats treated with a systemic inhibitor of $11\beta\text{-HSD-2}$ are consistent with such a mechanism.⁴⁰⁰ In addition, the cellular gain of function in the DCT would be expected to be associated with hypercalciuria, given the phenotype of pseudohypoaldosteronism type II and Gitelman syndrome (see “Hereditary Tubular Defects and Potassium Excretion” and “Familial Hypokalemic Alkalosis: Gitelman Syndrome”). Indeed, patients with AME are reported to exhibit nephrocalcinosis.³⁹⁰

Pharmacologic inhibition of $11\beta\text{-HSD-2}$ is also associated with hypokalemia and AME. The most infamous offender is licorice, in its multiple guises (e.g., licorice root, tea, candies, herbal remedies). The early observations that licorice required small amounts of cortisol to exert its kaliuretic effect, in the Addisonian absence of endogenous glucocorticoid,⁴⁰¹ presaged the observations that its active ingredients (glycyrrhetic acid, glycyrrhizinic acid, and carbenoxolone) inhibit $11\beta\text{-HSD-2}$ and related enzymes.³⁹⁰ Licorice intake remains considerable in European countries, particularly Iceland, the Netherlands, and Scandinavia.⁴⁰² Pontefract cakes, eaten both as a sweet and a laxative, are a continued source of licorice in England,⁴⁰² whereas it is an ingredient in several popular sweeteners and preservatives in Malaysia.⁴⁰³ Glycyrrhizinic acid is also a component of Chinese herbal remedies, prescribed for disorders such as for allergic rhinitis.⁴⁰⁴ Pharmacologic inhibition of $11\beta\text{-HSD-2}$ has also been tested in patients with ESKD as a novel mechanism to control hyperkalemia (see “[Management of Hyperkalemia](#)”).⁴⁰⁵ Carbenoxolone is, in turn, used in some countries in the management of peptic ulcer disease.³⁹⁰ More recently, the antifungals itraconazole and posaconazole have been shown to inhibit $11\beta\text{-HSD-2}$,³⁴⁰ with several reports of hypertension and hypokalemia in patients treated with these agents.^{341,342}

Finally, a mechanistically distinct form of AME has been reported, because of a gain-of-function mutation in the MR.⁴⁰⁶

A single kindred was thus described with autosomal dominant inheritance of severe hypertension and hypokalemia; the causative mutation involved a serine residue that is conserved in the MR from multiple species, yet differs from other nuclear steroid receptors. This mutation results in constitutive activation of the MR in the absence of ligand, also with significant affinity for progesterone.⁴⁰⁶ The MR is thus constitutively “on” in these patients, with a marked stimulation by progesterone. Of interest, pregnancies in affected female members of the family have been complicated by severe hypertension caused by marked increases in plasma progesterone levels induced by the gravid state.⁴⁰⁶

Liddle Syndrome

Liddle syndrome constitutes an autosomal dominant gain in function of ENaC, the amiloride-sensitive Na⁺ channel of the CNT and CCD.⁴⁰⁷ Patients manifest severe hypertension with hypokalemia, unresponsive to spironolactone yet sensitive to triamterene and amiloride. Liddle syndrome could therefore also be classified as a syndrome of apparent mineralocorticoid excess. Both hypertension and hypokalemia are variable aspects of the Liddle phenotype; consistent features include a blunted aldosterone response to ACTH and reduced urinary aldosterone excretion.^{100,408} The differential diagnosis for Liddle syndrome, as a cause of hereditary hypertension with hypokalemia and suppressed aldosterone, includes AME because of deficient 11 β -HSD-2; however, whereas the Liddle syndrome phenotype is resistant to blockade of the mineralocorticoid receptor with spironolactone and sensitive to amiloride, AME patients are sensitive to both drugs. Commercial genetic testing for both syndromes is available in the United States.

The vast majority of mutations target the C-terminus of either the beta or gamma ENaC subunit, with a very small minority of mutations in α -ENaC. ENaC channels containing Liddle syndrome mutations are constitutively overexpressed at the cell membrane^{101,409}; unlike wild-type ENaC channels, they are not sensitive to inhibition by intracellular Na⁺,⁴¹⁰ an important regulator of endogenous channel activity in the CCD.⁴¹¹ The mechanism whereby mutations in the C-terminus of ENaC subunits lead to this channel phenotype are discussed earlier in this chapter (see Fig. 17.5 and “Control of Potassium Secretion: Aldosterone”). In addition to effects on interaction with Nedd4-2–dependent retrieval from the plasma membrane, Liddle-associated mutations increase proteolytic cleavage of ENaC at the cell membrane⁴¹²; aldosterone-induced “channel-activating proteases” activate ENaC channels at the plasma membrane. This important result provides a mechanistic explanation for the long-standing observation that Liddle-associated mutations in ENaC appear to have a dual activating effect on both the open probability of the channel (i.e., on channel activity) and on expression at the cell membrane.¹⁰¹

Given the overlapping and synergistic mechanisms that regulate ENaC activity, it stands to reason that mutations in ENaC that give rise to Liddle syndrome might do so by a variety of means. Indeed, mutation of a residue within the extracellular domain of γ -ENaC, versus the usual C-terminal site of mutations in Liddle syndrome, increases open probability of the channel without changing surface expression; the patient with this mutation has a typical Liddle syndrome phenotype.⁴¹³ More recently, a missense mutation in the

extracellular domain of α -ENaC has also been described in a Liddle kindred, increasing the open probability of the channel without increasing susceptibility to activation by proteases.⁴¹⁴ Extensive searches for more common mutations and polymorphisms in ENaC subunits that correlate with blood pressure in the general population have essentially been negative, despite characterization of specific gain-of-function variants.⁴¹⁵ However, there are a few genetic studies that correlate specific variants in ENaC subunits with biochemical evidence of greater in vivo activity of the channel—that is, a suppressed PRA and aldosterone level and/or to-increased ratios of urinary K⁺-to-aldosterone/PRA.^{416,417}

Familial Hypokalemic Alkalosis: Bartter Syndrome

Bartter and Gitelman syndromes are the two major variants of familial hypokalemic alkalosis; Gitelman syndrome is a much more common cause of hypokalemia than Bartter.⁴¹⁸ Whereas a clinical subdivision of these syndromes has been used in the past, a genetic classification is increasingly in use, due in part to phenotypic overlap. Patients with classic Bartter syndrome (BS) typically suffer from polyuria and polydipsia and manifest a hypokalemic, hypochloremic alkalosis. They may have an increase in urinary calcium (Ca²⁺) excretion, and 20% are hypomagnesemic.⁴¹⁹ Other features include marked elevation of plasma Ang II, plasma aldosterone, and plasma renin levels. Patients with antenatal BS present earlier in life with a severe systemic disorder characterized by marked electrolyte wasting, polyhydramnios, and significant hypercalciuria with nephrocalcinosis. Prostaglandin synthesis and excretion are significantly increased and may account for many of the systemic symptoms. Decreasing prostaglandin synthesis by COX inhibition can improve polyuria in patients with BS by reducing the amplifying inhibition of urinary concentrating mechanisms by prostaglandins. Indomethacin also increases plasma K⁺ and decreases plasma renin activity, but does not correct the basic tubular defect; it does, however, appear to help increase the growth of BS patients.⁴²⁰ Of interest, COX-2 immunoreactivity is increased in the TAL and macula densa of patients with BS,⁴²¹ and reports have indicated a clinical benefit of COX-2 inhibitors.⁴²²

Early studies in Bartter syndrome have suggested that these patients have a defect in the function of the TAL.⁴²³ Many of the clinical features are mimicked by the administration of loop diuretics, to which at least a subset of patients with antenatal BS do not respond.⁴²⁴ The apical Na⁺-K⁺-2Cl[−] cotransporter (NKCC2, SLC12A1) of the mammalian TAL²⁷ (see Fig. 17.10) was thus an early candidate gene. In 1996, disease-associated mutations were found in the human NKCC2 gene in four kindreds with antenatal BS⁴²⁵; in the genetic classification of BS, these patients are considered to have BS type I. Although the functional consequences of disease-associated NKCC2 mutations have not been comprehensively studied, the first⁴²⁵ and subsequent reports²⁹ have included patients with frameshift mutations and premature stop codons that predict the absence of a functional NKCC2 protein.

BS is a genetically heterogeneous disease. Given the role of apical K⁺ permeability in the TAL, encoded at least in part by ROMK,^{78,426} this K⁺ channel was another early candidate gene. K⁺ recycling via the Na⁺-K⁺-2Cl[−] cotransporter and apical K⁺ channels generates a lumen-positive potential difference in the TAL, which drives the paracellular transport of Na⁺ and other cations⁴²⁷ (see also Fig. 17.10). Multiple disease-associated

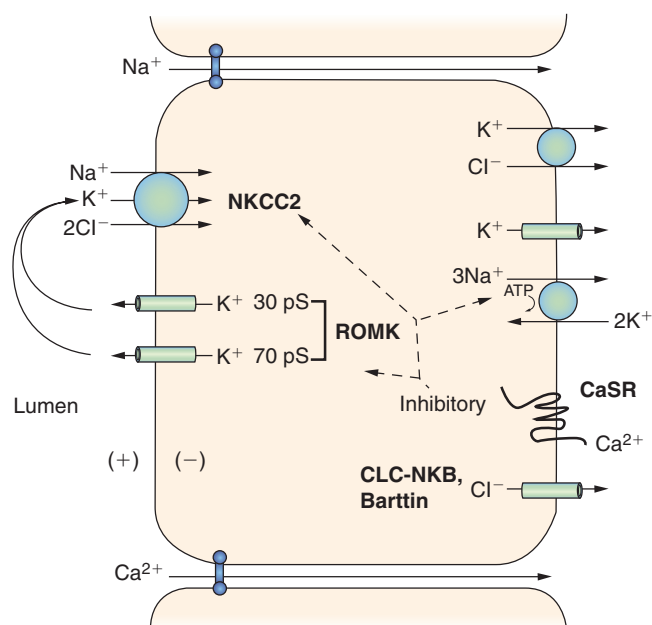


Fig. 17.10 Bartter syndrome and the thick ascending limb. Bartter syndrome can result from loss-of-function mutations in the Na⁺-K⁺-2Cl⁻ cotransporter NKCC2, the K⁺ channel subunit ROMK, or the Cl⁻ channel subunits CLC-NKB and Barttin (Bartter syndrome, types I–IV, respectively). Gain-of-function mutations in the calcium-sensing receptor (CaSR) can also cause a Bartter syndrome phenotype (type V); the CaSR has an inhibitory effect on salt transport by the thick ascending limb, targeting several transport pathways. ROMK encodes the low-conductance, 30-pS K⁺ channel in the apical membrane and also appears to function as a critical subunit of the higher conductance 70-pS channel. The loss of K⁺ channel activity Bartter syndrome type II leads to reduced apical K⁺ recycling and reduced Na⁺-K⁺-2Cl⁻ cotransport. Decreased apical K⁺ channels also lead to a decrease in the lumen-positive potential difference, which drives paracellular Na⁺, Ca²⁺, and Mg²⁺ transport. ROMK, Renal outer medullary K⁺ channel.

mutations in ROMK have been reported in patients with BS type II, most of whom exhibit the antenatal phenotype.^{420,428} Finally, mutations in BS type III have been reported in the chloride channel CLC-NKB,⁴²⁹ which is expressed at the basolateral membrane of at least the TAL and DCT.⁴³⁰ In a significant fraction of patients with BS, the NKCC2, ROMK, and CLC-NKB genes are not involved.⁴²⁹ For example, a subset of patients with associated sensorineural deafness exhibit linkage to chromosome 1p31²⁹; the gene for this syndrome, denoted Barttin, is an obligatory subunit for the CLC-NKB chloride channel.⁴³¹ The occurrence of deafness in these patients suggests that Barttin functions in the regulation or function of Cl⁻ channels in the inner ear. Notably, the CLC-NKB gene is immediately adjacent to another epithelial Cl⁻ channel, denoted CLC-NKA; inactivation of both genes was described in two siblings with deafness and BS,⁴³² suggesting that CLC-NKA plays an important role in Barttin-dependent Cl⁻ transport in the inner ear.

Patients with activating mutations in the CaSR have been described with autosomal dominant hypocalcemia and hypokalemic alkalosis.^{433,434} The CaSR is heavily expressed at the basolateral membrane of the TAL,⁴³⁵ where it is thought to play an important inhibitory role in regulating the transepithelial transport of both Na⁺-Cl⁻ and Ca²⁺. For example,

activation of the basolateral CaSR in the TAL is known to reduce apical K⁺ channel activity,⁴³⁶ which would induce a Bartter-like syndrome (see Fig. 17.10). Coexpression of NKCC2 with a Bartter syndrome gain-of-function mutant in the CaSR reveals reduced phosphorylation and reduced activity of NKCC2, dependent on the generation of inhibitory arachidonic acid-derived metabolites known to inhibit TAL function.⁴³⁷ Genetic activation of the CaSR by these mutations is also expected to increase urinary Ca²⁺ excretion by inhibiting the generation of the lumen-positive potential difference that drives paracellular Ca²⁺ transport in the TAL. In addition, the set point of the CaSR response to Ca²⁺ in the parathyroid is shifted to the left, inhibiting parathyroid hormone (PTH) secretion by this gland. No doubt the positional cloning of other BS genes will have a considerable impact on our mechanistic understanding of the TAL.

Despite the reasonable correlation between the disease gene involved and the associated subtype of familial alkalosis, there is significant phenotypic overlap and phenotypic variability in hereditary hypokalemic alkalosis. For example, patients with mutations in CLC-NKB most frequently (44.5%) exhibit classic BS, but can present with a more severe antenatal phenotype (29.5%) or even with a phenotype similar to GS (26.0%).^{29,438,439} With respect to BS because of mutations in NKCC2, a number of patients have been described with variant presentations, including an absence of hypokalemia.²⁹ Two brothers were described with a late onset of mild BS; they were found to be compound heterozygotes for a mutant form of NKCC2 that exhibits partial function, with a loss-of-function mutation on the other NKCC2 allele.⁴⁴⁰ Further genetic heterogeneity has also recently emerged, with the description of patients with hypokalemic alkalosis because of mutations in claudin 10, which presumably disrupt paracellular cation transport by the TAL.⁴⁴¹ Notably, the genetic heterogeneity in BS, GS, and related disorders is arguably not a major clinical issue in contemporary clinical practice; most commercial genetic analyses of patients with hereditary hypokalemic alkalosis now involve testing an extensive panel of relevant genes, rather than testing single genes.

BS type II is particularly relevant to K⁺ homeostasis, given that ROMK is the SK secretory channel of the CNT and CCD (see “Potassium Transport in the Distal Nephron”). Patients with BS type II typically have slightly higher serum K⁺ level than the other genetic forms of BS^{428,438}; patients with severe (9.0 mmol/L), transient, neonatal hyperkalemia have also been described.⁴⁴² It is likely that this reflects a transient developmental deficit in the other K⁺ channels involved in distal K⁺ secretion, including the apical maxi-K channel responsible for flow-dependent K⁺ secretion in the distal nephron.^{77,443} Distal K⁺ secretion in ROMK knockout mice is primarily mediated by maxi-K–BK channel activity⁴⁴⁴ so that developmental deficits in this channel would indeed lead to hyperkalemia in BS type II. The mammalian TAL has two major apical K⁺ conductances, the 30-pS channel corresponding to ROMK and a 70-pS channel⁴⁴⁵; both are thought to play a role in transepithelial salt transport by the TAL. ROMK is evidently a subunit of the 70-pS channel, given the absence of this conductance in TAL segments of ROMK knockout mice.⁴⁴⁶ The identity of the other putative subunit of this 70-pS channel is not as yet known; one would assume that deficiencies in this gene would also be a cause of BS.

Finally, BS must be clinically differentiated from the various causes of “pseudo-Bartter” syndrome; these commonly include laxative abuse, furosemide abuse, and bulimia (see “[Clinical Approach to Hypokalemia](#)”). Other reported causes include gentamicin nephrotoxicity,³³⁴ Sjögren syndrome,²⁹ and cystic fibrosis (CF).^{29,447} Augmented $\text{Na}^+\text{-Cl}^-$ loss in sweat is likely the dominant predisposing factor for hypokalemic alkalosis in patients with CF; patients with this presentation generally respond promptly to intravenous fluids and electrolyte replacement. However, the cystic fibrosis transmembrane conductance regulator (CFTR) protein co-associates with ROMK in the TAL and confers sensitivity to both ATP and glibenclamide to apical K^+ channels in this nephron segment.⁴⁴⁸ Lu and coworkers have proposed that this interaction serves to modulate the response of ROMK to cAMP and vasopressin, so that K^+ excretion in CFTR deficiency would not be appropriately reduced during water diuresis, thus predisposing such patients to the development of hypokalemic alkalosis.⁴⁴⁸

Familial Hypokalemic Alkalosis: Gitelman Syndrome

A major advance in the understanding of hereditary alkaloses was the realization that a subset of patients exhibit marked hypocalciuria, rather than the hypercalciuria typically seen in BS; patients in this hypocalciuric subset are universally hypomagnesemic.³²⁷ Such patients are now clinically classified as suffering from Gitelman syndrome.⁴⁴⁹ Although plasma renin activity may be increased, renal prostaglandin excretion is not elevated in these hypocalciuric patients,⁴⁵⁰ another distinguishing feature between Bartter and Gitelman syndromes. Gitelman syndrome (GS) is a milder disorder than BS; however, patients do report significant morbidity, mostly related to muscular symptoms and fatigue.^{449,451} The QT interval is frequently prolonged in GS, suggesting an increased risk of cardiac arrhythmia⁴⁵²; however, a more exhaustive cardiac evaluation of a large group of patients has failed to detect significant abnormalities of cardiac structure or rhythm.⁴⁵³ However, presyncope and/or ventricular tachycardia has been observed in at least two patients with GS,^{29,194} one with concomitant long QT syndrome because of a mutation in the cardiac *KCNQ1* K^+ channel.¹⁹⁴

The hypocalciuria detected in GS was an expected consequence of inactivating the thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransporter NCC (SLC12A2), and loss-of-function mutations in the human gene have been reported⁴⁵⁴; many of these mutations lead to a defect in cellular trafficking when introduced into the human NCC protein.⁴⁵⁵ GS is genetically homogeneous, except for patients with mutations in *CLC-NKB* and an overlapping phenotype.^{29,194,438,439} Loss-of-function mutations in the *KCNJ10* encoding the basolateral Kir 4.1 K^+ channel in the DCT have been implicated in a related syndrome of hypokalemic metabolic alkalosis with hypomagnesemia accompanied by seizures, sensorineural deafness, ataxia, and mental retardation (SeSAME or EAST syndrome).^{32,456} In Kir 4.1-deficient, *KCNJ10* knockout mice, the loss of basolateral K^+ conductance reduces basolateral Cl^- conductance, leading to reduced expression of SPAK kinase and reduced apical NCC expression.¹⁵⁰ Notably, the genetic heterogeneity in GS and related disorders is arguably not a major clinical issue in contemporary clinical practice; most commercial genetic analyses of patients with hereditary

hypokalemic alkalosis now involve testing an extensive panel of relevant genes, rather than testing for mutations in single candidate genes.

The NCC protein is localized to the apical membrane of epithelial cells in the DCT and CNT. A mouse strain with targeted deletion of the *Slc12a2* gene encoding NCC exhibits hypocalciuria and hypomagnesemia, with a mild alkalosis and marked increase in circulating aldosterone.⁴⁵⁷ These knockout mice exhibit marked morphologic defects in the early DCT,⁴⁵⁷ with both a reduction in absolute number of DCT cells and changes in ultrastructural appearance. That GS is a disorder of cellular development and/or cellular apoptosis should perhaps not be a surprise, given the observation that thiazide treatment promotes marked apoptosis of this nephron segment.⁴⁵⁸ This cellular deficit leads to downregulation of the DCT magnesium channel TRPM6,⁴⁵⁹ resulting in the magnesium wasting and hypomagnesemia seen in GS. The downstream CNT tubules are hypertrophied in NCC-deficient mice,⁴⁵⁷ reminiscent of the hypertrophic DCT and CNT segments seen in furosemide-treated animals.²⁹ These CNT cells also exhibit an increased expression of ENaC at their apical membranes versus littermate controls⁴⁵⁷; this is likely because of activation of SGK1-dependent trafficking of ENaC by the increase in circulating aldosterone (see “[Control of Potassium Secretion: Aldosterone](#)”).

Hypokalemia does not occur in *NCC*^{-/-} mice on a standard rodent diet, but emerges on a K^+ -restricted diet; plasma K^+ levels of these mice are about 1 mM lower than in K^+ -restricted littermate controls.⁴⁶⁰ Several mechanisms account for the hypokalemia seen in GS and *NCC*^{-/-} mice. The distal delivery of both Na^+ and fluid is decreased in *NCC*^{-/-} mice, at least on a normal diet; however, the increased circulating aldosterone and CNT hypertrophy likely compensate, leading to increased kaliuresis. As discussed previously for thiazides, decreased luminal Ca^{2+} in NCC deficiency may augment baseline ENaC activity,³²⁹ further exacerbating the kaliuresis. Of particular interest, NCC-deficient mice develop considerable polydipsia and polyuria on a K^+ -restricted diet⁴⁶⁰; this is reminiscent perhaps of the polydipsia that has been implicated in thiazide-associated hyponatremia.⁴⁶¹

Hypocalciuria in GS is not accompanied by changes in plasma calcium, phosphate, vitamin D, or PTH levels,⁴⁶² suggesting a direct effect on renal calcium transport. The late DCT is morphologically intact in NCC-deficient mice, with preserved expression of the epithelial calcium channel (ECAC1, or TRPV5) and the basolateral $\text{Na}^+\text{-Ca}^{2+}$ exchanger.⁴⁵⁷ Furthermore, the hypocalciuric effect of thiazides persists in mice deficient in TRPV5,⁴⁵⁹ arguing against the putative effects of this drug on distal Ca^{2+} absorption. Rather, several lines of evidence have argued that the hypocalciuria of GS and thiazide treatment is because of increased absorption of Na^+ by the proximal tubule,^{457,459} with secondary increases in proximal Ca^{2+} absorption. Regardless, reminiscent of the clinical effect of thiazides on bone, there are clear differences in bone density between affected and unaffected members of specific Gitelman kindreds. Thus, homozygous patients have much higher bone densities than unaffected wild type family members, whereas heterozygotes have intermediate values for both bone density and calcium excretion.⁴⁶² An interesting association has repeatedly been described between chondrocalcinosis, the abnormal deposition of calcium pyrophosphate dihydrate (CPPD) in joint cartilage, and GS.⁴⁶³

Patients have also been reported with ocular choroidal calcification.⁴⁶⁴

Treatment of GS encompasses liberalization of salt intake and lifelong supplementation with oral magnesium and potassium. Reasonable targets for serum potassium and serum magnesium are 3.0 mEq/L and 0.6 mmol/L (1.46 mg/dL), respectively.⁴⁴⁹ Organic anion salts of magnesium are preferred over magnesium chloride, magnesium hydroxide, or magnesium oxide. Patients with refractory hypokalemia may require treatment with potassium-sparing diuretics, RAAS inhibitors, and/or nonsteroidal inflammatory drugs (NSAIDs).⁴⁴⁹

Finally, as in BS, there have been reports of acquired tubular defects that mimic GS. These include patients with hypokalemic alkalosis, hypomagnesemia, and hypocalciuria after chemotherapy with cisplatin.⁴⁶⁵ Patients have also been described with acquired GS caused by Sjögren syndrome and tubulointerstitial nephritis,^{29,466} with a documented absence of coding sequence mutations in NCC.⁴⁶⁶

Renal Tubular Acidosis

Renal tubular acidosis (RTA) and related tubular defects can be associated with hypokalemia. Proximal RTA is characterized by a reduction in proximal bicarbonate absorption, with a reduced plasma bicarbonate concentration. Isolated proximal RTA is quite rare; genetic causes include loss of function by mutations in the basolateral Na⁺-bicarbonate transporter. More commonly, proximal RTA occurs in the context of multiple proximal tubular transport defects, encompassing the Fanconi syndrome.³³³ The cardinal features of Fanconi syndrome include hyperaminoaciduria, glycosuria with a normal plasma glucose concentration, and phosphate wasting. The finding of severe hypophosphatemia with metabolic acidosis should alert clinicians to this possibility, especially if acute kidney injury is also present. Associated defects include the proximal RTA, hypouricemia, hypercalciuria, hypokalemia, salt wasting, and increased excretion of low-molecular-weight proteins. Fanconi syndrome is usually drug-associated; currently, important causes include aristolochic acid, ifosfamide, and the acyclic nucleoside phosphonates (e.g., tenofovir, cidofovir, adefovir).³³³ Before treatment with bicarbonate, patients with a proximal RTA will typically demonstrate mild hypokalemia, due primarily to baseline hyperaldosteronism⁴⁶⁷; patients have, however, been described with profound hypokalemia on presentation before treatment.⁴⁶⁸ Regardless, treatment with oral sodium bicarbonate will markedly increase distal tubular Na⁺ and HCO₃⁻ delivery, causing a marked increase in renal potassium wasting.⁴⁶⁷ Patients will often require mixed base replacement with oral citrate and bicarbonate in addition to aggressive K⁺-Cl⁻ supplementation.

Hypokalemia is also associated with distal RTA, the so-called *type I RTA*. Hypokalemic distal RTA is usually caused by a secretory defect, with reduced H⁺-ATPase activity and decreased ability to acidify the urine. For example, hereditary defects in subunits of the H⁺-ATPase are associated with profound hypokalemia, in addition to acidosis and hypercalciuria.⁴⁶⁹ The pathophysiology of the associated hypokalemia is multifactorial because of the loss of electrogenic H⁺ secretion (with enhanced K⁺ secretion to maintain electroneutrality in the distal nephron), loss of resorptive H⁺/K⁺-ATPase activity, and increases in aldosterone.^{470,471} Notably, in hereditary distal RTA, hypokalemia is more common in those with H⁺-ATPase

defects than in patients with mutations in the basolateral Cl⁻-HCO₃⁻ exchanger.⁴⁷² Sjögren syndrome is perhaps the most common cause of hypokalemic distal RTA in adults; the associated hypokalemia can be truly profound, often resulting in marked weakness and respiratory arrest.⁴⁷¹

Magnesium Deficiency

Magnesium deficiency results in refractory hypokalemia, particularly if the plasma Mg²⁺ level is less than 0.5 mmol/L²⁵⁶; hypomagnesemic patients are thus refractory to K⁺ replacement in the absence of Mg²⁺ repletion.^{473,474} Magnesium deficiency is also a common concomitant of hypokalemia, in part because associated tubular disorders (e.g., aminoglycoside nephrotoxicity) may cause both a kaliuresis and magnesium wasting. Plasma Mg²⁺ levels must thus be checked on a routine basis in hypokalemia.^{255,475}

Several mechanisms appear to contribute to the effect of magnesium depletion on plasma K⁺ levels. Magnesium depletion has inhibitory effects on muscle Na⁺/K⁺-ATPase activity,⁴⁷⁶ resulting in significant efflux from muscle and a secondary kaliuresis. Distal K⁺ secretion also appears to be enhanced because of a reduction in the normal physiologic inward rectification of ROMK secretory K⁺ channels, with a subsequent increase in outward conductance.⁴⁷⁷ ROMK and other KIR channels are inward rectifying—that is, K⁺ flows inward more readily than outward, even though outward conductance is usually less than inward conductance, K⁺ efflux predominates in the CNT and CCD because the membrane potential is more positive than the equilibrium potential for K⁺. Intracellular Mg²⁺ plays a key role in inward rectification, binding and blocking the pore of the channel from the cytoplasmic side.⁴⁷⁷ The hypomagnesemia-associated reduction in cytoplasmic Mg²⁺ in principal cells reduces inward rectification of ROMK, increasing outward conductance and increasing K⁺ secretion; this has been confirmed in vivo.⁴⁷⁸ Finally, it has been suggested that the repletion of intracellular K⁺ is impaired in hypomagnesemia, even in normokalemic patients.⁴⁷⁵ Decreased intracellular Mg²⁺ enhances K⁺ efflux from the cytoplasm of cardiac and perhaps skeletal myocytes, likely because of reduced intracellular blockade of inward rectifying K⁺ channels (increased efflux) and inhibition of the Na⁺/K⁺-ATPase (decreased influx). Plasma K⁺ levels thus remain normal at the expense of intracellular K⁺.^{21,475,479} This phenomenon is particularly important in patients with cardiac disease taking both diuretics and digoxin. In such patients, hypokalemia and arrhythmias will respond to correction of magnesium deficiency and potassium supplementation.^{21,475}

CLINICAL APPROACH TO HYPOKALEMIA

The initial priority in the evaluation of hypokalemia is an assessment for signs and/or symptoms (e.g., muscle weakness, electrocardiographic changes) suggestive of an impending emergency that requires immediate treatment. The cause of hypokalemia is usually obvious from history, physical examination, and/or basic laboratory tests. However, persistent hypokalemia despite appropriate initial intervention requires a more rigorous workup; in most cases, a systematic approach reveals the underlying cause (Fig. 17.11).

The history should focus on medications (e.g., diuretics, laxatives, antibiotics, herbal medications), diet and dietary supplements (e.g., licorice), and associated symptoms (e.g.,

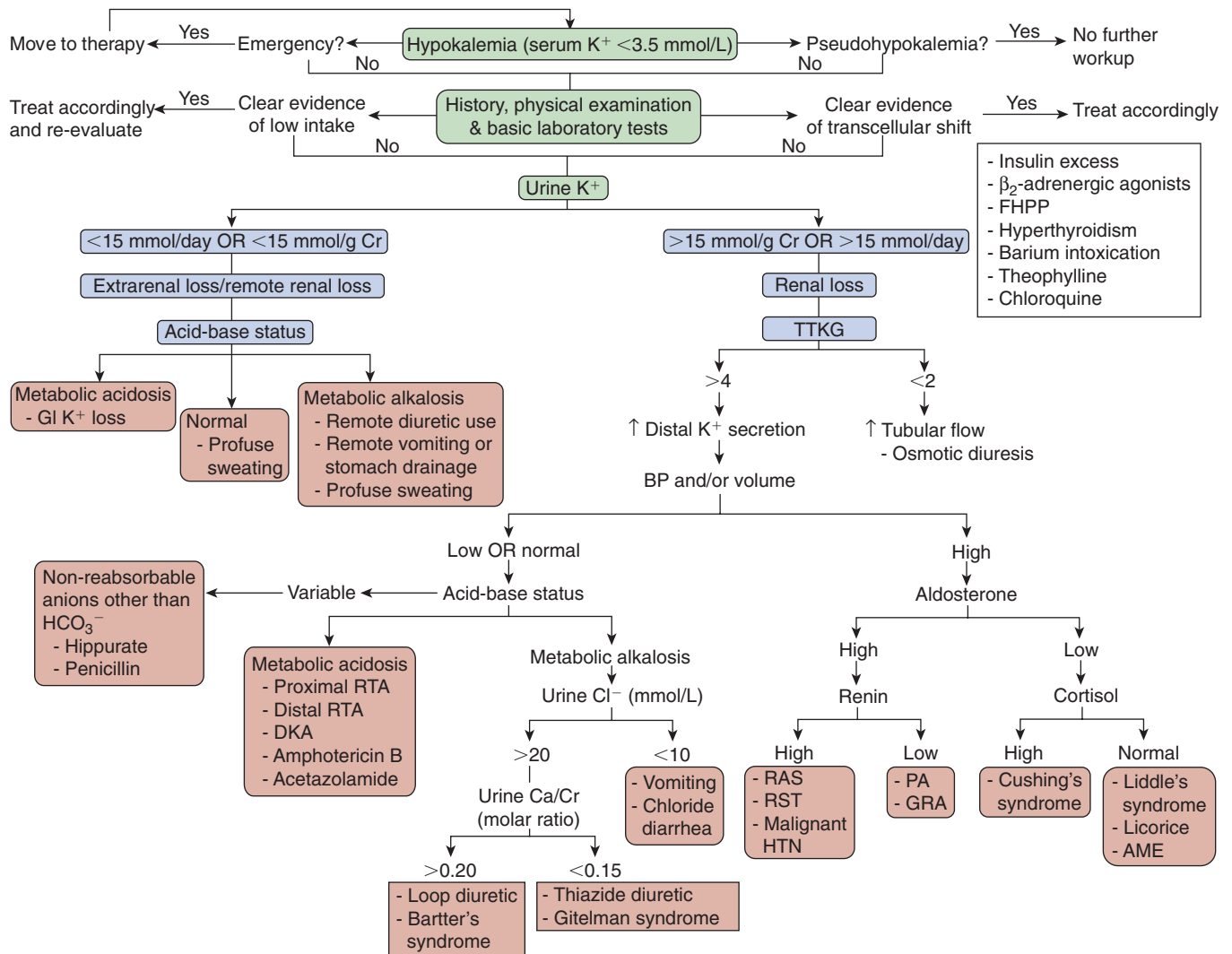


Fig. 17.11 The clinical approach to hypokalemia. See text for details. AME, Apparent mineralocorticoid excess; BP, blood pressure; CCD, cortical collecting duct; DKA, diabetic ketoacidosis; FHPP, familial hypokalemic periodic paralysis; GI, gastrointestinal; GRA, glucocorticoid-remediable aldosteronism; HTN, hypertension; PA, primary aldosteronism; RAS, renal artery stenosis; RST, renin-secreting tumor; RTA, renal tubular acidosis; TTKG, transtubular potassium gradient.

diarrhea). During the physical examination, particular attention should be paid to blood pressure, volume status, and signs suggestive of specific disorders associated with hypokalemia (e.g., hyperthyroidism, Cushing syndrome). Initial laboratory tests should include electrolytes, urea, creatinine, plasma osmolality, Mg^{2+} , and Ca^{2+} , a complete blood count, and urinary pH, osmolality, creatinine, and electrolytes. Plasma and urine osmolality are required for calculation of the TTKG¹⁸¹ and urinary K^{+} -to-creatinine ratio (see “Urinary Indices of Potassium Excretion”). A TTKG of less than 2 to 3 separates patients with redistributive hypokalemia from those with hypokalemia because of renal potassium wasting, who will have TTKG values that are more than 4.¹⁸⁶ Further tests such as urinary Mg^{2+} and Ca^{2+} and plasma renin and aldosterone levels may be necessary in specific cases (see Fig. 17.11). The timing and evolution of hypokalemia are also helpful in differentiating the cause, particularly in hospitalized patients—for example, hypokalemia because of transcellular shift usually occurs in a matter of hours.⁴⁸⁰

The most common causes of chronic, diagnosis-resistant hypokalemia are GS, surreptitious vomiting, and diuretic abuse.⁴⁸¹ Alternatively, an associated acidosis would suggest the diagnosis of hypokalemic distal or proximal renal tubular acidosis. Hypokalemia occurred in 5.5% of patients with eating disorders in an American study from the mid-1990s,⁴⁸² mostly in those with surreptitious vomiting (bulimia) or laxative abuse (the purging³¹³ subtype of anorexia nervosa). These patients may have a constellation of associated symptoms and signs, including dental erosion and depression.⁴⁸³ Hypokalemic patients with bulimia will have an associated metabolic alkalosis, with an obligatory natriuresis accompanying the loss of bicarbonate; urinary Cl^{-} is typically less than 10 mmol/L, and this clue can often yield the diagnosis.^{481,484} Urinary electrolyte levels are, however, generally unremarkable in unselected, mostly normokalemic patients with bulimia.⁴⁸³ Urinary excretion of Na^{+} , K^{+} , and Cl^{-} is high in patients who abuse diuretics, albeit not to the levels seen in GS. Marked variability in urinary electrolyte levels is an important

clue for diuretic abuse, which can be verified with urinary drug screens.

Clinically, nephrocalcinosis is very common in furosemide abuse because of the increase in urinary calcium excretion.⁴⁸⁵ Differentiation of GS from BS requires a 24-hour urine test to assess calcium excretion because hypocalciuria is a distinguishing feature for the former³²⁷; patients with GS are also invariably hypomagnesemic. BS must be differentiated from “pseudo-Barter” syndrome because of gentamicin toxicity,^{334,486} mutations in CFTR, the cystic fibrosis gene,^{447,487} or Sjögren syndrome with tubulointerstitial nephritis.⁴⁸⁸ Acquired forms of GS have, in turn, been reported after cisplatin therapy⁴⁶⁵ and in patients with Sjögren syndrome.^{29,466} Finally, although laxative abuse is perhaps a less common cause of chronic hypokalemia, an accompanying metabolic acidosis with a negative urinary anion gap should raise the diagnostic suspicion of this cause.¹⁸⁷

TREATMENT OF HYPOKALEMIA

The goals of therapy in hypokalemia are to prevent life-threatening conditions (e.g., diaphragmatic weakness, rhabdomyolysis, cardiac arrhythmias), replace any K^+ deficit, and diagnose and correct the underlying cause. The urgency of therapy depends on the severity of hypokalemia, associated conditions and settings (e.g., a patient with heart failure on digoxin, or a patient with hepatic encephalopathy), and the rate of decline in plasma K^+ . A rapid drop to less than 2.5 mmol/L poses a high risk of cardiac arrhythmias and calls for urgent replacement.⁴⁸⁹ Although replacement is usually limited to patients with a true deficit, it should be considered in patients with hypokalemia caused by redistribution (e.g., hypokalemic periodic paralysis) when serious complications such as muscle weakness, rhabdomyolysis, and cardiac arrhythmias are present or imminent.⁴⁹⁰ The risk of arrhythmia from hypokalemia is highest in older patients, patients with evidence of organic heart disease, and patients on digoxin or antiarrhythmic drugs.²⁵⁵ In these high-risk patients, an increased incidence of arrhythmias may occur at even mild to modest degrees of hypokalemia. The American Heart Association guidelines on the use of hospital telemetry have recommended monitoring in patients with hypokalemia and a prolonged QT interval.⁴⁹¹

It is also crucial to diagnose and eliminate the underlying cause to tailor therapy to the pathophysiology involved. For example, the risk of overcorrection or rebound hyperkalemia in hypokalemia caused by redistribution is particularly high, with the potential for fatal hyperkalemic arrhythmias.^{256,271,308,490,492} When increased sympathetic tone or increased sympathetic response is thought to play a dominant role in the hypokalemia, the use of nonspecific β -adrenergic blockade with short-lived propranolol generally avoids this complication and should be considered. The relevant causes of hypokalemia include thyrotoxic periodic paralysis,³⁰⁶ theophylline overdose,⁴⁹³ and acute head injury.²⁷¹

K^+ replacement is the mainstay of therapy in hypokalemia. However, hypomagnesemic patients can be refractory to K^+ replacement alone⁴⁷⁴ so that concomitant Mg^{2+} deficiency should always be addressed with oral or parenteral repletion. To prevent hyperkalemia caused by excessive supplementation, the deficit and the rate of correction should be estimated as accurately as possible. Renal function, medications, and

comorbid conditions such as diabetes (with a risk of both insulinopenia and autonomic neuropathy) should also be considered to gauge the risk of overcorrection. Arbitrary adjustments in the dose of administered K^+-Cl^- replacement based on the estimated GFR can potentially reduce the risk of hyperkalemia.⁴⁹⁴ The goal is to raise the plasma K^+ to a safe range rapidly and then replace the remaining deficit at a slower rate, over days to weeks.^{255,256,490} In the absence of abnormal K^+ redistribution, the total deficit correlates with serum K^+ ,^{257,491,496} so that serum K^+ drops by approximately 0.27 mmol/L for every 100-mmol reduction in total-body stores. Loss of 400 to 800 mmol of body K^+ results in a reduction in serum K^+ by approximately 2.0 mmol/L⁴⁹⁵; these parameters can be used to estimate replacement goals. However, such estimates are just an approximation of the amount of K^+ replacement required to normalize plasma K^+ , with as much as a 1 in 6 risk of overreplacement²⁶⁰; serum K^+ should also be closely monitored during replacement, withdrawing or adjusting K^+ replacement if necessary.

Although the treatment of asymptomatic patients with borderline or low normal serum K^+ remains controversial, supplementation is recommended in patients with a serum K^+ lower than 3 mmol/L. In high-risk patients (e.g., those with heart failure, cardiac arrhythmias, myocardial infarction, ischemic heart disease, or taking digoxin), the serum K^+ level should be maintained at 4.0 mmol/L or higher²⁵⁵ or even 4.5 mmol/L or more.²⁶⁴ Patients with severe hepatic disease may not be able to tolerate mild to moderate hypokalemia because of the associated augmentation in ammoniogenesis, so the serum K^+ should be maintained at approximately 4.0 mmol/L.^{496,497} In asymptomatic patients with mild to moderate hypertension, an attempt should be made to maintain serum K^+ above 4.0 mmol/L,²⁵⁵ and potassium supplementation should be considered when serum K^+ falls below 3.5 mmol/L.²⁵⁵ Notably, prospective studies have shown an inverse relationship between dietary potassium intake and both fatal and nonfatal stroke, independent of the associated antihypertensive effect.^{255,498,499}

Potassium is available in the form of potassium chloride, potassium phosphate, potassium bicarbonate or its precursors (potassium citrate, potassium acetate), and potassium gluconate.^{255,256,490} Potassium phosphate is indicated when a phosphate deficit accompanies K^+ depletion (e.g., in diabetic ketoacidosis).⁴⁹⁰ Potassium bicarbonate (or its precursors) should be considered in patients with hypokalemia and metabolic acidosis.^{255,490} Potassium chloride should otherwise be the default salt of choice in most patients, for several reasons. First, metabolic alkalosis typically accompanies chloride loss from a renal (e.g., diuretics) or upper gastrointestinal route (e.g., vomiting) and contributes significantly to renal K^+ wasting.²⁵⁶ In this setting, replacing chloride along with K^+ is essential in treating the alkalosis and preventing further kaliuresis; because dietary K^+ is mainly in the form of potassium phosphate or potassium citrate, it usually does not suffice. Second, potassium bicarbonate may offset the benefits of K^+ administration by aggravating concomitant alkalosis. Third, potassium chloride raises serum K^+ at a faster rate than potassium bicarbonate, a factor that is crucial in patients with marked hypokalemia and related symptoms. In all likelihood, this faster rise in plasma K^+ occurs because Cl^- is mainly an extracellular fluid anion that does not enter

cells to the same extent as bicarbonate, keeping the K^+ in the extracellular fluid compartment.⁵⁰⁰

Parenteral (intravenous) K^+ administration should be limited to patients unable to use the enteral route or when the patient is experiencing associated signs and symptoms. However, rapid correction of hypokalemia through oral supplementation is possible and may be faster than intravenous K^+ supplementation because of limitations in the rapidity of intravenous K^+ infusion. For example, serum K^+ can be increased by 1 to 1.4 mmol/L in 60 to 90 minutes, following the oral intake of 75 mmol of K^+ ⁵⁰¹; the ingestion of approximately 125 to 165 mmol of K^+ as a single oral dose can increase serum K^+ by approximately 2.5 to 3.5 mmol/L in 60 to 120 minutes.⁵⁰² The oral route is thus both effective and appropriate in patients with asymptomatic severe hypokalemia. If the patient is experiencing life-threatening signs and symptoms of hypokalemia, however, the maximum possible IV infusion of K^+ should be administered acutely for symptom control, followed by rapid oral supplementation.

The usual intravenous dose is 20 to 40 mmol of K^+Cl^- in 1 L of vehicle solution.⁴⁹⁰ The vehicle solution should be dextrose-free to prevent a transient reduction in the serum K^+ level of 0.2 to 1.4 mmol/L because of an enhanced endogenous insulin secretion induced by the dextrose.⁵⁰³ Higher concentrations of K^+Cl^- (up to 400 mmol/L, as 40 mmol in 100 mL of normal saline) have been used in life-threatening conditions.^{504,505} In these cases, the amount of K^+ per intravenous bag should be limited (e.g., 20 mmol in 100 mL of saline solution) to prevent inadvertent infusion of a large dose.^{505,506} These solutions are best given through a large central vein. Femoral veins are preferable because infusion through upper body central lines can acutely increase the local concentration of K^+ , with deleterious effects on cardiac conduction.^{505,506} As a general rule, and to avoid venous pain, irritation, and sclerosis, concentrations of more than 60 mmol/L should not be given through a peripheral vein.⁴⁹⁰

Although the recommended rate of administration is 10 to 20 mmol/hour, rates of 40 to 100 mmol/hour or even higher (for a short period) have been used in patients with life-threatening conditions.^{504,506–508} However, a rapid increase in serum K^+ associated with electrocardiographic changes may occur with higher rates of infusion (e.g., ≥ 80 mmol/hour).⁵⁰⁹ Intravenous administration of K^+ at a rate of more than 10 mmol/hour requires continuous electrocardiographic monitoring.⁴⁹⁰ In patients receiving K^+ at such a high infusion rate, close monitoring of the appropriate physiologic consequences of hypokalemia is essential; after these effects have abated, the rate of infusion should be decreased to the standard dose of 10 to 20 mmol/hour.⁵⁰⁶ It is important to remember that volume expansion in patients with moderate to severe hypokalemia and Cl^- -responsive metabolic alkalosis should be performed cautiously and with close follow-up of serum K^+ because bicarbonaturia associated with volume expansion may aggravate renal K^+ wasting and hypokalemia.⁴⁸⁹ In patients with combined severe hypokalemia and hypophosphatemia (e.g., diabetic ketoacidosis), intravenous K^+ phosphate can be used. However, this solution should be infused at a rate of less than 50 mmol over 8 hours to prevent the risk of hypocalcemia and metastatic calcification.⁴⁸⁹ A combination of potassium phosphate and potassium chloride may be necessary to correct hypokalemia effectively in these patients.

The easiest and most straightforward method of oral K^+ supplementation is to increase the dietary intake of potassium-rich foods²⁵⁶ (Box 17.1). One study has compared the effectiveness of diet versus medication supplementation in cardiac surgery patients receiving diuretics in hospital and found no difference between the two groups in respect to the maintenance of serum K^+ . However, limitations of this study included a small number of subjects, relatively short duration, and lack of information on acid-base status, making it less than conclusive and not generalizable.⁵¹⁰

Nevertheless, dietary K^+ is mainly in the form of potassium phosphate or potassium citrate and is inadequate in most patients who have combined K^+ and Cl^- deficiency.⁵¹¹ For example, bananas contain only 2 mEq of potassium/inch, with a dominance of nonchloride anions.⁵¹¹ Most patients will therefore need to combine a high- K^+ diet with a prescribed dose of K^+Cl^- .²⁵⁶ Salt substitutes are an inexpensive and potent source of K^+Cl^- ; each gram contains 10 to 13 mmol of K^+ ⁵¹³; however, patients, particularly those with an impaired ability to excrete potassium, need to be counseled regarding the

Box 17.1 Foods With High Potassium Content

Highest Content (>1000 mg [25 mmol]/100 g)

- Dried figs
- Molasses
- Seaweed

Very High Content (>500 mg [12.5 mmol]/100 g)

- Dried fruits (dates, prunes)
- Nuts
- Avocados
- Bran cereals
- Wheat germ
- Lima beans

High Content (>250 mg [6.2 mmol]/100 g)

- Vegetables
 - Spinach
 - Tomatoes
 - Broccoli
 - Winter squash
 - Beets
 - Carrots
 - Cauliflower
 - Potatoes
- Fruits
 - Bananas
 - Cantaloupe
 - Kiwis
 - Oranges
 - Mangos
- Meats
 - Ground beef
 - Steak
 - Pork
 - Veal
 - Lamb

From Gennari FJ: Hypokalemia. *N Engl J Med.* 1998;339:451–458.

Table 17.4 Oral Preparations of Potassium Chloride

Supplement	Characteristics
Controlled-release microencapsulated tablets	Disintegrate better in stomach than encapsulated microparticles; less adherent and less cohesive
Encapsulated, controlled-release, microencapsulated particles	Fewer gastrointestinal tract erosions than with wax matrix tablets
Potassium chloride elixir	Inexpensive, tastes bad, poor patient adherence; few gastrointestinal tract erosions; immediate effect
Potassium chloride (effervescent tablets) for solution	Convenient, but more expensive than elixir; immediate effect
Wax matrix extended-release tablets	Easier to swallow; more gastrointestinal tract erosions than with microencapsulated formulas

From Cohn JN, Kowey PR, Whelton PK, et al: New guidelines for potassium replacement in clinical practice: a contemporary review by the National Council on Potassium in Clinical Practice. *Arch Intern Med.* 2000;160:2429–2436.

appropriate amount and the potential for hyperkalemia.⁵¹² Potassium chloride is also available in liquid or tablet form (Table 17.4).²⁵⁵ In general, the available preparations are well absorbed.²⁵⁶ Liquid forms are less expensive but are less well tolerated. Slow-release forms are more palatable and better tolerated; however, they have been associated with gastrointestinal ulceration and bleeding, ascribed to local accumulation of high concentrations of K^+ .^{257,257} Notably, this risk is rather low and is lower still with the microencapsulated forms.²⁵⁶ The chance of overdose and hyperkalemia is higher with slow-release formulations; unlike the immediate-release forms, these tablets are less irritating to the stomach and less likely to induce vomiting.⁵¹³ The usual dose is 40 to 100 mmol of K^+ (as K^+Cl^-)/day, bid or tid, in patients taking diuretics.²⁵⁶ (K^+Cl^- can be toxic in doses > 2 mmol/Kg).⁵¹³ This dose is effective in maintaining serum K^+ in up to 90% of patients; however, in the 10% of patients who remain hypokalemic, increasing the oral dose or adding a K^+ -sparing diuretic is an appropriate choice.²⁵⁶

In addition to potassium supplementation, strategies to minimize K^+ losses should be considered. These measures may include minimizing the dose of non- K^+ -sparing diuretics, restricting Na^+ intake, and using a combination of non- K^+ -sparing and K^+ -sparing medications (e.g., ACE inhibitors, ARBs, K^+ -sparing diuretics, beta-blockers).^{225,255} The use of a K^+ -sparing diuretic is of particular importance in hypokalemia resulting from primary hyperaldosteronism and related disorders, such as Liddle syndrome and AME; K^+ supplementation alone may be ineffective in these settings.^{514–516} In patients with hypokalemia caused by loss through upper gastrointestinal secretion (e.g., with continuous nasogastric tube suction, continuous or self-induced vomiting), proton pump inhibitors

are reportedly useful in helping correct the metabolic alkalosis and reduce hypokalemia.⁵¹⁷

HYPERKALEMIA

EPIDEMIOLOGY

Hyperkalemia is usually defined as a potassium level of 5.5 mmol/L or higher,^{518,519} although in some studies 5.0 to 5.4 mmol/L qualifies for the diagnosis.⁵²⁰ Hyperkalemia has been reported in 1.1% to 10% of all hospitalized patients,^{232,518–521} with approximately 1.0% of patients (8%–10% of hyperkalemic patients) having significant hyperkalemia (≥ 6.0 mmol/L).⁵¹⁸ Hyperkalemia has been associated with a higher mortality rate (14.3%–41%),^{29,518,519} accounting for approximately 1 death/1000 patients in one case series from the mid-1980s.⁵²² In most hospitalized patients, the pathophysiology of hyperkalemia is multifactorial, with reduced renal function, medications, older age (≥ 60 years), and hyperglycemia being the most common contributing factors.^{232,518,519}

In patients with ESKD, the prevalence of hyperkalemia is 5% to 10%.^{523–525} Data from the NephroTest cohort have indicated that the prevalence of hyperkalemia increases from 2% to 42% as the GFR decreases from 60 to 90 to less than 20 mL/min/1.73 m².^{25,26} The risk of hyperkalemia is increased in males with chronic kidney disease (CKD) and tripled by treatment with ACE inhibitors or ARBs.⁵²⁶ Hyperkalemia accounts for or contributes to 1.9% to 5% of deaths among patients with ESKD.^{232,525} Notably, however, the risk of death from hyperkalemia is reduced as CKD progresses, presumably caused by as yet uncharacterized cardiac adaptation to chronic hyperkalemia.⁵²⁷ However, multiple studies have implicated hyperkalemia with increased risk of death in ESKD, as managed by both hemodialysis⁵²⁸ and peritoneal dialysis.⁵²⁹ Hyperkalemia is the reason for emergency hemodialysis in 24% of patients with ESKD on hemodialysis,⁵²⁵ and renal failure is the most common cause of hyperkalemia diagnosed in the emergency room.⁵²³ The prevalence of marked hyperkalemia ($K^+ \geq 5.8$ mmol/L) is approximately 1% in a general medicine outpatient setting. Alarmingly, the management of outpatient hyperkalemia is often suboptimal, with approximately 25% of patients lacking any follow-up, ECGs performed in only 36% of cases, and frequent delays in repeating serum K^+ .⁵³⁰

PSEUDOHYPERKALEMIA

Factitious or pseudohyperkalemia is an artifactual increase in the serum K^+ caused by the release of K^+ during or after venipuncture. There are several potential causes for pseudohyperkalemia⁵³¹:

1. Forearm contraction,⁵³² fist clenching,²¹ or tourniquet use⁵³¹ may increase K^+ efflux from local muscle and thus raise the measured serum K^+ .
2. Thrombocytosis,⁵³³ leukocytosis,⁵³⁴ and/or erythrocytosis⁵³⁵ may also cause pseudohyperkalemia because of release from these cellular elements.
3. Acute anxiety during venipuncture may provoke a respiratory alkalosis and hyperkalemia because of redistribution.^{55,56,58,66}

This hyperkalemic response is mediated by enhanced α -adrenergic activity and counterregulated, in part, by β -adrenergic stimulation. Increased catecholamine concentrations may be as a result of the decrease in plasma bicarbonate.

4. Sample contamination with K^+ EDTA, used as a sample anticoagulant for some laboratory assays, can cause spurious hyperkalemia.⁵³⁶

There are several mechanisms for sample contamination with K^+ EDTA during blood draws or sample handling.⁵³⁶

Gross contamination with K^+ EDTA usually results in spurious hypocalcemia and hypomagnesemia; lesser contamination is less obvious, leading to the practice in some laboratories to perform EDTA assays on samples with a plasma K^+ of more than 6 mmol/L with K^+ EDTA.

5. Mechanical and physical factors may induce pseudohyperkalemia after blood has been drawn.

For example, pneumatic tube transport has been shown to induce pseudohyperkalemia in one patient with leukemia and massive leukocytosis.⁵³⁷ Cooling of blood before the separation of cells from serum or plasma is also a well-recognized cause of artifactual hyperkalemia.⁵³⁸

The converse is the risk of increased uptake of K^+ by cells at high ambient temperatures, leading to normal values for hyperkalemic patients and/or to spurious hypokalemia in patients who are normokalemic.^{29,266}

This issue is particularly important for outpatient primary practice samples that are transported off site and analyzed at a central facility²⁹; this phenomenon leads to “seasonal pseudohyperkalemia and hypokalemia,”^{29,539} with fluctuations of outpatient samples as a function of season and ambient temperature.

6. There are several hereditary subtypes of pseudohyperkalemia, caused by increase in passive K^+ permeability of erythrocytes.

Abnormal red cell morphology, varying degrees of hemolysis, and/or perinatal edema can accompany hereditary pseudohyperkalemia, whereas in many kindreds there are no overt hematologic consequences. Serum K^+ increases in pseudohyperkalemic patient samples that have been left at room temperature because of abnormal K^+ permeability of erythrocytes. Several subtypes have been defined, based on differences in the temperature dependence curve of this red cell leak pathway.^{29,540} The disorder is genetically heterogeneous, with characterized genes on chromosome 17q21 (*SLC4A1*), 16q23-ter (*PIEZO1*), and 2q35-36 (*ABCB6*).^{29,541} Initially, 11 pedigrees of patients with autosomal dominant hemolysis, pseudohyperkalemia, and temperature-dependent loss of red cell K^+ were found to have heterozygous mutations in the *SLC4A1* gene on chromosome 17q21, which encodes the band 3 anion exchanger, AE1.⁵⁴⁰ The mutations that were detected all cluster within exon 17 of the gene,⁵⁴⁰ between transmembrane domains 8 and 10 of the AE1 protein. These mutations reduce anion transport in both red cells and *Xenopus* oocytes injected with AE1, with the novel acquisition of a nonselective transport pathway for both Na^+ and K^+ . Pseudohyperkalemia in these patients thus results from a genetic event that endows AE1 with the ability to transport K^+ ; the fact that single point mutations can convert an anion exchanger to a nonselective cation channel serves to underline the narrow boundaries that separate exchangers and transporters from

ion channels.⁵⁴⁰ Notably, however, other reports have suggested that these mutations in *SLC4A1* induce cation fluxes that are independent of the AE1 anion exchanger protein—that is, mediated by other transport pathways.^{542,543}

In another study, mutations in the red cell Rh-associated glycoprotein (RhAG) have been linked to the monovalent cation leak associated with overhydrated hereditary stomatocytosis.⁵⁴⁴ These mutations cause an exaggerated cation leak in the RhAG, thought to function as an NH_3 or NH_4^+ transporter RhAG. Exaggerated red cell cation leaks are also implicated in stomatocytosis because of mutations in the mechanically activated cation channel *PIEZO1*⁵⁴⁵; disease-associated mutations give rise to mechanically activated currents that inactivate more slowly than the wild-type channel. Physiologically, calcium influx via the *PIEZO1* channel collaborates with the calcium-activated Gardos K^+ channel (encoded by *KCNN4*) in controlling red cell volume.⁵⁴¹ Not surprisingly, mutations in the *KCNN4* that increase the calcium sensitivity of the Gardos channel are associated with stomatocytosis.⁵⁴⁶ Finally, mutations in the gene encoding the ATP-binding cassette family member 6 (*ABCB6*) transporter have also been associated with familial pseudohyperkalemia.⁵⁴¹

EXCESS INTAKE OF POTASSIUM AND TISSUE NECROSIS

Increased intake of even small amounts of K^+ may provoke severe hyperkalemia in patients with predisposing factors. For example, the oral administration of 32 millimoles to a diabetic patient with hyporeninemic hypoaldosteronism resulted in an increase in serum K^+ from 4.9 mmol/L to a peak of 7.3 mmol/L, within 3 hours.⁵⁴⁷ Increased intake or changes in intake of dietary sources rich in K^+ (Box 17.1) may also provoke hyperkalemia in susceptible patients. Very rarely, marked intake of K^+ —for example in sports beverages⁵⁴⁸—may provoke severe hyperkalemia in individuals free of predisposing factors. Other occult sources of K^+ must also be considered, including salt substitutes,⁵⁴⁹ alternative medicines,⁵⁵⁰ and alternative diets.⁵⁵¹ Geophagia with ingestion of K^+ -rich clay²⁹ and cautoypreophagia⁵⁵² (ingestion of burnt matchsticks) are two forms of pica that have been reported to cause hyperkalemia in dialysis patients. Sustained-release K^+-Cl^- tablets can cause hyperkalemia in suicidal overdoses.⁵¹³ Such pills are radiopaque and may thus be seen on radiographs; whole-bowel irrigation should be used for gastrointestinal decontamination.⁵¹³ Iatrogenic causes include simple overreplacement with K^+-Cl^- , as can occur commonly in hypokalemic patients,²⁶⁰ or administration of a potassium-containing medication, such as K^+ penicillin,⁵⁵³ to a susceptible patient.

Red cell transfusion is a well-described cause of hyperkalemia, typically seen in children or in those receiving a massive transfusion. Risk factors for transfusion-related hyperkalemia include the rate and volume of the transfusion, use of a central venous infusion and/or pressure pumping, use of irradiated blood, and age of the blood infused.^{21,554} Whereas 7-day-old blood has a free K^+ concentration of about 23 mmol/L, this rises to the 50-mmol/L range in 42-day-old blood.⁵⁵⁵ Hyperkalemia is a common occurrence in patients with severe trauma, with a period prevalence of 29% in massively traumatized patients at a US military combat support hospital in Iraq.⁵⁵⁶ Although red cell and/or blood product transfusion plays an important role, this and other studies

have indicated a complex pathophysiology for resuscitative hyperkalemia, with low cardiac output, acidosis, hypocalcemia, and other factors contributing to the risk of hyperkalemia in patients with severe trauma.^{554,556}

Tissue necrosis is an important cause of hyperkalemia. Hyperkalemia because of rhabdomyolysis is particularly common due the enormous store of K^+ in muscle (see Fig. 17.1). In many cases, volume depletion, medications (statins in particular), and a metabolic predisposition contribute to the genesis of rhabdomyolysis. Hypokalemia is an important metabolic predisposing factor in rhabdomyolysis (see “Effects of Hypokalemia”); others include hypophosphatemia, hyponatremia and hyponatremia, and hyperglycemia. Patients with hypokalemia-associated rhabdomyolysis in whom redistribution is the cause of hypokalemia are at particular risk of subsequent hyperkalemia as rhabdomyolysis evolves and renal function worsens.^{21,282} Finally, massive release of K^+ and other intracellular contents may occur as a result of acute tumor lysis.⁵⁴⁷

REDISTRIBUTION AND HYPERKALEMIA

Several different mechanisms can induce an efflux of intracellular K^+ , resulting in hyperkalemia. The infusion of hypertonic mannitol or saline, but not hypertonic bicarbonate, generates an increase in serum K^+ .⁵⁵⁷ Potential mechanisms include a dilutional acidosis with a subsequent shift in K^+ , increased passive exit of K^+ because of an increase in intracellular K^+ activity from intracellular water loss, acute hemolysis, and a solvent drag effect as water exits cells.^{558,559} Regardless, severe hyperkalemia, typically with an acute dilutional hyponatremia, is a well-described complication of mannitol for the treatment or prevention of cerebral edema.^{559–561} Diabetics are prone to severe hyperkalemia in response to intravenous hypertonic glucose in the absence of adequate coadministered insulin because of a similar osmotic effect.^{562,563} Finally, a retrospective report has documented considerable increases in serum K^+ after IV contrast dye in five patients with CKD, four on dialysis and one with stage IV CKD⁵⁶⁴; again, the acute osmolar load was the likely cause of the acute hyperkalemia in these patients. The implications of this provocative study are not entirely clear; however, one would expect the development or worsening of hyperkalemia in dialysis patients exposed to large volumes of hyperosmolar contrast dye.

Several reports have appeared regarding the risk of hyperkalemia with epsilon-aminocaproic acid (Amicar),^{565–567} a cationic amino acid that is structurally similar to lysine and arginine. Cationic but not anionic amino acids induce an efflux of K^+ from cells, although the transport pathways involved are unknown.²¹ A recent resurgence in the use of cationic amino acid infusions during endoradiotherapy has served to emphasize this risk, with the development of significant hyperkalemic acidosis.⁵⁶⁸

Muscle plays a dominant role in extrarenal K^+ homeostasis, primarily via regulated uptake by the Na^+/K^+ -ATPase. Although exercise is a well-described cause of acute hyperkalemia, this effect is usually transient, and the clinical relevance is difficult to judge. ESKD patients on dialysis do not have an exaggerated increase in plasma K^+ with maximal exercise, perhaps because of greater insulin, catecholamine, and aldosterone responses to exercise and/or to their preexisting hyperkalemia.⁵⁶⁹ The results and design of this and other studies of exercise-associated hyperkalemia in ESKD have been criticized by a

more recent report, which linked abnormal extrarenal K^+ homeostasis to increased fatigue in ESKD patients.⁵⁷⁰ Regardless, however, exercise-associated hyperkalemia is not a major clinical cause of hyperkalemia. Dialysis patients are, however, susceptible to modest increases in plasma K^+ after prolonged fasting as a result of the relative insulinopenia in this setting.⁵⁷¹ This may be clinically relevant in preoperative ESKD patients, for whom intravenous glucose infusions with or without insulin are appropriate preventive measures for the development of hyperkalemia.⁵⁷¹

Insulin stimulates the uptake of K^+ by several tissues, primarily via stimulation of Na^+/K^+ -ATPase activity.^{29,43,45,48} Reduction in circulating insulin is thus an important factor or cofactor in the generation of hyperkalemia in diabetic patients. Patients with DKA typically present with serum K^+ levels that are within normal limits or moderately elevated, but with profound whole-body potassium deficits. However, significant hyperkalemia (serum $K^+ > 6$ –6.5 mmol/L) is not uncommon in DKA^{572,573} because of a variety of potential factors—insulinopenia, renal dysfunction, and the hyperosmolar effect of severe hyperglycemia.^{572–574} Inhibition of insulin secretion by the somatostatin agonist octreotide can also cause significant hyperkalemia in both anephric patients⁴⁰ and patients with normal renal function.⁴¹

Digoxin inhibits Na^+/K^+ -ATPase and thus impairs the uptake of K^+ by skeletal muscle (see “Factors Affecting Internal Distribution of Potassium”) so that digoxin overdose can result in hyperkalemia. The skin and venom gland of the cane toad *Bufo marinus* contains high concentrations of bufadienolide, structurally similar to a glycoside. The direct ingestion of such toads⁵⁷⁵ or toad extracts can result in fatal hyperkalemia. In particular, certain herbal aphrodisiac pills contain appreciable amounts of toad venom and have led to several case reports in the United States.^{21,576} Patients may have detectable plasma levels using standard digoxin assays because bufadienolide is immunologically similar to digoxin. Moreover, treatment with a digoxin-specific Fab fragment, indicated for treatment of digoxin overdoses, may be effective and lifesaving in bufadienolide toxicity.^{21,576} Finally, fluoride ions also inhibit Na^+/K^+ -ATPase so that fluoride poisoning is typically associated with hyperkalemia.⁵⁷⁷

Succinylcholine depolarizes muscle cells, resulting in the efflux of K^+ through acetylcholine receptors (AChRs) and a rapid, but usually transient, hyperkalemia. The use of this agent is contraindicated in patients who have sustained thermal trauma, neuromuscular injury (upper or lower motor neuron), disuse atrophy, mucositis, or prolonged immobilization in an intensive care unit (ICU) setting; the efflux of K^+ induced by succinylcholine is enhanced in these patients and can result in significant hyperkalemia.⁵⁷⁸ These disorders share a 2- to 100-fold upregulation of AChRs at the plasma membrane of muscle cells, with loss of the normal clustering at the neuromuscular junction.⁵⁷⁸ Depolarization of these upregulated AChRs by succinylcholine results in an exaggerated efflux of K^+ through the receptor-associated cation channels that are spread throughout the muscle cell membrane (Fig. 17.12). Concomitant upregulation of the neuronal α_7 -AChR subunit has also been observed in denervated muscle; the α_7 -containing AChR is a homomeric, pentameric channel that depolarizes in response to both succinylcholine and choline, its metabolite.⁵⁷⁸ Depolarization α_7 -AChRs in response to choline is furthermore not subject to desensitization, and

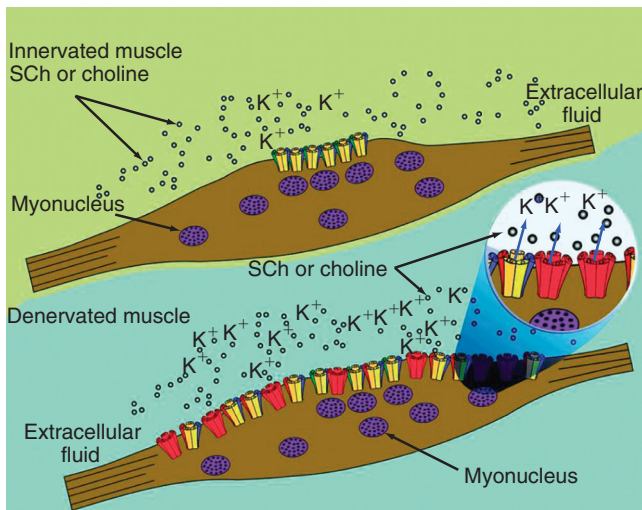


Fig. 17.12 Succinylcholine-induced efflux of potassium is increased in denervated muscle. In innervated muscle, succinylcholine (Sch) interacts with the entire plasma membrane, but depolarizes only the junctional (α_1 , β_1 , δ , and ϵ ; multicolored) acetylcholine receptors (AChRs); this leads to a modest transient hyperkalemia. With denervation, there is a considerable upregulation of muscle AChRs, with increased extrajunctional AChRs (α_1 , β_1 , δ , and ϵ ; multicolored) and acquisition of homomeric, neuronal-type α_7 -AChRs. Depolarization of denervated muscle leads to an exaggerated K^+ efflux as a result of the upregulation and redistribution of these AChRs. In addition, choline generated from metabolism of succinylcholine maintains the depolarization mediated via α_7 -AChRs, thus enhancing and prolonging the K^+ efflux after paralysis has subsided. (From Martyn JA, Richtsfeld M. Succinylcholine-induced hyperkalemia in acquired pathologic states: etiologic factors and molecular mechanisms. *Anesthesiology*. 2006;104:158–169.)

may explain in part the hyperkalemic effect that persists in some patients well after the paralytic effect of succinylcholine has subsided.⁵⁷⁸ Perhaps consistent with this neuromuscular pathophysiology, patients with renal failure do not appear to have an increased risk of succinylcholine-associated hyperkalemia.⁵⁷⁹

A report of three patients has suggested the possibility that drugs that share the ability to open K_{ATP} channels may have an underappreciated propensity to cause hyperkalemia in critically ill patients. The implicated drugs included cyclosporine, isoflurane, and nicorandil.⁵⁸⁰ These patients exhibited hyperkalemia that resisted usual therapies (insulin/dextrose $\pm$ hemofiltration), with a temporal hypokalemic response to the K_{ATP} inhibitor glibenclamide (glyburide). The daring, off-label use of glibenclamide was presumably instigated by the senior author's observation that cyclosporine activates K_{ATP} channels in vascular smooth muscle.⁵⁸¹ K_{ATP} channels are widely distributed, including in skeletal muscle, so that activation of these channels is a plausible cause of acute hyperkalemia. Other case reports have emerged of nicorandil-associated hyperkalemia.^{582,583}

Finally, beta-blockers cause hyperkalemia, in part by inhibiting cellular uptake, but also through hyporeninemic hypoaldosteronism induced by effects of these drugs on both renal renin release and adrenal aldosterone release (see “[Regulation of Renal Renin and Adrenal Aldosterone](#)”). Labetalol, a combined alpha and beta-blocker, is a particularly common cause of hyperkalemia in susceptible patients.^{21,584} However,

both nonspecific and cardiospecific beta-blockers have been shown to reduce PRA, Ang II, and aldosterone,⁵⁸⁵ so that beta blockade in general will increase susceptibility to hyperkalemia.

REDUCED RENAL EXCRETION OF POTASSIUM

Hypoaldosteronism

Aldosterone promotes kaliuresis by activating apical amiloride-sensitive Na^+ currents in the CNT and CCD, thus increasing the lumen-negative driving force for K^+ excretion (see “[Control of Potassium Excretion](#)”). Aldosterone release from the adrenals may be reduced by hyporeninemic hypoaldosteronism and its multiple causes, by medications, or because of an isolated deficiency of ACTH. The isolated loss in pituitary secretion of ACTH leads to a deficit in circulating cortisol; variable defects in other pituitary hormones are likely secondary to this reduction in cortisol.⁵⁸⁶ Concomitant hyporeninemic hypoaldosteronism is frequent²⁹; however, hyperkalemia is perhaps less common in secondary hypoaldosteronism than in Addison disease.⁵⁸⁶

Primary hypoaldosteronism may be genetic or acquired.⁵⁸⁷ The X-linked disorder adrenal hypoplasia congenita is caused by loss-of-function mutations in the transcriptional repressor *Dax-1*. Patients with adrenal hypoplasia congenita present with primary adrenal failure and hyperkalemia either shortly after birth or much later in childhood.⁵⁸⁸ This bimodal presentation pattern does not appear to be influenced by *Dax-1* genotype; rather, if patients survive the early neonatal period, they will then miss being diagnosed until much later in life, presenting with delayed puberty (see later) or with an adrenal crisis. The steroidogenic factor-1 (*SF-1*), a functional partner for *Dax-1*, is also required for adrenal development in mice and humans. Both genes are involved in gonadal development, with *Dax-1* deficiency leading to hypogonadotropic hypogonadism⁵⁸⁸ and *SF-1* deficiency causing male to female sex reversal in addition to adrenal insufficiency.

Reduced steroidogenesis causes two other important forms of primary hypoaldosteronism.⁵⁸⁷ Congenital lipoid adrenal hyperplasia (lipoid CAH) is a severe autosomal recessive syndrome characterized by impaired synthesis of mineralocorticoids, glucocorticoids, and gonadal steroids.²¹ Patients present in early infancy with adrenal crisis, including severe hyperkalemia.⁵⁸⁹ Genotypically male 46,XY patients with lipoid CAH have female external genitalia because of the developmental absence of testosterone. Lipoid CAH is caused by loss-of-function mutations in steroidogenic acute regulatory protein, a small mitochondrial protein that helps shuttle cholesterol from the outer to the inner mitochondrial membrane, thus initiating steroidogenesis⁵⁹⁰; some patients may alternatively have mutations in the side chain cleavage P450 enzyme.⁵⁹¹ The classic, salt-wasting form of CAH because of 21-hydroxylase deficiency is associated with marked reductions in both cortisol and aldosterone, leading to adrenal insufficiency.⁵⁹² Concomitant overproduction of androgenic steroids results in virilization in female patients with this form of CAH.

Isolated deficits in aldosterone synthesis with hyperreninemia are caused by loss-of-function mutations in aldosterone synthase, although genetic heterogeneity has been reported.⁵⁹³ Patients typically present in childhood with volume depletion and hyperkalemia.⁵⁹⁴ Much like pseudohypoaldosteronism

because of loss-of-function mutations in the MR (see later), patients tend to become asymptomatic in adulthood. Acquired hyperreninemic hypoaldosteronism has been described in critical illness,²¹ type 2 diabetes,⁵⁹⁵ amyloidosis because of familial Mediterranean fever,⁵⁹⁶ and after metastasis of carcinoma to the adrenal gland.²¹ Finally, aldosterone synthesis is selectively reduced by heparin, with a 7% incidence of hyperkalemia associated with heparin therapy.⁵⁹⁷ Both unfractionated⁵⁹⁷ and low-molecular-weight^{21,598} heparin can cause hyperkalemia. Hyperkalemia because of prophylactic subcutaneous unfractionated heparin (5000 U bid) has also been reported.⁵⁹⁹ Heparin reduces the adrenal aldosterone response to Ang II and hyperkalemia, resulting in hyperreninemic hyperaldosteronism. Histologic findings in experimental animals include a marked diminution in size of the zona glomerulosa and an attenuated hyperplastic response to salt depletion.⁵⁹⁷

Most primary adrenal insufficiency is as a result of autoimmunity in Addison disease or in the context of a polyglandular endocrinopathy.^{587,600} Adrenal insufficiency can be seen following adrenalectomy for primary hyperaldosteronism, with 14% developing postoperative hyperkalemia and 5% developing long-term insufficiency requiring fludrocortisone.⁶⁰¹ The antiphospholipid syndrome may also cause bilateral adrenal hemorrhage and adrenal insufficiency.⁶⁰² Another renal syndrome in which there should be a high index of suspicion for adrenal insufficiency is renal amyloidosis.⁶⁰³ Finally, HIV is a particularly important infectious cause of adrenal insufficiency. The most common cause of adrenalitis in HIV disease is cytomegalovirus (CMV); however, a long list of infectious, degenerative, and infiltrative processes may involve the adrenal glands in these patients.⁶⁰⁴ Although the adrenal involvement in HIV is usually subclinical, adrenal insufficiency may be precipitated by stress, drugs such as ketoconazole that inhibit steroidogenesis, or the acute withdrawal of steroid agents such as megestrol.

Current estimates of the risk of hyperkalemia with Addison disease are lacking, but the incidence is likely 50% to 60%.²¹ The interaction between dietary Na⁺ and K⁺ intake and sweat Na⁺ loss is especially clinically relevant in physically active individuals with Addison disease in humid environments. Mineralocorticoid supplementation may be necessary in these active individuals. The absence of hyperkalemia in such a high percentage of typical hypoadrenal patients underscores the importance of aldosterone-independent modulation of K⁺ excretion by the distal nephron. A high-K⁺ diet and high peritubular K⁺ serves to increase apical Na⁺ reabsorption and K⁺ secretion in the CNT and CCD (see “Control of Potassium Excretion”). In most patients with reductions in circulating aldosterone, this homeostatic mechanism would appear to be sufficient to regulate plasma K⁺ to within normal limits.

Hyporeninemic Hypoaldosteronism

Hyporeninemic hypoaldosteronism⁶⁰⁵ is a very common predisposing factor in several large, overlapping subsets of hyperkalemic patients—diabetics,⁶⁰⁶ older adults,^{21,179,607} and patients with renal insufficiency.²¹ Hyporeninemic hypoaldosteronism has also been described in systemic lupus erythematosus (SLE),^{21,608} multiple myeloma,²⁹ and acute glomerulonephritis.²⁹ Classically, patients should have suppressed plasma renin activity (PRA) and aldosterone, which cannot be activated by typical maneuvers such as furosemide

or sodium restriction.⁶⁰⁵ Approximately 50% have an associated acidosis, with a reduced renal excretion of NH₄⁺, a positive urinary anion gap, and urine pH less than 5.5.^{187,254} Although the generation of this acidosis is clearly multifactorial,⁶⁰⁹ strong clinical^{253,254,610} and experimental²⁴⁹ evidence has suggested that hyperkalemia per se is the dominant factor as a result of competitive inhibition of NH₄⁺ transport in the TAL and reduced distal excretion of NH₄⁺.⁶¹¹ (See also “Consequences of Hyperkalemia.”)

Several factors account for the reduced PRA in diabetic patients with hyporeninemic hypoaldosteronism.⁶⁰⁶ First, many patients have an associated autonomic neuropathy, with impaired release of renin during orthostatic challenges.²¹ Failure to respond to isoproterenol with an increase in PRA, despite an adequate cardiovascular response, suggests a postreceptor defect in the ability of the juxtaglomerular apparatus to respond to β-adrenergic stimuli²¹ (see also “Regulation of Renal Renin and Adrenal Aldosterone”). Second, the conversion of prorenin to active renin is impaired in some diabetics,⁶⁰⁶ despite adequate release of prorenin in response to furosemide²¹; this suggests a defect in the normal processing of prorenin. Third, as is the case with perhaps all patients with hyporeninemic hypoaldosteronism (see later), many diabetic patients appear to be volume-expanded, with subsequent suppression of PRA.

The most attractive unifying hypothesis for the suppression of PRA in hyporeninemic hypoaldosteronism is that primary volume expansion increases circulating ANP levels, which then exerts a negative effect on renal renin and adrenal aldosterone release (see also “Regulation of Renal Renin and Adrenal Aldosterone”). There is evidence that these patients are volume-expanded, and many will respond to Na⁺-Cl⁻ restriction or to furosemide with an increased PRA—that is, renin is physiologically rather than pathologically suppressed.^{612–614} Patients with hyporeninemic hypoaldosteronism caused by a diversity of underlying causes have elevated ANP levels,^{21,29,179,613,615} which is also an indicator of their underlying volume expansion. Patients who respond to furosemide with an increase in PRA exhibit a concomitant decrease in ANP.⁶¹³ Furthermore, the infusion of exogenous ANP can suppress the adrenal aldosterone response to hyperkalemia¹⁷⁹ and dietary Na⁺-Cl⁻ depletion.⁶¹⁶

Acquired Tubular Defects and Potassium Excretion

Unlike hyporeninemic hypoaldosteronism, hyperkalemic distal renal tubular acidosis is associated with a normal or increased aldosterone and/or PRA. Urine pH in these patients is more than 5.5, and they are unable to increase acid or K⁺ excretion in response to furosemide, Na⁺-SO₄²⁻, or fludrocortisone.^{617–619} Classic causes include SLE,⁶¹⁷ sickle cell anemia,^{21,619} and amyloidosis.²¹

Hereditary Tubular Defects and Potassium Excretion

Hereditary tubular causes of hyperkalemia have overlapping clinical features with hypoaldosteronism—hence, the shared label pseudohypoaldosteronism (PHA). PHA-I has both an autosomal recessive and autosomal dominant form. The autosomal dominant form is caused by loss-of-function mutations in the mineralocorticoid receptor.⁶²⁰ These patients require aggressive salt supplementation during early childhood; however, similar to the hypoaldosteronism caused by

mutations in aldosterone synthase, they typically become asymptomatic in adulthood.⁴⁰⁷ Of interest, the lifelong increases in circulating aldosterone, Ang II, and renin seen in this syndrome do not appear to have untoward cardiovascular consequences.⁶²⁰

The recessive form of PHA-I is caused by various combinations of mutations in all three subunits of ENaC, resulting in impairment in its channel activity.⁴⁰⁷ Patients with this syndrome present with severe neonatal salt wasting, hypotension, and hyperkalemia; in contrast to the autosomal dominant form of PHA-I, the syndrome does not improve in adulthood.⁴⁰⁷ One unexpected result in the physiologic characterization of ENaC was that mice with a targeted deletion of the α -ENaC subunit were found to die within 40 hours of birth caused by pulmonary edema.²⁹ Patients with recessive PHA-I may have pulmonary symptoms, which can occasionally be very severe.⁶²¹ However, it appears that unlike in ENaC-deficient mice, the modest residual activity associated with heteromeric PHA-I channels is generally sufficient to mediate pulmonary Na^+ and fluid clearance in humans with loss-of-function mutations in ENaC.⁶²²

Pseudohypoaldosteronism type II (PHA-II) (also known as Gordon syndrome and, more recently, as FHHT) is in every respect the mirror image of GS; the clinical phenotype includes hypertension, hyperkalemia, hyperchloremic metabolic acidosis, suppressed PRA and aldosterone, hypercalciuria, and reduced bone density.⁶²³ FHHT behaves like a gain of function in the thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransporter NCC, and treatment with thiazides typically results in resolution of the entire clinical picture.⁶²³ FHHT is an extreme form of hyporeninemic hypoaldosteronism caused by volume expansion; aggressive salt restriction decreases ANP levels and increases PRA, with resolution of the hypertension, hyperkalemia, and metabolic acidosis.⁶¹⁵

FHHT is an autosomal dominant syndrome, with four genetic loci. In an initial landmark paper, mutations in two related serine-threonine kinases were detected in various kindreds with FHHT.⁶²⁴ The catalytic sites of these kinases lack specific catalytic lysines conserved in other kinases—hence, the designation WNK. Whereas FHHT mutations in WNK4 affect the C-terminus of the coding sequence, large intronic deletions in the WNK1 gene result in increased expression. Both kinases are expressed in the distal nephron, in both DCT and CCD cells; whereas WNK1 localizes to the cytoplasm and basolateral membrane, WNK4 protein is found at the apical tight junctions.⁶²⁴ WNK-dependent phosphorylation and activation of the downstream SPAK (STE20/SPS1-related proline/alanine-rich kinase) and OSR1 (oxidative stress-responsive kinase 1) kinases lead to phosphorylation of a cluster of N-terminal threonines in NCC, resulting in an activation of $\text{Na}^+\text{-Cl}^-$ cotransport¹⁴⁴ (see also Fig. 17.6). However, coexpression of WNK4 with NCC reveals an additional inhibitory influence of the kinase on NCC, effects that are blocked by FHHT-associated point mutations in the kinase.⁶²⁵ In particular, the inhibitory effects of WNK4 appear to dominate in mouse models with overexpression of wild-type versus FHHT mutant WNK4.⁷³ Mutations in the *Cullin 3* and *kelch-like 3* (*KLHL3*) genes also cause FHHT; the proteins encoded by these genes are part of a ubiquitin ligase complex that regulates the WNK1 and WNK4 kinase proteins. Autosomal dominant mutations in both *Cullin 3* and *KLHL3* act through dominant negative effects.^{626,627}

A key insight from the mechanistic study of FHHT is that the activation of NCC in the DCT in this syndrome serves to reduce Na^+ delivery to principal cells in the downstream CNT and CCD, leading to hyperkalemia.⁷³ This and other effects of the WNK pathways on distal K^+ secretion are discussed earlier in this chapter (see “Control of Potassium Secretion: Effect of Potassium Intake”).

MEDICATION-RELATED HYPERKALEMIA

Cyclooxygenase Inhibitors

Hyperkalemia is a well-recognized complication of NSAIDs that inhibit cyclooxygenases. NSAIDs cause hyperkalemia by a variety of mechanisms, as would be predicted from the relevant physiology. By decreasing the GFR and increasing sodium retention, they decrease distal delivery of Na^+ and reduce the distal flow rate. Moreover, the flow-activated apical maxi-K channel in the CNT and CCD is activated by prostaglandins⁶²⁸; hence, NSAIDs will reduce its activity and the flow-dependent component of K^+ excretion.^{77,443} NSAIDs are also a classic cause of hyporeninemic hypoaldosteronism.^{629,630} The administration of indomethacin to normal volunteers thus attenuates furosemide-induced increases in PRA.^{165,631} Finally, NSAIDs would not cause hyperkalemia with such regularity if they did not also blunt the adrenal response to hyperkalemia, which is at least partially dependent on prostaglandins acting through prostaglandin EP2 receptors and cAMP.¹⁷⁸

The physiology reviewed earlier in this chapter (see “Regulation of Renal Renin and Adrenal Aldosterone”) would suggest that COX-2 inhibitors are equally likely to cause hyperkalemia. Indeed, COX-2 inhibitors can clearly cause sodium retention and a decrease in the GFR,^{632,633} suggesting NSAID-like effects on renal pathophysiology. COX-2–derived prostaglandins stimulate renal renin release,²¹ and COX-2 inhibitors reduce PRA in both dogs²⁹ and humans.¹⁶⁵ Salt restriction potentiates the hyperkalemia seen in dogs treated with COX-2 inhibitors²⁹ so that hypovolemic patients may be particularly prone to hyperkalemia in this setting. The COX-2 inhibitor celecoxib and the nonselective NSAID ibuprofen have equivalent negative effects on K^+ excretion after a defined oral load.⁶³⁴ Not surprisingly, hyperkalemia and acute kidney injury are associated with COX-2 inhibitors.^{21,635,636} Where the data have been reported, circulating PRA and/or aldosterone levels have been reduced in hyperkalemia associated with COX-2 inhibitors.^{29,636}

Cyclosporine and Tacrolimus

Both cyclosporine⁶³⁷ (CsA) and tacrolimus⁶³⁸ cause hyperkalemia; the risk of sustained hyperkalemia may be higher in renal transplant patients treated with tacrolimus than in those treated with CsA.⁶³⁹ CsA is perhaps the most versatile of all drugs in the variety of mechanisms whereby it causes hyperkalemia. It causes hyporeninemic hypoaldosteronism,⁶⁴⁰ due in part to its inhibitory effect on COX-2 expression in the macula densa.⁶⁴¹ CsA inhibits apical SK secretory K^+ channels in the distal nephron⁶⁴² in addition to basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$.²¹ Finally, CsA causes redistribution of K^+ and hyperkalemia, particularly when used in combination with beta-blockers.⁶⁴³ A provocative but preliminary report has linked acute hyperkalemia secondary to CsA to indirect activation of K_{ATP} channels (also see earlier).⁵⁸⁰

Epithelial Na⁺ Channel Inhibition

Inhibition of apical ENaC activity in the distal nephron by amiloride and other K⁺-sparing diuretics predictably results in hyperkalemia. Amiloride is structurally similar to the antibiotics TMP and pentamidine, which can also inhibit ENaC.⁶⁴⁴⁻⁶⁴⁶ TMP thus inhibits Na⁺ reabsorption and K⁺ secretion in perfused CCDs.⁶⁴⁷ Both TMP-SMX (Bactrim) and pentamidine were reported to cause hyperkalemia during high-dose treatment of *Pneumocystis* pneumonia in HIV patients,^{21,646} who are otherwise predisposed to hyperkalemia. However, this side effect is not restricted to high-dose intravenous therapy. In a study of hospitalized patients treated with standard doses of TMP, significant hyperkalemia occurred in more than 50%, with severe hyperkalemia (>5.5 mmol/L) in 21%.⁶⁴⁸ Risk factors for hyperkalemia because of normal-dose TMP include renal insufficiency,⁶⁴⁸ hyporeninemic hypoaldosteronism,⁶⁴⁹ and concomitant use of ACE inhibitors and ARBs.⁶⁵⁰ This is not a trivial association, in that TMP-SMX administration increases the risk of sudden death in patients treated with ACE inhibitors, ARBs, or spironolactone.⁶⁵¹

Whereas TMP and pentamidine directly inhibit ENaC, a novel indirect mechanism for ENaC inhibition—associated hyperkalemia has also been reported.^{21,112} Aldosterone induces expression of the membrane-associated proteases CAP1–3 (see “Control of Potassium Excretion”). Nafamostat, a protease inhibitor that is used for pancreatitis, especially in Japan, and during hemodialysis as an anticoagulant to prevent the proteolysis of fibrinogen into fibrin, is known to cause hyperkalemia.¹¹² Indirect evidence has suggested that the mechanism involves inhibition of amiloride-sensitive Na⁺ channels in the CCD.²¹ Treatment of rats with nafamostat was also shown to reduce the urinary excretion of CAP1/prostasin, in contrast to the reported effect of aldosterone.¹⁰⁶ Thus, inhibition of the protease activity of CAP1 by nafamostat appears to abrogate its activating effect on ENaC (Fig. 17.13) and may reduce expression of the protein in the CCD.¹¹³

Angiotensin-Converting Enzyme Inhibitors and Mineralocorticoid and Angiotensin Antagonists

Hyperkalemia is a predictable and common effect of ACE inhibition, direct renin inhibition, and antagonism of the mineralocorticoid and angiotensin receptors⁶⁵² (Fig. 17.14). The oral contraceptive agent Yasmin-28 and related products contain the progestin drospirenone, which inhibits the mineralocorticoid receptor⁶⁵³ and can potentially cause hyperkalemia in susceptible patients. As with many other causes of hyperkalemia, that induced by pharmacologic targeting of the RAAS axis depends on concomitant inhibition of the adrenal aldosterone release by hyperkalemia. The adrenal release of aldosterone because of increased K⁺ is clearly dependent on an intact adrenal RAAS, such that this response is abrogated by systemic ACE inhibitors and ARBs¹⁷³ (see “Regulation of Renal Renin and Adrenal Aldosterone”). Dual treatment with lisinopril and spironolactone in subjects with CKD is also associated with a reduction in extrarenal potassium disposition, given that reduced K⁺ excretion alone does not explain the substantial increase in serum K⁺ after a defined oral potassium load.⁶⁵⁴ Similarly, the addition of spironolactone to losartan in the treatment of diabetic nephropathy causes a significant increase in serum K⁺ without significant changes in urinary K⁺ excretion.⁶⁵⁵ ACE inhibitors

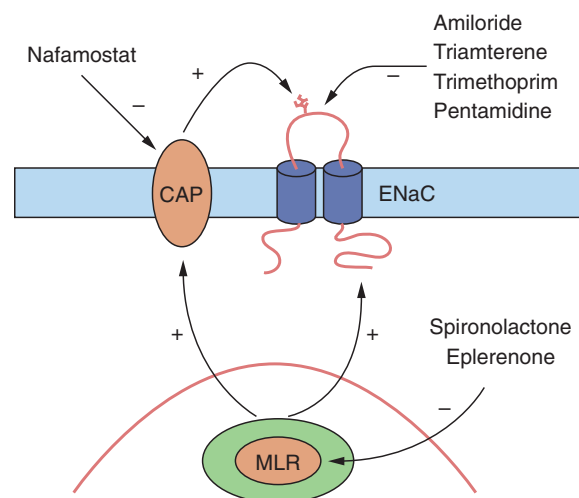


Fig. 17.13 Pharmacologic inhibition of the epithelial Na⁺ channel (ENaC). Whereas amiloride and related compounds directly inhibit the channel, the protease inhibitor nafamostat inhibits membrane-associated proteases such as channel-activating protease (CAP), thus indirectly inhibiting the channel. Spironolactone and related drugs inhibit the mineralocorticoid receptor (MLR), thus reducing transcription of the alpha subunit of ENaC, the ENaC-activating kinase SGK, and several other target genes (see text for details).

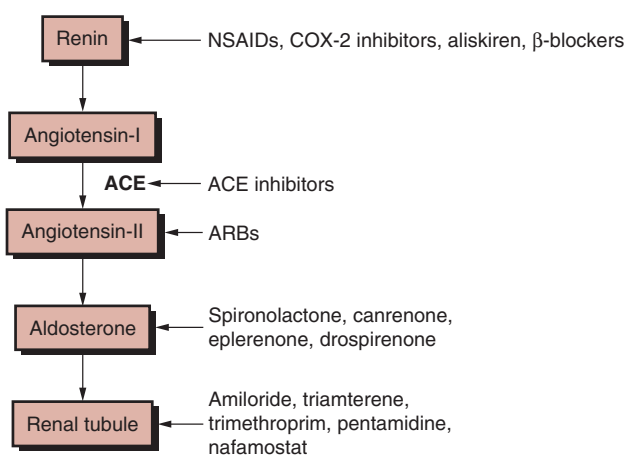


Fig. 17.14 Medications that target the renin-angiotensin-aldosterone axis are common causes of hyperkalemia, as are drugs that inhibit epithelial Na⁺ channels (ENaC) in the renal tubule (connecting segment [CNT] or cortical collecting duct [CCD]). ACE, Angiotensin-converting enzyme; ARB, angiotensin-receptor blocker; COX-2, cyclooxygenase-2; NSAIDs, nonsteroidal inflammatory drugs.

and ARBs have the additional potential to cause acute renal failure and acute hyperkalemia in patients with an angiotensin-dependent GFR; the renin inhibitor aliskiren has also been reported to cause acute renal failure with acute hyperkalemia, albeit in conjunction with spironolactone.⁶⁵⁶

RAAS inhibitors are an increasingly important cause of hyperkalemia, given the indications to combine spironolactone with ACE inhibitors and/or ARBs in renal and cardiac disease,⁶⁵⁷ in addition to the emergence of mineralocorticoid receptor antagonists with perhaps a greater potential for hyperkalemia.⁶⁵⁸ Hyperkalemia can occur within 1 week of starting angiotensin receptor blockade.⁶⁵⁹ Heart failure,

diabetes, and CKD increase the risk of hyperkalemia from these agents.^{652,660,661} The prevalence of hyperkalemia associated with the combined use of mineralocorticoid receptor antagonists and ACE inhibitors/ARBs appears to be much higher in clinical practice (~10%⁶⁶²) than what has been reported in large clinical trials,⁶⁵² in part because of the use of higher than recommended doses.²¹ Notably, Juurlink and colleagues have studied the correlation between the rate of spironolactone prescription for Canadian patients with heart failure on ACE inhibitors following the publication of the Randomized Aldactone Evaluation Study (RALES), with hyperkalemia and associated morbidity.^{663,664} This provocative study found an abrupt increase in the rate of prescribing for spironolactone after the release of RALES, with a temporal correlation to increases in the rate of admissions with hyperkalemia.⁶⁶⁴ The association remained statistically significant for admissions where hyperkalemia was the primary diagnosis.⁶⁶⁵ However, a study from the United Kingdom found a similar increase in spironolactone use after the publication of RALES, but without an increase in hyperkalemia or hyperkalemia-associated admissions to hospital.⁶⁶⁶ It should also be emphasized that the development of hyperkalemia, or for that matter the presence of predisposing factors for hyperkalemia, does not appear to mitigate the mortality benefits of eplerenone in heart failure.⁶⁶¹

Given the mounting evidence supporting the combined use of ACE inhibitors, ARBs, and/or mineralocorticoid receptor antagonists, it is prudent to adhere systematically to measures that will minimize the chance of associated hyperkalemia, therefore allowing patients to benefit from the cardiovascular and renal effects of these agents. The patients at risk for the development of hyperkalemia in response to drugs that target the RAAS axis, singly or in combination therapy, are those in whom the ability of kidneys to excrete the potassium load is markedly diminished because of one or a combination of the following: (1) decreased delivery of sodium to the CCD (e.g., as in congestive heart failure, volume depletion); (2) decreased circulating aldosterone (e.g., hyporeninemic hypoaldosteronism, drugs such as heparin or ketoconazole); (3) inhibition of amiloride-sensitive Na⁺ channels in the CNT and CCD by coadministration of TMP-SMX, pentamidine, or amiloride; (4) chronic tubulointerstitial disease, with associated dysfunction of the distal nephron; and (5) increased potassium intake (e.g., salt substitutes, diet). Overall, patients with diabetes, heart failure, and/or CKD are at particular risk for hyperkalemia from RAAS inhibition.^{652,660,661} In these susceptible patients, the following approach is recommended to prevent or minimize the occurrence of hyperkalemia in response to medications that interfere with the RAAS^{29,667}:

1. Estimate the GFR using the MDRD and/or related equations.
2. Inquire about diet and dietary supplements (e.g., salt substitutes, licorice) and prescribe a low-potassium diet, preferably with the assistance of a nutritionist.
3. Inquire about medications, particularly those that can interfere with renal K⁺ excretion (e.g., NSAIDs, COX-2 inhibitors, K⁺-sparing diuretics) and, if appropriate, discontinue these agents.
4. Continue or initiate loop or thiazide-like diuretics, if otherwise appropriate (e.g., hypertension, edema).

5. Correct acidosis with sodium bicarbonate.
6. Initiate treatment with a low dose of only one of the agents (e.g., ACE inhibitor, ARB, mineralocorticoid receptor antagonist).
7. Check serum K⁺ 3 to 5 days after initiation of therapy and each dose increment, at most within 1 week,⁶⁵⁹ followed by another measurement 1 week later.
8. If the serum K⁺ is more than 5.6 mmol/L, ACE inhibitors, ARBs, and/or mineralocorticoid receptor blockers should be stopped and the patient should be treated for hyperkalemia.
9. If the serum K⁺ is increased but less than 5.6 mmol/L, reduce the dose and reassess the possible contributing factors. If the patient is on a combination of ACE inhibitors, ARBs, and/or mineralocorticoid receptor blockers, all but one should be stopped and the potassium level rechecked.
10. A combination of a mineralocorticoid receptor blocker and an ACE inhibitor or ARB should be used with extreme caution in patients with CKD stage IV or V (eGFR < 30 mL/min/1.73 m²).
11. The dose of spironolactone in combination with ACE inhibitors or ARBs should be no more than 25 mg/day.

CLINICAL APPROACH TO HYPERKALEMIA

The first priority in the management of hyperkalemia is to assess the need for emergency treatment (electrocardiographic changes, K⁺ ≥ 6.5 mmol/L). This should be followed by a comprehensive workup to determine the cause (Fig. 17.15). The history and physical examination should focus on medications (e.g., ACE inhibitors, NSAIDs, TMP-SMX), diet and dietary supplements (e.g., salt substitute), risk factors for kidney failure, reduction in urine output, blood pressure, and volume status. Initial laboratory tests should include electrolytes, urea, creatinine, plasma osmolality, Mg²⁺, and Ca²⁺, a complete blood count, and urinary pH, osmolality, creatinine, and electrolytes. Plasma and urine osmolality are required for calculation of the TTKG. The determination of urinary electrolytes for calculation of the TTKG or urine K⁺-to-creatinine ratio also provides the opportunity for the measurement of urinary Na⁺, which will determine whether significant prerenal stimuli are limiting distal Na⁺ delivery and thus K⁺ excretion (see “Urinary Indices of Potassium Excretion”). Plasma renin activity, plasma aldosterone, and assessing the response in TTKG or urinary K⁺-to-creatinine ratio to fludrocortisone, after a few hours, may be necessary to determine the specific cause of an inappropriately low TTKG in hyperkalemia.

MANAGEMENT OF HYPERKALEMIA

Indications for the hospitalization of patients with hyperkalemia are poorly defined, in part because there is no universally accepted definition for mild, moderate, or severe hyperkalemia. The clinical sequelae of hyperkalemia, which are primarily cardiac and neuromuscular, depend on many other variables (e.g., plasma calcium level, acid-base status, plasma K⁺ rate of change²³⁴⁻²³⁷), in addition to the absolute value of the serum K⁺.^{521,668} These issues are likely to influence management decisions. Severe hyperkalemia (serum K⁺ ≥ 8.0 mmol/L), electrocardiographic changes other than peaked T waves,

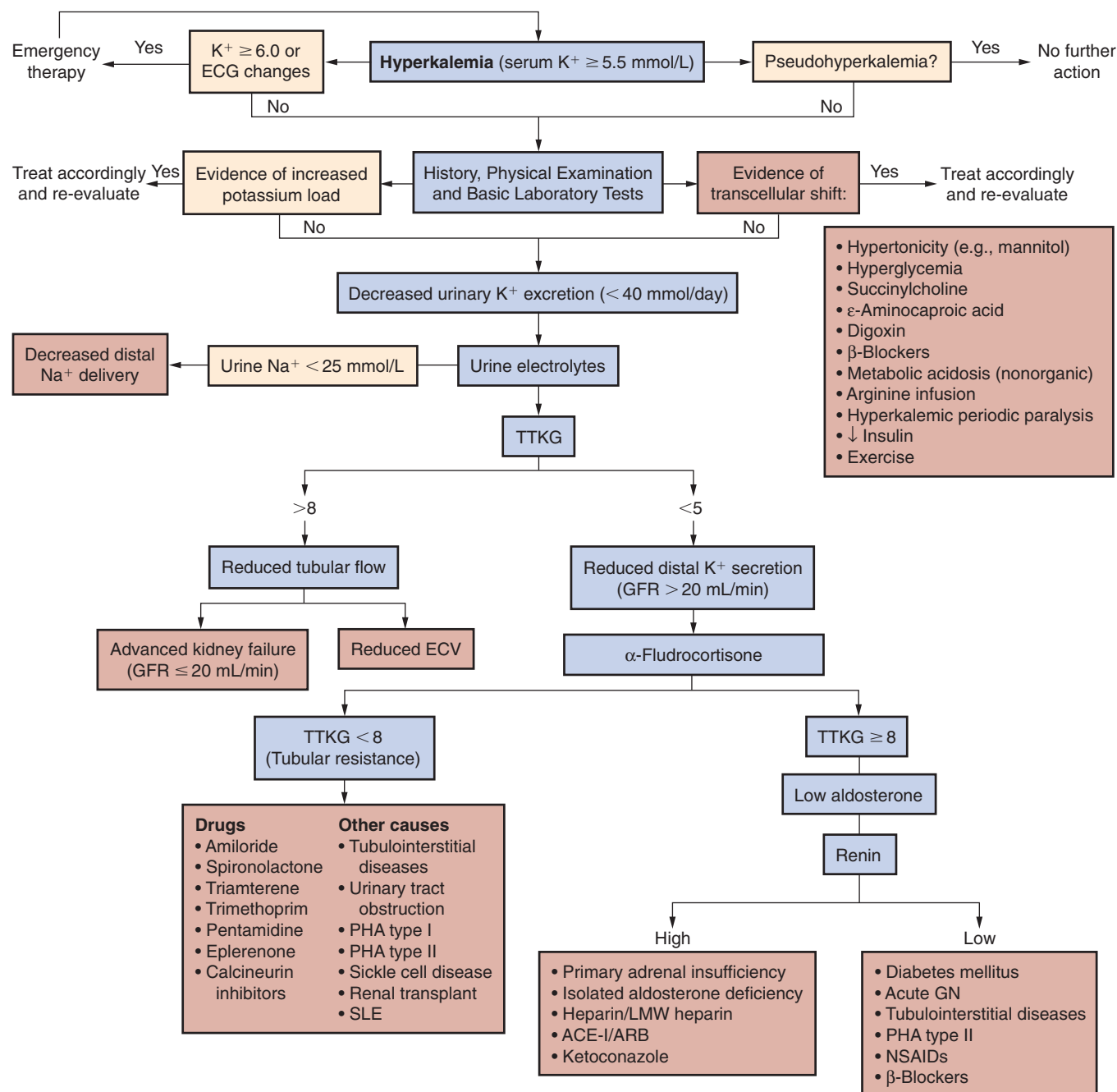


Fig. 17.15 The clinical approach to hyperkalemia. See text for details. ACE-I, Angiotensin-converting enzyme inhibitor; ARB, angiotensin-receptor blocker; ECG, electrocardiogram; ECV, effective circulatory volume; GFR, glomerular filtration rate; acute GN, acute glomerulonephritis; LMW heparin, low-molecular-weight heparin; NSAIDs, nonsteroidal antiinflammatory drugs; PHA, pseudohypoaldosteronism; SLE, systemic lupus erythematosus; TTKG, transtubular potassium gradient.

acute deterioration of renal function, and the existence of additional medical problems have been suggested as appropriate criteria for hospitalization.⁵²¹ However, hyperkalemia in patients with any electrocardiographic manifestation should be considered a true medical emergency and treated urgently^{229,520,669,670}; adequate management and serial monitoring of serum K^+ will generally require admission. Given the limitations of electrocardiographic changes as a predictor of cardiac toxicity (see “Consequences of Hyperkalemia”), patients with severe hyperkalemia ($K^+ \geq 6.5$ – 7.0 mmol/L) in

the absence of electrocardiographic changes should be aggressively managed.^{195,229,520,670–672}

Urgent management of hyperkalemia constitutes a 12-lead ECG, admission to the hospital, continuous electrocardiographic monitoring, and immediate treatment. The treatment of hyperkalemia is generally divided into three categories: (1) antagonism of the cardiac effects of hyperkalemia; (2) rapid reduction in K^+ by redistribution into cells; and (3) removal of K^+ from the body. The necessary measures to treat the underlying conditions causing hyperkalemia should

be undertaken to minimize the factors that are contributing to hyperkalemia and to prevent future episodes.²²⁹ Dietary restriction (usually 60 mEq/day) with emphasis on K^+ content of total parenteral nutrition (TPN) solutions and enteral feeding products (typically 25–50 mmol/L) and adjustment of medications and intravenous fluids are necessary; hidden sources of K^+ , such as intravenous antibiotics,⁵⁵³ should not be overlooked.

ANTAGONISM OF CARDIAC EFFECTS

Intravenous calcium is the first-line drug in the emergency management of hyperkalemia, even in patients with normal calcium levels. The mutually antagonistic effects of calcium and K^+ on the myocardium and the protective role of Ca^{2+} in hyperkalemia have long been known.⁶⁷³ Calcium raises the action potential threshold to a less negative value, without changing the resting membrane potential; by restoring the usual 15-mV difference between resting and threshold potentials, myocyte excitability is reduced.^{195,674} Administration of calcium also alters the relationship between V_{max} and the resting membrane potential, maintaining a more normal V_{max} at less negative resting membrane potentials and thus restoring myocardial conduction.¹⁹⁵

Calcium is available as calcium chloride or calcium gluconate (10-mL ampules of 10% solutions) for intravenous infusion. Each milliliter of 10% calcium gluconate or calcium chloride has 8.9 mg (0.22 mmol) and 27.2 mg (0.68 mmol) of elemental calcium, respectively.⁶⁷⁵ Calcium gluconate⁶⁷⁶ is less irritating to the veins and can be used through a peripheral intravenous (IV) line; calcium chloride can cause tissue necrosis if it extravasates and requires a central line. The efficacy and onset of action are equivalent for calcium chloride and calcium gluconate, despite the higher molar calcium concentration of calcium chloride ampules.⁶⁷⁷ A study of patients undergoing cardiac surgery with extracorporeal perfusion (with concomitant high gluconate infusion) has suggested that the increase in the ionized calcium level is significantly lower with calcium gluconate.⁶⁷⁸ This finding was attributed to a requirement for hepatic metabolism in the release of ionized calcium from calcium gluconate, so that less ionized calcium would be bioavailable in cases of liver failure or diminished hepatic perfusion.⁶⁷⁸ However, further in vitro studies in animals, in humans with normal hepatic function, and during the anhepatic stage of liver transplantation have shown equal and rapid dissociation of ionized calcium from equal doses of calcium chloride and calcium gluconate, indicating that the release of ionized calcium from calcium gluconate is independent of hepatic metabolism.^{29,677}

The recommended dose is 10 mL of 10% calcium gluconate (3–4 mL of calcium chloride), infused intravenously over 2 to 3 minutes and under continuous electrocardiographic monitoring. The effect of the infusion starts in 1 to 3 minutes and lasts 30 to 60 minutes.^{525,672} The dose should be repeated if there is no change in electrocardiographic findings or if they recur after initial improvement.^{525,672,677} However, calcium should be used with extreme caution in patients taking digitalis because hypercalcemia potentiates the toxic effects of digitalis on the myocardium.⁶⁷⁶ In this case, 10 mL of 10% calcium gluconate should be added to 100 mL of D5W and infused over 20 to 30 minutes to avoid hypercalcemia and allow for an even distribution of calcium in the extracellular

compartment.^{523,671,674} To prevent the precipitation of calcium carbonate, calcium should not be administered in solutions containing bicarbonate.

REDISTRIBUTION OF K^+ INTO CELLS

Sodium bicarbonate, beta-2 agonists, and insulin with glucose are all used in the treatment of hyperkalemia to induce a redistribution of K^+ . Of these treatments, insulin with glucose is the most constant and reliable, whereas bicarbonate is the most controversial. However, they are all temporary measures and should not be substituted for the definitive therapy of hyperkalemia, which is removal of K^+ from the body.

Insulin and Glucose

Insulin has the ability to lower the plasma K^+ level by shifting K^+ into cells, particularly into skeletal myocytes and hepatocytes (see “Factors Affecting Internal Distribution of Potassium”). This effect is reliable, reproducible, dose-dependent,⁵²⁵ and effective, even in patients with CKD and ESKD^{679–681} and in the anhepatic stage of liver transplantation.⁶⁸² The effect of insulin on plasma K^+ levels is independent of age, adrenergic activity,⁶⁸³ and its hypoglycemic effect, which may be impaired in patients with CKD and/or ESKD.²⁹

Insulin can be administered with glucose as a constant infusion or as a bolus injection.^{680,681} One recommended dose for insulin with glucose infusion is 10 U of regular insulin in 500 mL of 10% dextrose, given over 60 minutes (there is no further drop in plasma K^+ after 90 minutes of insulin infusion).^{674,683} However, a bolus injection is easier to administer, particularly under emergency conditions.²²⁹ The recommended dose is 10 U of regular insulin administered intravenously followed immediately by 50 mL of 50% dextrose (25 g of glucose).^{669,680,684,685} The effect of insulin on the K^+ level begins in 10 to 20 minutes, peaks at 30 to 60 minutes, and lasts for 4 to 6 hours.^{525,671,680,686} In almost all patients, the serum K^+ drops by 0.5 to 1.2 mmol/L after this treatment.^{681,682,685,686} The dose can be repeated as necessary.

Despite glucose administration, hypoglycemia may occur in up to 75% of patients treated with the bolus regimen described previously, typically 1 hour after the infusion.⁶⁸⁰ The likelihood of hypoglycemia is greater when the dose of glucose given is less than 30 g.⁵²³ To prevent this, infusion of 10% dextrose at 50 to 75 mL/hour and close monitoring of the blood glucose levels is recommended.^{669,684} Administration of glucose without insulin is not recommended because the endogenous insulin release may be variable.⁵⁷¹ Glucose in the absence of insulin may increase the plasma K^+ by increasing plasma osmolality.^{562,563,684} In hyperglycemic patients with glucose levels of 200 to 250 mg/dL or more, insulin should be administered without glucose and with close monitoring of the plasma glucose level.⁶⁷⁴ Combined treatment with beta-2 agonists, in addition to their synergism with insulin in lowering plasma K^+ , may reduce the level of hypoglycemia.⁶⁸⁰ Of note, the combined regimen may increase the heart rate by 15.1 ± 6.0 beats per minute.⁶⁸⁰

There are theoretic reasons to consider the use of newer, rapid-acting forms of insulin that do not form insulin hexamers, particularly in patients with renal insufficiency.⁶⁸⁷ At present, however, in the absence of rigorous clinical trials that directly compare different regimens,⁶⁸⁸ there is no perceived need to alter the “typical” recommended dose of

10 U of regular insulin administered intravenously followed immediately by 50 mL of 50% dextrose (25 g of glucose).

β_2 -Adrenergic Agonists

β_2 -Agonists are an important but underused group of agents for the acute management of hyperkalemia. They exert their effect by activating Na^+/K^+ -ATPase and the NKCC1 $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter, shifting K^+ into hepatocytes and skeletal myocytes (see also “[Factors Affecting Internal Distribution of Potassium](#)”). Albuterol (Salbutamol), a selective beta-2 agonist, is the most widely studied and used. It is available in oral, inhaled, and intravenous forms; both the intravenous and inhaled or nebulized forms are effective.⁶⁸⁹

The recommended dose for intravenous administration, which is not available in the United States, is 0.5 mg of albuterol in 100 mL of 5% dextrose, given over 10 to 15 minutes.^{674,689,690} Its K^+ -lowering effect starts in a few minutes, is maximal at about 30 to 40 minutes,^{689,690} and lasts for 2 to 6 hours.⁵²³ It reduces serum K^+ levels by approximately 0.9 to 1.4 mmol/L.⁵²³

The recommended dose for inhaled albuterol, also known as salbutamol, is 10 to 20 mg of nebulized albuterol in 4 mL of normal saline, inhaled over 10 minutes⁶⁸⁰ (nebulized levalbuterol is as effective as albuterol).⁶⁹¹ This dose is distinct from the 100- μg dose administered by each puffer spray. Its kaliopenic effect starts at about 30 minutes, reaches its peak at about 90 minutes,^{680,689} and lasts for 2 to 6 hours.^{523,689} Inhaled albuterol reduces serum K^+ levels by approximately 0.5 to 1.0 mmol/L⁵²³; albuterol administered by a metered-dose inhaler with spacer reduces serum K^+ level by approximately 0.4 mmol/L.⁶⁹² Albuterol (in inhaled or parenteral form) and insulin with glucose have an additive effect on reducing serum K^+ levels by approximately 1.2 to 1.5 mmol/L in total.^{523,680,686} However, a subset of patients with ESKD (~20%–40%) are not responsive to the K^+ -lowering effect of albuterol ($\Delta\text{K} \leq 0.4$ mmol/L); albuterol (or other beta-2 agonists) should not be used as a single agent in the treatment of hyperkalemia.^{525,571} In an attempt to reduce pharmacokinetic variability, one study tested the effects of weight-based dosing on serum K^+ levels, using 7 $\mu\text{g}/\text{kg}$ of subcutaneous terbutaline (a beta-2 agonist) in a group of ESKD patients.⁶⁹³ The results showed a significant decline in serum K^+ levels in almost all patients (mean, 1.31 ± 0.5 mmol/L; range, 0.5 to 2.3 mmol/L) in 30 to 90 minutes; of note, the heart rate increased by an average of 25.8 ± 10.5 beats per minute (range, 6.5–48 beats per minute).⁶⁹³

Treatment with albuterol may result in an increase in plasma glucose (~2–3 mmol/L) and heart rate. The increase in heart rate is more pronounced with the intravenous form (~20 beats per minute) than with the inhaled form (~6–10 beats per minute).^{571,689} There is no significant increase in systolic or diastolic blood pressure with nebulized or intravenous administration of albuterol.⁶⁸⁹ However, it is prudent to use these agents with caution in patients with ischemic heart disease.⁵²³

Sodium Bicarbonate

Bicarbonate has prevailed as a preferred treatment modality of hyperkalemia for decades. For example, in a survey of nephrology training program directors in 1989, it was ranked as the second-line treatment, after Ca^{2+} .⁶⁹⁴ Its use to treat acute hyperkalemia was mainly based on small, older,

uncontrolled clinical studies with a very limited number of patients,^{58,695,696} in which bicarbonate was typically administered as a long infusion over many hours (contrary to an IV push, which later became the routine).⁶⁹⁷ One of these studies, which is frequently quoted, concluded that the K^+ -lowering effect of bicarbonate is independent of changes in pH.⁶⁹⁵ However, confounding variables included the duration of infusion, use of glucose-containing solutions, and infrequent monitoring of serum K^+ .^{695,698}

The role of bicarbonate in the acute treatment of hyperkalemia has been challenged.^{681,697,699} Blumberg and colleagues have compared different K^+ -lowering modalities (Fig. 17.16) and showed that bicarbonate infusion (isotonic or hypertonic) for up to 60 minutes has no effect on serum K^+ in their cohort of ESKD patients on hemodialysis⁶⁸¹; there is, however, an effect of isotonic bicarbonate at 4 to 6 hours (see later).⁶⁹⁷ These observations were later confirmed by others, who failed to show any acute (60–20 minutes) K^+ -lowering effects for bicarbonate.^{697–699} A few studies have shown that metabolic acidosis may attenuate the physiologic responses to insulin and beta-2 agonists.^{525,699} The combined effect of bicarbonate and insulin with glucose has been studied, with conflicting results.⁶⁹⁹ In addition, bicarbonate and albuterol coadministration failed to show any additional benefit over albuterol alone.⁶⁹⁹

In summary, bicarbonate administration, especially as a single agent without dilution, has no role in the contemporary treatment of acute hyperkalemia. Prolonged infusion of isotonic bicarbonate in ESKD patients does, however, reduce serum K^+ at 5 to 6 hours by up to 0.7 mmol/L; approximately half of this effect can be attributed to volume expansion.⁶⁹⁷ Regardless of the mechanism, bicarbonate

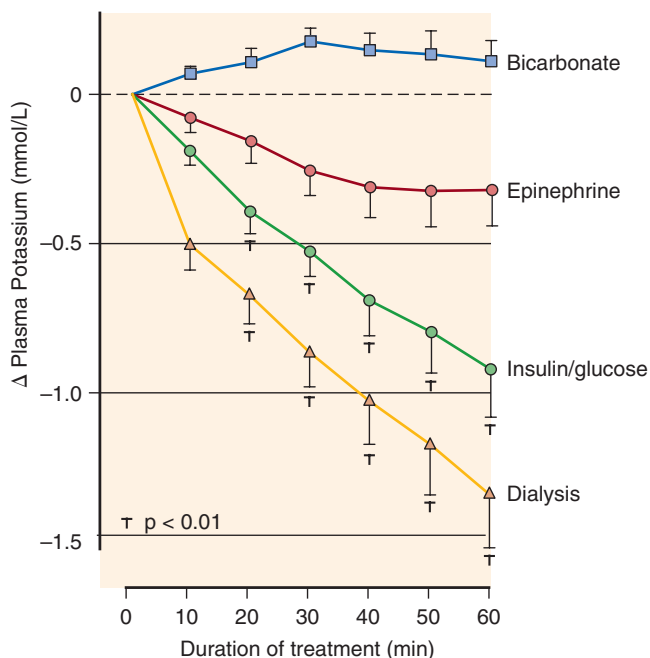


Fig. 17.16 Changes in serum K^+ during intravenous infusion of bicarbonate, epinephrine, or insulin in glucose and during hemodialysis. (From Blumberg A, Weidmann P, Shaw S, Gnadinger M. Effect of various therapeutic approaches on plasma potassium and major regulating factors in terminal renal failure. *Am J Med.* 1988;85:507–512.)

infusion may thus have a limited role in the subacute control of hyperkalemia—for example in the nondialytic management of patients with severe hyperkalemia.⁷⁰⁰ The acute effect of bicarbonate infusion on serum K^+ in severely acidemic patients is not clear; however, it may be of some benefit in this setting.^{229,669} Of note, the infusion of sodium bicarbonate may reduce serum ionized calcium levels and cause volume overload, issues of relevance in acidemic patients with renal failure.^{525,671} When bicarbonate administration is used for hyperkalemia, we recommend isotonic infusion of sodium bicarbonate; although hypertonic sodium bicarbonate does not increase the serum K^+ level,⁵⁵⁸ it has been reported to cause hyponatremia.⁶⁸¹

Removal of Potassium

Diuretics. Diuretics have a relatively modest effect on urinary K^+ excretion in patients with CKD,⁷⁰¹ particularly in an acute setting.⁶⁷² However, diuretics are reasonable therapy for hyperkalemia in patients who have sufficient residual renal function to respond with an increase in urine output, particularly if successful control of serum K^+ may avoid the need for dialysis. In patients with impaired renal function, use of the following agents is recommended for acute control of serum K^+ : (1) oral diuretics with the highest bioavailability (e.g., torsemide) and the least renal metabolism, (e.g., torsemide, bumetanide) to minimize the chance of accumulation and toxicity; (2) intravenous agents (short-term treatment) with the least hepatic metabolism (e.g., furosemide rather than bumetanide); (3) combinations of loop and thiazide-like diuretics for better efficacy, although this may decrease the GFR because of activation of tubuloglomerular feedback⁷⁰²; and (4) the maximal effective “ceiling” dose.^{701,702} To avoid causing hypovolemia, it may be necessary to increase the rate of IV hydration with normal saline or isotonic bicarbonate. This will also serve to help increase distal tubular delivery of Na^+ , augmenting K^+ excretion.

For the chronic management of hyperkalemia, diuretic therapy is not recommended in patients who are euvolemic or who otherwise lack indications for diuretic therapy, given the potential for inducing hypovolemia. However, with liberalization of fluid and salt intake, if appropriate, diuretics can be useful for correcting hyperkalemia in patients with the syndrome of hyporeninemic hypoaldosteronism⁷⁰³ and selective renal K^+ secretory problems (e.g., after transplantation or administration of TMP).^{704,705}

Mineralocorticoids. Limited data are available on the role of mineralocorticoids in the management of acute hyperkalemia.^{29,706} However, these agents have been used to treat chronic hyperkalemia in patients with hypoaldosteronism, with or without hyporeninism, those with SLE,⁷⁰⁷ kidney transplant patients on cyclosporine,⁷⁰⁸ and ESKD patients on hemodialysis with interdialytic hyperkalemia.^{709,710} The recommended dose is 0.1 to 0.3 mg/day of fludrocortisone, a synthetic glucocorticoid with potent mineralocorticoid activity and modest glucocorticoid activity (0.3 mg of fludrocortisone is equal to 1 mg of prednisone with regard to glucocorticoid activity).^{29,708–710} In patients with ESKD on hemodialysis, this regimen reduces serum K^+ by 0.5 to 0.7 mmol/L and has not been associated with significant changes in blood pressure or weight (as a surrogate for fluid retention).⁷⁰⁹ However, other studies of 0.1 mg/day of

fludrocortisone in patients on chronic hemodialysis have resulted in statistically significant but clinically inconsequential effects on serum K^+ .^{711,712} The long-term safety of fludrocortisone in ESKD has not been established and, given the minimal effect on serum K^+ , I do not recommend its use for the management of interdialytic hyperkalemia.

Pharmacologic inhibition of 11β -HSD-2 with glycyrrhetic acid (GA) has also been tested as a novel mechanism to control hyperkalemia in ESKD.⁴⁰⁵ As in other aldosterone-sensitive epithelia, the 11β -HSD-2 enzyme protects colonic epithelial cells from illicit activation of the mineralocorticoid receptor by cortisol. Hypothesizing that GA would activate extrarenal potassium secretion by the colon and other tissues, Farese and associates have tested the effect of the drug in a double-blind, placebo-controlled trial in 10 ESKD patients. Treatment with GA significantly increased the serum ratio of cortisol to cortisone, consistent with successful inhibition of 11β -HSD-2. This effect was associated with a significant reduction in mean serum K^+ , with 70% of predialysis values in the normal range (3.5–4.7 mmol/L) in the GA phase of the trial, versus 24% in the placebo phase.⁴⁰⁵ Plasma renin activity and aldosterone levels also dropped, perhaps because of the lower median plasma K^+ level. GA is thus a promising agent for long-term management of plasma K^+ in ESKD; clearly, however, more extensive clinical testing is required before widespread use.

Cation Exchange Resins. Ion exchange resins are cross-linked polymers containing acidic or basic structural units that can exchange anions or cations on contact with a solution. They are capable of binding to a variety of monovalent and divalent cations. Cation exchange resins are classified based on the cation (e.g., hydrogen, ammonium, sodium, potassium, calcium) that is cycled during the synthesis of the resin to saturate sulfonic or carboxylic groups. In 1950, Elkinton and colleagues successfully used a carboxylic resin in the ammonium cycle in three patients with hyperkalemia.⁷¹³ However, hydrogen- or ammonium-cycled resins were associated with metabolic acidosis⁷¹⁴ and mouth ulcers,⁷¹⁵ making the sodium-cycled resins preferable.⁷¹⁶ Calcium-cycled resins may have other potential benefits, including a phosphate-lowering effect; however, this requires large, potentially toxic, doses of resin⁷¹⁷; moreover, these resins have been associated with hypercalcemia.⁷¹⁸ The dominant resin clinically available in the United States is sodium polystyrene sulfonate (SPS).

SPS exchanges Na^+ for K^+ in the gastrointestinal tract, mainly in the colon,^{669,715,719} and has been shown to increase the fecal excretion of K^+ .⁷¹⁵ Occasional constipation easily controlled with enema or cathartics has been reported with the oral administration of SPS in water.⁷¹⁵ To prevent constipation and facilitate the passage of the resin through the gastrointestinal tract, Flinn and coworkers added sorbitol to the resin,⁷²⁰ despite an absence of prior constipation or impaction in the relevant, cited reference.⁷¹⁶ It has since become routine to administer SPS with sorbitol, with approximately 5 million annual doses administered in the United States alone.⁷²¹ Notably, although SPS is used herein to denote sodium polystyrene sulfonate, SPS is a brand name for sodium polystyrene sulfonate in sorbitol, illustrating the frequency with which these agents are administered together.⁶⁷⁰

The effect of ingested SPS on serum K^+ is slow; it may take from 4 to 24 hours to see a significant effect on the serum

K⁺.^{672,673,715} The oral dose is usually 15 to 30 g, which can be repeated every 4 to 6 hours. Each gram of resin binds 0.5 to 1.2 mEq of K⁺ in exchange for 2 to 3 mEq of Na⁺.^{673,715,722,723} The discrepancy is caused, in part, by the binding of small amounts of other cations.⁷¹⁵ The hypokalemic effect may be as a result of the coadministered laxative. One study of healthy subjects has compared the rate of fecal excretion of K⁺ by different laxatives, with or without SPS, and found that the combination of phenolphthalein-docusate with resin produced greater fecal excretion of K⁺ (49 mmol in 12 hours) than did phenolphthalein/docusate alone (37 mmol in 12 hours) or other laxative-resin combinations.⁷²²

Earlier studies with SPS, mostly before the era of chronic hemodialysis, used multiple doses of the exchange resin orally or rectally as an enema and were associated with declines in the serum K⁺ of 1 mmol/L and 0.8 mmol/L in 24 hours, respectively.⁷¹⁵ However, with the advent of routine hemodialysis, it has become common to order only a single dose of resin cathartic in the management of acute hyperkalemia. One study has addressed the efficacy of this practice, evaluating the effect of four single-dose resin cathartic regimens on serum K⁺ levels of six patients with CKD on maintenance hemodialysis; none of the regimens used reduced the serum K⁺ below the initial baseline.⁷²³ Notably, the subjects in this study were normokalemic. Regardless, if SPS is judged to be appropriate in the management of hyperkalemia (see later), repeated doses are usually required for an adequate effect.

SPS can be administered rectally as a retention enema in patients unable to take or tolerate the oral form. The recommended dose is 30 to 50 g of resin as an emulsion in 100 mL of an aqueous vehicle (e.g., 20% dextrose in water) every 6 hours. It should be administered warm (body temperature), after a cleansing enema with body temperature tap water, through a rubber tube placed at about 20 cm from the rectum, with the tip well into the sigmoid colon. The emulsion should be introduced by gravity, flushed with an additional 50 to 100 mL of non-sodium-containing fluid, retained for at least 30 to 60 minutes, and followed by a cleansing enema (250–1000 mL of body temperature tap water).⁷²⁴ SPS in sorbitol should not be used for enemas, given the risk of colonic necrosis.^{672,725}

An increasing concern with SPS in sorbitol is intestinal necrosis because of the administration of this preparation.^{721,725–728}; this is frequently a fatal complication.^{721,725,727} Studies in experimental animals have suggested that sorbitol is required for the intestinal injury⁷²⁵; however, a recent animal study has indicated that SPS alone can cause intestinal necrosis, with no requirement for sorbitol.⁷²⁹ Additionally, SPS crystals can often be detected in human pathologic specimens, adherent to the injured mucosa.^{727,728} Several cases of colonic necrosis following oral SPS alone, without sorbitol, have also been reported,^{730,731} directly implicating SPS in the intestinal injury. The risk of intestinal necrosis appears to be greatest when SPS is given with sorbitol within the first week after surgery. For example, of 117 patients who received SPS with sorbitol within 2 weeks of surgery, two patients developed intestinal necrosis.⁷²⁶ Notably, however, in a case series of SPS in sorbitol-associated intestinal necrosis, only 2 of 11 confirmed cases occurred in the postoperative setting.⁷²⁷ Although most cases of intestinal necrosis have occurred in patients receiving SPS in 70% sorbitol, this has also been reported in patients receiving SPS in 33% sorbitol.⁷²¹

In response to these reports, the US Food and Drug Administration (FDA) removed recommendations for the concomitant or postdosing use of sorbitol from the labeling of SPS in 2005.⁷²¹ However, the FDA allowed continuous marketing of the most frequently used ready-made SPS in sorbitol suspension, given that it contained only 33% sorbitol. Subsequently, more cases of intestinal necrosis were reported, with some reportedly associated with SPS in 33% sorbitol.⁷²¹ As a result, in September 2009, the FDA changed safety labeling for SPS powder, stating that concomitant administration of sorbitol is no longer recommended.⁷²¹ As noted, several cases of colonic necrosis following oral SPS alone have also been reported.^{730,731}

Given these serious concerns, clinicians must carefully consider whether emergency treatment with SPS is actually necessary for the treatment of hyperkalemia.^{670,721,732} Additionally, patiomer and ZS-9, newly available alternatives, should be used in preference to SPS. There are minimal data on the efficacy of SPS within the first 24 hours of administration for hyperkalemia⁷²¹; at best, the effect occurs within 4 to 6 hours of administration.^{671,672,715} This temporal limitation should be taken into consideration when deciding whether or not to administer SPS in acute hyperkalemia. In patients with intact renal function, alternative measures such as hydration, to increase distal tubular delivery of Na⁺ and distal tubular flow rate, and/or diuretics are often sufficient for potassium removal. In patients with advanced renal failure, the use of SPS is reasonable as a temporizing maneuver while awaiting hemodialysis but, if hemodialysis is available within 1 to 4 hours, the need for SPS should be questioned, given the delayed hypokalemic response and the risk of potentially fatal intestinal necrosis. Furthermore, if a patient has an existing vascular access for hemodialysis, the risk of intestinal necrosis outweighs that of the dialysis procedure.

If SPS is administered, the preparation should ideally not contain sorbitol; SPS without sorbitol is typically available as a powder that must be reconstituted with water. If a laxative other than sorbitol is coadministered, it should not contain potassium or other cations such as magnesium or calcium, which can compete with potassium for binding to the resin. In patients with renal insufficiency, the laxative should not contain phosphorus. Reasonable laxatives for this purpose include lactulose and some preparations of polyethylene glycol 3350. However, data demonstrating the efficacy and safety of these laxatives with SPS are not available.⁶⁷⁰ It should also be noted that administering SPS without sorbitol might not eliminate the risk of intestinal necrosis, given the role for the SPS resin itself in this complication.^{727,728,730,731} Note that SPS without sorbitol is rarely available in the United States; many pharmacies and hospitals only stock SPS premixed with sorbitol.⁷²¹ Although there are indications that SPS preparations with 33% sorbitol have a lesser risk, cases of intestinal necrosis have also been described with this preparation. Clinicians will have to weigh the relative risk of using this preparation in the management of acute hyperkalemia.⁷²¹ Regardless, SPS with sorbitol should not be used in patients at higher risk for intestinal necrosis, including postoperative patients, patients with a history of bowel obstruction, patients with slow intestinal transit, patients with ischemic bowel disease, and renal transplant recipients.

Finally, oral SPS with sorbitol can also injure the upper gastrointestinal tract, although the clinical significance of

these findings is not known.⁷³³ Other potential complications include reduction of serum calcium levels,⁷³⁴ volume overload,⁷¹⁴ interference with lithium absorption,⁷³⁵ and iatrogenic hypokalemia.⁷²⁴

Novel Intestinal Potassium Binders. Two new K⁺ binders, patiomer and ZS-9, have recently become available for the treatment of hyperkalemia. Given the clinical need and serious limitations of SPS, there is considerable interest in these drugs. At this time, the drugs are expensive and limited in availability at many US-based hospital settings.

Patiomer is a nonabsorbed polymer, provided as a powder for suspension, which binds K⁺ in exchange for Ca²⁺. In a study of 237 patients with CKD and hyperkalemia, the mean change in serum K⁺ after treatment with patiomer was -1.01 ± 0.03 mM.⁷³⁶ Approximately 75% of patients achieved the target serum K⁺ level of 3.8 to 5.0 mEq/L. The 107 patients whose baseline serum K⁺ was 5.5 mEq/L or higher and who achieved the target serum K⁺ during the initial 4-week treatment period were randomly assigned to patiomer or placebo for another 8 weeks. Serum K⁺ remained the same in patients who continued on patiomer and increased by 0.7 mEq/L in those assigned placebo. The incidence of hyperkalemia (≥ 5.5 mEq/L) was significantly higher in the placebo group (60% vs. 15%). Serious adverse events were rare. However, a serum magnesium level less than 1.4 mg/dL occurred in 8 patients (3%) during the initial phase, and 9 patients were initiated on magnesium replacement, suggesting that impaired intestinal absorption of magnesium may be a limitation in long-term use. Patiomer also binds to ingested medications, with characterized interactions with ciprofloxacin, thyroxine, and metformin; these three drugs need to be administered more than 3 hours before or after patiomer.⁷³⁷

Sodium zirconium cyclosilicate (ZS-9) is an inorganic, nonabsorbable crystalline compound that exchanges sodium and hydrogen ions for K⁺ and NH₄⁺ in the intestine. The binding of K⁺ to ZS-9 has some similarity to the selectivity filter of K⁺ channels, with a more than 25-fold selectivity for K⁺ over Ca²⁺ and Mg²⁺.⁷³⁸ The efficacy of ZS-9 in hyperkalemic outpatients was evaluated in two nearly identical phase III, randomized, placebo-controlled trials.

In the Hyperkalemia Randomized Intervention Multidose ZS-9 Maintenance (HARMONIZE) study, 258 adult patients with persistent hyperkalemia entered a 48-hour, open-label, run-in during which they received 10 g of ZS-9 tid.⁷³⁹ Of the 258 patients who entered the open-label run-in, 237 (92%) achieved a normal serum K⁺ (3.5–5.0 mEq/L) at 48 hours and were then randomly assigned to placebo or 5, 10, or 15 g of ZS-9 once daily for 4 weeks. During randomized therapy, the mean serum K⁺ was significantly lower with ZS-9 (4.8, 4.5, and 4.4 mEq/L with 5-, 10-, and 15-g dosing, respectively) as compared with placebo (5.1 mEq/L). Similarly, the proportion of normokalemic patients at the end of the study was significantly greater with ZS-9 (71%–85% as compared with 48% with placebo). Serious adverse events were uncommon and were not significantly increased with ZS-9. However, edema was more common with the 10-g and 15-g doses compared with placebo, as was hypokalemia.

In a second ZS-9 trial,⁷⁴⁰ 753 adult patients with a serum K⁺ of 5.0 to 6.5 mM were randomly assigned to receive 1.25, 2.5, 5, or 10 g of ZS-9 tid daily for 48 hours. A normal serum K⁺ (3.5–4.9 mEq/L) at 48 hours was attained by 543 patients

(72%), and they were then reassigned to receive placebo or 1.25, 2.5, 5, or 10 g of ZS-9 once daily for 2 weeks. The serum potassium level at 2 weeks was significantly lower in patients receiving 5 and 10 g doses of ZS-9 as compared with placebo (by ≈ 0.3 and 0.5 mEq/L, respectively), but not with 1.25- and 2.5-g doses. Adverse events were similar with placebo and ZS-9.

In these trials, the steepest decline in serum potassium with ZS-9 occurred during the first 4 hours of therapy.^{739,740} This suggests an acute effect on intestinal potassium secretion, rather than simply a reduction in intestinal potassium absorption.

Dialysis. All modes of acute renal replacement therapies are effective in removing K⁺. Continuous hemodiafiltration has been increasingly used in the management of critically ill and hemodynamically unstable patients.⁷⁴¹ Peritoneal dialysis, although not very effective in an acute setting, has been used effectively in cardiac arrest complicating acute hyperkalemia.⁷⁴² Peritoneal dialysis is capable of removing significant amounts of K⁺ (5 mmol/hour [or 240 mmol in 48 hours]) using 2-L exchanges, with each exchange taking almost 1 hour.⁶⁷⁴ However, hemodialysis is the preferred mode when rapid correction of a hyperkalemic episode is desired.⁷⁴³

An average 3- to 5-hour hemodialysis session removes approximately 40 to 120 mmol of K⁺.^{29,743-750} Approximately 15% of the total K⁺ removal results from ultrafiltration, with the remaining clearance from dialysis.^{744,745} Of the total K⁺ removed, about 40% is from extracellular spaces, and the remainder is from intracellular compartments.^{744,746,747} In most patients, the greatest decline in serum K⁺ (1.2–1.5 mmol/L) and the largest amount of K⁺ removed occur during the first hour; the serum K⁺ usually reaches its nadir at about 3 hours. Despite a relatively constant serum K⁺, K⁺ removal continues until the end of the hemodialysis session, although at a significantly lower rate.^{29,744,748}

The amount of K⁺ removed depends primarily on the type and surface area of the dialyzer used, blood flow rate, dialysate flow rate, dialysis duration, and serum-to-dialysate K⁺ gradient. However, about 40% of the difference in removal cannot be explained by the abovementioned factors and may instead be related to the relative distribution of K⁺ between the intracellular and extracellular spaces.⁷⁴⁶ Glucose-free dialysates are more efficient in removing K⁺.^{745,748} This effect may be caused by alterations in endogenous insulin levels, with concomitant intracellular shift of K⁺; the insulin level is 50% lower when glucose-free dialysates are used.⁷⁴⁶ Furthermore, these findings imply that K⁺ removal may be greater if hemodialysis is performed with the patient in a fasting state.⁷⁴⁵ Treatment with beta-2 agonists also reduces the total K⁺ removal by approximately 40%.⁷⁴³ The change in pH during dialysis has been thought to have no significant effect on K⁺ removal.^{743,745} One study evaluated this issue in detail, examining the effect of dialysate bicarbonate concentration on both serum K⁺ and K⁺ removal. Dialysates with a bicarbonate concentration of 39 mmol/L (high), 35 mmol/L (standard), and 27 mmol/L (low) were used. The use of a high concentration of bicarbonate was associated with a more rapid decline in serum K⁺; this was statistically significant for high versus standard and low-bicarbonate dialysates, at 60 and 240 minutes, respectively. However, the total amount of K⁺ removed was higher with the low-bicarbonate dialysate ($116.4 \pm$

21.6 mmol/dialysis) in comparison to standard (73.2 ± 12.8 mmol/dialysis) and high (80.9 ± 15.4 mmol/dialysis) bicarbonate dialysates, all statistically insignificant.⁷⁴⁹ Therefore, whereas high-bicarbonate dialysis *may acutely* have a more rapid effect on serum K^+ , this advantage is potentially mitigated by a lesser total removal of the ion over the course of a typical treatment session.

One of the major determinants of total K^+ removal is the K^+ gradient between the plasma and dialysate. Dialysates with a lower K^+ concentration are more effective at reducing plasma K^+ .^{746,750} Many nephrologists use the so-called rule of 7s to set the dialysate K^+ concentration; the plasma K^+ plus the dialysate K^+ should equal approximately 7. However, this then entails dialysis with a 0- or 1.0-mEq/L K^+ dialysate (0 K and 1 K bath) in patients with serum potassium concentrations that exceed 6 to 7.0 mmol/L. However, a rapid decline in plasma K^+ caused by 0 K or 1 K dialysates can be deleterious because of several mechanisms. First, an acute decrease in plasma K^+ can be associated with rebound hypertension (i.e., a significant increase in blood pressure 1 hour after dialysis),⁷⁵⁰ which is attributed in part to the peripheral vasoconstriction that is a direct result of the change in plasma K^+ .⁷⁴⁴ Second, a low plasma K^+ can alter the rate of tissue metabolism, the so-called Solandt effect,⁷⁵¹ and decrease tissue oxygen consumption, promoting arteriolar constriction.⁷⁵⁰ This vasoconstriction, in turn, may reduce the efficiency of dialysis²⁹; a randomized, prospective study did not, however, confirm this finding.⁷⁴⁸ The difference may have been as a result of the glucose content of the dialysate (i.e., 200 mg/dL in the former and 0 mg/dL in the latter study); differences in circulating insulin may have had additional, unrelated effects on muscle blood flow.⁷⁵² Finally, dialysates with a very low K^+ concentration may increase the risk of significant arrhythmia and sudden cardiac death.^{753–756}

Several studies have found an increased incidence of significant arrhythmia with hemodialysis, occurring during and immediately after treatment^{755,757,758}; an incidence of up to 76% has been reported.⁷⁵⁹ In addition to ventricular arrhythmias, hemodialysis can provoke atrial fibrillation, particularly in patients with a lower dialysate K^+ concentration.⁷⁶⁰ Historically, some investigators did not consider the hemodialysis procedure to be significantly arrhythmogenic,^{761–763} whereas others had suggested that a relationship exists among decreases in K^+ , dialysate K^+ , and the incidence of significant arrhythmias.⁷⁵⁵ Given the more recent data,^{754,756} it seems prudent to recommend that dialysates with a very low K^+ (0 or 1 mmol/L) be used cautiously, if at all. Higher risk patients include patients receiving digoxin, those with a history of arrhythmia, coronary artery disease, left ventricular hypertrophy, or high systolic blood pressure, and those of an advanced age. Continuous cardiac monitoring for all patients dialyzed against a 0 or 1 mmol/L K^+ bath is strongly recommended.⁵²⁵

Given the risk of inducing arrhythmias with very low potassium dialysates, an alternative approach has been proposed for the treatment of significant hyperkalemia.^{525,764} In this regimen, dialysis is initiated with a 3 to 4 mEq/L potassium bath, which will immediately lower the plasma potassium concentration in a slower and perhaps safer manner.⁷⁵⁵ The potassium concentration in the dialysate may then be lowered in a stepwise fashion with each subsequent hour. A more sophisticated approach uses potassium profiling

to maintain a constant potassium gradient during dialysis.^{764–766} Potassium profiling results in a more sustained, even removal of potassium⁷⁶⁵ than dialysis against a fixed-potassium bath (2.5 mEq/L), with less effect on ventricular ectopy.^{764–766}

For the management of severe hyperkalemia (serum $K^+ \geq 7.0$ mEq/L), I favor the use of stepped reduction or potassium profiling of dialysate potassium concentrations. I rarely encounter the need to use 0- or 1-K dialysate baths, which I also avoid at the beginning of dialysis sessions for acute hyperkalemia. I recommend restricting the up-front use of these low-potassium baths to patients with life-threatening hyperkalemic arrhythmias and/or life-threatening conduction abnormalities.

A rebound increase in plasma K^+ can occur after hemodialysis. This phenomenon can be especially marked in cases of massive release from devitalized tissues (e.g., tumor lysis, rhabdomyolysis), requiring frequent monitoring of serum K^+ and further hemodialysis. However, a rebound increase may also occur in ESKD patients during regular maintenance hemodialysis, despite technically adequate treatment,⁷⁴⁴ particularly in those patients with a high predialysis K^+ . Factors attenuating K^+ removal and thus increasing the risk and magnitude of postdialysis rebound include pretreatment with beta-2 agonists,⁷⁴³ pretreatment with insulin and glucose, eating early during the dialysis treatment,⁷⁴⁵ a high predialysis plasma K^+ ,⁷⁴⁷ and higher dialysate Na^+ concentrations.⁷⁶⁷ Notably, a recent study of emergency hemodialysis treatments for hyperkalemia has failed to detect an association among the predialysis K^+ , administration of pretreatment expected to cause a shift in serum K^+ , and either recurrent hyperkalemia or the need for repeat dialysis within 24 hours.⁷⁶⁸

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

4. Terker AS, et al. Potassium modulates electrolyte balance and blood pressure through effects on distal cell voltage and chloride. *Cell Metab*. 2015;21:39–50.
22. McDonough AA, Thompson CB, Youn JH. Skeletal muscle regulates extracellular potassium. *Am J Physiol Renal Physiol*. 2002;282:F967–F974.
31. Guo J, et al. Extracellular K^+ concentration controls cell surface density of IKr in rabbit hearts and of the HERG channel in human cell lines. *J Clin Invest*. 2009;119:2745–2757.
57. Aronson PS, Giebisch G. Effects of pH on potassium: new explanations for old observations. *J Am Soc Nephrol*. 2011;22:1981–1989.
73. Lalioti MD, et al. Wnk4 controls blood pressure and potassium homeostasis via regulation of mass and activity of the distal convoluted tubule. *Nat Genet*. 2006;38:1124–1132.
82. Mamenko MV, et al. The renal TRPV4 channel is essential for adaptation to increased dietary potassium. *Kidney Int*. 2017;91:1398–1409.
114. Frindt G, Shah A, Edvinsson J, et al. Dietary K regulates ROMK channels in connecting tubule and cortical collecting duct of rat kidney. *Am J Physiol Renal Physiol*. 2009;296:F347–F354.
108. Kleyman TR, Carattino MD, Hughey RP. ENaC at the cutting edge: regulation of epithelial sodium channels by proteases. *J Biol Chem*. 2009;284:20447–20451.
115. Palmer LG, Frindt G. Regulation of apical K channels in rat cortical collecting tubule during changes in dietary K intake. *Am J Physiol*. 1999;277:F805–F812.
143. Vallon V, Schlöth J, Lang F, et al. Expression and phosphorylation of the Na^+ -Cl[−] cotransporter NCC in vivo is regulated by dietary salt, potassium, and SGK1. *Am J Physiol Renal Physiol*. 2009;297:F704–F712.
174. Mazzocchi G, Malendowicz LK, Markowska A, et al. Role of adrenal renin-angiotensin system in the control of aldosterone secretion in sodium-restricted rats. *Am J Physiol Endocrinol Metab*. 2000;278:E1027–E1030.

179. Clark BA, Brown RS, Epstein FH. Effect of atrial natriuretic peptide on potassium-stimulated aldosterone secretion: potential relevance to hypoaldosteronism in man. *J Clin Endocrinol Metab.* 1992;75:399–403.
186. Lin SH, et al. Laboratory tests to determine the cause of hypokalemia and paralysis. *Arch Intern Med.* 2004;164:1561–1566.
187. Battle DC, Hizon M, Cohen E, et al. The use of the urinary anion gap in the diagnosis of hyperchloremic metabolic acidosis. *N Engl J Med.* 1988;594:599.
195. Parham WA, Mehdirad AA, Biermann KM, et al. Hyperkalemia revisited. *Tex Heart Inst J.* 2006;33:40–47.
197. Goldstein SA. K2P potassium channels, mysterious and paradoxically exciting. *Sci Signal.* 2011;4:pe35.
222. Ray PE, Suga S, Liu XH, et al. Chronic potassium depletion induces renal injury, salt sensitivity, and hypertension in young rats. *Kidney Int.* 2001;59:1850–1858.
223. Sun Y, et al. Dietary potassium regulates vascular calcification and arterial stiffness. *JCI Insight.* 2017;2.
241. Evers S, Engelien A, Karsch V, et al. Secondary hyperkalaemic paralysis. *J Neurol Neurosurg Psychiatry.* 1998;64:249–252.
249. DuBose TD Jr, Good DW. Chronic hyperkalemia impairs ammonium transport and accumulation in the inner medulla of the rat. *J Clin Invest.* 1992;90:1443–1449.
272. Tse HF, Yeung CK. From profound hypokalemia to fatal rhabdomyolysis after severe head injury. *Am J Med.* 2000;109:599–600.
294. Puwanant A, Ruff RL. INa and IKir are reduced in Type 1 hypokalemic and thyrotoxic periodic paralysis. *Muscle Nerve.* 2010;42:315–327.
319. Blondon H, Bechade D, Desrame J, et al. Secretory diarrhoea with high faecal potassium concentrations: a new mechanism of diarrhoea associated with colonic pseudo-obstruction? Report of five patients. *Gastroenterol Clin Biol.* 2008;32:401–404.
340. Beck KR, et al. Inhibition of 11beta-hydroxysteroid dehydrogenase 2 by the fungicides itraconazole and posaconazole. *Biochem Pharmacol.* 2017;130:93–103.
368. Nanba K, Vaidya A, Rainey WE. Aging and adrenal aldosterone production. *Hypertension.* 2018;71:218–223.
396. Kotelevtsev Y, et al. Hypertension in mice lacking 11beta-hydroxysteroid dehydrogenase type 2. *J Clin Invest.* 1999;103:683–689.
437. Carmosino M, et al. NKCC2 activity is inhibited by the Bartter's syndrome type 5 gain-of-function CaR-A843E mutant in renal cells. *Biol Cell.* 2015;107:98–110.
439. Seys E, et al. Clinical and genetic spectrum of Bartter syndrome type 3. *J Am Soc Nephrol.* 2017;28:2540–2552.
441. Bongers E, et al. A novel Hypokalemic-alkalotic salt-losing tubulopathy in patients with CLDN10 mutations. *J Am Soc Nephrol.* 2017;28:3118–3128.
449. Blanchard A, et al. Gitelman syndrome: consensus and guidance from a Kidney Disease: improving Global Outcomes (KDIGO) Controversies Conference. *Kidney Int.* 2017;91:24–33.
456. Bockenbauer D, et al. Epilepsy, ataxia, sensorineural deafness, tubulopathy, and KCNJ10 mutations. *N Engl J Med.* 2009;360:1960–1970.
471. Comer DM, Droogan AG, Young IS, et al. Hypokalaemic paralysis precipitated by distal renal tubular acidosis secondary to Sjogren's syndrome. *Ann Clin Biochem.* 2008;45:221–225.
478. Yang L, Frindt G, Palmer LG. Magnesium modulates ROMK channel-mediated potassium secretion. *J Am Soc Nephrol.* 2010;21:2109–2116.
481. Reimann D, Gross P. Chronic, diagnosis-resistant hypokalaemia. *Nephrol Dial Transplant.* 1999;14:2957–2961.
495. Sterns RH, Cox M, Feig PU, et al. Internal potassium balance and the control of the plasma potassium concentration. *Medicine (Baltimore).* 1981;60:339–354.
501. Nicolis GL, Kahn T, Sanchez A, et al. Glucose-induced hyperkalemia in diabetic subjects. *Arch Intern Med.* 1981;141:49–53.
503. Kunin AS, Surawicz B, Sims EAH. Decrease in serum potassium concentrations and appearance of cardiac arrhythmias during infusion of potassium with glucose in potassium-depleted patients. *N Engl J Med.* 1962;266:228–233.
526. Moranne O, et al. Timing of onset of CKD-related metabolic complications. *J Am Soc Nephrol.* 2009;20:164–171.
528. Kovesdy CP, et al. Serum and dialysate potassium concentrations and survival in hemodialysis patients. *Clin J Am Soc Nephrol.* 2007;2:999–1007.
562. Goldfarb S, Strunk B, Singer I, et al. Paradoxical glucose-induced hyperkalemia. Combined aldosterone-insulin deficiency. *Am J Med.* 1975;59:744–750.
565. Perazella MA, Biswas P. Acute hyperkalemia associated with intravenous epsilon-aminocaproic acid therapy. *Am J Kidney Dis.* 1999;33:782–785.
578. Martyn JA, Richtsfeld M. Succinylcholine-induced hyperkalemia in acquired pathologic states: etiologic factors and molecular mechanisms. *Anesthesiology.* 2006;104:158–169.
597. Oster JR, Singer I, Fishman LM. Heparin-induced aldosterone suppression and hyperkalemia. *Am J Med.* 1995;98:575–586.
605. Schambelan M, Stockigt JR, Biglieri EG. Isolated hypoaldosteronism in adults. A renin-deficiency syndrome. *N Engl J Med.* 1972;287:573–578.
630. Tan SY, Shapiro R, Franco R, et al. Indomethacin-induced prostaglandin inhibition with hyperkalemia. A reversible cause of hyporeninemic hypoaldosteronism. *Ann Intern Med.* 1979;90:783–785.
646. Velazquez H, Perazella MA, Wright FS, et al. Renal mechanism of trimethoprim-induced hyperkalemia. *Ann Intern Med.* 1993;119:296–301.
680. Allon M, Copkney C. Albuterol and insulin for treatment of hyperkalemia in hemodialysis patients. *Kidney Int.* 1990;38:869–872.
681. Blumberg A, Weidmann P, Shaw S, et al. Effect of various therapeutic approaches on plasma potassium and major regulating factors in terminal renal failure. *Am J Med.* 1988;85:507–512.
697. Blumberg A, Weidmann P, Ferrari P. Effect of prolonged bicarbonate administration on plasma potassium in terminal renal failure. *Kidney Int.* 1992;41:369–374.
729. Ayoub I, et al. Colon necrosis due to sodium polystyrene sulfonate with and without sorbitol: an experimental study in rats. *PLoS ONE.* 2015;10:e0137636.
739. Kosiborod M, et al. Effect of sodium zirconium cyclosilicate on potassium lowering for 28 days among outpatients with hyperkalemia: the HARMONIZE randomized clinical trial. *JAMA.* 2014;312:2223–2233.
754. Hung AM, Hakim RM. Dialysate and serum potassium in hemodialysis. *Am J Kidney Dis.* 2015;66:125–132.
765. Santoro A, et al. Patients with complex arrhythmias during and after haemodialysis suffer from different regimens of potassium removal. *Nephrol Dial Transplant.* 2008;23:1415–1421.

REFERENCES

- Teo AE, et al. Pregnancy, primary aldosteronism, and adrenal CTNNB1 mutations. *N Engl J Med*. 2015;373:1429–1436.
- Choi M, et al. K⁺ channel mutations in adrenal aldosterone-producing adenomas and hereditary hypertension. *Science*. 2011;331:768–772.
- Snyder PM, Olson DR, Thomas BC. Serum and glucocorticoid-regulated kinase modulates Nedd4-2-mediated inhibition of the epithelial Na⁺ channel. *J Biol Chem*. 2002;277:5–8.
- Terker AS, et al. Potassium modulates electrolyte balance and blood pressure through effects on distal cell voltage and chloride. *Cell Metab*. 2015;21:39–50.
- Shibata S, et al. Angiotensin II signaling via protein kinase C phosphorylates Kelch-like 3, preventing WNK4 degradation. *Proc Natl Acad Sci USA*. 2014;111:15556–15561.
- Shibata S, Zhang J, Puthumana J, et al. Kelch-like 3 and Cullin 3 regulate electrolyte homeostasis via ubiquitination and degradation of WNK4. *Proc Natl Acad Sci USA*. 2013;110:7838–7843.
- Boyden LM, et al. Mutations in kelch-like 3 and cullin 3 cause hypertension and electrolyte abnormalities. *Nature*. 2012;482:98–102.
- Scortegagna M, et al. The E3 ubiquitin ligase Siah1 regulates adrenal gland organization and aldosterone secretion. *JCI Insight*. 2017;2.
- Williams ME, Rosa RM, Silva P, et al. Impairment of extrarenal potassium disposal by alpha-adrenergic stimulation. *N Engl J Med*. 1984;311:145–149.
- Rosa RM, Epstein FH. Extrarenal potassium metabolism. In: Seldin DW, Giebisch G, eds. *The Kidney: Physiology and Pathophysiology*. Vol. 2. Philadelphia: Lippincott Williams & Williams; 2000:1552–1573.
- Knochel JP, Dotin LN, Hamburger RJ. Pathophysiology of intense physical conditioning in a hot climate. I. Mechanisms of potassium depletion. *J Clin Invest*. 1972;51:242–255.
- Chen P, et al. Modest dietary K⁺ restriction provokes insulin resistance of cellular K⁺ uptake and phosphorylation of renal outer medulla K⁺ channel without fall in plasma K⁺ concentration. *Am J Physiol Cell Physiol*. 2006;290:C1355–C1363.
- Therien AG, Blostein R. Mechanisms of sodium pump regulation. *Am J Physiol Cell Physiol*. 2000;279:C541–C566.
- Lingrel JB, Croyle ML, Woo AL, et al. Ligand binding sites of Na,K-ATPase. *Acta Physiol Scand Suppl*. 1998;643:69–77.
- McDonough AA, Wang J, Farley RA. Significance of sodium pump isoforms in digitalis therapy. *J Mol Cell Cardiol*. 1995;27:1001–1009.
- Crambert G, et al. Transport and pharmacological properties of nine different human Na, K-ATPase isozymes. *J Biol Chem*. 2000;275:1976–1986.
- James PF, et al. Identification of a specific role for the Na,K-ATPase alpha 2 isoform as a regulator of calcium in the heart. *Mol Cell*. 1999;3:555–563.
- Lorenz JN, et al. ACTH-induced hypertension is dependent on the ouabain-binding site of the alpha2-Na⁺K⁺-ATPase subunit. *Am J Physiol Heart Circ Physiol*. 2008;295:H273–H280.
- Akimova O, Tremblay J, Hamet P, et al. The Na⁺/K⁺-ATPase as [K⁺]_o sensor: role in cardiovascular disease pathogenesis and augmented production of endogenous cardiogenic steroids. *Pathophysiology*. 2006.
- Juel C, Pilegaard H, Nielsen JJ, et al. Interstitial K⁺ in human skeletal muscle during and after dynamic graded exercise determined by microdialysis. *Am J Physiol Regul Integr Comp Physiol*. 2000;278:R400–R406.
- Mount DB, Zandi-Nejad K. Disorders of potassium balance. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: W.B. Saunders Co.; 2004:997–1040.
- McDonough AA, Thompson CB, Youn JH. Skeletal muscle regulates extracellular potassium. *Am J Physiol Renal Physiol*. 2002;282:F967–F974.
- Clausen T. Clinical and therapeutic significance of the Na⁺K⁺ pump. *Clin Sci*. 1998;95:3–17.
- Bundgaard H, Schmidt TA, Larsen JS, et al. K⁺ supplementation increases muscle [Na⁺K⁺-ATPase] and improves extrarenal K⁺ homeostasis in rats. *J Appl Physiol*. 1997;82:1136–1144.
- McKenna MJ. Effects of training on potassium homeostasis during exercise. *J Mol Cell Cardiol*. 1995;27:941–949.
- Nielsen JJ, et al. Effects of high-intensity intermittent training on potassium kinetics and performance in human skeletal muscle. *J Physiol*. 2004;554:857–870.
- Hebert SC, Mount DB, Gamba G. Molecular physiology of cation-coupled Cl[−] cotransport: the SLC12 family. *Pflugers Arch*. 2004;447:580–593.
- Gosmanov AR, Wong JA, Thomason DB. Duality of G protein-coupled mechanisms for beta-adrenergic activation of NKCC activity in skeletal muscle. *Am J Physiol Cell Physiol*. 2002;283:C1025–C1032.
- Mount DB, Zandi-Nejad K. Disorders of potassium balance. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: W.B. Saunders Co.; 2008:547–587.
- Ryan DP, et al. Mutations in potassium channel Kir2.6 cause susceptibility to thyrotoxic hypokalemic periodic paralysis. *Cell*. 2010;140:88–98.
- Guo J, et al. Extracellular K⁺ concentration controls cell surface density of IKr in rabbit hearts and of the HERG channel in human cell lines. *J Clin Invest*. 2009;119:2745–2757.
- Scholl UI, et al. Seizures, sensorineural deafness, ataxia, mental retardation, and electrolyte imbalance (SeSAME syndrome) caused by mutations in KCNJ10. *Proc Natl Acad Sci USA*. 2009;106:5842–5847.
- Briggs AP, Koechig I, Doisy EA, et al. Some changes in the composition of blood due the injection of insulin. *J Biol Chem*. 1924;58:721–730.
- Ferrannini E, et al. Independent stimulation of glucose metabolism and Na⁺K⁺ exchange by insulin in the human forearm. *Am J Physiol*. 1988;255:E953–E958.
- Nguyen TQ, Maalouf NM, Sakhaee K, et al. Comparison of insulin action on glucose versus potassium uptake in humans. *Clin J Am Soc Nephrol*. 2011;6:1533–1539.
- Rossetti L, Klein-Robbenhaar G, Giebisch G, et al. Effect of insulin on renal potassium metabolism. *Am J Physiol*. 1987;252:F60–F64.
- Dluhy RG, Axelrod L, Williams GH. Serum immunoreactive insulin and growth hormone response to potassium infusion in normal man. *J Appl Physiol*. 1972;33:22–26.
- Choi CS, Thompson CB, Leong PK, et al. Short-term K⁺ deprivation provokes insulin resistance of cellular K⁺ uptake revealed with the K⁺ clamp. *Am J Physiol Renal Physiol*. 2001;280:F95–F102.
- DeFronzo RA, et al. Influence of basal insulin and glucagon secretion on potassium and sodium metabolism. Studies with somatostatin in normal dogs and in normal and diabetic human beings. *J Clin Invest*. 1978;61:472–479.
- Adabala M, Jhaveri KD, Gitman M. Severe hyperkalemia resulting from octreotide use in a haemodialysis patient. *Nephrol Dial Transplant*. 2010.
- Sargent AI, Overton CC, Kuwik RJ, et al. Octreotide-induced hyperkalemia. *Pharmacotherapy*. 1994;14:497–501.
- DeFronzo RA, Felig P, Ferrannini E, et al. Effect of graded doses of insulin on splanchnic and peripheral potassium metabolism in man. *Am J Physiol*. 1980;238:E421–E427.
- Clausen T, Everts ME. Regulation of the Na,K-pump in skeletal muscle. *Kidney Int*. 1989;35:1–13.
- Li D, Sweeney G, Wang Q, et al. Participation of PI3K and atypical PKC in Na⁺K⁺-pump stimulation by IGF-I in VSMC. *Am J Physiol*. 1999;276:H2109–H2116.
- Al-Khalili L, Yu M, Chibalin AV. Na⁺(+),K⁺-ATPase trafficking in skeletal muscle: insulin stimulates translocation of both alpha(1)- and alpha(2)-subunit isoforms. *FEBS Lett*. 2003;536:198–202.
- Yudowski GA, et al. Phosphoinositide-3 kinase binds to a proline-rich motif in the Na⁺, K⁺-ATPase alpha subunit and regulates its trafficking. *Proc Natl Acad Sci USA*. 2000;97:6556–6561.
- Feraile E, et al. Insulin-induced stimulation of Na⁺K⁺-ATPase activity in kidney proximal tubule cells depends on phosphorylation of the alpha-subunit at Tyr-10. *Mol Biol Cell*. 1999;10:2847–2859.
- Boini KM, Graf D, Kuhl D, et al. SGK1 dependence of insulin induced hypokalemia. *Pflugers Arch*. 2009;457:955–961.
- D'Silva JL. The action of adrenaline on serum potassium. *J Physiol*. 1934;82:393–398.
- Olsson AM, Persson S, Schoder R. Effects of terbutaline and isoproterenol on hyperkalemia in nephrectomized rabbits. *Scand J Urol Nephrol*. 1978;12:35–38.
- Ballanyi K, Grafe P. Changes in intracellular ion activities induced by adrenaline in human and rat skeletal muscle. *Pflugers Arch*. 1988;411:283–288.
- Scheid CR, Fay FS. Beta-adrenergic stimulation of 42K influx in isolated smooth muscle cells. *Am J Physiol*. 1984;246:C415–C421.
- Williams ME, et al. Catecholamine modulation of rapid potassium shifts during exercise. *N Engl J Med*. 1985;312:823–827.

54. Fenn WO, Cobb DM. The potassium equilibrium in muscle. *J Gen Physiol*. 1934;17:629–656.
55. Simmons DH, Avedon M. Acid-base alterations and plasma potassium concentration. *Am J Physiol*. 1959;197:319–326.
56. Adrogué HJ, Madias NE. Changes in plasma potassium concentration during acute acid-base disturbances. *Am J Med*. 1981;71:456–467.
57. Aronson PS, Giebisch G. Effects of pH on potassium: new explanations for old observations. *J Am Soc Nephrol*. 2011;22:1981–1989.
58. Burnell JM, Villamill ME, Uyeno BT, et al. The effect in humans of extracellular PH change on the relationship between serum potassium concentration and intracellular potassium. *J Clin Invest*. 1956;35:935–939.
59. Oster JR, Perez G, Castro A, et al. Plasma potassium response to acute metabolic acidosis induced by mineral and nonmineral acids. *Miner Electrolyte Metab*. 1980;4:28–36.
60. Abrams WB, Lewis DW, Bellet S. The effect of acidosis and alkalosis on the plasma potassium concentration and the electrocardiogram of normal and potassium depleted dogs. *Am J Med Sci*. 1951;222:506–515.
61. Fulop M. Serum potassium in lactic acidosis and ketoacidosis. *N Engl J Med*. 1979;300:1087–1089.
62. Orringer CE, Eustace JC, Wunsch CD, et al. Natural history of lactic acidosis after grand-mal seizures. A model for the study of an anion-gap acidosis not associated with hyperkalemia. *N Engl J Med*. 1977;297:796–799.
63. Wiederseiner JM, Muser J, Lutz T, et al. Acute metabolic acidosis: characterization and diagnosis of the disorder and the plasma potassium response. *J Am Soc Nephrol*. 2004;15:1589–1596.
64. Adrogué HJ, Madias NE. PCO₂ and [K⁺]_p in metabolic acidosis: certainty for the first and uncertainty for the other. *J Am Soc Nephrol*. 2004;15:1667–1668.
65. Sonikian M, Metaxaki P, Vlassopoulos D, et al. Long-term management of sevelamer hydrochloride-induced metabolic acidosis aggravation and hyperkalemia in hemodialysis patients. *Ren Fail*. 2006;28:411–418.
66. Krapf R, Caduff P, Wagdi P, et al. Plasma potassium response to acute respiratory alkalosis. *Kidney Int*. 1995;47:217–224.
67. Natalini G, et al. Acute respiratory acidosis does not increase plasma potassium in normokalemic anaesthetized patients. A controlled randomized trial. *Eur J Anaesthesiol*. 2001;18:394–400.
68. Malnic G, Klose RM, Giebisch G. Micropuncture study of renal potassium excretion in the rat. *Am J Physiol*. 1964;206:674–686.
69. Canessa CM, et al. Amiloride-sensitive epithelial Na⁺ channel is made of three homologous subunits. *Nature*. 1994;367:463–467.
70. Giebisch G. Renal potassium transport: mechanisms and regulation. *Am J Physiol*. 1998;274:F817–F833.
71. Muto S. Potassium transport in the mammalian collecting duct. *Physiol Rev*. 2001;81:85–116.
72. Stokes JB. Potassium secretion by cortical collecting tubule: relation to sodium absorption, luminal sodium concentration, and transepithelial voltage. *Am J Physiol*. 1981;241:F395–F402.
73. Lalioti MD, et al. Wnk4 controls blood pressure and potassium homeostasis via regulation of mass and activity of the distal convoluted tubule. *Nat Genet*. 2006;38:1124–1132.
74. Meneton P, Loffing J, Warnock DG. Sodium and potassium handling by the aldosterone-sensitive distal nephron: the pivotal role of the distal and connecting tubule. *Am J Physiol Renal Physiol*. 2004;287:F593–F601.
75. Frindt G, Palmer LG. Apical potassium channels in the rat connecting tubule. *Am J Physiol Renal Physiol*. 2004;287:F1030–F1037.
76. Frindt G, Palmer LG. Na channels in the rat connecting tubule. *Am J Physiol Renal Physiol*. 2004;286:F669–F674.
77. Pluznick JL, Sansom SC. BK channels in the kidney: role in K⁺ secretion and localization of molecular components. *Am J Physiol Renal Physiol*. 2006;291:F517–F529.
78. Lu M, et al. Absence of small conductance K⁺ channel (SK) activity in apical membranes of thick ascending limb and cortical collecting duct in ROMK (Bartter's) knockout mice. *J Biol Chem*. 2002;277:37881–37887.
79. Grimm PR, Foutz RM, Brenner R, et al. Identification and localization of BK-beta subunits in the distal nephron of the mouse kidney. *Am J Physiol Renal Physiol*. 2007;293:F350–F359.
80. Rieg T, et al. The role of the BK channel in potassium homeostasis and flow-induced renal potassium excretion. *Kidney Int*. 2007.
81. Grimm PR, Irsik DL, Settles DC, et al. Hypertension of Kcnmb1^{-/-} is linked to deficient K secretion and aldosteronism. *Proc Natl Acad Sci USA*. 2009;106:11800–11805.
82. Mamenko MV, et al. The renal TRPV4 channel is essential for adaptation to increased dietary potassium. *Kidney Int*. 2017;91:1398–1409.
83. Amorim JB, Bailey MA, Musa-Aziz R, et al. Role of luminal anion and pH in distal tubule potassium secretion. *Am J Physiol Renal Physiol*. 2003;284:F381–F388.
84. Mount DB, Gamba G. Renal K-Cl cotransporters. *Curr Opin Nephrol Hypertens*. 2001;10:685–692.
85. Frindt G, Palmer LG. K⁺ secretion in the rat kidney: Na⁺ channel-dependent and -independent mechanisms. *Am J Physiol Renal Physiol*. 2009;297:F389–F396.
86. Wingo CS, Armitage FE. Rubidium absorption and proton secretion by rabbit outer medullary collecting duct via H-K-ATPase. *Am J Physiol*. 1992;263:F849–F857.
87. Okusa MD, Unwin RJ, Velazquez H, et al. Active potassium absorption by the renal distal tubule. *Am J Physiol*. 1992;262:F488–F493.
88. Palmer LG, Frindt G. Aldosterone and potassium secretion by the cortical collecting duct. *Kidney Int*. 2000;57:1324–1328.
89. Giebisch GH. A trail of research on potassium. *Kidney Int*. 2002;62:1498–1512.
90. Gennari FJ, Segal AS. Hyperkalemia: an adaptive response in chronic renal insufficiency. *Kidney Int*. 2002;62:1–9.
91. Palmer LG, Antonian L, Frindt G. Regulation of apical K and Na channels and Na/K pumps in rat cortical collecting tubule by dietary K. *J Gen Physiol*. 1994;104:693–710.
92. Mick VE, et al. The alpha-subunit of the epithelial sodium channel is an aldosterone-induced transcript in mammalian collecting ducts, and this transcriptional response is mediated via distinct cis-elements in the 5'-flanking region of the gene. *Mol Endocrinol*. 2001;15:575–588.
93. Reisenauer MR, et al. AF17 competes with AF9 for binding to Dot1a to up-regulate transcription of epithelial Na⁺ channel alpha. *J Biol Chem*. 2009;284:35659–35669.
94. Snyder PM. Minireview: regulation of epithelial Na⁺ channel trafficking. *Endocrinology*. 2005;146:5079–5085.
95. Loffing J, et al. Differential subcellular localization of ENaC subunits in mouse kidney in response to high- and low-Na diets. *Am J Physiol Renal Physiol*. 2000;279:F252–F258.
96. Loffing J, et al. Aldosterone induces rapid apical translocation of ENaC in early portion of renal collecting system: possible role of SGK. *Am J Physiol Renal Physiol*. 2001;280:F675–F682.
97. Nielsen J, et al. Sodium transporter abundance profiling in kidney: effect of spironolactone. *Am J Physiol Renal Physiol*. 2002;283:F923–F933.
98. Naray-Fejes-Toth A, Canessa C, Cleaveland ES, et al. Sgk is an aldosterone-induced kinase in the renal collecting duct. Effects on epithelial Na⁺ channels. *J Biol Chem*. 1999;274:16973–16978.
99. Kamynina E, Tauxe C, Staub O. Distinct characteristics of two human Nedd4 proteins with respect to epithelial Na⁺ channel regulation. *Am J Physiol Renal Physiol*. 2001;281:F469–F477.
100. Findling JW, Raff H, Hansson JH, et al. Liddle's syndrome: prospective genetic screening and suppressed aldosterone secretion in an extended kindred. *J Clin Endocrinol Metab*. 1997;82:1071–1074.
101. Firsov D, et al. Cell surface expression of the epithelial Na channel and a mutant causing Liddle syndrome: a quantitative approach. *Proc Natl Acad Sci USA*. 1996;93:15370–15375.
102. Huang DY, et al. Impaired regulation of renal K⁺ elimination in the sgk1-knockout mouse. *J Am Soc Nephrol*. 2004;15:885–891.
103. Wulff P, et al. Impaired renal Na⁺ retention in the sgk1-knockout mouse. *J Clin Invest*. 2002;110:1263–1268.
104. Yang L, et al. SGK1-dependent ENaC processing and trafficking in mice with high dietary K intake and elevated aldosterone. *Am J Physiol Renal Physiol*. 2017;312:F65–F76.
105. Vallet V, Chraïbi A, Gaeggeler HP, et al. An epithelial serine protease activates the amiloride-sensitive sodium channel. *Nature*. 1997;389:607–610.
106. Narikeyo T, et al. Regulation of prostaticin by aldosterone in the kidney. *J Clin Invest*. 2002;109:401–408.
107. Vuagniaux G, Vallet V, Jaeger NE, et al. Synergistic activation of ENaC by three membrane-bound channel-activating serine proteases (mCAP1, mCAP2, and mCAP3) and serum- and glucocorticoid-regulated kinase (Sgk1) in Xenopus Oocytes. *J Gen Physiol*. 2002;120:191–201.
108. Kleyman TR, Carattino MD, Hughey RP. ENaC at the cutting edge: regulation of epithelial sodium channels by proteases. *J Biol Chem*. 2009;284:20447–20451.

109. Kleyman TR, Myerburg MM, Hughey RP. Regulation of ENaCs by proteases: an increasingly complex story. *Kidney Int.* 2006;70:1391–1392.
110. Carattino MD, et al. The epithelial Na⁺ channel is inhibited by a peptide derived from proteolytic processing of its alpha subunit. *J Biol Chem.* 2006;281:18901–18907.
111. Kashlan OB, Blobner BM, Zuzek Z, et al. Na⁺ inhibits the epithelial Na⁺ channel by binding to a site in an extracellular acidic cleft. *J Biol Chem.* 2015;290:568–576.
112. Kitagawa H, Chang H, Fujita T. Hyperkalemia due to nafamostat mesylate. *N Engl J Med.* 1995;332:687.
113. Iwashita K, et al. Inhibition of prostatic secretion by serine protease inhibitors in the kidney. *J Am Soc Nephrol.* 2003;14:11–16.
114. Frindt G, Shah A, Edvinsson J, et al. Dietary K regulates ROMK channels in connecting tubule and cortical collecting duct of rat kidney. *Am J Physiol Renal Physiol.* 2009;296:F347–F354.
115. Palmer LG, Frindt G. Regulation of apical K channels in rat cortical collecting tubule during changes in dietary K intake. *Am J Physiol.* 1999;277:F805–F812.
116. Lin DH, et al. Protein tyrosine kinase is expressed and regulates ROMK1 location in the cortical collecting duct. *Am J Physiol Renal Physiol.* 2004;286:F881–F892.
117. Najjar F, et al. Dietary K⁺ regulates apical membrane expression of maxi-K channels in rabbit cortical collecting duct. *Am J Physiol Renal Physiol.* 2005;289:F922–F932.
118. Estilo G, et al. Effect of aldosterone on BK channel expression in mammalian cortical collecting duct. *Am J Physiol Renal Physiol.* 2008;295:F780–F788.
119. Kahle KT, et al. WNK4 regulates the balance between renal NaCl reabsorption and K⁺ secretion. *Nat Genet.* 2003;35:372–376.
120. Delaloy C, et al. Multiple promoters in the WNK1 gene: one controls expression of a kidney-specific kinase-defective isoform. *Mol Cell Biol.* 2003;23:9208–9221.
121. Wade JB, et al. WNK1 kinase isoform switch regulates renal potassium excretion. *Proc Natl Acad Sci USA.* 2006;103:8558–8563.
122. Lazrak A, Liu Z, Huang CL. Antagonistic regulation of ROMK by long and kidney-specific WNK1 isoforms. *Proc Natl Acad Sci USA.* 2006;103:1615–1620.
123. Wu P, et al. Role of WNK4 and kidney-specific WNK1 in mediating the effect of high dietary K(+) intake on ROMK channel in the distal convoluted tubule. *Am J Physiol Renal Physiol.* 2018;315:F223–F230.
124. O'Reilly M, et al. Dietary electrolyte-driven responses in the renal WNK kinase pathway in vivo. *J Am Soc Nephrol.* 2006;17:2402–2413.
125. Zhuang J, et al. WNK4 kinase inhibits Maxi K channel activity by a kinase-dependent mechanism. *Am J Physiol Renal Physiol.* 2011;301:F410–F419.
126. Yue P, Zhang C, Lin DH, et al. WNK4 inhibits Ca(2+)-activated big-conductance potassium channels (BK) via mitogen-activated protein kinase-dependent pathway. *Biochim Biophys Acta.* 2013;1833:2101–2110.
127. Wang Z, et al. Regulation of large-conductance Ca²⁺-activated K⁺ channels by WNK4 kinase. *Am J Physiol Cell Physiol.* 2013;305:C846–C853.
128. Lin DH, et al. K depletion increases protein tyrosine kinase-mediated phosphorylation of ROMK. *Am J Physiol Renal Physiol.* 2002;283:F671–F677.
129. Sterling H, et al. Inhibition of protein-tyrosine phosphatase stimulates the dynamin-dependent endocytosis of ROMK1. *J Biol Chem.* 2002;277:4317–4323.
130. Mount DB, Yu AS. Transport of inorganic solutes: sodium, potassium, calcium, magnesium, and phosphate. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: W.B. Saunders Co.; 2008:156–213.
131. Wei Y, Bloom P, Lin D, et al. Effect of dietary K intake on apical small-conductance K channel in CCD: role of protein tyrosine kinase. *Am J Physiol Renal Physiol.* 2001;281:F206–F212.
132. Babilonia E, et al. Superoxide anions are involved in mediating the effect of low K intake on c-Src expression and renal K secretion in the cortical collecting duct. *J Biol Chem.* 2005;280:10790–10796.
133. Babilonia E, et al. Role of gp91phox-containing NADPH oxidase in mediating the effect of K restriction on ROMK channels and renal K excretion. *J Am Soc Nephrol.* 2007;18:2037–2045.
134. Wang ZJ, et al. Decrease in dietary K intake stimulates the generation of superoxide anions in the kidney and inhibits K secretory channels in the CCD. *Am J Physiol Renal Physiol.* 2010;298:F1515–F1522.
135. Wei Y, Zavilowitz B, Satlin LM, et al. Angiotensin II inhibits the ROMK-like small conductance K channel in renal cortical collecting duct during dietary potassium restriction. *J Biol Chem.* 2007;282:6455–6462.
136. Jin Y, Wang Y, Wang ZJ, et al. Inhibition of angiotensin type 1 receptor impairs renal ability of K conservation in response to K restriction. *Am J Physiol Renal Physiol.* 2009;296:F1179–F1184.
137. Tomita K, Pisano JJ, Burg MB, et al. Effects of vasopressin and bradykinin on anion transport by the rat cortical collecting duct. Evidence for an electroneutral sodium chloride transport pathway. *J Clin Invest.* 1986;77:136–141.
138. Terada Y, Knepper MA. Thiazide-sensitive NaCl absorption in rat cortical collecting duct. *Am J Physiol.* 1990;259:F519–F528.
139. Levell F, et al. The Na⁺-dependent chloride-bicarbonate exchanger SLC4A8 mediates an electroneutral Na⁺ reabsorption process in the renal cortical collecting ducts of mice. *J Clin Invest.* 2010;120:1627–1635.
140. San-Cristobal P, et al. Angiotensin II signaling increases activity of the renal Na-Cl cotransporter through a WNK4-SPAK-dependent pathway. *Proc Natl Acad Sci USA.* 2009;106:4384–4389.
141. Sandberg MB, Riquier AD, Pihakski-Maunsbach K, et al. ANG II provokes acute trafficking of distal tubule Na⁺-Cl(-) cotransporter to apical membrane. *Am J Physiol Renal Physiol.* 2007;293:F662–F669.
142. Sealey JE, Clark I, Bull MB, et al. Potassium balance and the control of renin secretion. *J Clin Invest.* 1970;49:2119–2127.
143. Vallon V, Schroth J, Lang F, et al. Expression and phosphorylation of the Na⁺-Cl⁻ cotransporter NCC in vivo is regulated by dietary salt, potassium, and SGK1. *Am J Physiol Renal Physiol.* 2009;297:F704–F712.
144. Welling PA, Chang YP, Delpire E, et al. Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int.* 2010;77:1063–1069.
145. Al-Qusairi L, et al. Renal tubular Ubiquitin-Protein ligase NEDD4-2 is required for renal adaptation during long-term potassium depletion. *J Am Soc Nephrol.* 2017;28:2431–2442.
146. Ring AM, et al. An SGK1 site in WNK4 regulates Na⁺ channel and K⁺ channel activity and has implications for aldosterone signaling and K⁺ homeostasis. *Proc Natl Acad Sci USA.* 2007;104:4025–4029.
147. Sterling D, Brown NJ, Supuran CT, et al. The functional and physical relationship between the DRA bicarbonate transporter and carbonic anhydrase II. *Am J Physiol Cell Physiol.* 2002;283:C1522–F1529.
148. Lin DH, Sterling H, Wang WH. The protein tyrosine kinase-dependent pathway mediates the effect of K intake on renal K secretion. *Physiology (Bethesda).* 2005;20:140–146.
149. Yue P, et al. Src family protein tyrosine kinase (PTK) modulates the effect of SGK1 and WNK4 on ROMK channels. *Proc Natl Acad Sci USA.* 2009;106:15061–15066.
150. Lin DH, et al. Src-family protein tyrosine kinase phosphorylates WNK4 and modulates its inhibitory effect on KCNJ1 (ROMK). *Proc Natl Acad Sci USA.* 2015;112:4495–4500.
151. Skott O, Briggs JP. Direct demonstration of macula densa-mediated renin secretion. *Science.* 1987;237:1618–1620.
152. Lorenz JN, Weihprecht H, Schnermann J, et al. Renin release from isolated juxtaglomerular apparatus depends on macula densa chloride transport. *Am J Physiol.* 1991;260:F486–F493.
153. Friis UG, et al. Regulation of renin secretion by renal juxtaglomerular cells. *Pflugers Arch.* 2013;465:25–37.
154. Wagner C, Jensen BL, Kramer BK, et al. Control of the renal renin system by local factors. *Kidney Int Suppl.* 1998;67:S78–S83.
155. Schweda F, Wagner C, Kramer BK, et al. Preserved macula densa-dependent renin secretion in A1 adenosine receptor knockout mice. *Am J Physiol Renal Physiol.* 2003;284:F770–F777.
156. Henrich WL, McAlister EA, Smith PB, et al. Direct inhibitory effect of atriopeptin III on renin release in primate kidney. *Life Sci.* 1987;41:259–264.
157. Todorov V, Muller M, Schweda F, et al. Tumor necrosis factor-alpha inhibits renin gene expression. *Am J Physiol Regul Integr Comp Physiol.* 2002;283:R1046–R1051.
158. Li YC, et al. 1,25-Dihydroxyvitamin D(3) is a negative endocrine regulator of the renin-angiotensin system. *J Clin Invest.* 2002;110:229–238.
159. Wagner C, Pfeifer A, Ruth P, et al. Role of cGMP-kinase II in the control of renin secretion and renin expression. *J Clin Invest.* 1998;102:1576–1582.
160. Peti-Peterdi J, Harris RC. Macula densa sensing and signaling mechanisms of renin release. *J Am Soc Nephrol.* 2010.
161. Jensen BL, Kramer BK, Kurtz A. Adrenomedullin stimulates renin release and renin mRNA in mouse juxtaglomerular granular cells. *Hypertension.* 1997;29:1148–1155.

162. Boivin V, et al. Immunofluorescent imaging of beta 1- and beta 2-adrenergic receptors in rat kidney. *Kidney Int.* 2001;59:515–531.
163. Vargas SL, Toma I, Kang JJ, et al. Activation of the succinate receptor GPR91 in macula densa cells causes renin release. *J Am Soc Nephrol.* 2009;20:1002–1011.
164. Cheng HF, Wang JL, Zhang MZ, et al. Role of p38 in the regulation of renal cortical cyclooxygenase-2 expression by extracellular chloride. *J Clin Invest.* 2000;106:681–688.
165. Stichtenoeth DO, Marhauer V, Tsikas D, et al. Effects of specific COX-2-inhibition on renin release and renal and systemic prostanoid synthesis in healthy volunteers. *Kidney Int.* 2005;68:2197–2207.
166. Yang Y, et al. Salt restriction leads to activation of adult renal mesenchymal stromal cell-like cells via prostaglandin E2 and E-prostanoid receptor 4. *Hypertension.* 2015;65:1047–1054.
167. Dluhy RG, Axelrod L, Underwood RH, et al. Studies of the control of plasma aldosterone concentration in normal man. II. Effect of dietary potassium and acute potassium infusion. *J Clin Invest.* 1972;51:1950–1957.
168. Davies LA, et al. TASK channel deletion in mice causes primary hyperaldosteronism. *Proc Natl Acad Sci USA.* 2008;105:2203–2208.
169. Chen XL, Bayliss DA, Fern RJ, et al. A role for T-type Ca²⁺ channels in the synergistic control of aldosterone production by ANG II and K. *Am J Physiol.* 1999;276:F674–F683.
170. Somekawa S, et al. Regulation of aldosterone and cortisol production by the transcriptional repressor neuron restrictive silencer factor. *Endocrinology.* 2009;150:3110–3117.
171. Condon JC, Pezzi V, Drummond BM, et al. Calmodulin-dependent kinase I regulates adrenal cell expression of aldosterone synthase. *Endocrinology.* 2002;143:3651–3657.
172. Boulkroun S, et al. Prevalence, clinical, and molecular correlates of KCNJ5 mutations in primary aldosteronism. *Hypertension.* 2012;59:592–598.
173. Hilbers U, et al. Local renin-angiotensin system is involved in K⁺-induced aldosterone secretion from human adrenocortical NCI-H295 cells. *Hypertension.* 1999;33:1025–1030.
174. Mazzocchi G, Malendowicz LK, Markowska A, et al. Role of adrenal renin-angiotensin system in the control of aldosterone secretion in sodium-restricted rats. *Am J Physiol Endocrinol Metab.* 2000;278:E1027–E1030.
175. Saruta T, Kaplan NM. Adrenocortical steroidogenesis: the effects of prostaglandins. *J Clin Invest.* 1972;51:2246–2251.
176. Gordon RD, Kuchel O, Liddle GW, et al. Role of the sympathetic nervous system in regulating renin and aldosterone production in man. *J Clin Invest.* 1967;46:599–605.
177. Gupta P, Franco-Saenz R, Mulrow PJ. Regulation of the adrenal renin angiotensin system in cultured bovine zona glomerulosa cells: effect of catecholamines. *Endocrinology.* 1992;130:2129–2134.
178. Csukas S, Hanke CJ, Rewolinski D, et al. Prostaglandin E2-induced aldosterone release is mediated by an EP2 receptor. *Hypertension.* 1998;31:575–581.
179. Clark BA, Brown RS, Epstein FH. Effect of atrial natriuretic peptide on potassium-stimulated aldosterone secretion: potential relevance to hypoaldosteronism in man. *J Clin Endocrinol Metab.* 1992;75:399–403.
180. Cherradi N, et al. Atrial natriuretic peptide inhibits calcium-induced steroidogenic acute regulatory protein gene transcription in adrenal glomerulosa cells. *Mol Endocrinol.* 1998;12:962–972.
181. Choi MJ, Ziyadeh FN. The utility of the transtubular potassium gradient in the evaluation of hyperkalemia. *J Am Soc Nephrol.* 2008;19:424–426.
182. Weinstein AM. A mathematical model of rat cortical collecting duct: determinants of the transtubular potassium gradient. *Am J Physiol Renal Physiol.* 2001;280:F1072–F1092.
183. Cil O, et al. Salt-sparing diuretic action of a water-soluble urea analog inhibitor of urea transporters UTA and UT-B in rats. *Kidney Int.* 2015;88:311–320.
184. Chacko M, Fordtran JS, Emmett M. Effect of mineralocorticoid activity on transtubular potassium gradient, urinary [K⁺]/[Na⁺] ratio, and fractional excretion of potassium. *Am J Kidney Dis.* 1998;32:47–51.
185. Joo KW, et al. Transtubular potassium concentration gradient (TTKG) and urine ammonium in differential diagnosis of hypokalemia. *J Nephrol.* 2000;13:120–125.
186. Lin SH, et al. Laboratory tests to determine the cause of hypokalemia and paralysis. *Arch Intern Med.* 2004;164:1561–1566.
187. Battle DC, Hizon M, Cohen E, et al. The use of the urinary anion gap in the diagnosis of hyperchloremic metabolic acidosis. *N Engl J Med.* 1988;594–599.
188. Cohen JD, Neaton JD, Prineas RJ, et al. Diuretics, serum potassium and ventricular arrhythmias in the Multiple Risk Factor Intervention Trial. *Am J Cardiol.* 1987;60:548–554.
189. Wahr JA, et al. Preoperative serum potassium levels and perioperative outcomes in cardiac surgery patients. Multicenter Study of Perioperative Ischemia Research Group. *JAMA.* 1999;281:2203–2210.
190. Modesto KM, et al. Safety of exercise stress testing in patients with abnormal concentrations of serum potassium. *Am J Cardiol.* 2006;97:1247–1249.
191. Slovis C, Jenkins R. ABC of clinical electrocardiography: conditions not primarily affecting the heart. *BMJ.* 2002;324:1320–1323.
192. Yelamanchi VP, Molnar J, Ranade V, et al. Influence of electrolyte abnormalities on interlead variability of ventricular repolarization times in 12-lead electrocardiography. *Am J Ther.* 2001;8:117–122.
193. Roden DM. A practical approach to torsade de pointes. *Clin Cardiol.* 1997;20:285–290.
194. Darbar D, Sile S, Fish FA, et al. Congenital long QT syndrome aggravated by salt-wasting nephropathy. *Heart Rhythm.* 2005;2:304–306.
195. Parham WA, Mehdirad AA, Biermann KM, et al. Hyperkalemia revisited. *Tex Heart Inst J.* 2006;33:40–47.
196. Christe G. Effects of low [K⁺]_o on the electrical activity of human cardiac ventricular and Purkinje cells. *Cardiovasc Res.* 1983;17:243–250.
197. Goldstein SA. K_{2P} potassium channels, mysterious and paradoxically exciting. *Sci Signal.* 2011;4:pe35.
198. Ma L, Zhang X, Chen H. TWIK-1 two-pore domain potassium channels change ion selectivity and conduct inward leak sodium currents in hypokalemia. *Sci Signal.* 2011;4:ra37.
199. Pezhouman A, et al. Molecular basis of hypokalemia-induced ventricular fibrillation. *Circulation.* 2015;132:1528–1537.
200. Aronsen JM, et al. Hypokalaemia induces Ca²⁺(+) overload and Ca²⁺(+) waves in ventricular myocytes by reducing Na⁺(+),K⁺(+)-ATPase alpha(2) activity. *J Physiol.* 2015;593:1509–1521.
201. Weiss JN, Qu Z, Shivkumar K. Electrophysiology of Hypokalemia and Hyperkalemia. *Circ Arrhythm Electrophysiol.* 2017;10.
202. Tang NL, et al. Severe hypokalemic myopathy in Gitelman's syndrome. *Muscle Nerve.* 1999;22:545–547.
203. Wong KM, et al. Hypokalemic metabolic acidosis attributed to cough mixture abuse. *Am J Kidney Dis.* 2001;38:390–394.
204. Tattersall RB. A paper which changed clinical practice (slowly). Jacob Holler on potassium deficiency in diabetic acidosis (1946). *Diabet Med.* 1999;16:978–984.
205. Comi G, Testa D, Cornelio F, et al. Potassium depletion myopathy: a clinical and morphological study of six cases. *Muscle Nerve.* 1985;8:17–21.
206. Mujais SK, Katz AI. Potassium deficiency. In: Seldin DW, Giebisch G, eds. *The Kidney: Physiology and Pathophysiology*. Vol. 2. Lippincott Williams & Wilkins; 2000:1615–1646.
207. Schwartz WB, Relman AS. Effects of electrolyte disorders on renal structure and function. *N Engl J Med.* 1967;276:383–389. contd.
208. Elitok S, Bieringer M, Schneider W, et al. Kaliopenic nephropathy revisited. *Clin Kidney J.* 2016;9:543–546.
209. Cremer W, Bock KD. Symptoms and course of chronic hypokalemic nephropathy in man. *Clin Nephrol.* 1977;7:112–119.
210. Torres VE, Young WF Jr, Offord KP, et al. Association of hypokalemia, aldosteronism, and renal cysts. *N Engl J Med.* 1990;322:345–351.
211. Yasuhara D, et al. "End-stage kidney" in longstanding bulimia nervosa. *Int J Eat Disord.* 2005;38:383–385.
212. Menahem SA, Perry GJ, Dowling J, et al. Hypokalaemia-induced acute renal failure. *Nephrol Dial Transplant.* 1999;14:2216–2218.
213. Suga SI, et al. Hypokalemia induces renal injury and alterations in vasoactive mediators that favor salt sensitivity. *Am J Physiol Renal Physiol.* 2001;281:F620–F629.
214. Reungjui S, et al. Hypokalemic nephropathy is associated with impaired angiogenesis. *J Am Soc Nephrol.* 2008;19:125–134.
215. Levi M, McDonald LA, Preisig PA, et al. Chronic K depletion stimulates rat renal brush-border membrane Na-citrate cotransporter. *Am J Physiol.* 1991;261:F767–F773.
216. Elkjaer ML, et al. Altered expression of renal NHE3, TSC, BSC-1, and ENaC subunits in potassium-depleted rats. *Am J Physiol Renal Physiol.* 2002;283:F1376–F1388.
217. Berl T, Linas SL, Aisenbrey GA, et al. On the mechanism of polyuria in potassium depletion. The role of polydipsia. *J Clin Invest.* 1977;60:620–625.

218. Marples D, Frokiaer J, Dorup J, et al. Hypokalemia-induced downregulation of aquaporin-2 water channel expression in rat kidney medulla and cortex. *J Clin Invest*. 1996;97:1960–1968.
219. AmLal H, Krane CM, Chen Q, et al. Early polyuria and urinary concentrating defect in potassium deprivation. *Am J Physiol Renal Physiol*. 2000;279:F655–F663.
220. Khositseth S, et al. Autophagic degradation of aquaporin-2 is an early event in hypokalemia-induced nephrogenic diabetes insipidus. *Sci Rep*. 2015;5:18311.
221. Coca SG, Perazella MA, Buller GK. The cardiovascular implications of hypokalemia. *Am J Kidney Dis*. 2005;45:233–247.
222. Ray PE, Suga S, Liu XH, et al. Chronic potassium depletion induces renal injury, salt sensitivity, and hypertension in young rats. *Kidney Int*. 2001;59:1850–1858.
223. Sun Y, et al. Dietary potassium regulates vascular calcification and arterial stiffness. *JCI Insight*. 2017;2.
224. He FJ, MacGregor GA. Fortnightly review: beneficial effects of potassium. *BMJ*. 2001;323:497–501.
225. Kaplan NM, Carnegie A, Raskin P, et al. Potassium supplementation in hypertensive patients with diuretic-induced hypokalemia. *N Engl J Med*. 1985;312:746–749.
226. Cohen HW, Madhavan S, Alderman MH. High and low serum potassium associated with cardiovascular events in diuretic-treated patients. *J Hypertens*. 2001;19:1315–1323.
227. Shafi T, Appel LJ, Miller ER 3rd, et al. Changes in serum potassium mediate thiazide-induced diabetes. *Hypertension*. 2008;52:1022–1029.
228. Mattu A, Brady WJ, Robinson DA. Electrocardiographic manifestations of hyperkalemia. *Am J Emerg Med*. 2000;18:721–729.
229. Greenberg A. Hyperkalemia: treatment options. *Semin Nephrol*. 1998;18:46–57.
230. McLean SA, Paul ID, Spector PS. Lidocaine-induced conduction disturbance in patients with systemic hyperkalemia. *Ann Emerg Med*. 2000;36:615–618.
231. Littmann L, Monroe MH, Taylor L 3rd, et al. The hyperkalemic Brugada sign. *J Electrocardiol*. 2007;40:53–59.
232. Acker CG, Johnson JP, Palevsky PM, et al. Hyperkalemia in hospitalized patients: causes, adequacy of treatment, and results of an attempt to improve physician compliance with published therapy guidelines. *Arch Intern Med*. 1998;158:917–924.
233. Surawicz B, Chlebus H, Mazzoleni A. Hemodynamic and electrocardiographic effects of hyperpotassemia. Differences in response to slow and rapid increases in concentration of plasma K. *Am Heart J*. 1967;73:647–664.
234. Paice B, Gray JM, McBride D, et al. Hyperkalaemia in patients in hospital. *Br Med J (Clin Res Ed)*. 1983;286:1189–1192.
235. Montague BT, Ouellette JR, Buller GK. Retrospective review of the frequency of ECG changes in hyperkalemia. *Clin J Am Soc Nephrol*. 2008;3:324–330.
236. Aslam S, Friedman EA, Ifudu O. Electrocardiography is unreliable in detecting potentially lethal hyperkalaemia in haemodialysis patients. *Nephrol Dial Transplant*. 2002;17:1639–1642.
237. Szerlip HM, Weiss J, Singer I. Profound hyperkalemia without electrocardiographic manifestations. *Am J Kidney Dis*. 1986;7:461–465.
238. Somers MP, Brady WJ, Perron AD, et al. The prominent T wave: electrocardiographic differential diagnosis. *Am J Emerg Med*. 2002;20:243–251.
239. Green D, Green HD, New DI, et al. The clinical significance of hyperkalaemia-associated repolarization abnormalities in end-stage renal disease. *Nephrol Dial Transplant*. 2013;28:99–105.
240. Freeman SJ, Fale AD. Muscular paralysis and ventilatory failure caused by hyperkalaemia. *Br J Anaesth*. 1993;70:226–227.
241. Evers S, Engelen A, Karsch V, et al. Secondary hyperkalaemic paralysis. *J Neurol Neurosurg Psychiatry*. 1998;64:249–252.
242. Miller TM, et al. Correlating phenotype and genotype in the periodic paralyses. *Neurology*. 2004;63:1647–1655.
243. Lehmann-Horn F, Jurkat-Rott K. Voltage-gated ion channels and hereditary disease. *Physiol Rev*. 1999;79:1317–1372.
244. Raja Rayan DL, Hanna MG. Skeletal muscle channelopathies: nondystrophic myotonias and periodic paralysis. *Curr Opin Neurol*. 2010.
245. Abbott GW, et al. MiRP2 forms potassium channels in skeletal muscle with Kv3.4 and is associated with periodic paralysis. *Cell*. 2001;104:217–231.
246. Tannen RL, Wedell E, Moore R. Renal adaptation to a high potassium intake. The role of hydrogen ion. *J Clin Invest*. 1973;52:2089–2101.
247. DuBose TD Jr, Good DW. Effects of chronic hyperkalemia on renal production and proximal tubule transport of ammonium in rats. *Am J Physiol*. 1991;260:F680–F687.
248. Good DW. Ammonium transport by the thick ascending limb of Henle's loop. *Annu Rev Physiol*. 1994;56:623–647.
249. DuBose TD Jr, Good DW. Chronic hyperkalemia impairs ammonium transport and accumulation in the inner medulla of the rat. *J Clin Invest*. 1992;90:1443–1449.
250. Attmane-Elakeb A, et al. Stimulation by in vivo and in vitro metabolic acidosis of expression of rBSC1, the Na-K(NH4)-2Cl cotransporter of the rat medullary thick ascending limb. *J Biol Chem*. 1998;273:33681–33691.
251. Bourgeois S, et al. NHE4 is critical for the renal handling of ammonia in rodents. *J Clin Invest*. 2010;120:1895–1904.
252. Harris AN, et al. Mechanism of hyperkalemia-induced metabolic acidosis. *J Am Soc Nephrol*. 2018.
253. Matsuda O, et al. Primary role of hyperkalemia in the acidosis of hyporeninemic hypoaldosteronism. *Nephron*. 1988;49:203–209.
254. Szymlan P, Better OS, Chaimowitz C, et al. Role of hyperkalemia in the metabolic acidosis of isolated hypoaldosteronism. *N Engl J Med*. 1976;294:361–365.
255. Cohn JN, Kowey PR, Whelton PK, et al. New guidelines for potassium replacement in clinical practice: a contemporary review by the National Council on Potassium in Clinical Practice. *Arch Intern Med*. 2000;160:2429–2436.
256. Gennari FJ. Hypokalemia. *N Engl J Med*. 1998;339:451–458.
257. Greco A, et al. Hypokalaemia in hospitalised patients. *Swiss Med Wkly*. 2016;146:w14320.
258. Jensen HK, Brabrand M, Vinholt PJ, et al. Hypokalemia in acute medical patients: risk factors and prognosis. *Am J Med*. 2015;128:60–67.e1.
259. Paltiel O, Salakhov E, Ronen I, et al. Management of severe hypokalemia in hospitalized patients: a study of quality of care based on computerized databases. *Arch Intern Med*. 2001;161:1089–1095.
260. Crop MJ, Hoorn EJ, Lindemans J, et al. Hypokalaemia and subsequent hyperkalaemia in hospitalized patients. *Nephrol Dial Transplant*. 2007;22:3471–3477.
261. Schnaper HW, et al. Potassium restoration in hypertensive patients made hypokalemic by hydrochlorothiazide. *Arch Intern Med*. 1989;149:2677–2681.
262. Rosenberg J, Gustafsson F, Galatius S, et al. Combination therapy with metolazone and loop diuretics in outpatients with refractory heart failure: an observational study and review of the literature. *Cardiovasc Drugs Ther*. 2005;19:301–306.
263. Tziviskou E, et al. Prevalence and pathogenesis of hypokalemia in patients on chronic peritoneal dialysis: one center's experience and review of the literature. *Int Urol Nephrol*. 2003;35:429–434.
264. Macdonald JE, Struthers AD. What is the optimal serum potassium level in cardiovascular patients? *J Am Coll Cardiol*. 2004;43:155–161.
265. Masters PW, Lawson N, Marenah CB, et al. High ambient temperature: a spurious cause of hypokalaemia. *BMJ*. 1996;312:1652–1653.
266. Ulahannan TJ, McVittie J, Keenan J. Ambient temperatures and potassium concentrations. *Lancet*. 1998;352:1680–1681.
267. Heller SR, Robinson RT. Hypoglycaemia and associated hypokalaemia in diabetes: mechanisms, clinical implications and prevention. *Diabetes Obes Metab*. 2000;2:75–82.
268. Fuentebella J, Kerner JA. Refeeding syndrome. *Pediatr Clin North Am*. 2009;56:1201–1210.
269. Laso FJ, Gonzalez-Buitrago JM, Martin-Ruiz C, et al. Inter-relationship between serum potassium and plasma catecholamines and 3':5' cyclic monophosphate in alcohol withdrawal. *Drug Alcohol Depend*. 1990;26:183–188.
270. Madias JE, Shah B, Chintalapally G, et al. Admission serum potassium in patients with acute myocardial infarction: its correlates and value as a determinant of in-hospital outcome. *Chest*. 2000;118:904–913.
271. Schaefer M, Link J, Hannemann L, et al. Excessive hypokalemia and hyperkalemia following head injury. *Intensive Care Med*. 1995;21:235–237.
272. Tse HF, Yeung CK. From profound hypokalemia to fatal rhabdomyolysis after severe head injury. *Am J Med*. 2000;109:599–600.
273. Braden GL, von Oeyen PT, Germain MJ, et al. Ritodrine- and terbutaline-induced hypokalemia in preterm labor: mechanisms and consequences. *Kidney Int*. 1997;51:1867–1875.
274. Hoffman RS, Kirrane BM, Marcus SM. A descriptive study of an outbreak of clenbuterol-containing heroin. *Ann Emerg Med*. 2008;52:548–553.

275. de Wijkerslooth LR, Koch BC, Malingre MM, et al. Life-threatening hypokalaemia and lactate accumulation after autointoxication with Stacker 2, a 'powerful slimming agent'. *Br J Clin Pharmacol*. 2008;66:728–731.
276. Amitai Y, Lovejoy FH Jr. Hypokalemia in acute theophylline poisoning. *Am J Emerg Med*. 1988;6:214–218.
277. Rice JE, Faunt JD. Excessive cola consumption as a cause of hypokalaemic myopathy. *Intern Med J*. 2001;31:317–318.
278. Whyte KF, Reid C, Addis GJ, et al. Salbutamol induced hypokalaemia: the effect of theophylline alone and in combination with adrenaline. *Br J Clin Pharmacol*. 1988;25:571–578.
279. Bhoelan BS, Stevering CH, van der Boog AT, et al. Barium toxicity and the role of the potassium inward rectifier current. *Clin Toxicol (Phila)*. 2014;52:584–593.
280. Ahlawat SK, Sachdev A. Hypokalaemic paralysis. *Postgrad Med J*. 1999;75:193–197.
281. Downs JC, Milling D, Nichols CA. Suicidal ingestion of barium-sulfide-containing shaving powder. *Am J Forensic Med Pathol*. 1995;16:56–61.
282. Sigue G, et al. From profound hypokalemia to life-threatening hyperkalemia: a case of barium sulfide poisoning. *Arch Intern Med*. 2000;160:548–551.
283. Jacobs IA, Taddeo J, Kelly K, et al. Poisoning as a result of barium styphnate explosion. *Am J Ind Med*. 2002;41:285–288.
284. Wells JA, Wood KE. Acute barium poisoning treated with hemodialysis. *Am J Emerg Med*. 2001;19:175–177.
285. Horn S, Naidus E, Alper SL, et al. Cesium-associated hypokalemia successfully treated with amiloride. *Clin Kidney J*. 2015;8:335–338.
286. Bradberry SM, Vale JA. Disturbances of potassium homeostasis in poisoning. *J Toxicol Clin Toxicol*. 1995;33:295–310.
287. Jurkat-Rott K, et al. Voltage-sensor sodium channel mutations cause hypokalemic periodic paralysis type 2 by enhanced inactivation and reduced current. *Proc Natl Acad Sci USA*. 2000;97:9549–9554.
288. Plaster NM, et al. Mutations in Kir2.1 cause the developmental and episodic electrical phenotypes of Andersen's syndrome. *Cell*. 2001;105:511–519.
289. Jurkat-Rott K, et al. K⁺-dependent paradoxical membrane depolarization and Na⁺ overload, major and reversible contributors to weakness by ion channel leaks. *Proc Natl Acad Sci USA*. 2009;106:4036–4041.
290. Wu F, et al. A sodium channel knockin mutant (NaV1.4-R669H) mouse model of hypokalemic periodic paralysis. *J Clin Invest*. 2011;121:4082–4094.
291. Ruff RL. Skeletal muscle sodium current is reduced in hypokalemic periodic paralysis. *Proc Natl Acad Sci USA*. 2000;97:9832–9833.
292. Ruff RL. Insulin acts in hypokalemic periodic paralysis by reducing inward rectifier K⁺ current. *Neurology*. 1999;53:1556–1563.
293. Tricarico D, Servidei S, Tonali P, et al. Impairment of skeletal muscle adenosine triphosphate-sensitive K⁺ channels in patients with hypokalemic periodic paralysis. *J Clin Invest*. 1999;103:675–682.
294. Puwanant A, Ruff RL. INa and IKir are reduced in Type I hypokalemic and thyrotoxic periodic paralysis. *Muscle Nerve*. 2010;42:315–327.
295. Renaud JM. Modulation of force development by Na⁺, K⁺, Na⁺ K⁺ pump and KATP channel during muscular activity. *Can J Appl Physiol*. 2002;27:296–315.
296. Tricarico D, Capriulo R, Conte Camerino D. Insulin modulation of ATP-sensitive K⁺ channel of rat skeletal muscle is impaired in the hypokalaemic state. *Pflugers Arch*. 1999;437:235–240.
297. Cheng NL, Kao MC, Hsu YD, et al. Novel thiazide-sensitive Na-Cl cotransporter mutation in a Chinese patient with Gitelman's syndrome presenting as hypokalaemic paralysis. *Nephrol Dial Transplant*. 2003;18:1005–1008.
298. Yang SS, Chu P, Lin YF, et al. Aristolochic acid-induced Fanconi's syndrome and nephropathy presenting as hypokalemic paralysis. *Am J Kidney Dis*. 2002;39:E14.
299. Feldman M, et al. Molecular investigation and long-term clinical progress in Greek Cypriot families with recessive distal renal tubular acidosis and sensorineural deafness due to mutations in the ATP6V1B1 gene. *Clin Genet*. 2006;69:135–144.
300. Manoukian MA, Foote JA, Crapo LM. Clinical and metabolic features of thyrotoxic periodic paralysis in 24 episodes. *Arch Intern Med*. 1999;159:601–606.
301. Cheung CL, et al. Genome-wide association study identifies a susceptibility locus for thyrotoxic periodic paralysis at 17q24.3. *Nat Genet*. 2012;44:1026–1029.
302. Goh SH. Thyrotoxic periodic paralysis: reports of seven patients presenting with weakness in an Asian emergency department. *Emerg Med J*. 2002;19:78–79.
303. Ko GT, et al. Thyrotoxic periodic paralysis in a Chinese population. *QJM*. 1996;89:463–468.
304. Chan A, Shinde R, Chow CC, et al. In vivo and in vitro sodium pump activity in subjects with thyrotoxic periodic paralysis. *BMJ*. 1991;303:1096–1099.
305. Azuma KK, Hensley CB, Tang MJ, et al. Thyroid hormone specifically regulates skeletal muscle Na⁺(+)-K⁺(+)-ATPase alpha 2- and beta 2-isoforms. *Am J Physiol*. 1993;265:C680–C687.
306. Lin SH, Lin YF. Propranolol rapidly reverses paralysis, hypokalemia, and hypophosphatemia in thyrotoxic periodic paralysis. *Am J Kidney Dis*. 2001;37:620–623.
307. Birkhahn RH, Gaeta TJ, Melniker L. Thyrotoxic periodic paralysis and intravenous propranolol in the emergency setting. *J Emerg Med*. 2000;18:199–202.
308. Ahmed I, Chilimuri SS. Fatal dysrhythmia following potassium replacement for hypokalemic periodic paralysis. *West J Emerg Med*. 2010;11:57–59.
309. Cheng CJ, Kuo E, Huang CL. Extracellular potassium homeostasis: insights from hypokalemic periodic paralysis. *Semin Nephrol*. 2013;33:237–247.
310. Kuhn M, Jurkat-Rott K, Lehmann-Horn F. Rare KCNJ18 variants do not explain hypokalaemic periodic paralysis in 263 unrelated patients. *J Neurol Neurosurg Psychiatry*. 2016;87:49–52.
311. Dassau L, Conti LR, Radeke CM, et al. Kir2.6 regulates the surface expression of Kir2.x inward rectifier potassium channels. *J Biol Chem*. 2011;286:9526–9541.
312. Kassirer JP, Schwartz WB. The response of normal man to selective depletion of hydrochloric acid. Factors in the genesis of persistent gastric alkalosis. *Am J Med*. 1966;40:10–18.
313. Coghill NF, McAllen PM, Edwards F. Electrolyte losses associated with the taking of purges investigated with aid of sodium and potassium radioisotopes. *BMJ*. 1959;1:14–19.
314. Orman RA, Lewis JB Jr. Flaccid quadriplegia associated with Yersinia enterocolitis-induced hypokalemia. *Arch Intern Med*. 1989;149:1193–1194.
315. Ho JM, Juurlink DN, Cavalcanti RB. Hypokalemia following polyethylene glycol-based bowel preparation for colonoscopy in older hospitalized patients with significant comorbidities. *Ann Pharmacother*. 2010;44:466–470.
316. Reumkens A, et al. Prevalence of hypokalemia before and after bowel preparation for colonoscopy in high-risk patients. *Gastrointest Endosc*. 2017;86:673–679.
317. Wolf I, Mouallem M, Farfel Z. Adult celiac disease presented with celiac crisis: severe diarrhea, hypokalemia, and acidosis. *J Clin Gastroenterol*. 2000;30:324–326.
318. Simon M, et al. Over-expression of colonic K⁺ channels associated with severe potassium secretory diarrhoea after haemorrhagic shock. *Nephrol Dial Transplant*. 2008;23:3350–3352.
319. Blondon H, Bechade D, Desrame J, et al. Secretory diarrhoea with high faecal potassium concentrations: a new mechanism of diarrhoea associated with colonic pseudo-obstruction? Report of five patients. *Gastroenterol Clin Biol*. 2008;32:401–404.
320. van Dinter TG Jr, et al. Stimulated active potassium secretion in a patient with colonic pseudo-obstruction: a new mechanism of secretory diarrhea. *Gastroenterology*. 2005;129:1268–1273.
321. Sandle GI, Hunter M. Apical potassium (BK) channels and enhanced potassium secretion in human colon. *QJM*. 2010;103:85–89.
322. Sorensen MV, et al. Adrenaline-induced colonic K⁺ secretion is mediated by KCa1.1 (BK) channels. *J Physiol*. 2010;588:1763–1777.
323. Sorensen MV, et al. Aldosterone increases KCa1.1 (BK) channel-mediated colonic K⁺ secretion. *J Physiol*. 2008;586:4251–4264.
324. Ram P, Goyal A, Lu M, et al. Use of Aldosterone Antagonist to Treat Diarrhea and Hypokalemia of Ogilvie's Syndrome. *Case Rep Gastrointest Med*. 2016;2016:1207240.
325. Sandle GI, et al. Altered cryptal expression of luminal potassium (BK) channels in ulcerative colitis. *J Pathol*. 2007;212:66–73.
326. Elriz K, Savoye G, Tamion F, et al. Severe hypokalemia and metabolic alkalosis in a Crohn's disease patient after budesonide treatment: evidence for intestinal secretion of potassium. *Dig Liver Dis*. 2008;40:149–150.
327. Bettinelli A, et al. Use of calcium excretion values to distinguish two forms of primary renal tubular hypokalemic alkalosis: Bartter and Gitelman syndromes [see comments]. *J Pediatr*. 1992;120:38–43.
328. Suki WN, et al. Acute treatment of hypercalcemia with furosemide. *N Engl J Med*. 1970;283:836–840.

329. Okusa MD, Velazquez H, Ellison DH, et al. Luminal calcium regulates potassium transport by the renal distal tubule. *Am J Physiol*. 1990;258:F423–F428.
330. Sands JM, et al. Apical extracellular calcium/polyvalent cation-sensing receptor regulates vasopressin-elicited water permeability in rat kidney inner medullary collecting duct. *J Clin Invest*. 1997;99:1399–1405.
331. Waring WS, Stephen AF, Malkowska AM, et al. Acute acetaminophen overdose is associated with dose-dependent hypokalaemia: a prospective study of 331 patients. *Basic Clin Pharmacol Toxicol*. 2008;102:325–328.
332. Johnson DW, Kay TD, Hawley CM. Severe hypokalaemia secondary to dicloxacillin. *Intern Med J*. 2002;32:357–358.
333. Izzedine H, Launay-Vacher V, Isnard-Bagnis C, et al. Drug-induced Fanconi's syndrome. *Am J Kidney Dis*. 2003;41:292–309.
334. Landau D, Kher KK. Gentamicin-induced Bartter-like syndrome. *Pediatr Nephrol*. 1997;11:737–740.
335. Malin A, Miller RF. Fosarnet-induced hypokalaemia. *J Infect*. 1992;25:329–330.
336. Milonis HJ, Bourantas CL, Siamopoulos KC, et al. Acid-base and electrolyte abnormalities in patients with acute leukemia. *Am J Hematol*. 1999;62:201–207.
337. Husband DJ, Watkin SW. Fatal hypokalaemia associated with ifosfamide/mesna chemotherapy. *Lancet*. 1988;1:1116.
338. Cao Y, Liu L, Liao C, et al. Meta-analysis of incidence and risk of hypokalemia with cetuximab-based therapy for advanced cancer. *Cancer Chemother Pharmacol*. 2010;66:37–42.
339. Groenestegte WM, et al. Impaired basolateral sorting of pro-EGF causes isolated recessive renal hypomagnesemia. *J Clin Invest*. 2007;117:2260–2267.
340. Beck KR, et al. Inhibition of 11beta-hydroxysteroid dehydrogenase 2 by the fungicides itraconazole and posaconazole. *Biochem Pharmacol*. 2017;130:93–103.
341. Hoffmann WJ, McHardy I, Thompson GR 3rd. Itraconazole induced hypertension and hypokalemia: mechanistic evaluation. *Mycoses*. 2018.
342. Boughton C, Taylor D, Ghataore L, et al. Mineralocorticoid hypertension and hypokalaemia induced by posaconazole. *Endocrinol Diabetes Metab Case Rep*. 2018;2018.
343. Bunchman TE, Sinaiko AR. Renovascular hypertension presenting with hypokalemic metabolic alkalosis. *Pediatr Nephrol*. 1990;4:169–170.
344. Pinter TJ, Zimmerman S. Hyperreninemic hypertension secondary to a subcapsular perinephric hematoma in a patient with polyarteritis nodosa. *Am J Kidney Dis*. 1998;32:503–507.
345. Ringrose TR, Phillips PA, Lindop GB. Renin-secreting adenocarcinoma of the colon. *Ann Intern Med*. 1999;131:794–795.
346. Corvol P, Pinet F, Plouin PF, et al. Renin-secreting tumors. *Endocrinol Metab Clin North Am*. 1994;23:255–270.
347. Nicholls MG. Unilateral renal ischemia causing the hyponatremic hypertensive syndrome in children—more common than we think? *Pediatr Nephrol*. 2006;21:887–890.
348. White PC. Steroid 11 beta-hydroxylase deficiency and related disorders. *Endocrinol Metab Clin North Am*. 2001;30:61–79, vi.
349. Goldsmith O, Solomon DH, Horton R. Hypogonadism and mineralocorticoid excess. The 17-hydroxylase deficiency syndrome. *N Engl J Med*. 1967;277:673–677.
350. Dluhy RG, Lifton RP. Glucocorticoid-remediable aldosteronism. *J Clin Endocrinol Metab*. 1999;84:4341–4344.
351. Sukor N, et al. Further evidence for linkage of familial hyperaldosteronism type II at chromosome 7p22 in Italian as well as Australian and South American families. *J Hypertens*. 2008;26:1577–1582.
352. Geller DS, et al. A novel form of human mendelian hypertension featuring nonglucocorticoid-remediable aldosteronism. *J Clin Endocrinol Metab*. 2008;93:3117–3123.
353. Scholl UI, et al. Recurrent gain of function mutation in calcium channel CACNA1H causes early-onset hypertension with primary aldosteronism. *Elife*. 2015;4:e06315.
354. Daniil G, et al. CACNA1H mutations are associated with different forms of primary aldosteronism. *EBioMedicine*. 2016;13:225–236.
355. Litchfield WR, New MI, Coolidge C, et al. Evaluation of the dexamethasone suppression test for the diagnosis of glucocorticoid-remediable aldosteronism. *J Clin Endocrinol Metab*. 1997;82:3570–3573.
356. Lifton RP, et al. A chimaeric 11 beta-hydroxylase/aldosterone synthase gene causes glucocorticoid-remediable aldosteronism and human hypertension. *Nature*. 1992;355:262–265.
357. Rich GM, et al. Glucocorticoid-remediable aldosteronism in a large kindred: clinical spectrum and diagnosis using a characteristic biochemical phenotype. *Ann Intern Med*. 1992;116:813–820.
358. Pascoe L, et al. Glucocorticoid-suppressible hyperaldosteronism and adrenal tumors occurring in a single French pedigree. *J Clin Invest*. 1995;96:2236–2246.
359. Funder JW, et al. Case detection, diagnosis, and treatment of patients with primary aldosteronism: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2008;93:3266–3281.
360. Litchfield WR, Anderson BF, Weiss RJ, et al. Intracranial aneurysm and hemorrhagic stroke in glucocorticoid-remediable aldosteronism. *Hypertension*. 1998;31:445–450.
361. Jamieson A, et al. Clinical, biochemical and genetic features of five extended kindred's with glucocorticoid-suppressible hyperaldosteronism. *Endocr Res*. 1995;21:463–469.
362. Litchfield WR, et al. Impaired potassium-stimulated aldosterone production: a possible explanation for normokalemic glucocorticoid-remediable aldosteronism. *J Clin Endocrinol Metab*. 1997;82:1507–1510.
363. Young WF. Primary aldosteronism: renaissance of a syndrome. *Clin Endocrinol (Oxf)*. 2007;66:607–618.
364. Mulatero P, Rabbia F, Veglio F. Paraneoplastic hyperaldosteronism associated with non-Hodgkin's lymphoma. *N Engl J Med*. 2001;344:1558–1559.
365. Seccia TM, et al. Somatic mutations in the KCNJ5 gene raise the lateralization index: implications for the diagnosis of primary aldosteronism by adrenal vein sampling. *J Clin Endocrinol Metab*. 2012;97:E2307–E2313.
366. Scholl UI, et al. Somatic and germLine CACNA1D calcium channel mutations in aldosterone-producing adenomas and primary aldosteronism. *Nat Genet*. 2013;45:1050–1054.
367. Fernandes-Rosa FL, et al. Genetic spectrum and clinical correlates of somatic mutations in aldosterone-producing adenoma. *Hypertension*. 2014;64:354–361.
368. Nanba K, Vaidya A, Rainey WE. Aging and adrenal aldosterone production. *Hypertension*. 2018;71:218–223.
369. Akerstrom T, et al. Activating mutations in CTNNB1 in aldosterone producing adenomas. *Sci Rep*. 2016;6:19546.
370. Wu VC, et al. The prevalence of CTNNB1 mutations in primary aldosteronism and consequences for clinical outcomes. *Sci Rep*. 2017;7:39121.
371. Nishimoto K, et al. Adrenocortical zonation in humans under normal and pathological conditions. *J Clin Endocrinol Metab*. 2010;95:2296–2305.
372. Nishimoto K, et al. Aldosterone-stimulating somatic gene mutations are common in normal adrenal glands. *Proc Natl Acad Sci USA*. 2015;112:E4591–E4599.
373. Inoue K, et al. Evaluation of cortisol production in aldosterone-producing adenoma. *Horm Metab Res*. 2017;49:847–853.
374. Arlt W, et al. Steroid metabolome analysis reveals prevalent glucocorticoid excess in primary aldosteronism. *JCI Insight*. 2017;2.
375. Montori VM, Young WF Jr. Use of plasma aldosterone concentration-to-plasma renin activity ratio as a screening test for primary aldosteronism. A systematic review of the literature. *Endocrinol Metab Clin North Am*. 2002;31:619–632, xi.
376. Williams JS, et al. Prevalence of primary hyperaldosteronism in mild to moderate hypertension without hypokalaemia. *J Hum Hypertens*. 2006;20:129–136.
377. Brown JM, et al. The spectrum of subclinical primary aldosteronism and incident hypertension: a cohort study. *Ann Intern Med*. 2017;167:630–641.
378. Baudrand R, et al. Continuum of renin-independent aldosteronism in normotension. *Hypertension*. 2017;69:950–956.
379. Funder JW, et al. The management of primary aldosteronism: case detection, diagnosis, and treatment: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2016;101:1889–1916.
380. Kline GA, Pasieka JL, Harvey A, et al. High-probability features of primary aldosteronism may obviate the need for confirmatory testing without increasing false-positive diagnoses. *J Clin Hypertens (Greenwich)*. 2014;16:488–496.
381. Liu SY, et al. Aldosterone-producing adenoma in primary aldosteronism: CT-guided radiofrequency ablation-long-term results and recurrence rate. *Radiology*. 2016;281:625–634.
382. Blumenfeld JD, et al. Diagnosis and treatment of primary hyperaldosteronism. *Ann Intern Med*. 1994;121:877–885.

383. Eiam-Ong S, Kurtzman NA, Sabatini S. Regulation of collecting tubule adenosine triphosphatases by aldosterone and potassium. *J Clin Invest.* 1993;91:2385–2392.
384. Howlett TA, et al. Diagnosis and management of ACTH-dependent Cushing's syndrome: comparison of the features in ectopic and pituitary ACTH production. *Clin Endocrinol (Oxf).* 1986;24:699–713.
385. Torpy DJ, Mullen N, Ilias I, et al. Association of hypertension and hypokalemia with Cushing's syndrome caused by ectopic ACTH secretion: a series of 58 cases. *Ann N Y Acad Sci.* 2002;970:134–144.
386. Isidori AM, et al. The ectopic adrenocorticotropin syndrome: clinical features, diagnosis, management, and long-term follow-up. *J Clin Endocrinol Metab.* 2006;91:371–377.
387. Stewart PM, Walker BR, Holder G, et al. 11 beta-Hydroxysteroid dehydrogenase activity in Cushing's syndrome: explaining the mineralocorticoid excess state of the ectopic adrenocorticotropin syndrome. *J Clin Endocrinol Metab.* 1995;80:3617–3620.
388. Koren W, et al. Enhanced Na⁺/H⁺ exchange in Cushing's syndrome reflects functional hypermineralocorticoidism. *J Hypertens.* 1998;16:1187–1191.
389. Mendonca BB, et al. Female pseudohermaphroditism caused by a novel homozygous missense mutation of the GR gene. *J Clin Endocrinol Metab.* 2002;87:1805–1809.
390. White PC, Mune T, Agarwal AK. 11 beta-Hydroxysteroid dehydrogenase and the syndrome of apparent mineralocorticoid excess. *Endocr Rev.* 1997;18:135–156.
391. Bostanjoglo M, et al. 11Beta-hydroxysteroid dehydrogenase, mineralocorticoid receptor, and thiazide-sensitive Na-Cl cotransporter expression by distal tubules [published erratum appears in *J Am Soc Nephrol* 1998 Nov;9(11):2179]. *J Am Soc Nephrol.* 1998;9:1347–1358.
392. Funder JW, Pearce JT, Smith R, et al. Mineralocorticoid action: target tissue specificity is enzyme, not receptor, mediated. *Science.* 1988;242:583–585.
393. Li A, et al. Molecular basis for hypertension in the “type II variant” of apparent mineralocorticoid excess. *Am J Hum Genet.* 1998;63:370–379.
394. Wilson RC, et al. A genetic defect resulting in mild low-renin hypertension. *Proc Natl Acad Sci USA.* 1998;95:10200–10205.
395. Nunez BS, et al. Mutants of 11beta-hydroxysteroid dehydrogenase (11-HSD2) with partial activity: improved correlations between genotype and biochemical phenotype in apparent mineralocorticoid excess. *Hypertension.* 1999;34:638–642.
396. Kotelevtsev Y, et al. Hypertension in mice lacking 11beta-hydroxysteroid dehydrogenase type 2. *J Clin Invest.* 1999;103:683–689.
397. Loffing J, et al. Distribution of transcellular calcium and sodium transport pathways along mouse distal nephron. *Am J Physiol Renal Physiol.* 2001;281:F1021–F1027.
398. Kim GH, et al. The thiazide-sensitive Na-Cl cotransporter is an aldosterone-induced protein. *Proc Natl Acad Sci USA.* 1998;95:14552–14557.
399. Masilamani S, Kim GH, Mitchell C, et al. Aldosterone-mediated regulation of ENaC alpha, beta, and gamma subunit proteins in rat kidney. *J Clin Invest.* 1999;104:R19–R23.
400. Biller KJ, Unwin RJ, Shirley DG. Distal tubular electrolyte transport during inhibition of renal 11beta-hydroxysteroid dehydrogenase. *Am J Physiol Renal Physiol.* 2001;280:F172–F179.
401. Borst JGG, Ten Holt SP, de Vries LA, et al. Synergistic action of liquorice and cortisone in Addison's and Simmond's disease. *Lancet.* 1953;1:657–663.
402. Woywodt A, Herrmann A, Haller H, et al. Severe hypokalaemia: is one reason enough? *Nephrol Dial Transplant.* 2004;19:2914–2917.
403. Hamidon BB, Jeyabalan V. Exogenously-induced apparent hypermineralocorticoidism associated with ingestion of “asam boi”. *Singapore Med J.* 2006;47:156–158.
404. Iida R, Otsuka Y, Matsumoto K, et al. Pseudoaldosteronism due to the concurrent use of two herbal medicines containing glycyrrhizin: interaction of glycyrrhizin with angiotensin-converting enzyme inhibitor. *Clin Exp Nephrol.* 2006;10:131–135.
405. Farese S, et al. Glycyrrhetic acid food supplementation lowers serum potassium concentration in chronic hemodialysis patients. *Kidney Int.* 2009;76:877–884.
406. Geller DS, et al. Activating mineralocorticoid receptor mutation in hypertension exacerbated by pregnancy. *Science.* 2000;289:119–123.
407. Lifton RP, Gharavi AG, Geller DS. Molecular mechanisms of human hypertension. *Cell.* 2001;104:545–556.
408. Botero-Velez M, Curtis JJ, Warnock DG. Brief report: Liddle's syndrome revisited—a disorder of sodium reabsorption in the distal tubule. *N Engl J Med.* 1994;330:178–181.
409. Pradervand S, et al. Dysfunction of the epithelial sodium channel expressed in the kidney of a mouse model for Liddle syndrome. *J Am Soc Nephrol.* 2003;14:2219–2228.
410. Kellenberger S, Gautschi I, Rossier BC, et al. Mutations causing Liddle syndrome reduce sodium-dependent downregulation of the epithelial sodium channel in the *Xenopus* oocyte expression system. *J Clin Invest.* 1998;101:2741–2750.
411. Frindt G, Silver RB, Windhager EE, et al. Feedback regulation of Na channels in rat CCT. II. Effects of inhibition of Na entry. *Am J Physiol.* 1993;264:F565–F574.
412. Knight KK, Olson DR, Zhou R, et al. Liddle's syndrome mutations increase Na⁺ transport through dual effects on epithelial Na⁺ channel surface expression and proteolytic cleavage. *Proc Natl Acad Sci USA.* 2006;103:2805–2808.
413. Hiltunen TP, et al. Liddle's syndrome associated with a point mutation in the extracellular domain of the epithelial sodium channel gamma subunit. *J Hypertens.* 2002;20:2383–2390.
414. Salih M, et al. A missense mutation in the extracellular domain of alphaENaC causes Liddle syndrome. *J Am Soc Nephrol.* 2017;28:3291–3299.
415. Ray EC, et al. Human epithelial Na⁺ channel missense variants identified in the GenSalt study alter channel activity. *Am J Physiol Renal Physiol.* 2016;311:F908–F914.
416. Hannila-Handelberg T, et al. Common variants of the beta and gamma subunits of the epithelial sodium channel and their relation to plasma renin and aldosterone levels in essential hypertension. *BMC Med Genet.* 2005;6:4.
417. Ambrosius WT, et al. Genetic variants in the epithelial sodium channel in relation to aldosterone and potassium excretion and risk for hypertension. *Hypertension.* 1999;34:631–637.
418. Gladziwa U, et al. Chronic hypokalaemia of adults: Gitelman's syndrome is frequent but classical Bartter's syndrome is rare. *Nephrol Dial Transplant.* 1995;10:1607–1613.
419. Guay-Woodford LM. Bartter syndrome: unraveling the pathophysiologic enigma. *Am J Med.* 1998;105:151–161.
420. Brochard K, et al. Phenotype-genotype correlation in antenatal and neonatal variants of Bartter syndrome. *Nephrol Dial Transplant.* 2009;24:1455–1464.
421. Komhoff M, et al. Cyclooxygenase-2 expression is associated with the renal macula densa of patients with Bartter-like syndrome. *Kidney Int.* 2000;58:2420–2424.
422. Reinalter SC, et al. Role of cyclooxygenase-2 in hyperprostaglandin E syndrome/antenatal Bartter syndrome. *Kidney Int.* 2002;62:253–260.
423. Gill JR Jr, Bartter FC. Evidence for a prostaglandin-independent defect in chloride reabsorption in the loop of Henle as a proximal cause of Bartter's syndrome. *Am J Med.* 1978;65:766–772.
424. Kockerling A, Reinalter SC, Seyberth HW. Impaired response to furosemide in hyperprostaglandin E syndrome: evidence for a tubular defect in the loop of Henle. *J Pediatr.* 1996;129:519–528.
425. Simon DB, et al. Bartter's syndrome, hypokalaemic alkalosis with hypercalciuria, is caused by mutations in the Na-K-2Cl cotransporter NKCC2. *Nat Genet.* 1996;13:183–188.
426. Xu JZ, et al. Localization of the ROMK protein on apical membranes of rat kidney nephron segments. *Am J Physiol.* 1997;F739–F748.
427. Sun A, Grossman EB, Lombardi M, et al. Vasopressin alters the mechanism of apical Cl[−] entry from Na⁺:Cl[−] to Na⁺:K⁺:2Cl[−] cotransport in mouse medullary thick ascending limb. *J Membr Biol.* 1991;120:83–94.
428. Simon DB, et al. Genetic heterogeneity of Bartter's syndrome revealed by mutations in the K⁺ channel, ROMK. *Nat Genet.* 1996;14:152–156.
429. Simon DB, et al. Mutations in the chloride channel gene, CLCNKB, cause Bartter's syndrome type III. *Nat Genet.* 1997;17:171–178.
430. Vandewalle A, et al. Localization and induction by dehydration of ClC-K chloride channels in the rat kidney. *Am J Physiol.* 1997;272:F678–F688.
431. Estevez R, et al. Barttin is a Cl[−] channel beta-subunit crucial for renal Cl[−] reabsorption and inner ear K⁺ secretion. *Nature.* 2001;414:558–561.
432. Schlingmann KP, et al. Salt wasting and deafness resulting from mutations in two chloride channels. *N Engl J Med.* 2004;350:1314–1319.
433. Watanabe S, et al. Association between activating mutations of calcium-sensing receptor and Bartter's syndrome. *Lancet.* 2002;360:692–694.

434. Vargas-Poussou R, et al. Functional characterization of a calcium-sensing receptor mutation in severe autosomal dominant hypocalcemia with a Bartter-like syndrome. *J Am Soc Nephrol*. 2002;13:2259–2266.
435. Riccardi D, et al. Localization of the extracellular Ca²⁺/polyvalent cation-sensing protein in rat kidney. *Am J Physiol*. 1998;274:F611–F622.
436. Wang WH, Lu M, Hebert SC. Cytochrome P-450 metabolites mediate extracellular Ca(2+)-induced inhibition of apical K⁺ channels in the TAL. *Am J Physiol*. 1996;271:C103–C111.
437. Carmosino M, et al. NKCC2 activity is inhibited by the Bartter's syndrome type 5 gain-of-function CaR-A843E mutant in renal cells. *Biol Cell*. 2015;107:98–110.
438. Peters M, et al. Clinical presentation of genetically defined patients with hypokalemic salt-losing tubulopathies. *Am J Med*. 2002;112:183–190.
439. Seys E, et al. Clinical and genetic spectrum of Bartter syndrome type 3. *J Am Soc Nephrol*. 2017;28:2540–2552.
440. Pressler CA, et al. Late-onset manifestation of antenatal Bartter syndrome as a result of residual function of the mutated renal Na⁺+K⁺+2Cl[−] Co-transporter. *J Am Soc Nephrol*. 2006;17:2136–2142.
441. Bongers E, et al. A novel hypokalemic-alkalotic salt-losing tubulopathy in patients with CLDN10 mutations. *J Am Soc Nephrol*. 2017;28:3118–3128.
442. Finer G, et al. Transient neonatal hyperkalemia in the antenatal (ROMK defective) Bartter syndrome. *J Pediatr*. 2003;142:318–323.
443. Woda CB, Bragin A, Kleyman TR, et al. Flow-dependent K⁺ secretion in the cortical collecting duct is mediated by a maxi-K channel. *Am J Physiol Renal Physiol*. 2001;280:F786–F793.
444. Bailey MA, et al. Maxi-K channels contribute to urinary potassium excretion in the ROMK-deficient mouse model of Type II Bartter's syndrome and in adaptation to a high-K diet. *Kidney Int*. 2006;70:51–59.
445. Lu M, Wang W. Two types of K(+) channels are present in the apical membrane of the thick ascending limb of the mouse kidney. *Kidney Blood Press Res*. 2000;23:75–82.
446. Lu M, et al. ROMK is required for expression of the 70-pS K channel in the thick ascending limb. *Am J Physiol Renal Physiol*. 2004;286:F490–F495.
447. Dave S, Honney S, Raymond J, et al. An unusual presentation of cystic fibrosis in an adult. *Am J Kidney Dis*. 2005;45:e41–e44.
448. Lu M, et al. CFTR is required for PKA-regulated ATP sensitivity of Kir1.1 potassium channels in mouse kidney. *J Clin Invest*. 2006;116:797–807.
449. Blanchard A, et al. Gitelman syndrome: consensus and guidance from a Kidney Disease: improving Global Outcomes (KDIGO) Controversies Conference. *Kidney Int*. 2017;91:24–33.
450. Luthy C, et al. Normal prostaglandinuria E2 in Gitelman's syndrome, the hypocalciuric variant of Bartter's syndrome. *Am J Kidney Dis*. 1995;25:824–828.
451. Cruz DN, Shaer AJ, Bia MJ, et al. Gitelman's syndrome revisited: an evaluation of symptoms and health-related quality of life. *Kidney Int*. 2001;59:710–717.
452. Bettinelli A, et al. Electrocardiogram with prolonged QT interval in Gitelman disease. *Kidney Int*. 2002;62:580–584.
453. Foglia PE, et al. Cardiac work up in primary renal hypokalaemia-hypomagnesaemia (Gitelman syndrome). *Nephrol Dial Transplant*. 2004;19:1398–1402.
454. Simon DB, et al. Gitelman's variant of Bartter's syndrome, inherited hypokalaemic alkalosis, is caused by mutations in the thiazide-sensitive Na-Cl cotransporter. *Nat Genet*. 1996;12:24–30.
455. De Jong JC, et al. Functional expression of mutations in the human NaCl cotransporter: evidence for impaired routing mechanisms in Gitelman's syndrome. *J Am Soc Nephrol*. 2002;13:1442–1448.
456. Bockenhauer D, et al. Epilepsy, ataxia, sensorineural deafness, tubulopathy, and KCNJ10 mutations. *N Engl J Med*. 2009;360:1960–1970.
457. Loffing J, et al. Altered renal distal tubule structure and renal Na(+) and Ca(2+) handling in a mouse model for Gitelman's syndrome. *J Am Soc Nephrol*. 2004;15:2276–2288.
458. Loffing J, et al. Thiazide treatment of rats provokes apoptosis in distal tubule cells. *Kidney Int*. 1996;50:1180–1190.
459. Nijenhuis T, et al. Enhanced passive Ca²⁺ reabsorption and reduced Mg²⁺ channel abundance explains thiazide-induced hypocalciuria and hypomagnesemia. *J Clin Invest*. 2005;115:1651–1658.
460. Morris RG, Hoorn EJ, Knepper MA. Hypokalemia in a mouse model of Gitelman's syndrome. *Am J Physiol Renal Physiol*. 2006;290:F1416–F1420.
461. Friedman E, Shadel M, Halkin H, et al. Thiazide-induced hyponatremia. Reproducibility by single dose rechallenge and an analysis of pathogenesis. *Ann Intern Med*. 1989;110:24–30.
462. Cruz D, Simon D, Lifton RP. Inactivating mutations in the Na-Cl cotransporter is associated with high bone density. *J Am Soc Nephrol*. 1999;10:597A.
463. Cobeta-Garcia JC, Gascon A, Iglesias E, et al. Chondrocalcinosis and Gitelman's syndrome. A new association? *Ann Rheum Dis*. 1998;57:748–749.
464. Vezzoli G, Soldati L, Jansen A, et al. Choroidal calcifications in patients with Gitelman's syndrome. *Am J Kidney Dis*. 2000;36:855–858.
465. Panichpisal K, Angulo-Pernett F, Selhi S, et al. Gitelman-like syndrome after cisplatin therapy: a case report and literature review. *BMC Nephrol*. 2006;7:10.
466. Chen YC, et al. Primary Sjogren's syndrome associated with Gitelman's syndrome presenting with muscular paralysis. *Am J Kidney Dis*. 2003;42:586–590.
467. Sebastian A, McSherry E, Morris RC Jr. On the mechanism of renal potassium wasting in renal tubular acidosis associated with the Fanconi syndrome (type 2 RTA). *J Clin Invest*. 1971;50:231–243.
468. Tsai CS, Chen YC, Chen HH, et al. An unusual cause of hypokalemic paralysis: aristolochic acid nephropathy with Fanconi syndrome. *Am J Med Sci*. 2005;330:153–155.
469. Karet FE, et al. Mutations in the gene encoding B1 subunit of H⁺-ATPase cause renal tubular acidosis with sensorineural deafness. *Nat Genet*. 1999;21:84–90.
470. Sebastian A, McSherry E, Morris RC Jr. Renal potassium wasting in renal tubular acidosis (RTA): its occurrence in types 1 and 2 RTA despite sustained correction of systemic acidosis. *J Clin Invest*. 1971;50:667–678.
471. Comer DM, Droogan AG, Young IS, et al. Hypokalaemic paralysis precipitated by distal renal tubular acidosis secondary to Sjogren's syndrome. *Ann Clin Biochem*. 2008;45:221–225.
472. Alonso-Varela M, et al. Distal renal tubular acidosis. Clinical manifestations in patients with different underlying gene mutations. *Pediatr Nephrol*. 2018.
473. Rodriguez M, Solanki DL, Whang R. Refractory potassium repletion due to cisplatin-induced magnesium depletion. *Arch Intern Med*. 1989;149:2592–2594.
474. Whang R, et al. Magnesium depletion as a cause of refractory potassium repletion. *Arch Intern Med*. 1985;145:1686–1689.
475. Whang R, Whang DD, Ryan MP. Refractory potassium repletion. A consequence of magnesium deficiency. *Arch Intern Med*. 1992;152:40–45.
476. Dorup I, Clausen T. Correlation between magnesium and potassium contents in muscle: role of Na(+)-K⁺ pump. *Am J Physiol*. 1993;264:C457–C463.
477. Huang CL, Kuo E. Mechanism of hypokalemia in magnesium deficiency. *J Am Soc Nephrol*. 2007;18:2649–2652.
478. Yang L, Frindt G, Palmer LG. Magnesium modulates ROMK channel-mediated potassium secretion. *J Am Soc Nephrol*. 2010;21:2109–2116.
479. Vandenberg CA. Inward rectification of a potassium channel in cardiac ventricular cells depends on internal magnesium ions. *Proc Natl Acad Sci USA*. 1987;84:2560–2564.
480. Groeneveld JH, Sijpkens YW, Lin SH, et al. An approach to the patient with severe hypokalaemia: the potassium quiz. *QJM*. 2005;98:305–316.
481. Reimann D, Gross P. Chronic, diagnosis-resistant hypokalaemia. *Nephrol Dial Transplant*. 1999;14:2957–2961.
482. Greenfeld D, Mickley D, Quinlan DM, et al. Hypokalemia in outpatients with eating disorders. *Am J Psychiatry*. 1995;152:60–63.
483. Crow SJ, Rosenberg ME, Mitchell JE, et al. Urine electrolytes as markers of bulimia nervosa. *Int J Eat Disord*. 2001;30:279–287.
484. Woywodt A, Herrmann A, Eisenberger U, et al. The tell-tale urinary chloride. *Nephrol Dial Transplant*. 2001;16:1066–1068.
485. Kim YG, et al. Medullary nephrocalcinosis associated with long-term furosemide abuse in adults. *Nephrol Dial Transplant*. 2001;16:2303–2309.
486. Holmes AM, Hesling CM, Wilson TM. Drug-induced secondary hyperaldosteronism in patients with pulmonary tuberculosis. *QJ Med*. 1970;39:299–315.
487. Bates CM, Baum M, Quigley R. Cystic fibrosis presenting with hypokalemia and metabolic alkalosis in a previously healthy adolescent. *J Am Soc Nephrol*. 1997;8:352–355.

488. Casatta L, Ferraccioli GF, Bartoli E. Hypokalaemic alkalosis, acquired Gitelman's and Bartter's syndrome in chronic sialoadenitis. *Br J Rheumatol*. 1997;36:1125–1128.
489. Kone BC. Hypokalemia. In: DuBose TD Jr, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner & Rector's the Kidney*. Saunders; 2002:381–394.
490. Kim GH, Han JS. Therapeutic approach to hypokalemia. *Nephron*. 2002;92(suppl 1):28–32.
491. Drew BJ, et al. Practice standards for electrocardiographic monitoring in hospital settings: an American Heart Association scientific statement from the Councils on Cardiovascular Nursing, Clinical Cardiology, and Cardiovascular Disease in the Young; endorsed by the International Society of Computerized Electrocardiology and the American Association of Critical-Care Nurses. *Circulation*. 2004;110:2721–2746.
492. Zydlewski AW, Hasbargen JA. Hypothermia-induced hypokalemia. *Mil Med*. 1998;163:719–721.
493. Kearney TE, Manoguerra AS, Curtis GP, et al. Theophylline toxicity and the beta-adrenergic system. *Ann Intern Med*. 1985;102:766–769.
494. Chapman SA, Kaufenberg AJ, Anderson P, et al. Safety and effectiveness of a modification of diet in renal disease equation-based potassium replacement protocol. *Ann Pharmacother*. 2009;43:436–443.
495. Sterns RH, Cox M, Feig PU, et al. Internal potassium balance and the control of the plasma potassium concentration. *Medicine (Baltimore)*. 1981;60:339–354.
496. Gabduzda CJ, Hall PW 3rd. Relation of potassium depletion to renal ammonium metabolism and hepatic coma. *Medicine (Baltimore)*. 1966;45:481–490.
497. Jaeger P, Karlmark B, Giebisch G. Ammonium transport in rat cortical tubule: relationship to potassium metabolism. *Am J Physiol*. 1983;245:F593–F600.
498. Khaw KT, Barrett-Connor E. Dietary potassium and stroke-associated mortality. A 12-year prospective population study. *N Engl J Med*. 1987;316:235–240.
499. Ascherio A, et al. Intake of potassium, magnesium, calcium, and fiber and risk of stroke among US men. *Circulation*. 1998;98:1198–1204.
500. Villamil MF, Deland EC, Henney RP, et al. Anion effects on cation movements during correction of potassium depletion. *Am J Physiol*. 1975;229:161–166.
501. Nicolis GL, Kahn T, Sanchez A, et al. Glucose-induced hyperkalemia in diabetic subjects. *Arch Intern Med*. 1981;141:49–53.
502. Keith NM, Osterberg AE, Burchell HB. Some effects of potassium salts in man. *Ann Intern Med*. 1942;16:879.
503. Kunin AS, Surawicz B, Sims EAH. Decrease in serum potassium concentrations and appearance of cardiac arrhythmias during infusion of potassium with glucose in potassium-depleted patients. *N Engl J Med*. 1962;266:228–233.
504. Hamill RJ, Robinson LM, Wexler HR, et al. Efficacy and safety of potassium infusion therapy in hypokalemic critically ill patients. *Crit Care Med*. 1991;19:694–699.
505. Kruse JA, Carlson RW. Rapid correction of hypokalemia using concentrated intravenous potassium chloride infusions. *Arch Intern Med*. 1990;150:613–617.
506. Rose BD, Post TW. Hypokalemia. In: Rose BD, Post TW, eds. *Clinical Physiology of Acid-Base and Electrolyte Disorders*. Vol. 1. McGraw-Hill; 2001:836–887.
507. Pullen H, Doig A, Lambie AT. Intensive intravenous potassium replacement therapy. *Lancet*. 1967;2:809–811.
508. Abramson E, Arky R. Diabetic acidosis with initial hypokalemia. Therapeutic implications. *JAMA*. 1966;196:401–403.
509. Seftel HC, Kew MC. Early and intensive potassium replacement in diabetic acidosis. *Diabetes*. 1966;15:694–696.
510. Norris W, Kunzelman KS, Bussell S, et al. Potassium supplementation, diet vs pills: a randomized trial in postoperative cardiac surgery patients. *Chest*. 2004;125:404–409.
511. Kopyt N, Dalal F, Narins RG. Renal retention of potassium in fruit. *N Engl J Med*. 1985;313:582–583.
512. Doorenbos CJ, Vermeij CG. Danger of salt substitutes that contain potassium in patients with renal failure. *BMJ*. 2003;326:35–36.
513. Su M, et al. Sustained-release potassium chloride overdose. *J Toxicol Clin Toxicol*. 2001;39:641–648.
514. Ganguly A, Weinberger MH. Triamterene-thiazide combination: alternative therapy for primary aldosteronism. *Clin Pharmacol Ther*. 1981;30:246–250.
515. Brown JJ, et al. Comparison of surgery and prolonged spironolactone therapy in patients with hypertension, aldosterone excess, and low plasma renin. *Br Med J*. 1972;2:729–734.
516. Griffing GT, et al. Amiloride in primary hyperaldosteronism. *Clin Pharmacol Ther*. 1982;31:56–61.
517. Eiro M, Katoh T, Watanabe T. Use of a proton-pump inhibitor for metabolic disturbances associated with anorexia nervosa. *N Engl J Med*. 2002;346:140.
518. Stevens MS, Dunlay RW. Hyperkalemia in hospitalized patients. *Int Urol Nephrol*. 2000;32:177–180.
519. Moore ML, Bailey RR. Hyperkalaemia in patients in hospital. *N Z Med J*. 1989;102:557–558.
520. Rastegar A, Soleimani M, Rastegar A. Hypokalaemia and hyperkalaemia. *Postgrad Med J*. 2001;77:759–764.
521. Charytan D, Goldfarb DS. Indications for hospitalization of patients with hyperkalemia. *Arch Intern Med*. 2000;160:1605–1611.
522. Ponce SP, Jennings AE, Madias NE, et al. Drug-induced hyperkalemia. *Medicine (Baltimore)*. 1985;64:357–370.
523. Ahee P, Crowe AV. The management of hyperkalaemia in the emergency department. *J Accid Emerg Med*. 2000;17:188–191.
524. Allon M, Dunlay R, Copkney C. Nebulized albuterol for acute hyperkalemia in patients on hemodialysis. *Ann Intern Med*. 1989;110:426–429.
525. Ahmed J, Weisberg LS. Hyperkalemia in dialysis patients. *Semin Dial*. 2001;14:348–356.
526. Moranne O, et al. Timing of onset of CKD-related metabolic complications. *J Am Soc Nephrol*. 2009;20:164–171.
527. Einhorn LM, et al. The frequency of hyperkalemia and its significance in chronic kidney disease. *Arch Intern Med*. 2009;169:1156–1162.
528. Kovesdy CP, et al. Serum and dialysate potassium concentrations and survival in hemodialysis patients. *Clin J Am Soc Nephrol*. 2007;2:999–1007.
529. Torlen K, Kalantar-Zadeh K, Molnar MZ, et al. Serum potassium and cause-specific mortality in a large peritoneal dialysis cohort. *Clin J Am Soc Nephrol*. 2012;7:1272–1284.
530. Moore CR, Lin JJ, O'Connor N, et al. Follow-up of markedly elevated serum potassium results in the ambulatory setting: implications for patient safety. *Am J Med Qual*. 2006;21:115–124.
531. Wiederkehr MR, Moe OW. Factitious hyperkalemia. *Am J Kidney Dis*. 2000;36:1049–1053.
532. Skinner S. A cause of erroneous potassium levels. *Lancet*. 1961;277:478–480.
533. Graber M, Subramani K, Corish D, et al. Thrombocytosis elevates serum potassium. *Am J Kidney Dis*. 1988;12:116–120.
534. Bellevue R, Dosik H, Spergel G, et al. Pseudohyperkalemia and extreme leukocytosis. *J Lab Clin Med*. 1975;85:660–664.
535. Sevastos N, et al. Pseudohyperkalemia in patients with increased cellular components of blood. *Am J Med Sci*. 2006;331:17–21.
536. Cornes MP, Ford C, Gama R. Spurious hyperkalaemia due to EDTA contamination: common and not always easy to identify. *Ann Clin Biochem*. 2008;45:601–603.
537. Kellerman PS, Thornberry JM. Pseudohyperkalemia due to pneumatic tube transport in a leukemic patient. *Am J Kidney Dis*. 2005;46:746–748.
538. Oliver TK Jr, Young GA, Bates GD, et al. Factitious hyperkalemia due to icing before analysis. *Pediatrics*. 1966;38:900–902.
539. Sodi R, Davison AS, Holmes E, et al. The phenomenon of seasonal pseudohypokalemia: effects of ambient temperature, plasma glucose and role for sodium-potassium-exchanging-ATPase. *Clin Biochem*. 2009;42:813–818.
540. Bruce LJ, et al. Monovalent cation leaks in human red cells caused by single amino-acid substitutions in the transport domain of the band 3 chloride-bicarbonate exchanger, AE1. *Nat Genet*. 2005;37:1258–1263.
541. Flatt JF, Bruce LJ. The molecular basis for altered cation permeability in hereditary stomatocytic human red blood cells. *Front Physiol*. 2018;9:367.
542. Stewart AK, et al. The GPA-dependent, spherostomatocytosis mutant AE1 E758K induces GPA-independent, endogenous cation transport in amphibian oocytes. *Am J Physiol Cell Physiol*. 2010;298:C283–C297.
543. Bogdanova A, et al. Cryohydrocytosis: increased activity of cation carriers in red cells from a patient with a band 3 mutation. *Hematologica*. 2010;95:189–198.
544. Bruce LJ, et al. The monovalent cation leak in overhydrated stomatocytic red blood cells results from amino acid substitutions in the Rh-associated glycoprotein. *Blood*. 2009;113:1350–1357.
545. Albuisson J, et al. Dehydrated hereditary stomatocytosis linked to gain-of-function mutations in mechanically activated PIEZO1 ion channels. *Nat Commun*. 2013;4:1884.

546. Rapetti-Mauss R, et al. Red blood cell Gardos channel (KCNN4): the essential determinant of erythrocyte dehydration in hereditary xerocytosis. *Haematologica*. 2017;102:e415–e418.
547. Arrambide K, Toto RD. Tumor lysis syndrome. *Semin Nephrol*. 1993;13:273–280.
548. Parisi A, et al. Complex ventricular arrhythmia induced by overuse of potassium supplementation in a young male football player. Case report. *J Sports Med Phys Fitness*. 2002;42:214–216.
549. Sopko JA, Freeman RM. Salt substitutes as a source of potassium. *JAMA*. 1977;238:608–610.
550. Mueller BA, Scott MK, Sowinski KM, et al. Noni juice (*Morinda citrifolia*): hidden potential for hyperkalemia? *Am J Kidney Dis*. 2000;35:310–312.
551. Nagasaki A, Takamine W, Takasu N. Severe hyperkalemia associated with “alternative” nutritional cancer therapy. *Clin Nutr*. 2005;24:864–865.
552. Abu-Hamdan DK, Sondheimer JH, Mahajan SK. Cautopyreliophagia. Cause of life-threatening hyperkalemia in a patient undergoing hemodialysis. *Am J Med*. 1985;79:517–519.
553. Thiele A, Rehman HU. Hyperkalemia caused by penicillin. *Am J Med*. 2008;121:e1–e2.
554. Smith HM, Farrow SJ, Ackerman JD, et al. Cardiac arrests associated with hyperkalemia during red blood cell transfusion: a case series. *Anesth Analg*. 2008;106:1062–1069. table of contents.
555. Baz EM, Kanazi GE, Mahfouz RA, et al. An unusual case of hyperkalaemia-induced cardiac arrest in a paediatric patient during transfusion of a ‘fresh’ 6-day-old blood unit. *Transfus Med*. 2002;12:383–386.
556. Perkins RM, Aboudara MC, Abbott KC, et al. Resuscitative hyperkalemia in noncrush trauma: a prospective, observational study. *Clin J Am Soc Nephrol*. 2007;2:313–319.
557. Moreno M, Murphy C, Goldsmith C. Increase in serum potassium resulting from the administration of hypertonic mannitol and other solutions. *J Lab Clin Med*. 1969;73:291–298.
558. Makoff DL, da Silva JA, Rosenbaum BJ, et al. Hypertonic expansion: acid-base and electrolyte changes. *Am J Physiol*. 1970;218:1201–1207.
559. Flynn BC. Hyperkalemic cardiac arrest with hypertonic mannitol infusion: the strong ion difference revisited. *Anesth Analg*. 2007;104:225–226.
560. Hirota K, et al. Two cases of hyperkalemia after administration of hypertonic mannitol during craniotomy. *J Anesth*. 2005;19:75–77.
561. Hassan ZU, Kruer JJ, Fuhrman TM. Electrolyte changes during craniotomy caused by administration of hypertonic mannitol. *J Clin Anesth*. 2007;19:307–309.
562. Goldfarb S, Strunk B, Singer I, et al. Paradoxical glucose-induced hyperkalemia. Combined aldosterone-insulin deficiency. *Am J Med*. 1975;59:744–750.
563. Magnus Nzerue C, Jackson E. Intractable life-threatening hyperkalaemia in a diabetic patient. *Nephrol Dial Transplant*. 2000;15:113–114.
564. Sirken G, Raja R, Garces J, et al. Contrast-induced translocational hyponatremia and hyperkalemia in advanced kidney disease. *Am J Kidney Dis*. 2004;43:e31–e35.
565. Perazella MA, Biswas P. Acute hyperkalemia associated with intravenous epsilon-aminocaproic acid therapy. *Am J Kidney Dis*. 1999;33:782–785.
566. Nzerue CM, Falana B. Refractory hyperkalaemia associated with use of epsilon-aminocaproic acid during coronary bypass in a dialysis patient. *Nephrol Dial Transplant*. 2002;17:1150–1151.
567. Banerjee A, Stoica C, Walia A. Acute hyperkalemia as a complication of intravenous therapy with epsilon-aminocaproic acid. *J Clin Anesth*. 2011;23:565–568.
568. Pfof CH, et al. Hyperkalemia in patients treated with endoradiotherapy combined with amino acid infusion is associated with severe metabolic acidosis. *EJNMMI Res*. 2018;8:17.
569. Clark BA, Shannon C, Brown RS, et al. Extrarenal potassium homeostasis with maximal exercise in end-stage renal disease. *J Am Soc Nephrol*. 1996;7:1223–1227.
570. Sangkabutra T, et al. Impaired K⁺ regulation contributes to exercise limitation in end-stage renal failure. *Kidney Int*. 2003;63:283–290.
571. Allon M. Hyperkalemia in end-stage renal disease: mechanisms and management. *J Am Soc Nephrol*. 1995;6:1134–1142.
572. Fulop M. Hyperkalemia in diabetic ketoacidosis. *Am J Med Sci*. 1990;299:164–169.
573. Van Gaal LF, De Leeuw IH, Bekaert JL. Diabetic ketoacidosis-induced hyperkalemia. Prevalence and possible origin. *Intensive Care Med*. 1986;12:416–418.
574. Milionis HJ, Dimos G, Elisaf MS. Severe hyperkalaemia in association with diabetic ketoacidosis in a patient presenting with severe generalized muscle weakness. *Nephrol Dial Transplant*. 2003;18:198–200.
575. Chi HT, Hung DZ, Hu WH, et al. Prognostic implications of hyperkalemia in toad toxin intoxication. *Hum Exp Toxicol*. 1998;17:343–346.
576. Gowda RM, Cohen RA, Khan IA. Toad venom poisoning: resemblance to digoxin toxicity and therapeutic implications. *Heart*. 2003;89:e14.
577. Baltazar RF, Mower MM, Reider R, et al. Acute fluoride poisoning leading to fatal hyperkalemia. *Chest*. 1980;78:660–663.
578. Martyn JA, Richtsfeld M. Succinylcholine-induced hyperkalemia in acquired pathologic states: etiologic factors and molecular mechanisms. *Anesthesiology*. 2006;104:158–169.
579. Thapa S, Brull SJ. Succinylcholine-induced hyperkalemia in patients with renal failure: an old question revisited. *Anesth Analg*. 2000;91:237–241.
580. Singer M, Coluzzi F, O’Brien A, et al. Reversal of life-threatening, drug-related potassium-channel syndrome by glibenclamide. *Lancet*. 2005;365:1873–1875.
581. Wilson AJ, Jabr RI, Clapp LH. Calcium modulation of vascular smooth muscle ATP-sensitive K(+) channels: role of protein phosphatase-2B. *Circ Res*. 2000;87:1019–1025.
582. Lee HH, Hsu PC, Lin TH, et al. Nicorandil-induced hyperkalemia in a uremic patient. *Case Rep Med*. 2012;2012:812178.
583. Chen HT, Wu YT, Yang WC, et al. Nicorandil-induced hyperkalemia in a hemodialysis patient. *Am J Med Sci*. 2017;353:411–412.
584. McCauley J, et al. Labetalol-induced hyperkalemia in renal transplant recipients. *Am J Nephrol*. 2002;22:347–351.
585. Blumenfeld JD, et al. Beta-adrenergic receptor blockade as a therapeutic approach for suppressing the renin-angiotensin-aldosterone system in normotensive and hypertensive subjects. *Am J Hypertens*. 1999;12:451–459.
586. Yamamoto T, Fukuyama J, Hasegawa K, et al. Isolated corticotropin deficiency in adults. Report of 10 cases and review of literature. *Arch Intern Med*. 1992;152:1705–1712.
587. Fujieda K, Tajima T. Molecular basis of adrenal insufficiency. *Pediatr Res*. 2005;57:62R–69R.
588. Achermann JC, Meeks JJ, Jameson JL. Phenotypic spectrum of mutations in DAX-1 and SF-1. *Mol Cell Endocrinol*. 2001;185:17–25.
589. Fujieda K, et al. Spontaneous puberty in 46,XX subjects with congenital lipoid adrenal hyperplasia. Ovarian steroidogenesis is spared to some extent despite inactivating mutations in the steroidogenic acute regulatory protein (StAR) gene. *J Clin Invest*. 1997;99:1265–1271.
590. Bose HS, Sugawara T, Strauss JF 3rd, et al. The pathophysiology and genetics of congenital lipoid adrenal hyperplasia. International Congenital Lipoid Adrenal Hyperplasia Consortium. *N Engl J Med*. 1996;335:1870–1878.
591. Hiort O, et al. Homozygous disruption of P450 side-chain cleavage (CYP11A1) is associated with prematurity, complete 46,XY sex reversal, and severe adrenal failure. *J Clin Endocrinol Metab*. 2005;90:538–541.
592. Speiser PW. Congenital adrenal hyperplasia owing to 21-hydroxylase deficiency. *Endocrinol Metab Clin North Am*. 2001;30:31–59, vi.
593. Kayes-Wandover KM, et al. Congenital hyperreninemic hypoaldosteronism unlinked to the aldosterone synthase (CYP11B2) gene. *J Clin Endocrinol Metab*. 2001;86:5379–5382.
594. Kayes-Wandover KM, Schindler RE, Taylor HC, et al. Type 1 aldosterone synthase deficiency presenting in a middle-aged man. *J Clin Endocrinol Metab*. 2001;86:1008–1012.
595. Morimoto S, et al. Selective hypoaldosteronism with hyperreninemia in a diabetic patient. *J Clin Endocrinol Metab*. 1979;49:742–747.
596. Agmon D, Green J, Platau E, et al. Isolated adrenal mineralocorticoid deficiency due to amyloidosis associated with familial Mediterranean fever. *Am J Med Sci*. 1984;288:40–43.
597. Oster JR, Singer I, Fishman LM. Heparin-induced aldosterone suppression and hyperkalemia. *Am J Med*. 1995;98:575–586.
598. Koren-Michowitz M, et al. Early onset of hyperkalemia in patients treated with low molecular weight heparin: a prospective study. *Pharmacoevidenciol Drug Saf*. 2004;13:299–302.
599. Liu AA, Bui T, Nguyen HV, et al. Subcutaneous unfractionated heparin-induced hyperkalaemia in an elderly patient. *Australas J Ageing*. 2009;28:97.
600. Oelkers W. Adrenal insufficiency. *N Engl J Med*. 1996;335:1206–1212.
601. Fischer E, et al. Prolonged zona glomerulosa insufficiency causing hyperkalemia in primary aldosteronism after adrenalectomy. *J Clin Endocrinol Metab*. 2012;97:3965–3973.

602. Espinosa G, et al. Adrenal involvement in the antiphospholipid syndrome: clinical and immunologic characteristics of 86 patients. *Medicine (Baltimore)*. 2003;82:106–118.
603. Danby P, Harris KP, Williams B, et al. Adrenal dysfunction in patients with renal amyloid. *Q J Med*. 1990;76:915–922.
604. Mayo J, Collazos J, Martinez E, et al. Adrenal function in the human immunodeficiency virus-infected patient. *Arch Intern Med*. 2002;162:1095–1098.
605. Schambelan M, Stockigt JR, Biglieri EG. Isolated hypoaldosteronism in adults. A renin-deficiency syndrome. *N Engl J Med*. 1972;287:573–578.
606. Lush DJ, King JA, Fray JC. Pathophysiology of low renin syndromes: sites of renal renin secretory impairment and prorenin overexpression. *Kidney Int*. 1993;43:983–999.
607. Michelis MF. Hyperkalemia in the elderly. *Am J Kidney Dis*. 1990;16:296–299.
608. Lee FO, et al. Mechanisms of hyperkalemia in systemic lupus erythematosus. *Arch Intern Med*. 1988;148:397–401.
609. DuBose TD Jr, Caffisch CR. Effect of selective aldosterone deficiency on acidification in nephron segments of the rat inner medulla. *J Clin Invest*. 1988;82:1624–1632.
610. Sebastian A, Schambelan M, Lindenfeld S, et al. Amelioration of metabolic acidosis with fludrocortisone therapy in hyporeninemic hypoaldosteronism. *N Engl J Med*. 1977;297:576–583.
611. DuBose TD. Hyperkalemic hyperchloremic metabolic acidosis: pathophysiological insights. *Kidney Int*. 1997;51:591–602.
612. Sebastian A, Schambelan M. Amelioration of type 4 renal tubular acidosis in chronic renal failure with furosemide. *Kidney Int*. 1977;12:534.
613. Chan R, et al. Renin-aldosterone system can respond to furosemide in patients with hyperkalemic hyporeninism. *J Lab Clin Med*. 1998;132:229–235.
614. Oh MS, et al. A mechanism for hyporeninemic hypoaldosteronism in chronic renal disease. *Metabolism*. 1974;23:1157–1166.
615. Klemm SA, Gordon RD, Tunny TJ, et al. Biochemical correction in the syndrome of hypertension and hyperkalemia by severe dietary salt restriction suggests renin-aldosterone suppression critical in pathophysiology. *Clin Exp Pharmacol Physiol*. 1990;17:191–195.
616. Tuchelt H, et al. Role of atrial natriuretic factor in changes in the responsiveness of aldosterone to angiotensin II secondary to sodium loading and depletion in man. *Clin Sci*. 1990;79:57–65.
617. DeFronzo RA, et al. Impaired renal tubular potassium secretion in systemic lupus erythematosus. *Ann Intern Med*. 1977;86:268–271.
618. DeFronzo RA, Taufield PA, Black H, et al. Impaired renal tubular potassium secretion in sickle cell disease. *Ann Intern Med*. 1979;90:310–316.
619. Battle D, Itsarayoungyuen K, Arruda JA, et al. Hyperkalemic hyperchloremic metabolic acidosis in sickle cell hemoglobinopathies. *Am J Med*. 1982;72:188–192.
620. Geller DS, et al. Autosomal dominant pseudohypoaldosteronism type I: mechanisms, evidence for neonatal lethality, and phenotypic expression in adults. *J Am Soc Nephrol*. 2006;17:1429–1436.
621. Akcay A, Yavuz T, Semiz S, et al. Pseudohypoaldosteronism type I and respiratory distress syndrome. *J Pediatr Endocrinol Metab*. 2002;15:1557–1561.
622. Bonny O, et al. Functional expression of a pseudohypoaldosteronism type I mutated epithelial Na⁺ channel lacking the pore-forming region of its alpha subunit. *J Clin Invest*. 1999;104:967–974.
623. Mayan H, et al. Pseudohypoaldosteronism type II: marked sensitivity to thiazides, hypercalciuria, normomagnesemia, and low bone mineral density. *J Clin Endocrinol Metab*. 2002;87:3248–3254.
624. Wilson FH, et al. Human hypertension caused by mutations in WNK kinases. *Science*. 2001;293:1107–1112.
625. Ko B, Hoover RS. Molecular physiology of the thiazide-sensitive sodium-chloride cotransporter. *Curr Opin Nephrol Hypertens*. 2009;18:421–427.
626. Sasaki E, et al. KLHL3 knockout mice reveal the physiological role of KLHL3 and the pathophysiology of pseudohypoaldosteronism type II caused by mutant KLHL3. *Mol Cell Biol*. 2017;37.
627. Ferdaus MZ, et al. Mutant Cullin 3 causes familial hyperkalemic hypertension via dominant effects. *JCI Insight*. 2017;2.
628. Ling BN, Webster SK, Eaton DC. Eicosanoids modulate apical Ca(2+)-dependent K⁺ channels in cultured rabbit principal cells. *Am J Physiol*. 1992;263:F116–F126.
629. Mactier RA, Khanna R. Hyperkalemia induced by indomethacin and naproxen and reversed by fludrocortisone. *South Med J*. 1988;81:799–801.
630. Tan SY, Shapiro R, Franco R, et al. Indomethacin-induced prostaglandin inhibition with hyperkalemia. A reversible cause of hyporeninemic hypoaldosteronism. *Ann Intern Med*. 1979;90:783–785.
631. Tan SY, Mulrow PJ. Inhibition of the renin-aldosterone response to furosemide by indomethacin. *J Clin Endocrinol Metab*. 1977;45:174–176.
632. Harris RC Jr. Cyclooxygenase-2 inhibition and renal physiology. *Am J Cardiol*. 2002;89:10D–17D.
633. Swan SK, et al. Effect of cyclooxygenase-2 inhibition on renal function in elderly persons receiving a low-salt diet. A randomized, controlled trial. *Ann Intern Med*. 2000;133:1–9.
634. Preston RA, Afshartous D, Alonso AB. Effects of selective vs. non-selective cyclooxygenase inhibition on dynamic renal potassium excretion: a randomized trial. *Clin Pharmacol Ther*. 2008;84:208–211.
635. Braden GL, O'Shea MH, Mulhern JG, et al. Acute renal failure and hyperkalemia associated with cyclooxygenase-2 inhibitors. *Nephrol Dial Transplant*. 2004;19:1149–1153.
636. Lam Q, Schneider HG. Hyperkalemia with cyclooxygenase-2 inhibition and hypoaldosteronism. *Intern Med J*. 2005;35:572–573.
637. Perazella MA. Drug-induced hyperkalemia: old culprits and new offenders. *Am J Med*. 2000;109:307–314.
638. Oishi M, et al. A case of hyperkalemic distal renal tubular acidosis secondary to tacrolimus in living donor liver transplantation. *Transplant Proc*. 2000;32:2225–2226.
639. Higgins R, et al. Hyponatraemia and hyperkalemia are more frequent in renal transplant recipients treated with tacrolimus than with cyclosporin. Further evidence for differences between cyclosporin and tacrolimus nephrotoxicities. *Nephrol Dial Transplant*. 2004;19:444–450.
640. Bantle JP, Nath KA, Sutherland DE, et al. Effects of cyclosporine on the renin-angiotensin-aldosterone system and potassium excretion in renal transplant recipients. *Arch Intern Med*. 1985;145:505–508.
641. Hocherl K, Dreher F, Vitzthum H, et al. Cyclosporine A suppresses cyclooxygenase-2 expression in the rat kidney. *J Am Soc Nephrol*. 2002;13:2427–2436.
642. Ling BN, Eaton DC. Cyclosporin A inhibits apical secretory K⁺ channels in rabbit cortical collecting tubule principal cells. *Kidney Int*. 1993;44:974–984.
643. Pei Y, Richardson R, Greenwood C, et al. Extrarenal effect of cyclosporine A on potassium homeostasis in renal transplant recipients. *Am J Kidney Dis*. 1993;22:314–319.
644. Choi MJ, et al. Brief report: trimethoprim-induced hyperkalemia in a patient with AIDS. *N Engl J Med*. 1993;328:703–706.
645. Kleyman TR, Roberts C, Ling BN. A mechanism for pentamidine-induced hyperkalemia: inhibition of distal nephron sodium transport. *Ann Intern Med*. 1995;122:103–106.
646. Velazquez H, Perazella MA, Wright FS, et al. Renal mechanism of trimethoprim-induced hyperkalemia. *Ann Intern Med*. 1993;119:296–301.
647. Muto S, Tsuruoka S, Miyata Y, et al. Effect of trimethoprim-sulfamethoxazole on Na and K⁺ transport properties in the rabbit cortical collecting duct perfused in vitro. *Nephron Physiol*. 2006;102:p51–p60.
648. Alappan R, Perazella MA, Buller GK. Hyperkalemia in hospitalized patients treated with trimethoprim-sulfamethoxazole. *Ann Intern Med*. 1996;124:316–320.
649. Elisaf M, Terrovitou C, Tomos P, et al. Severe hyperkalemia after cotrimoxazole administration in a patient with hyporeninemic hypoaldosteronism. *Nephrol Dial Transplant*. 1997;12:1254–1255.
650. Antoniou T, et al. Trimethoprim-sulfamethoxazole-induced hyperkalemia in patients receiving inhibitors of the renin-angiotensin system: a population-based study. *Arch Intern Med*. 2010;170:1045–1049.
651. Antoniou T, et al. Trimethoprim-sulfamethoxazole and risk of sudden death among patients taking spironolactone. *CMAJ*. 2015;187:E138–E143.
652. Weir MR, Rolfe M. Potassium homeostasis and renin-angiotensin-aldosterone system inhibitors. *Clin J Am Soc Nephrol*. 2010;5:531–548.
653. Genazzani AR, Mannella P, Simoncini T. Drospirenone and its antialdosterone properties. *Climacteric*. 2007;10(suppl 1):11–18.
654. Preston RA, et al. Mechanisms of impaired potassium handling with dual renin-angiotensin-aldosterone blockade in chronic kidney disease. *Hypertension*. 2009;53:754–760.
655. Van Buren PN, Adams-Huet B, Nguyen M, et al. Potassium handling with dual renin-angiotensin system inhibition in diabetic nephropathy. *Clin J Am Soc Nephrol*. 2014;9:295–301.

656. Venzin RM, Cohen CD, Maggiorini M, et al. Aliskiren-associated acute renal failure with hyperkalemia. *Clin Nephrol*. 2009;71:326–328.
657. Mehdi UF, Adams-Huet B, Raskin P, et al. Addition of angiotensin receptor blockade or mineralocorticoid antagonism to maximal angiotensin-converting enzyme inhibition in diabetic nephropathy. *J Am Soc Nephrol*. 2009;20:2641–2650.
658. Pitt B, et al. Eplerenone, a selective aldosterone blocker, in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med*. 2003;348:1309–1321.
659. Park IW, et al. Onset time of hyperkalaemia after angiotensin receptor blocker initiation: when should we start serum potassium monitoring? *J Clin Pharm Ther*. 2014;39:61–68.
660. Johnson ES, et al. Predicting the risk of hyperkalemia in patients with chronic kidney disease starting lisinopril. *Pharmacoepidemiol Drug Saf*. 2010;19:266–272.
661. Pitt B, Bakris G, Ruilope LM, et al. Serum potassium and clinical outcomes in the Eplerenone Post-Acute Myocardial Infarction Heart Failure Efficacy and Survival Study (EPHESUS). *Circulation*. 2008;118:1643–1650.
662. Gross P, Pistrosch F. Hyperkalaemia: again. *Nephrol Dial Transplant*. 2004;19:2163–2166.
663. Pitt B, et al. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. Randomized Aldactone Evaluation Study Investigators. *N Engl J Med*. 1999;341:709–717.
664. Juurlink DN, et al. Rates of hyperkalemia after publication of the Randomized Aldactone Evaluation Study. *N Engl J Med*. 2004;351:543–551.
665. Goldfarb DS. Hyperkalemia after the publication of RALES. *N Engl J Med*. 2004;351:2448–2450, author reply 2448–2450.
666. Wei L, Struthers AD, Fahey T, et al. Spironolactone use and renal toxicity: population based longitudinal analysis. *BMJ*. 2010;340:c1768.
667. Palmer BF. Managing hyperkalemia caused by inhibitors of the renin-angiotensin-aldosterone system. *N Engl J Med*. 2004;351:585–592.
668. Levinsky NG. Management of emergencies. VI. Hyperkalemia. *N Engl J Med*. 1966;274:1076–1077.
669. Allon M. Treatment and prevention of hyperkalemia in end-stage renal disease. *Kidney Int*. 1993;43:1197–1209.
670. Mount DB. *Treatment and Prevention of Hyperkalemia*. 18.2 ed. (ed. Basow DS). Waltham, MA: UpToDate; 2010.
671. Kim HJ, Han SW. Therapeutic approach to hyperkalemia. *Nephron*. 2002;92(suppl 1):33–40.
672. Evans KJ, Greenberg A. Hyperkalemia: a review. *J Intensive Care Med*. 2005;20:272–290.
673. Winkler AW, Hoff HE, Smith PK. Factors affecting the toxicity of potassium. *Am J Physiol*. 1939;127:430–436.
674. Pergola PE, DeFronzo R. Clinical disorders of hyperkalemia. In: Seldin DW, Giebisch G, eds. *The Kidney: Physiology and Pathophysiology*. Vol. 2. Lippincott Williams & Wilkins; 2000:1647–1700.
675. Davey M, Caldicott D. Calcium salts in management of hyperkalaemia. *Emerg Med J*. 2002;19:92–93.
676. Bower JO, Mengle HAK. The additive effects of calcium and digitalis. *JAMA*. 1936;106:1151–1153.
677. Long B, Warix JR, Koyfman A. Controversies in management of hyperkalemia. *J Emerg Med*. 2018;55:192–205.
678. White RD, Goldsmith RS, Rodriguez R, et al. Plasma ionic calcium levels following injection of chloride, gluconate, and gluceptate salts of calcium. *J Thorac Cardiovasc Surg*. 1976;71:609–613.
679. Alvestrand A, Wahren J, Smith D, et al. Insulin-mediated potassium uptake is normal in uremic and healthy subjects. *Am J Physiol*. 1984;246:E174–E180.
680. Allon M, Copkney C. Albuterol and insulin for treatment of hyperkalemia in hemodialysis patients. *Kidney Int*. 1990;38:869–872.
681. Blumberg A, Weidmann P, Shaw S, et al. Effect of various therapeutic approaches on plasma potassium and major regulating factors in terminal renal failure. *Am J Med*. 1988;85:507–512.
682. De Wolf A, Frenette L, Kang Y, et al. Insulin decreases the serum potassium concentration during the anhepatic stage of liver transplantation. *Anesthesiology*. 1993;78:677–682.
683. Minaker KL, Rowe JW. Potassium homeostasis during hyperinsulinemia: effect of insulin level, beta-blockade, and age. *Am J Physiol*. 1982;242:E373–E377.
684. Perazella MA. Approach to hyperkalemic end-stage renal disease patients in the emergency department. *Conn Med*. 1999;63:131–136.
685. Emmett M. Non-dialytic treatment of acute hyperkalemia in the dialysis patient. *Semin Dial*. 2000;13:279–280.
686. Lens XM, Montoliu J, Cases A, et al. Treatment of hyperkalaemia in renal failure: salbutamol v. insulin. *Nephrol Dial Transplant*. 1989;4:228–232.
687. Sterns RH, Grieff M, Bernstein PL. Treatment of hyperkalemia: something old, something new. *Kidney Int*. 2016;89:546–554.
688. Harel Z, Kamel KS. Optimal dose and method of administration of intravenous insulin in the management of emergency hyperkalemia: a systematic review. *PLoS ONE*. 2016;11:e0154963.
689. Liou HH, et al. Hypokalemic effects of intravenous infusion or nebulization of salbutamol in patients with chronic renal failure: comparative study. *Am J Kidney Dis*. 1994;23:266–271.
690. Montoliu J, Lens XM, Revert L. Potassium-lowering effect of albuterol for hyperkalemia in renal failure. *Arch Intern Med*. 1987;147:713–717.
691. Ostovar H, Jones J, Brown M. Best evidence topic report. Nebulised levalbuterol or albuterol for lowering serum potassium. *Emerg Med J*. 2005;22:366–367.
692. Mandelberg A, et al. Salbutamol metered-dose inhaler with spacer for hyperkalemia: how fast? How safe? *Chest*. 1999;115:617–622.
693. Sowinski KM, Cronin D, Mueller BA, et al. Subcutaneous terbutaline use in CKD to reduce potassium concentrations. *Am J Kidney Dis*. 2005;45:1040–1045.
694. Iqbal Z, Friedman EA. Preferred therapy of hyperkalemia in renal insufficiency: survey of nephrology training-program directors. *N Engl J Med*. 1989;320:60–61.
695. Fraley DS, Adler S. Correction of hyperkalemia by bicarbonate despite constant blood pH. *Kidney Int*. 1977;12:354–360.
696. Schwarz KC, Cohen BD, Lubash GD, et al. Severe acidosis and hyperpotasemia treated with sodium bicarbonate infusion. *Circulation*. 1959;19:215–220.
697. Blumberg A, Weidmann P, Ferrari P. Effect of prolonged bicarbonate administration on plasma potassium in terminal renal failure. *Kidney Int*. 1992;41:369–374.
698. Gutierrez R, Schlessinger F, Oster JR, et al. Effect of hypertonic versus isotonic sodium bicarbonate on plasma potassium concentration in patients with end-stage renal disease. *Miner Electrolyte Metab*. 1991;17:297–302.
699. Allon M, Shanklin N. Effect of bicarbonate administration on plasma potassium in dialysis patients: interactions with insulin and albuterol. *Am J Kidney Dis*. 1996;28:508–514.
700. Carvalhana V, Burry L, Lapinsky SE. Management of severe hyperkalemia without hemodialysis: case report and literature review. *J Crit Care*. 2006;21:316–321.
701. Suki WN. Use of diuretics in chronic renal failure. *Kidney Int Suppl*. 1997;59:S33–S35.
702. Wilcox CS. New insights into diuretic use in patients with chronic renal disease. *J Am Soc Nephrol*. 2002;13:798–805.
703. Sebastian A, Schambelan M, Sutton JM. Amelioration of hyperchloremic acidosis with furosemide therapy in patients with chronic renal insufficiency and type 4 renal tubular acidosis. *Am J Nephrol*. 1984;4:287–300.
704. DeFronzo RA, et al. Investigations into the mechanisms of hyperkalemia following renal transplantation. *Kidney Int*. 1977;11:357–365.
705. Reiser IW, Chou SY, Brown MI, et al. Reversal of trimethoprim-induced antidiuresis. *Kidney Int*. 1996;50:2063–2069.
706. Sherman DS, Kass CL, Fish DN. Fludrocortisone for the treatment of heparin-induced hyperkalemia. *Ann Pharmacother*. 2000;34:606–610.
707. Dreyling KW, Wanner C, Schollmeyer P. Control of hyperkalemia with fludrocortisone in a patient with systemic lupus erythematosus. *Clin Nephrol*. 1990;33:179–183.
708. Petersen KC, Silberman H, Berne TV. Hyperkalaemia after cyclosporin therapy. *Lancet*. 1984;1:1470.
709. Furuya R, Kumagai H, Sakao T, et al. Potassium-lowering effect of mineralocorticoid therapy in patients undergoing hemodialysis. *Nephron*. 2002;92:576–581.
710. Imbriano LJ, Durham JH, Maesaka JK. Treating interdialytic hyperkalemia with fludrocortisone. *Semin Dial*. 2003;16:5–7.
711. Kaiser MO, et al. A randomized controlled trial of fludrocortisone for the treatment of hyperkalemia in hemodialysis patients. *Am J Kidney Dis*. 2006;47:809–814.
712. Kim DM, Chung JH, Yoon SH, et al. Effect of fludrocortisone acetate on reducing serum potassium levels in patients with end-stage renal disease undergoing haemodialysis. *Nephrol Dial Transplant*. 2007;22:3273–3276.
713. Elkinton JR, Clark JK, Squires RD, et al. Treatment of potassium retention in uremia with cation exchange resin: preliminary report. *Am J Med Sci*. 1950;220:547–552.

714. Berlyne GM, Janabi K, Shaw AB. Dangers of resonium A in the treatment of hyperkalemia in renal failure. *Lancet*. 1966;1:167–169.
715. Scherr L, Ogden DA, Mead AW, et al. Management of hyperkalemia with cation-exchange resin. *N Engl J Med*. 1961;264:115–119.
716. Evans BM, Jones NC, Milne MD, et al. Ion-exchange resins in the treatment of anuria. *Lancet*. 1953;265:791–795.
717. Monzu B, Caramelo C, Traba ML, et al. Effect of potassium-chelating resins on phosphorus absorption. *Nephron*. 1994;68:148.
718. Papadimitriou M, Gingell JC, Chisholm GD. Hypercalcaemia from calcium ion-exchange resin in patients on regular haemodialysis. *Lancet*. 1968;2:948–950.
719. Agarwal R, Afzalpurkar R, Fordtran JS. Pathophysiology of potassium absorption and secretion by the human intestine. *Gastroenterology*. 1994;107:548–571.
720. Flinn RB, Merrill JP, Welzant WR. Treatment of the oliguric patient with a new sodium-exchange resin and sorbitol. *N Engl J Med*. 1961;264:111–115.
721. Sterns RH, Rojas M, Bernstein P, et al. Ion-exchange resins for the treatment of hyperkalemia: are they safe and effective? *J Am Soc Nephrol*. 2010.
722. Emmett M, et al. Effect of three laxatives and a cation exchange resin on fecal sodium and potassium excretion. *Gastroenterology*. 1995;108:752–760.
723. Gruy-Kapral C, et al. Effect of single dose resin-cathartic therapy on serum potassium concentration in patients with end-stage renal disease. *J Am Soc Nephrol*. 1998;9:1924–1930.
724. Gales MA, Gales BJ, Dyer ME, et al. Rectally administered sodium polystyrene sulfonate. *Am J Health Syst Pharm*. 1995;52:2813–2815.
725. Lillemoe KD, et al. Intestinal necrosis due to sodium polystyrene (Kayexalate) in sorbitol enemas: clinical and experimental support for the hypothesis. *Surgery*. 1987;101:267–272.
726. Gerstman BB, Kirkman R, Platt R. Intestinal necrosis associated with postoperative orally administered sodium polystyrene sulfonate in sorbitol. *Am J Kidney Dis*. 1992;20:159–161.
727. McGowan CE, Saha S, Chu G, et al. Intestinal necrosis due to sodium polystyrene sulfonate (Kayexalate) in sorbitol. *South Med J*. 2009;102:493–497.
728. Rashid A, Hamilton SR. Necrosis of the gastrointestinal tract in uremic patients as a result of sodium polystyrene sulfonate (Kayexalate) in sorbitol: an underrecognized condition. *Am J Surg Pathol*. 1997;21:60–69.
729. Ayoub I, et al. Colon necrosis due to sodium polystyrene sulfonate with and without sorbitol: an experimental study in rats. *PLoS ONE*. 2015;10:e0137636.
730. Cheng ES, Stringer KM, Pegg SP. Colonic necrosis and perforation following oral sodium polystyrene sulfonate (Resonium A/Kayexalate) in a burn patient. *Burns*. 2002;28:189–190.
731. Goutorbe P, et al. Intestinal necrosis associated with orally administered calcium polystyrene sulfonate without sorbitol. *Ann Pharmacother*. 2011;45:e13.
732. Kamel KS, Wei C. Controversial issues in the treatment of hyperkalemia. *Nephrol Dial Transplant*. 2003;18:2215–2218.
733. Abraham SC, Bhagavan BS, Lee LA, et al. Upper gastrointestinal tract injury in patients receiving kayexalate (sodium polystyrene sulfonate) in sorbitol: clinical, endoscopic, and histopathologic findings. *Am J Surg Pathol*. 2001;25:637–644.
734. Ng YY, et al. Reduction of serum calcium by sodium sulfonated polystyrene resin. *J Formos Med Assoc*. 1990;89:399–402.
735. Linakis JG, et al. Sodium polystyrene sulfonate treatment for lithium toxicity: effects on serum potassium concentrations. *Acad Emerg Med*. 1996;3:333–337.
736. Bakris GL, et al. Effect of patiomer on serum potassium level in patients with hyperkalemia and diabetic kidney disease: the AMETHYST-DN randomized clinical trial. *JAMA*. 2015;314:151–161.
737. Lesko LJ, et al. Evaluation of the Potential for Drug Interactions With Patiomer in Healthy Volunteers. *J Cardiovasc Pharmacol Ther*. 2017;22:434–446.
738. Stavros F, Yang A, Leon A, et al. Characterization of structure and function of ZS-9, a K⁺ selective ion trap. *PLoS ONE*. 2014;9:e114686.
739. Kosiborod M, et al. Effect of sodium zirconium cyclosilicate on potassium lowering for 28 days among outpatients with hyperkalemia: the HARMONIZE randomized clinical trial. *JAMA*. 2014;312:2223–2233.
740. Kosiborod M, Peacock WF, Packham DK. Sodium zirconium cyclosilicate for urgent therapy of severe hyperkalemia. *N Engl J Med*. 2015;372:1577–1578.
741. Amaya F, Fukui M, Tsuruta H, et al. Simulation of potassium extraction by continuous haemodiafiltration. *Anaesth Intensive Care*. 2002;30:198–201.
742. Jackson MA, Lodwick R, Hutchinson SG. Hyperkalaemic cardiac arrest successfully treated with peritoneal dialysis. *BMJ*. 1996;312:1289–1290.
743. Allon M, Shanklin N. Effect of albuterol treatment on subsequent dialytic potassium removal. *Am J Kidney Dis*. 1995;26:607–613.
744. Blumberg A, Roser HW, Zehnder C, et al. Plasma potassium in patients with terminal renal failure during and after haemodialysis; relationship with dialytic potassium removal and total body potassium. *Nephrol Dial Transplant*. 1997;12:1629–1634.
745. Allon M. Medical and dialytic management of hyperkalemia in hemodialysis patients. *Int J Artif Organs*. 1996;19:697–699.
746. Sherman RA, Hwang ER, Bernholz AS, et al. Variability in potassium removal by hemodialysis. *Am J Nephrol*. 1986;6:284–288.
747. Ward RA, Wathen RL, Williams TE, et al. Hemodialysate composition and intradialytic metabolic, acid-base and potassium changes. *Kidney Int*. 1987;32:129–135.
748. Zehnder C, Gutzwiller JP, Huber A, et al. Low-potassium and glucose-free dialysis maintains urea but enhances potassium removal. *Nephrol Dial Transplant*. 2001;16:78–84.
749. Heguilen RM, et al. The faster potassium-lowering effect of high dialysate bicarbonate concentrations in chronic haemodialysis patients. *Nephrol Dial Transplant*. 2005;20:591–597.
750. Dolson GM, Ellis KJ, Bernardo MV, et al. Acute decreases in serum potassium augment blood pressure. *Am J Kidney Dis*. 1995;26:321–326.
751. Solandt DY. The effect of potassium on the excitability and resting metabolism of frog's muscle. *J Physiol (Lond)*. 1936;86:162–170.
752. Baron AD. Hemodynamic actions of insulin. *Am J Physiol*. 1994;267:E187–E202.
753. Hou S, McElroy PA, Nootens J, et al. Safety and efficacy of low-potassium dialysate. *Am J Kidney Dis*. 1989;13:137–143.
754. Hung AM, Hakim RM. Dialysate and serum potassium in hemodialysis. *Am J Kidney Dis*. 2015;66:125–132.
755. Morrison G, Michelson EL, Brown S, et al. Mechanism and prevention of cardiac arrhythmias in chronic hemodialysis patients. *Kidney Int*. 1980;17:811–819.
756. Karnik JA, et al. Cardiac arrest and sudden death in dialysis units. *Kidney Int*. 2001;60:350–357.
757. Kimura K, Tabei K, Asano Y, et al. Cardiac arrhythmias in hemodialysis patients. A study of incidence and contributory factors. *Nephron*. 1989;53:201–207.
758. Ramirez G, Brueggemeyer CD, Newton JL. Cardiac arrhythmias on hemodialysis in chronic renal failure patients. *Nephron*. 1984;36:212–218.
759. Rombola G, Colussi G, De Ferrari ME, et al. Cardiac arrhythmias and electrolyte changes during haemodialysis. *Nephrol Dial Transplant*. 1992;7:318–322.
760. Buiten MS, et al. The dialysis procedure as a trigger for atrial fibrillation: new insights in the development of atrial fibrillation in dialysis patients. *Heart*. 2014;100:685–690.
761. Weber H, et al. Chronic hemodialysis: high risk patients for arrhythmias? *Nephron*. 1984;37:180–185.
762. Wizemann V, Kramer W, Funke T, et al. Dialysis-induced cardiac arrhythmias: fact or fiction? Importance of preexisting cardiac disease in the induction of arrhythmias during renal replacement therapy. *Nephron*. 1985;39:356–360.
763. Kyriakidis M, et al. Cardiac arrhythmias in chronic renal failure? Holter monitoring during dialysis and everyday activity at home. *Nephron*. 1984;38:26–29.
764. Redaelli B, et al. Effect of a new model of hemodialysis potassium removal on the control of ventricular arrhythmias. *Kidney Int*. 1996;50:609–617.
765. Santoro A, et al. Patients with complex arrhythmias during and after haemodialysis suffer from different regimens of potassium removal. *Nephrol Dial Transplant*. 2008;23:1415–1421.
766. Munoz RI, et al. Effect of acetate-free biofiltration with a potassium-profiled dialysate on the control of cardiac arrhythmias in patients at risk: a pilot study. *Hemodial Int*. 2008;12:108–113.
767. De Nicola L, et al. Effect of dialysate sodium concentration on interdialytic increase of potassium. *J Am Soc Nephrol*. 2000;11:2337–2343.
768. Driver BE, Klein LR, Chittineni C, et al. Is transcellular potassium shifting with insulin, albuterol, or sodium bicarbonate in emergency department patients with hyperkalemia associated with recurrent hyperkalemia after dialysis? *J Emerg Med*. 2018;55:15–22.e3.